

## Project Data Page:





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## **DESIGN CALCULATION**

*In Accordance with ASME Section VIII Division 1*

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A150NRL11EQ0022nd Ammonia Chiller.pvdb

Date of Analysis : Sep 5, 2026 2:57pm

**PV Elite 28, January 2026**

### Vessel Design Summary:

#### ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1750.000 x 3250.000 mm. ID  
 Vessel Design Length, Tangent to Tangent 7843.30 mm.  
 Specified Datum Line Distance 0.00 mm.  
 Shell Side Design Temperature 85 °C  
 Channel Side Design Temperature 85 °C  
 Shell Side Design Pressure 25.000 bars  
 Channel Side Design Pressure 191.000 bars  
 Shop Shell Side Test Pressure 35.099 bars  
 Shop Channel Side Test Pressure 250.212 bars  
 Wind Design Code ASCE/SEI 7-22  
 Earthquake Design Code ASCE/SEI 7-22

### Materials of Construction:

| Component Type | Material   | Class | Thickness | UNS #  | Material Treatment Options |
|----------------|------------|-------|-----------|--------|----------------------------|
| Shell          | SA-516 70  | ...   | ...       | K02700 | +N                         |
| Head           | SA-516 70  | ...   | ...       | K02700 | +Q+IT                      |
| Head           | SA-516 70  | ...   | ...       | K02700 | +N+IT                      |
| Cone           | SA-516 70  | ...   | ...       | K02700 | +Q+IT                      |
| Flange         | SA-350 LF2 | 1     | ...       | K03011 | +IT                        |
| Nozzle         | SA-350 LF2 | 1     | ...       | K03011 | +IT                        |
| Nozzle Flg     | SA-350 LF2 | 1     | ...       | K03011 |                            |
| Nozzle Flg     | SA-105     | ...   | ...       | K03504 | +IT                        |
| Tubes          | SA-334 6   | ...   | ...       | K03006 | +IT                        |
| Tubesheet      | SA-350 LF2 | 1     | ...       | K03011 | +IT                        |
| Flg Bolting    | SA-320 L7  | ...   | <= 65     | G41400 |                            |
| Hrz Bolting    | SA-193 B7  | ...   | <= 64     | G41400 |                            |
| Saddles        | SA-516 70  | ...   | ...       | K02700 | +Q+IT                      |

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized  
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

### Element Pressures and MAWP (bars & mm.):

| Element Description or Type | Design Pressure + Stat. head | Ext. Press. | Element M.A.W.P | Total Corrosion Allowance | Str. Flg. Gov. | In Creep Range |
|-----------------------------|------------------------------|-------------|-----------------|---------------------------|----------------|----------------|
| CH. COVER                   | 191.112                      | 0.00        | 200.000         | 1.5000                    | N/A            | No             |
| CH. COVER                   | 191.103                      | 0.00        | 192.000         | 1.5000                    | N/A            | No             |
| SH. FLG.                    | 25.103                       | 0.00        | 27.000          | 1.5000                    | N/A            | No             |
| SHELL-1                     | 25.103                       | 0.00        | 68.000          | 1.5000                    | N/A            | No             |
| SHELL-2                     | 25.103                       | 1.03        | 32.000          | 1.5000                    | N/A            | No             |
| K-SHELL                     | 25.103                       | 1.03        | 28.000          | 1.5000                    | N/A            | No             |
| SH. COVER                   | 25.103                       | 1.03        | 28.000          | 1.5000                    | No             | No             |

Liquid Level: 1750.00 mm. Dens.: 0.001 kg./cm<sup>3</sup> Sp. Gr.: 0.655

### Element Types and Properties:

| Element Type | "To" Elev mm. | Element Length mm. | Nominal Thickness mm. | Finished Thickness mm. | Reqd Thk Internal mm. | Reqd Thk External mm. | Long Eff | Circ Eff |
|--------------|---------------|--------------------|-----------------------|------------------------|-----------------------|-----------------------|----------|----------|
| Sphere       | 0.0           | 0.0                | 78.0                  | 66.0                   | 63.0                  | ...                   | 1.00     | 1.00     |
| Cylinder     | 50.0          | 50.0               | 135.0                 | 135.0                  | 133.9                 | ...                   | 1.00     | 1.00     |



|          |        |        |       |      |      |      |      |      |
|----------|--------|--------|-------|------|------|------|------|------|
| Body Flg | 659.5  | 165.0  | 100.0 | 95.0 | 87.1 | 87.1 | 1.00 | 1.00 |
| Cylinder | 1009.5 | 350.0  | 46.0  | 46.0 | 17.6 | ...  | 1.00 | 1.00 |
| Conical  | 3607.5 | 2598.0 | 46.0  | 46.0 | 36.0 | 11.6 | 1.00 | 1.00 |
| Cylinder | 7792.5 | 4185.0 | 35.0  | 35.0 | 31.4 | 13.6 | 1.00 | 1.00 |
| Ellipse  | 7843.3 | 50.8   | 41.0  | 35.0 | 31.1 | 10.1 | 1.00 | 1.00 |

### Local Stress Analysis Results:

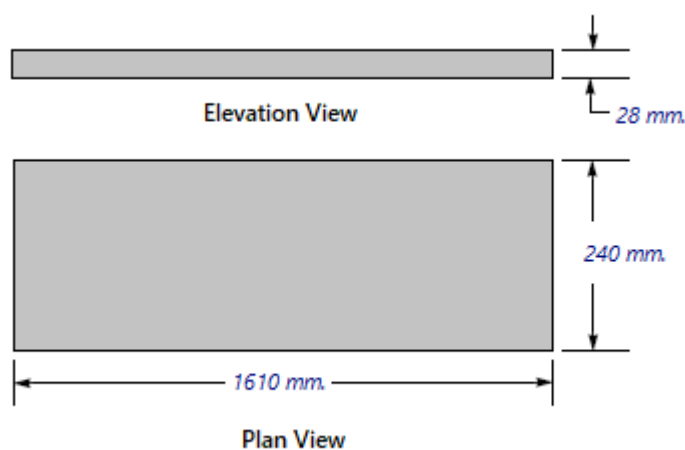
| Description     | Analysis Type | Max Stress Ratio | High Stress Location | Pass Fail |
|-----------------|---------------|------------------|----------------------|-----------|
| CH. IN NOZ. A1  | WRC-107/537   | 0.944            | n/a                  | Passed    |
| CH. OUT NOZ. B1 | WRC-107/537   | 0.944            | n/a                  | Passed    |
| SS IN NOZ. A2   | WRC-297       | 0.884            | Shell                | Passed    |
| SS OUT NOZ. B2  | WRC-297       | 0.884            | Shell                | Passed    |

### Foundation/Support Summary

#### Saddle Parameters:

|                                         |              |
|-----------------------------------------|--------------|
| Saddle Width                            | 200.000 mm.  |
| Saddle Bearing Angle                    | 120.000 deg. |
| Centerline Dimension                    | 1380.000 mm. |
| Wear Pad Width                          | 260.000 mm.  |
| Wear Pad Thickness                      | 16.000 mm.   |
| Wear Pad Bearing Angle                  | 132.000 deg. |
| Distance from Saddle to Tangent         | 150.000 mm.  |
| Baseplate Length                        | 1610.000 mm. |
| Baseplate Thickness                     | 28.000 mm.   |
| Baseplate Width                         | 240.000 mm.  |
| Number of Ribs (including outside ribs) | 4            |
| Rib Thickness                           | 16.000 mm.   |
| Web Thickness                           | 16.000 mm.   |
| Height of Center Web                    | 306.000 mm.  |
| Number of Bolts in Baseplate            | 3            |

### Baseplate Sketch



### Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

|                                        |             |
|----------------------------------------|-------------|
| Maximum Vertical Saddle Load           | 51367.00 Kg |
| Maximum Transverse Saddle Shear Load   | 3848.05 Kg  |
| Maximum Longitudinal Saddle Shear Load | 11544.14 Kg |

### Summary of Maximum Saddle Loads, Pressure Test Case:

|                                        |             |
|----------------------------------------|-------------|
| Maximum Vertical Saddle Load           | 44577.55 Kg |
| Maximum Transverse Saddle Shear Load   | 385.24 Kg   |
| Maximum Longitudinal Saddle Shear Load | 795.47 Kg   |



**Weights:**

|            |                                     |              |
|------------|-------------------------------------|--------------|
| Fabricated | - Bare w/o removable internals      | 77325.1 kg.  |
| Shop Test  | - Fabricated + water ( full )       | 130442.7 kg. |
| Shipping   | - Fab. + rem. intls.+ shipping app. | 77594.6 kg.  |
| Erected    | - Fab. + rem. intls.+ insul. (etc)  | 77594.6 kg.  |
| Empty      | - Fab. + intls. + details + wghts.  | 77594.6 kg.  |
| Operating  | - Empty + operating liquid (no ca)  | 121899.6 kg. |

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### Nozzle Calculation Summary (bars & mm.):

| Description     | MAWP | Ext | MAPNC | UG-45 | [tr]  | Weld Path | Areas or Stresses |
|-----------------|------|-----|-------|-------|-------|-----------|-------------------|
| CH. IN NOZ A1   | 195  | ... | ...   | OK    | 47.74 | OK        | Passed            |
| CH. OUT NOZ B1  | 196  | ... | ...   | OK    | 40.09 | OK        | Passed            |
| MANHOLE M1      | 196  | ... | ...   | OK    | 40.09 | OK        | Passed            |
| LIQ OUT NOZ B3  | 32   | OK  | ...   | OK    | 6.30  | OK        | Passed            |
| SS IN NOZ. A2   | 27   | OK  | ...   | OK    | 9.60  | OK        | Passed            |
| SS OUT NOZ. B2  | 27   | OK  | ...   | OK    | 9.83  | OK        | Passed            |
| VENT NOZ V1     | 28   | ... | ...   | OK    | 4.92  | OK        | No Calc[*]        |
| PR. IN P1       | 28   | ... | ...   | OK    | 4.92  | OK        | No Calc[*]        |
| PR. IN NOZ. P2  | 28   | ... | ...   | OK    | 4.92  | OK        | No Calc[*]        |
| SAFETY VALVE S1 | 28   | ... | ...   | OK    | 4.92  | OK        | No Calc[*]        |
| LEVEL GUAGE L1  | 28   | ... | ...   | OK    | 4.92  | OK        | No Calc[*]        |
| LEVEL GUAGE L2  | 28   | ... | ...   | OK    | 4.92  | OK        | No Calc[*]        |
| LEVEL GUAGE L3  | 28   | ... | ...   | OK    | 4.92  | OK        | No Calc[*]        |
| LEVEL GUAGE L4  | 28   | ... | ...   | OK    | 4.92  | OK        | No Calc[*]        |
| TEMP IN. T1     | 28   | ... | ...   | OK    | 4.92  | OK        | No Calc[*]        |

### Nozzle MAWP Summary:

Minimum MAWP Nozzles : 27 Nozzle : SS OUT NOZ. B2 [Shellside]  
 Minimum MAWP Nozzles : 195 Nozzle : CH. IN NOZ A1 [Tubeside]

[\*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

### Check the spatial relationship between the nozzles:

| From Node | Nozzle Description | X Coordinate mm. | Layout Angle deg | Dia. Limit mm. |
|-----------|--------------------|------------------|------------------|----------------|
| 10        | CH. IN NOZ A1      | 0.000            | 90.000           | 1225.200       |
| 10        | CH. OUT NOZ B1     | 0.000            | -30.050          | 1022.000       |
| 10        | MANHOLE M1         | 0.000            | -146.310         | 1022.000       |
| 50        | LIQ OUT NOZ B3     | 1259.500         | -90.000          | 205.650        |
| 60        | SS IN NOZ. A2      | 7357.500         | 90.000           | 491.750        |
| 60        | SS OUT NOZ. B2     | 5307.500         | 90.000           | 1225.200       |
| 60        | VENT NOZ V1        | 4007.500         | 90.000           | 127.325        |
| 60        | PR. IN P1          | 3757.500         | 90.000           | 127.325        |
| 60        | PR. IN NOZ. P2     | 6857.500         | 90.000           | 127.325        |
| 60        | SAFETY VALVE S1    | 6607.500         | 90.000           | 127.325        |
| 60        | LEVEL GUAGE L1     | 4407.500         | 30.481           | 127.325        |
| 60        | LEVEL GUAGE L2     | 4407.500         | -30.481          | 127.325        |
| 60        | LEVEL GUAGE L3     | 5107.500         | 30.481           | 127.325        |
| 60        | LEVEL GUAGE L4     | 5107.500         | -30.481          | 127.325        |
| 70        | TEMP IN. T1        | 0.000            | 270.000          | 127.325        |

The nozzle spacing is computed by the following:

$$= \sqrt{(l^2 + lc^2)} \text{ where}$$

l - Arc length along the inside vessel surface in the long. direction.  
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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#### Nozzle Schedule:

| Description     | Nominal or Actual Size | Schd or FVC Type | Flg Type | Nozzle O/Dia in | Wall Thk mm | Reinforcing Pad Diameter | Pad Thk mm | Cut Length mm | Flg Class |
|-----------------|------------------------|------------------|----------|-----------------|-------------|--------------------------|------------|---------------|-----------|
| VENT NOZ V1     | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 335.28        | 300       |
| PR. IN P1       | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 335.28        | 300       |
| PR. IN NOZ. P2  | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 335.28        | 300       |
| SAFETY VALVE S1 | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 335.28        | 300       |
| LEVEL GUAGE L1  | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 353.47        | 300       |
| LEVEL GUAGE L2  | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 353.47        | 300       |
| LEVEL GUAGE L3  | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 353.47        | 300       |
| LEVEL GUAGE L4  | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 353.47        | 300       |
| TEMP IN. T1     | 2.000 in               | 160              | LWN      | 2.375           | 8.738       | ...                      | ...        | 343.54        | 300       |
| LIQ OUT NOZ B3  | 3.000 in               | 160              | WNF      | 3.500           | 11.125      | ...                      | ...        | 347.80        | 300       |
| SS IN NOZ. A2   | 10.000 in              | 80               | WNF      | 10.750          | 15.088      | ...                      | ...        | 344.07        | 300       |
| CH. OUT NOZ B1  | 20.000 in Actual       | None             | RTJWn    | 23.937          | 50.000      | ...                      | ...        | 459.73        | ...       |
| MANHOLE M1      | 20.000 in Actual       | None             | RTJWn    | 23.937          | 50.000      | ...                      | ...        | 459.73        | 1500      |
| CH. IN NOZ A1   | 24.000 in Actual       | None             | WNF      | 28.688          | 59.540      | ...                      | ...        | 502.82        | ...       |
| SS OUT NOZ. B2  | 24.000 in Actual       | None             | WNF      | 28.688          | 59.540      | ...                      | ...        | 391.51        | 300       |

#### General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

#### Nozzle Material and Weld Fillet Leg Size Details (mm.):

| Description | Material   | Shl Grve Weld | Noz Shl/Pad Weld | Pad OD Weld | Pad Grve Weld | Inside Weld |
|-------------|------------|---------------|------------------|-------------|---------------|-------------|
| VENT NOZ V1 | SA-350 LF2 | 30.000        | 10.000           | ...         | ...           | ...         |
| PR. IN P1   | SA-350 LF2 | 30.000        | 10.000           | ...         | ...           | ...         |
| PR. IN NOZ. | SA-350 LF2 | 30.000        | 10.000           | ...         | ...           | ...         |
| SAFETY VALV | SA-350 LF2 | 30.000        | 10.000           | ...         | ...           | ...         |
| LEVEL GUAGE | SA-350 LF2 | 30.000        | 10.000           | ...         | ...           | ...         |
| LEVEL GUAGE | SA-350 LF2 | 30.000        | 10.000           | ...         | ...           | ...         |
| LEVEL GUAGE | SA-350 LF2 | 30.000        | 10.000           | ...         | ...           | ...         |
| LEVEL GUAGE | SA-350 LF2 | 30.000        | 10.000           | ...         | ...           | ...         |
| TEMP IN. T1 | SA-350 LF2 | 30.000        | 10.000           | ...         | ...           | ...         |
| LIQ OUT NOZ | SA-350 LF2 | 32.000        | 10.000           | ...         | ...           | ...         |
| SS IN NOZ.  | SA-350 LF2 | 30.000        | 10.000           | ...         | ...           | ...         |
| CH. OUT NOZ | SA-350 LF2 | 60.000        | 10.000           | ...         | ...           | ...         |
| MANHOLE M1  | SA-350 LF2 | 60.000        | 10.000           | ...         | ...           | ...         |
| CH. IN NOZ  | SA-350 LF2 | 60.000        | 10.000           | ...         | ...           | ...         |
| SS OUT NOZ. | SA-350 LF2 | 30.000        | 10.000           | ...         | ...           | ...         |

Note: The Outside projections below do not include the flange thickness.

#### Nozzle Miscellaneous Data:

| Description     | Elev/Distance From Datum mm. | Layout Angle deg | Proj Outside mm. | Proj Inside mm. | Installed in Component |
|-----------------|------------------------------|------------------|------------------|-----------------|------------------------|
| VENT NOZ V1     | 4007.500                     | 90.0             | 300.00           | 0.00            | K-SHELL                |
| PR. IN P1       | 3757.500                     | 90.0             | 300.00           | 0.00            | K-SHELL                |
| PR. IN NOZ. P2  | 6857.500                     | 90.0             | 300.00           | 0.00            | K-SHELL                |
| SAFETY VALVE S1 | 6607.500                     | 90.0             | 300.00           | 0.00            | K-SHELL                |
| LEVEL GUAGE L1  | 4407.500                     | 30.5             | 300.00           | 0.00            | K-SHELL                |
| LEVEL GUAGE L2  | 4407.500                     | -30.5            | 300.00           | 0.00            | K-SHELL                |
| LEVEL GUAGE L3  | 5107.500                     | 30.5             | 300.00           | 0.00            | K-SHELL                |



|                |          |        |        |      |           |
|----------------|----------|--------|--------|------|-----------|
| LEVEL GUAGE L4 | 5107.500 | -30.5  | 300.00 | 0.00 | K-SHELL   |
| TEMP IN. T1    | ...      | 270.0  | 300.00 | 0.00 | SH. COVER |
| LIQ OUT NOZ B3 | 1259.500 | -90.0  | 300.00 | 0.00 | SHELL-2   |
| SS IN NOZ. A2  | 7357.500 | 90.0   | 300.00 | 0.00 | K-SHELL   |
| CH. OUT NOZ B1 | ...      | -30.0  | 300.00 | 0.00 | CH. COVER |
| MANHOLE M1     | ...      | -146.3 | 300.00 | 0.00 | CH. COVER |
| CH. IN NOZ A1  | ...      | 90.0   | 300.00 | 0.00 | CH. COVER |
| SS OUT NOZ. B2 | 5307.500 | 90.0   | 300.00 | 0.00 | K-SHELL   |

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**Bill of Materials:**

| QTY  | DESCRIPTION                                                      | MATERIAL        |
|------|------------------------------------------------------------------|-----------------|
| 1    | SPHERICAL HEAD: 78.0mm. THK X 1750.0mm. ID X 0.0mm.              | SA-516 70       |
| 1    | WELD NECK FLANGE: 100.0mm. THK X 2020.0mm. OD                    | SA-350 LF2      |
| 1    | CYLINDER: 46.0mm. THK X 1750.0mm. ID X 350.0mm.                  | SA-516 70       |
| 1    | ELLIPTICAL HEAD: 2.0 X 1, 41.0mm. THK X 3250.0mm. ID X 50.8mm.   | SA-516 70       |
| 1    | INSULATION: 874mm X 70mm THK                                     | COLD INSULATION |
| 1    | INSULATION: 50mm X 70mm THK                                      | COLD INSULATION |
| 5    | BAFFLES: 1726.0mm DIA X 16.0mm THK                               | SA-516 70       |
| 14   | TIE RODS: 16mm DIA X 5400mm                                      | SA-193 B7M      |
| 1    | TUBESHEET: 2030.0mm X 440.0mm THK                                | SA-350 LF2      |
| 2332 | TUBES: 5400.0mm X 25.4mm DIA X 2.8mm THK                         | SA-334 6        |
| 1    | INSULATION: 235mm X 70mm THK                                     | COLD INSULATION |
| 1    | SADDLE: 200.0mm X 120 DEG                                        | SA-516 70       |
| 1    | INSULATION: 350mm X 70mm THK                                     | COLD INSULATION |
| 1    | INSULATION: 2598mm X 70mm THK                                    | COLD INSULATION |
| 2    | SADDLE: 350.0mm X 120 DEG                                        | SA-516 70       |
| 1    | INSULATION: 4185mm X 70mm THK                                    | COLD INSULATION |
| 1    | INSULATION: 863mm X 70mm THK                                     | COLD INSULATION |
| 1    | CLASS 1500 GR 1.1, 20.0" BLIND FLANGE(S)                         | SA-350 LF2      |
| 1    | CLASS 300 GR 1.1, 3.0" BLIND FLANGE(S)                           | SA-350 LF2      |
| 8    | CLASS 300 GR 1.1, 2.0" BLIND FLANGE(S)                           | SA-350 LF2      |
| 1    | CLASS 300 GR 1.1, 2.0" BLIND FLANGE(S)                           | SA-105          |
| 1    | GASKET: 1788.0mm. OD X 1779.0mm. ID                              | ...             |
| 56   | BODY FLANGE BOLTS: 36.0mm. DIA                                   | SA-320 L7       |
| 56   | NUTS FOR BODY FLANGE BOLTS: 36.0mm. DIA                          | ...             |
| 1    | CYLINDRICAL SEGMENT 135.0mm. THK X 3333.7mm. X 50.0mm.           | SA-516 70       |
| 1    | CYLINDRICAL SEGMENT 135.0mm. THK X 2164.0mm. X 50.0mm.           | SA-516 70       |
| 1    | ECC. CONICAL SEGMENT 46.0mm. THK X 6182.7mm. X 2598.0mm. X 3329. | SA-516 70       |
| 1    | ECC. CONICAL SEGMENT 46.0mm. THK X 4027.3mm. X 2598.0mm. X 2168. | SA-516 70       |
| 1    | CYLINDRICAL SEGMENT 35.0mm. THK X 3658.6mm. X 4185.0mm.          | SA-516 70       |
| 1    | CYLINDRICAL SEGMENT 35.0mm. THK X 3686.9mm. X 4185.0mm.          | SA-516 70       |
| 1    | CYLINDRICAL SEGMENT 35.0mm. THK X 2864.5mm. X 4185.0mm.          | SA-516 70       |
| 1    | NAMEPLATE                                                        | ...             |





### Minimum Design Metal Temperature Results Summary (°C):

| Description      | Notes         | Curve | Basic MDMT    | Reduced MDMT | UG-20(f) MDMT | Thickness ratio | Gov Thk mm. | E*   | PWHT reqd |
|------------------|---------------|-------|---------------|--------------|---------------|-----------------|-------------|------|-----------|
| SH. FLG. Bolting | [11] [21]     | !     | -46           | -46          |               | 0.397           | 46.000      | 1.00 | Yes       |
| SHELL-1          | [8]           | B     | 15            | -38          |               | 0.390           | 46.000      | 1.00 | Yes       |
| SHELL-2          | [8]           | !     | -35           | -44          |               | 0.836           | 46.000      | 1.00 | Yes       |
| K-SHELL          | [8]           | !     | -35           | -37          |               | 0.961           | 35.000      | 1.00 | Yes       |
| SH. COVER        | [10]          | !     | -35           | -38          |               | 0.951           | 35.000      | 1.00 | Yes       |
| SH. COVER        | [7]           | !     | -35           | -45          |               | 0.815           | 41.000      | 1.00 | Yes       |
| FIXED SADDLE     | [24]          | B     | -31           | -48          |               | 0.390           | 16.000      | 1.00 | No        |
| SLIDING SADDL    | [24]          | B     | -31           | -33          |               | 0.961           | 16.000      | 1.00 | No        |
| LIQ OUT NOZ B3   | [1]           | !     | -46           | -104         |               | 0.078           | 9.735       | 1.00 | No        |
| Nozzle Flg       | [5]           | !     | -46           | -79          |               |                 |             |      |           |
| SS IN NOZ. A2    | [1]           | !     | -46           | -46          |               | 0.889           | 35.000      | 1.00 | No        |
| Nozzle Flg       | [5]           | !     | -46           | -79          |               |                 |             |      |           |
| SS OUT NOZ. B2   | [1]           | !     | -46           | -46          |               | 0.889           | 35.000      | 1.00 | No        |
| Nozzle Flg       | [5]           | !     | -46           | -79          |               |                 |             |      |           |
| VENT NOZ V1      | [1]           | !     | -46           | -104         |               | 0.068           | 7.645       | 1.00 | No        |
| Nozzle Flg       | [5]           | !     | -46           | -46          |               |                 |             |      |           |
| PR. IN P1        | [1]           | !     | -46           | -104         |               | 0.068           | 7.645       | 1.00 | No        |
| Nozzle Flg       | [5]           | !     | -46           | -46          |               |                 |             |      |           |
| PR. IN NOZ. P2   | [1]           | !     | -46           | -104         |               | 0.068           | 7.645       | 1.00 | No        |
| Nozzle Flg       | [5]           | !     | -46           | -46          |               |                 |             |      |           |
| SAFETY VALVE S   | [1]           | !     | -46           | -104         |               | 0.068           | 7.645       | 1.00 | No        |
| Nozzle Flg       | [5]           | !     | -46           | -46          |               |                 |             |      |           |
| LEVEL GUAGE L1   | [1]           | !     | -46           | -104         |               | 0.068           | 7.645       | 1.00 | No        |
| Nozzle Flg       | [5]           | !     | -46           | -46          |               |                 |             |      |           |
| LEVEL GUAGE L2   | [1]           | !     | -46           | -104         |               | 0.068           | 7.645       | 1.00 | No        |
| Nozzle Flg       | [5]           | !     | -46           | -46          |               |                 |             |      |           |
| LEVEL GUAGE L3   | [1]           | !     | -46           | -104         |               | 0.068           | 7.645       | 1.00 | No        |
| Nozzle Flg       | [5]           | !     | -46           | -46          |               |                 |             |      |           |
| LEVEL GUAGE L4   | [1]           | !     | -46           | -104         |               | 0.068           | 7.645       | 1.00 | No        |
| Nozzle Flg       | [5]           | !     | -46           | -46          |               |                 |             |      |           |
| TEMP IN. T1      | [1]           | !     | -46           | -104         |               | 0.068           | 7.645       | 1.00 | No        |
| Nozzle Flg       | [5]           | A     | -18           | -48          |               |                 |             |      |           |
| Tubesheet: SS    | [13]          | !     | -46           | -46          |               | 0.912           | 110.000     | 1.00 | No        |
| Warmest MDMT:    |               |       | 15            | -33          |               |                 |             |      |           |
| CH. COVER        | [9]           | !     | -35           | -37          |               | 0.961           | 66.000      | 1.00 | Yes       |
| CH. IN NOZ A1    | [1]           | !     | -46           | -46          |               | 0.954           | 66.000      | 1.00 | No        |
| CH. OUT NOZ B1   | [1]           | !     | -46           | -46          |               | 0.954           | 66.000      | 1.00 | No        |
| MANHOLE M1       | [1]           | !     | -46           | -46          |               | 0.796           | 50.000      | 1.00 | Yes       |
| Nozzle Flg       | [5]           | !     | -46           | -60          |               |                 |             |      |           |
| Warmest MDMT:    |               |       | -35           | -37          |               |                 |             |      |           |
| Exchanger Side   | Computed MDMT |       | Required MDMT |              | Pass/Fail     |                 |             |      |           |
| Shell            | -33.0         |       | -35.0         |              | Fail          |                 |             |      |           |

### Notes:

- [ ! ] - This was an impact tested material.
- [ 1 ] - Governing Nozzle Weld.
- [ 4 ] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [ 5 ] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [ 6 ] - MDMT Calculations at the Shell/Head Joint.
- [ 7 ] - MDMT Calculations for the Straight Flange.
- [ 8 ] - Cylinder/Cone/Flange Junction MDMT.
- [ 9 ] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable



- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

**Notes:**

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

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Class From To: Check of Additional Element Data

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There were no geometry errors or warnings.

**PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.**

### Information Regarding Nozzle Loads

*Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.*



*One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.*



*The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.*



*The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.*

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**Units used in this Analysis (Kgfcmm2mm):**

| Name           | System Unit | Constant   | User Unit           |
|----------------|-------------|------------|---------------------|
| Length         | Feet        | 304.8000   | mm.                 |
| Force          | Pounds      | 0.4536     | Kgf                 |
| Mass           | Pounds      | 0.4536     | kg.                 |
| Area           | sq. inches  | 6.4516     | cm <sup>2</sup>     |
| Moment         | ft. lbs.    | 0.1383     | Kg-m.               |
| Stress         | lbs./sq.in. | 0.0069     | N./mm <sup>2</sup>  |
| Temperature    | Degrees F   | 0.5556     | °C                  |
| Pressure       | psig        | 0.0689     | bars                |
| Elast. Modulus | lbs./sq.in. | 6.8946     | KPa.                |
| Pipe Density   | lbs./cu.in. | 0.0277     | kg./cm <sup>3</sup> |
| Ins. Density   | lbs./cu.ft. | 0.1600E-04 | kg./cm <sup>3</sup> |
| Fluid Density  | lbs./cu.ft. | 0.1600E-04 | kg./cm <sup>3</sup> |
| Wind Speed     | miles/hr    | 1.6093     | Km/hr               |
| Tray Weight    | lbs./sq.ft. | 0.0005     | Kgs/cm <sup>2</sup> |
| Inertia        | in.4        | 41.6231    | cm4                 |
| G Load         | G's         | 1.0000     | g's                 |
| Wind Load      | lbs./sq.ft. | 4.8825     | Kgs/m <sup>2</sup>  |
| Elevation      | Feet        | 304.8000   | mm.                 |
| Volume         | in.3        | 16.3900    | cm <sup>3</sup>     |
| Diameter       | inches      | 25.4000    | mm.                 |
| Thickness      | inches      | 25.4000    | mm.                 |

**PV Elite Vessel Analysis Program: Input Data**

**Exchanger Design Pressures and Temperatures**

|                                       |       |      |
|---------------------------------------|-------|------|
| Shell Side Design Pressure            | 25    | bars |
| Channel Side Design Pressure          | 191   | bars |
| Shell Side Design Temperature         | 85.0  | °C   |
| Channel Side Design Temperature       | 85.0  | °C   |
| Radiography, Shell Side               | RT-1  |      |
| Radiography, Channel Side             | RT-1  |      |
| Service Type, Shell Side              | Sour  |      |
| Service Type, Channel Side            | Sour  |      |
| MDMT (CET), Shell Side                | -35.0 | °C   |
| MDMT (CET), Tube Side                 | -35.0 | °C   |
| User defined MAWP, Shell Side         | 0     | bars |
| User defined MAWP, Channel Side       | 0     | bars |
| User defined MAPnc, Shell Side        | 0     | bars |
| User defined MAPnc, Channel Side      | 0     | bars |
| User defined Test Pres., Shell Side   | 0     | bars |
| User defined Test Pres., Channel Side | 0     | bars |

|                                         |        |     |
|-----------------------------------------|--------|-----|
| Projection of Nozzle from Vessel Top    | 0      | mm. |
| Projection of Nozzle from Vessel Bottom | 0      | mm. |
| Type of Construction                    | Welded |     |
| Use Higher Longitudinal Stresses (Flag) | Y      |     |
| Select t for Internal Pressure (Flag)   | N      |     |
| Select t for External Pressure (Flag)   | N      |     |
| Select t for Axial Stress (Flag)        | N      |     |
| Select Location for Stiff. Rings (Flag) | N      |     |
| Consider Vortex Shedding                | N      |     |

**Shop Pressure Test:**

|                                      |            |
|--------------------------------------|------------|
| Type of Pressure Test                | UG-99(b)   |
| Pressure Test Position               | Horizontal |
| Test Performed in Corroded Condition | Yes        |

|             |                |
|-------------|----------------|
| Load Case 1 | NP+EW+WI+FW+BW |
| Load Case 2 | NP+EW+EE+FS+BS |
| Load Case 3 | NP+OW+WI+FW+BW |
| Load Case 4 | NP+OW+EQ+FS+BS |



|              |                |
|--------------|----------------|
| Load Case 5  | NP+HW+HI       |
| Load Case 6  | NP+HW+HE       |
| Load Case 7  | IP+OW+WI+FW+BW |
| Load Case 8  | IP+OW+EQ+FS+BS |
| Load Case 9  | EP+OW+WI+FW+BW |
| Load Case 10 | EP+OW+EQ+FS+BS |
| Load Case 11 | HP+HW+HI       |
| Load Case 12 | HP+HW+HE       |
| Load Case 13 | IP+WE+EW       |
| Load Case 14 | IP+WF+CW       |
| Load Case 15 | IP+VO+OW       |
| Load Case 16 | IP+VE+EW       |
| Load Case 17 | NP+VO+OW       |
| Load Case 18 | FS+BS+IP+OW    |
| Load Case 19 | FS+BS+EP+OW    |
| Load Case 20 | BL+IP+OW       |

|                                         |                       |
|-----------------------------------------|-----------------------|
| Wind Design Code                        | ASCE/SEI 7-22         |
| Wind Load Reduction Scale Factor        | 1.100                 |
| Basic Wind Speed                        | [V] 180 Km/hr         |
| Surface Roughness Category              | D: Flat, unobstructed |
| Type of Surface                         | 0                     |
| Base Elevation                          | 300 mm.               |
| Percent Wind for Hydrotect              | 33.0                  |
| Using User defined Wind Press. Vs Elev. | N                     |
| Height of Hill or Escarpment            | H or Hh 0 mm.         |
| Distance Upwind of Crest                | Lh 0 mm.              |
| Distance from Crest to the Vessel       | x 0 mm.               |
| Type of Terrain ( Hill, Escarpment )    | Flat                  |
| Damping Factor (Beta) for Wind (Ope)    | 0.0100                |
| Damping Factor (Beta) for Wind (Empty)  | 0.0000                |
| Damping Factor (Beta) for Wind (Filled) | 0.0000                |
| Site Elevation above Sea Level          | 0 mm.                 |

|                                  |                |
|----------------------------------|----------------|
| Seismic Design Code              | ASCE/SEI 7-22  |
| Importance Factor                | 1.500          |
| Force Modification Factor R      | 3.000          |
| Site Class                       | D              |
| Component Elevation Ratio        | z/h 0.000      |
| Amplification Factor             | Ap 0.000       |
| Force Factor                     | 0.000          |
| Consider Vertical Acceleration   | Yes            |
| Minimum Acceleration Multiplier  | 0.000          |
| User Value of Sds (used if > 0 ) | 0.000          |
| User Value of Sd1 (used if > 0 ) | 0.000          |
| Component resonance duct. fact.  | [CAR] 0.000    |
| Component strength factor        | [Rpo] 0.000    |
| Overstrength factor              | [Omega0] 0.000 |
| Response Acceleration Factor     | [SMS] 0.326    |
| Response Acceleration Factor     | [SM1] 0.139    |
| Building Importance Factor       | [Ie] 1.000     |
| Site Latitude                    | 0.000 deg      |
| Site Longitude                   | 0.000 deg      |

|                                          |   |
|------------------------------------------|---|
| Design Pressure + Static Head            | Y |
| Consider MAP New and Cold in Noz. Design | N |
| Consider External Loads for Nozzle Des.  | Y |
| Use ASME VIII-1 Appendix 1-9             | N |

|                             |                                |
|-----------------------------|--------------------------------|
| Perform Blast Load Analysis | No                             |
| Material Database Year      | Current w/Addenda or Code Year |

#### Configuration Directives:

|                                                    |     |
|----------------------------------------------------|-----|
| g0 is taken as min( g0, attached shell thk )       | Yes |
| Do not use Nozzle MDMT Interpretation VIII-1 01-37 | No  |
| Use Table G instead of exact equation for "A"      | Yes |
| Shell Head Joints are Tapered                      | Yes |
| Compute "K" in corroded condition                  | Yes |
| Use Code Case 2286                                 | No  |



Use the MAWP to compute the MDMT Yes  
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes  
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No  
 Using Metric Material Databases, ASME II D Yes  
 Calculate B31.3 type stress for Nozzles with Loads Yes  
 Reduce the MDMT due to lower membrane stress Yes  
 Consider Longitudinal Stress in MDMT Calculations Yes

#### Complete Listing of Vessel Elements and Details:

|                                      |                             |
|--------------------------------------|-----------------------------|
| Element#                             | 1/7                         |
| Element From Node                    | 10                          |
| Element To Node                      | 20                          |
| Element Type                         | Spherical                   |
| Description                          | CH. COVER                   |
| Distance "FROM" to "TO"              | 0 mm.                       |
| Inside Diameter                      | 1750 mm.                    |
| Element Thickness                    | 66 mm.                      |
| Internal Corrosion Allowance         | 1.5 mm.                     |
| Nominal Thickness                    | 78 mm.                      |
| External Corrosion Allowance         | 0 mm.                       |
| Design Internal Pressure             | 191 bars                    |
| Design Temperature Internal Pressure | 85 °C                       |
| Design External Pressure             | 0 bars                      |
| Design Temperature External Pressure | 85 °C                       |
| Effective Diameter Multiplier        | 1.2                         |
| Material Name                        | SA-516 70 +N+IT             |
| Allowable Stress, Ambient            | 138 N./mm <sup>2</sup>      |
| Allowable Stress, Operating          | 138 N./mm <sup>2</sup>      |
| Allowable Stress, Hydrotest          | 234 N./mm <sup>2</sup>      |
| Material Density                     | 0.00775 kg./cm <sup>3</sup> |
| P Number Thickness                   | 32.004 mm.                  |
| Yield Stress, Operating              | 242 N./mm <sup>2</sup>      |
| UCS-66 Chart Curve Designation       | Impact Tested               |
| External Pressure Chart Name         | CS-2                        |
| UNS Number                           | K02700                      |
| Product Form                         | Plate                       |
| Efficiency, Longitudinal Seam        | 1.0                         |
| Efficiency, Circumferential Seam     | 1.0                         |
| Weld is pre-Heated                   | No                          |

|                                      |                              |
|--------------------------------------|------------------------------|
| Element From Node                    | 10                           |
| Detail Type                          | Liquid                       |
| Detail ID                            | Liquid 10                    |
| Dist. from "FROM" Node / Offset dist | 0 mm.                        |
| Height/Length of Liquid              | 1750 mm.                     |
| Liquid Density                       | 0.000654 kg./cm <sup>3</sup> |

|                                      |                                 |
|--------------------------------------|---------------------------------|
| Element From Node                    | 10                              |
| Detail Type                          | Insulation                      |
| Detail ID                            | Ins: 20                         |
| Dist. from "FROM" Node / Offset dist | -875 mm.                        |
| Height/Length of Insulation          | 875 mm.                         |
| Thickness of Insulation              | 70 mm.                          |
| Density                              | 0.40000E-04 kg./cm <sup>3</sup> |

|                                      |                |
|--------------------------------------|----------------|
| Element From Node                    | 10             |
| Detail Type                          | Nozzle         |
| Detail ID                            | CH. IN NOZ A1  |
| Dist. from "FROM" Node / Offset dist | 600 mm.        |
| Nozzle Diameter                      | 24 in.         |
| Nozzle Schedule                      | None           |
| Nozzle Class                         | None           |
| Layout Angle                         | 90.0           |
| Blind Flange (Y/N)                   | N              |
| Weight of Nozzle ( Used if > 0 )     | 874.31 Kg      |
| Grade of Attached Flange             | None           |
| Nozzle Matl                          | SA-350 LF2 +IT |



Element From Node 10  
 Detail Type Nozzle  
 Detail ID CH. OUT NOZ B1  
 Dist. from "FROM" Node / Offset dist 698.95 mm.  
 Nozzle Diameter 20 in.  
 Nozzle Schedule None  
 Nozzle Class None  
 Layout Angle -30.0499  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 623.23 Kg  
 Grade of Attached Flange None  
 Nozzle Matl SA-350 LF2 +IT

Element From Node 10  
 Detail Type Nozzle  
 Detail ID MANHOLE M1  
 Dist. from "FROM" Node / Offset dist 721.11 mm.  
 Nozzle Diameter 20 in.  
 Nozzle Schedule None  
 Nozzle Class 1500  
 Layout Angle -146.31  
 Blind Flange (Y/N) Y  
 Weight of Nozzle ( Used if > 0 ) 2533 Kg  
 Grade of Attached Flange GR 1.1  
 Nozzle Matl SA-350 LF2 +IT

Element From Node 10  
 Detail Type Weight  
 Detail ID PP PLATE  
 Dist. from "FROM" Node / Offset dist -400 mm.  
 Miscellaneous Weight 850 Kg  
 Offset from Element Centerline 0 mm.

Element# 2/7  
 Element From Node 20  
 Element To Node 30  
 Element Type Cylinder  
 Description CH. COVER  
 Distance "FROM" to "TO" 50 mm.  
 Inside Diameter 1750 mm.  
 Element Thickness 135 mm.  
 Internal Corrosion Allowance 1.5 mm.  
 Nominal Thickness 135 mm.  
 External Corrosion Allowance 0 mm.  
 Design Internal Pressure 191 bars  
 Design Temperature Internal Pressure 85 °C  
 Design External Pressure 0 bars  
 Design Temperature External Pressure 85 °C  
 Effective Diameter Multiplier 1.2  
 Material Name SA-516 70 +N  
 Efficiency, Longitudinal Seam 1.0  
 Efficiency, Circumferential Seam 1.0  
 Weld is pre-Heated No

Element From Node 20  
 Detail Type Liquid  
 Detail ID Liquid 20  
 Dist. from "FROM" Node / Offset dist 0 mm.  
 Height/Length of Liquid 1600 mm.  
 Liquid Density 0.000654 kg./cm³

Element From Node 20  
 Detail Type Insulation  
 Detail ID Ins: 20  
 Dist. from "FROM" Node / Offset dist 0 mm.  
 Height/Length of Insulation 50 mm.  
 Thickness of Insulation 70 mm.  
 Density 0.40000E-04 kg./cm³



|                                         |                                 |
|-----------------------------------------|---------------------------------|
| Element#                                | 3/7                             |
| Element From Node                       | 30                              |
| Element To Node                         | 40                              |
| Element Type                            | Flange                          |
| Description                             | SH. FLG.                        |
| Distance "FROM" to "TO"                 | 165 mm.                         |
| Flange Inside Diameter                  | 1750 mm.                        |
| Element Thickness                       | 95 mm.                          |
| Internal Corrosion Allowance            | 1.5 mm.                         |
| Nominal Thickness                       | 100 mm.                         |
| External Corrosion Allowance            | 0 mm.                           |
| Design Internal Pressure                | 25 bars                         |
| Design Temperature Internal Pressure    | 85 °C                           |
| Design External Pressure                | 0 bars                          |
| Design Temperature External Pressure    | 85 °C                           |
| Effective Diameter Multiplier           | 1.2                             |
| Material Name                           | SA-350 LF2 +IT                  |
| Allowable Stress, Ambient               | 138 N./mm <sup>2</sup>          |
| Allowable Stress, Operating             | 138 N./mm <sup>2</sup>          |
| Allowable Stress, Hydrotest             | 225 N./mm <sup>2</sup>          |
| Material Density                        | 0.00775 kg./cm <sup>3</sup>     |
| P Number Thickness                      | 32.004 mm.                      |
| Yield Stress, Operating                 | 229.57 N./mm <sup>2</sup>       |
| UCS-66 Chart Curve Designation          | Impact Tested                   |
| External Pressure Chart Name            | CS-2                            |
| UNS Number                              | K03011                          |
| Class / Thickness / Grade               | 1 ::                            |
| Product Form                            | Forgings                        |
| Perform Flange Stress Calculation (Y/N) | Y                               |
| Weld is pre-Heated                      | No                              |
|                                         |                                 |
| Element From Node                       | 30                              |
| Detail Type                             | Liquid                          |
| Detail ID                               | Liquid 30                       |
| Dist. from "FROM" Node / Offset dist    | 0 mm.                           |
| Height/Length of Liquid                 | 1600 mm.                        |
| Liquid Density                          | 0.000654 kg./cm <sup>3</sup>    |
|                                         |                                 |
| Element From Node                       | 30                              |
| Detail Type                             | Insulation                      |
| Detail ID                               | Ins: 20                         |
| Dist. from "FROM" Node / Offset dist    | 0 mm.                           |
| Height/Length of Insulation             | 235 mm.                         |
| Thickness of Insulation                 | 70 mm.                          |
| Density                                 | 0.40000E-04 kg./cm <sup>3</sup> |

|                                      |                        |
|--------------------------------------|------------------------|
| Element#                             | 4/7                    |
| Element From Node                    | 40                     |
| Element To Node                      | 50                     |
| Element Type                         | Cylinder               |
| Description                          | SHELL-1                |
| Distance "FROM" to "TO"              | 350 mm.                |
| Inside Diameter                      | 1750 mm.               |
| Element Thickness                    | 46 mm.                 |
| Internal Corrosion Allowance         | 1.5 mm.                |
| Nominal Thickness                    | 46 mm.                 |
| External Corrosion Allowance         | 0 mm.                  |
| Design Internal Pressure             | 25 bars                |
| Design Temperature Internal Pressure | 85 °C                  |
| Design External Pressure             | 0 bars                 |
| Design Temperature External Pressure | 85 °C                  |
| Effective Diameter Multiplier        | 1.2                    |
| Material Name                        | SA-516 70 +Q           |
| Allowable Stress, Ambient            | 138 N./mm <sup>2</sup> |
| Allowable Stress, Operating          | 138 N./mm <sup>2</sup> |



|                                  |         |                     |
|----------------------------------|---------|---------------------|
| Allowable Stress, Hydrotest      | 234     | N./mm <sup>2</sup>  |
| Material Density                 | 0.00775 | kg./cm <sup>3</sup> |
| P Number Thickness               | 32.004  | mm.                 |
| Yield Stress, Operating          | 262     | N./mm <sup>2</sup>  |
| UCS-66 Chart Curve Designation   | B       |                     |
| External Pressure Chart Name     | CS-2    |                     |
| UNS Number                       | K02700  |                     |
| Product Form                     | Plate   |                     |
| Efficiency, Longitudinal Seam    | 1.0     |                     |
| Efficiency, Circumferential Seam | 1.0     |                     |
| Weld is pre-Heated               | No      |                     |

|                                      |              |     |
|--------------------------------------|--------------|-----|
| Element From Node                    | 40           |     |
| Detail Type                          | Saddle       |     |
| Detail ID                            | FIXED SADDLE |     |
| Dist. from "FROM" Node / Offset dist | 175          | mm. |
| Width of Saddle                      | 200          | mm. |
| Height of Saddle at Bottom           | 1380         | mm. |
| Saddle Contact Angle                 | 120.0        |     |
| Height of Composite Ring Stiffener   | 0            | mm. |
| Width of Wear Plate                  | 260          | mm. |
| Thickness of Wear Plate              | 16           | mm. |
| Contact Angle, Wear Plate (degrees)  | 132.0        |     |
| Friction coefficient                 | 0.60000002   |     |
| Moment Factor                        | 3.0          |     |
| Dimension E at base (optional)       | 0            | mm. |
| Circumferential Eff. over Saddle     | 1.0          |     |
| Circumferential Eff. at Midspan      | 1.0          |     |
| Tangent to Tangent dist. (optional)  | 0            | mm. |

|                                      |           |                     |
|--------------------------------------|-----------|---------------------|
| Element From Node                    | 40        |                     |
| Detail Type                          | Liquid    |                     |
| Detail ID                            | Liquid 40 |                     |
| Dist. from "FROM" Node / Offset dist | 0         | mm.                 |
| Height/Length of Liquid              | 1600      | mm.                 |
| Liquid Density                       | 0.000654  | kg./cm <sup>3</sup> |

|                                      |             |                     |
|--------------------------------------|-------------|---------------------|
| Element From Node                    | 40          |                     |
| Detail Type                          | Insulation  |                     |
| Detail ID                            | Ins: 20     |                     |
| Dist. from "FROM" Node / Offset dist | 0           | mm.                 |
| Height/Length of Insulation          | 350         | mm.                 |
| Thickness of Insulation              | 70          | mm.                 |
| Density                              | 0.40000E-04 | kg./cm <sup>3</sup> |

|                                      |                 |         |
|--------------------------------------|-----------------|---------|
| Element#                             | 5/7             |         |
| Element From Node                    | 50              |         |
| Element To Node                      | 60              |         |
| Element Type                         | Conical         |         |
| Description                          | SHELL-2         |         |
| Distance "FROM" to "TO"              | 2598            | mm.     |
| Inside Diameter                      | 1750            | mm.     |
| Element Thickness                    | 46              | mm.     |
| Internal Corrosion Allowance         | 1.5             | mm.     |
| Nominal Thickness                    | 46              | mm.     |
| External Corrosion Allowance         | 0               | mm.     |
| Design Internal Pressure             | 25              | bars    |
| Design Temperature Internal Pressure | 85              | °C      |
| Design External Pressure             | 1.0342          | bars    |
| Design Temperature External Pressure | 85              | °C      |
| Effective Diameter Multiplier        | 1.2             |         |
| Material Name                        | SA-516 70 +Q+IT |         |
| Efficiency, Longitudinal Seam        | 1.0             |         |
| Efficiency, Circumferential Seam     | 1.0             |         |
| Cone Diameter at "To" End            | 3250            | mm.     |
| Design Length of Cone                | 2598            | mm.     |
| Half Apex Angle of Cone              | 30.0            | degrees |
| Toriconical (Y/N)                    | N               |         |



|                                      |                                 |
|--------------------------------------|---------------------------------|
| Weld is pre-Heated                   | No                              |
| Element From Node                    | 50                              |
| Detail Type                          | Liquid                          |
| Detail ID                            | Liquid 50                       |
| Dist. from "FROM" Node / Offset dist | 0 mm.                           |
| Height/Length of Liquid              | 1600 mm.                        |
| Liquid Density                       | 0.000654 kg./cm <sup>3</sup>    |
| Element From Node                    | 50                              |
| Detail Type                          | Insulation                      |
| Detail ID                            | Ins: 20                         |
| Dist. from "FROM" Node / Offset dist | 0 mm.                           |
| Height/Length of Insulation          | 2598 mm.                        |
| Thickness of Insulation              | 70 mm.                          |
| Density                              | 0.40000E-04 kg./cm <sup>3</sup> |
| Element From Node                    | 50                              |
| Detail Type                          | Nozzle                          |
| Detail ID                            | LIQ OUT NOZ B3                  |
| Dist. from "FROM" Node / Offset dist | 250 mm.                         |
| Nozzle Diameter                      | 3 in.                           |
| Nozzle Schedule                      | 160                             |
| Nozzle Class                         | 300                             |
| Layout Angle                         | -90.0                           |
| Blind Flange (Y/N)                   | Y                               |
| Weight of Nozzle ( Used if > 0 )     | 26.876 Kg                       |
| Grade of Attached Flange             | GR 1.1                          |
| Nozzle Matl                          | SA-350 LF2 +IT                  |

|                                      |                 |
|--------------------------------------|-----------------|
| Element#                             | 6/7             |
| Element From Node                    | 60              |
| Element To Node                      | 70              |
| Element Type                         | Cylinder        |
| Description                          | K-SHELL         |
| Distance "FROM" to "TO"              | 4185 mm.        |
| Inside Diameter                      | 3250 mm.        |
| Element Thickness                    | 35 mm.          |
| Internal Corrosion Allowance         | 1.5 mm.         |
| Nominal Thickness                    | 35 mm.          |
| External Corrosion Allowance         | 0 mm.           |
| Design Internal Pressure             | 25 bars         |
| Design Temperature Internal Pressure | 85 °C           |
| Design External Pressure             | 1.0342 bars     |
| Design Temperature External Pressure | 85 °C           |
| Effective Diameter Multiplier        | 1.2             |
| Material Name                        | SA-516 70 +Q+IT |
| Efficiency, Longitudinal Seam        | 1.0             |
| Efficiency, Circumferential Seam     | 1.0             |
| Weld is pre-Heated                   | No              |

|                                      |                 |
|--------------------------------------|-----------------|
| Element From Node                    | 60              |
| Detail Type                          | Saddle          |
| Detail ID                            | SLIDING SADDLE- |
| Dist. from "FROM" Node / Offset dist | 400 mm.         |
| Width of Saddle                      | 350 mm.         |
| Height of Saddle at Bottom           | 2130 mm.        |
| Saddle Contact Angle                 | 120.0           |
| Height of Composite Ring Stiffener   | 0 mm.           |
| Width of Wear Plate                  | 480 mm.         |
| Thickness of Wear Plate              | 16 mm.          |
| Contact Angle, Wear Plate (degrees)  | 132.0           |
| Friction coefficient                 | 0.1             |
| Moment Factor                        | 3.0             |
| Dimension E at base (optional)       | 0 mm.           |
| Circumferential Eff. over Saddle     | 1.0             |
| Circumferential Eff. at Midspan      | 1.0             |
| Tangent to Tangent dist. (optional)  | 0 mm.           |



|                                      |                |                     |
|--------------------------------------|----------------|---------------------|
| Element From Node                    | 60             |                     |
| Detail Type                          | Saddle         |                     |
| Detail ID                            | SLIDING SADDLE |                     |
| Dist. from "FROM" Node / Offset dist | 3785           | mm.                 |
| Width of Saddle                      | 350            | mm.                 |
| Height of Saddle at Bottom           | 2130           | mm.                 |
| Saddle Contact Angle                 | 120.0          |                     |
| Height of Composite Ring Stiffener   | 0              | mm.                 |
| Width of Wear Plate                  | 480            | mm.                 |
| Thickness of Wear Plate              | 16             | mm.                 |
| Contact Angle, Wear Plate (degrees)  | 132.0          |                     |
| Friction coefficient                 | 0.1            |                     |
| Moment Factor                        | 3.0            |                     |
| Dimension E at base (optional)       | 0              | mm.                 |
| Circumferential Eff. over Saddle     | 1.0            |                     |
| Circumferential Eff. at Midspan      | 1.0            |                     |
| Tangent to Tangent dist. (optional)  | 0              | mm.                 |
|                                      |                |                     |
| Element From Node                    | 60             |                     |
| Detail Type                          | Liquid         |                     |
| Detail ID                            | Liquid 60      |                     |
| Dist. from "FROM" Node / Offset dist | 0              | mm.                 |
| Height/Length of Liquid              | 1600           | mm.                 |
| Liquid Density                       | 0.000654       | kg./cm <sup>3</sup> |
|                                      |                |                     |
| Element From Node                    | 60             |                     |
| Detail Type                          | Insulation     |                     |
| Detail ID                            | Ins: 20        |                     |
| Dist. from "FROM" Node / Offset dist | 0              | mm.                 |
| Height/Length of Insulation          | 4185           | mm.                 |
| Thickness of Insulation              | 70             | mm.                 |
| Density                              | 0.40000E-04    | kg./cm <sup>3</sup> |
|                                      |                |                     |
| Element From Node                    | 60             |                     |
| Detail Type                          | Nozzle         |                     |
| Detail ID                            | SS IN NOZ. A2  |                     |
| Dist. from "FROM" Node / Offset dist | 3750           | mm.                 |
| Nozzle Diameter                      | 10             | in.                 |
| Nozzle Schedule                      | 80             |                     |
| Nozzle Class                         | 300            |                     |
| Layout Angle                         | 90.0           |                     |
| Blind Flange (Y/N)                   | N              |                     |
| Weight of Nozzle ( Used if > 0 )     | 171.52         | Kgf                 |
| Grade of Attached Flange             | GR 1.1         |                     |
| Nozzle Matl                          | SA-350 LF2 +IT |                     |
|                                      |                |                     |
| Element From Node                    | 60             |                     |
| Detail Type                          | Nozzle         |                     |
| Detail ID                            | SS OUT NOZ. B2 |                     |
| Dist. from "FROM" Node / Offset dist | 1700           | mm.                 |
| Nozzle Diameter                      | 24             | in.                 |
| Nozzle Schedule                      | None           |                     |
| Nozzle Class                         | 300            |                     |
| Layout Angle                         | 90.0           |                     |
| Blind Flange (Y/N)                   | N              |                     |
| Weight of Nozzle ( Used if > 0 )     | 802.3          | Kgf                 |
| Grade of Attached Flange             | GR 1.1         |                     |
| Nozzle Matl                          | SA-350 LF2 +IT |                     |
|                                      |                |                     |
| Element From Node                    | 60             |                     |
| Detail Type                          | Nozzle         |                     |
| Detail ID                            | VENT NOZ V1    |                     |
| Dist. from "FROM" Node / Offset dist | 400            | mm.                 |
| Nozzle Diameter                      | 2              | in.                 |
| Nozzle Schedule                      | 160            |                     |
| Nozzle Class                         | 300            |                     |
| Layout Angle                         | 90.0           |                     |
| Blind Flange (Y/N)                   | Y              |                     |
| Weight of Nozzle ( Used if > 0 )     | 7.8513         | Kgf                 |



|                                      |                 |
|--------------------------------------|-----------------|
| Grade of Attached Flange             | GR 1.1          |
| Nozzle Matl                          | SA-350 LF2 +IT  |
| Element From Node                    | 60              |
| Detail Type                          | Nozzle          |
| Detail ID                            | PR. IN P1       |
| Dist. from "FROM" Node / Offset dist | 150 mm.         |
| Nozzle Diameter                      | 2 in.           |
| Nozzle Schedule                      | 160             |
| Nozzle Class                         | 300             |
| Layout Angle                         | 90.0            |
| Blind Flange (Y/N)                   | Y               |
| Weight of Nozzle ( Used if > 0 )     | 11.265 Kg       |
| Grade of Attached Flange             | GR 1.1          |
| Nozzle Matl                          | SA-350 LF2 +IT  |
| Element From Node                    | 60              |
| Detail Type                          | Nozzle          |
| Detail ID                            | PR. IN NOZ. P2  |
| Dist. from "FROM" Node / Offset dist | 3250 mm.        |
| Nozzle Diameter                      | 2 in.           |
| Nozzle Schedule                      | 160             |
| Nozzle Class                         | 300             |
| Layout Angle                         | 90.0            |
| Blind Flange (Y/N)                   | Y               |
| Weight of Nozzle ( Used if > 0 )     | 11.265 Kg       |
| Grade of Attached Flange             | GR 1.1          |
| Nozzle Matl                          | SA-350 LF2 +IT  |
| Element From Node                    | 60              |
| Detail Type                          | Nozzle          |
| Detail ID                            | SAFETY VALVE S1 |
| Dist. from "FROM" Node / Offset dist | 3000 mm.        |
| Nozzle Diameter                      | 2 in.           |
| Nozzle Schedule                      | 160             |
| Nozzle Class                         | 300             |
| Layout Angle                         | 90.0            |
| Blind Flange (Y/N)                   | Y               |
| Weight of Nozzle ( Used if > 0 )     | 11.265 Kg       |
| Grade of Attached Flange             | GR 1.1          |
| Nozzle Matl                          | SA-350 LF2 +IT  |
| Element From Node                    | 60              |
| Detail Type                          | Nozzle          |
| Detail ID                            | LEVEL GUAGE L1  |
| Dist. from "FROM" Node / Offset dist | 800 mm.         |
| Nozzle Diameter                      | 2 in.           |
| Nozzle Schedule                      | 160             |
| Nozzle Class                         | 300             |
| Layout Angle                         | 30.4806         |
| Blind Flange (Y/N)                   | Y               |
| Weight of Nozzle ( Used if > 0 )     | 11.265 Kg       |
| Grade of Attached Flange             | GR 1.1          |
| Nozzle Matl                          | SA-350 LF2 +IT  |
| Element From Node                    | 60              |
| Detail Type                          | Nozzle          |
| Detail ID                            | LEVEL GUAGE L2  |
| Dist. from "FROM" Node / Offset dist | 800 mm.         |
| Nozzle Diameter                      | 2 in.           |
| Nozzle Schedule                      | 160             |
| Nozzle Class                         | 300             |
| Layout Angle                         | -30.4806        |
| Blind Flange (Y/N)                   | Y               |
| Weight of Nozzle ( Used if > 0 )     | 11.265 Kg       |
| Grade of Attached Flange             | GR 1.1          |
| Nozzle Matl                          | SA-350 LF2 +IT  |
| Element From Node                    | 60              |
| Detail Type                          | Nozzle          |



|                                      |                |     |
|--------------------------------------|----------------|-----|
| Detail ID                            | LEVEL GUAGE L3 |     |
| Dist. from "FROM" Node / Offset dist | 1500           | mm. |
| Nozzle Diameter                      | 2              | in. |
| Nozzle Schedule                      | 160            |     |
| Nozzle Class                         | 300            |     |
| Layout Angle                         | 30.4806        |     |
| Blind Flange (Y/N)                   | Y              |     |
| Weight of Nozzle ( Used if > 0 )     | 11.265         | Kgf |
| Grade of Attached Flange             | GR 1.1         |     |
| Nozzle Matl                          | SA-350 LF2 +IT |     |

|                                      |                |     |
|--------------------------------------|----------------|-----|
| Element From Node                    | 60             |     |
| Detail Type                          | Nozzle         |     |
| Detail ID                            | LEVEL GUAGE L4 |     |
| Dist. from "FROM" Node / Offset dist | 1500           | mm. |
| Nozzle Diameter                      | 2              | in. |
| Nozzle Schedule                      | 160            |     |
| Nozzle Class                         | 300            |     |
| Layout Angle                         | -30.4806       |     |
| Blind Flange (Y/N)                   | Y              |     |
| Weight of Nozzle ( Used if > 0 )     | 11.265         | Kgf |
| Grade of Attached Flange             | GR 1.1         |     |
| Nozzle Matl                          | SA-350 LF2 +IT |     |

|                                      |                |     |
|--------------------------------------|----------------|-----|
| Element From Node                    | 60             |     |
| Detail Type                          | Weight         |     |
| Detail ID                            | VAPOR GET. DEV |     |
| Dist. from "FROM" Node / Offset dist | 1700           | mm. |
| Miscellaneous Weight                 | 900            | Kgf |
| Offset from Element Centerline       | 0              | mm. |

|                                      |                 |     |
|--------------------------------------|-----------------|-----|
| Element From Node                    | 60              |     |
| Detail Type                          | Weight          |     |
| Detail ID                            | Datsheet weight |     |
| Dist. from "FROM" Node / Offset dist | 0               | mm. |
| Miscellaneous Weight                 | 14550           | Kgf |
| Offset from Element Centerline       | 0               | mm. |

|                                      |                 |      |
|--------------------------------------|-----------------|------|
| Element#                             | 7/7             |      |
| Element From Node                    | 70              |      |
| Element To Node                      | 80              |      |
| Element Type                         | Elliptical      |      |
| Description                          | SH. COVER       |      |
| Distance "FROM" to "TO"              | 50.8            | mm.  |
| Inside Diameter                      | 3250            | mm.  |
| Element Thickness                    | 35              | mm.  |
| Internal Corrosion Allowance         | 1.5             | mm.  |
| Nominal Thickness                    | 41              | mm.  |
| External Corrosion Allowance         | 0               | mm.  |
| Design Internal Pressure             | 25              | bars |
| Design Temperature Internal Pressure | 85              | °C   |
| Design External Pressure             | 1.0342          | bars |
| Design Temperature External Pressure | 85              | °C   |
| Effective Diameter Multiplier        | 1.2             |      |
| Material Name                        | SA-516 70 +Q+IT |      |
| Efficiency, Longitudinal Seam        | 1.0             |      |
| Efficiency, Circumferential Seam     | 1.0             |      |
| Elliptical Head Factor               | 2.0             |      |
| Weld is pre-Heated                   | No              |      |

|                                      |           |         |
|--------------------------------------|-----------|---------|
| Element From Node                    | 70        |         |
| Detail Type                          | Liquid    |         |
| Detail ID                            | Liquid 70 |         |
| Dist. from "FROM" Node / Offset dist | 0         | mm.     |
| Height/Length of Liquid              | 1600      | mm.     |
| Liquid Density                       | 0.000654  | kg./cm³ |

|                   |    |  |
|-------------------|----|--|
| Element From Node | 70 |  |
|-------------------|----|--|



|                                      |                                 |
|--------------------------------------|---------------------------------|
| Detail Type                          | Insulation                      |
| Detail ID                            | Ins: 20                         |
| Dist. from "FROM" Node / Offset dist | 0 mm.                           |
| Height/Length of Insulation          | 863.3 mm.                       |
| Thickness of Insulation              | 70 mm.                          |
| Density                              | 0.40000E-04 kg./cm <sup>3</sup> |
| Element From Node                    | 70                              |
| Detail Type                          | Nozzle                          |
| Detail ID                            | TEMP IN. T1                     |
| Dist. from "FROM" Node / Offset dist | 700 mm.                         |
| Nozzle Diameter                      | 2 in.                           |
| Nozzle Schedule                      | 160                             |
| Nozzle Class                         | 300                             |
| Layout Angle                         | 270.0                           |
| Blind Flange (Y/N)                   | Y                               |
| Weight of Nozzle ( Used if > 0 )     | 11.265 Kgf                      |
| Grade of Attached Flange             | GR 1.1                          |
| Nozzle Matl                          | SA-350 LF2 +IT                  |



**XY Coordinate Calculations:**

| From      | To | X (Horiz.)<br>mm. | Y (Vert.)<br>mm. | DX (Horiz.)<br>mm. | DY (Vert.)<br>mm. |
|-----------|----|-------------------|------------------|--------------------|-------------------|
| CH. COVER |    | ...               | ...              | ...                | ...               |
| CH. COVER |    | 50                | ...              | 50                 | ...               |
| SH. FLG.  |    | 659.5             | ...              | 165                | ...               |
| SHELL-1   |    | 1009.5            | ...              | 350                | ...               |
| SHELL-2   |    | 3607.5            | ...              | 2598               | ...               |
| K-SHELL   |    | 7792.5            | ...              | 4185               | ...               |
| SH. COVER |    | 7843.3            | ...              | 50.8               | ...               |

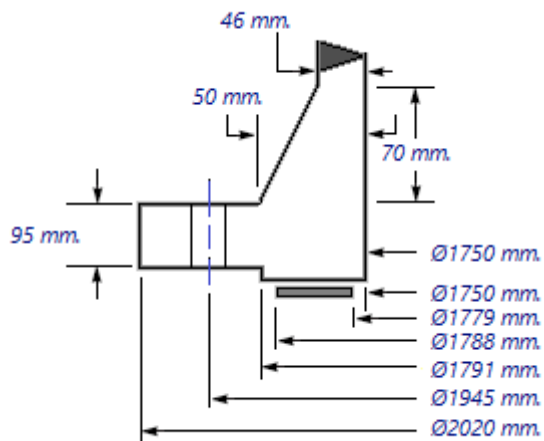
The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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**Flange Input Data Values** **Description: SH. FLG** :

**SH. FLG.**

|                                             |                    |          |                    |
|---------------------------------------------|--------------------|----------|--------------------|
| Flange Type                                 | Integral Weld Neck |          |                    |
| Design Pressure                             | P                  | 25.10    | bars               |
| Design Temperature                          |                    | 85       | °C                 |
| Internal Corrosion Allowance                | ci                 | 1.5000   | mm.                |
| External Corrosion Allowance                | ce                 | 0.0000   | mm.                |
| Use Corrosion Allowance in Thickness Calcs. |                    | Yes      |                    |
| Flange Inside Diameter                      | B                  | 1750.000 | mm.                |
| Flange Outside Diameter                     | A                  | 2020.000 | mm.                |
| Flange Thickness                            | t                  | 95.0000  | mm.                |
| Thickness of Hub at Small End               | go                 | 46.0000  | mm.                |
| Thickness of Hub at Large End               | gl                 | 50.0000  | mm.                |
| Length of Hub                               | h                  | 70.0000  | mm.                |
| Flange Material                             | SA-350 LF2 +IT     |          |                    |
| Flange Material UNS number                  | K03011             |          |                    |
| Flange Allowable Stress At Temperature      | Sfo                | 138.00   | N./mm <sup>2</sup> |
| Flange Allowable Stress At Ambient          | Sfa                | 138.00   | N./mm <sup>2</sup> |
| Bolt Material                               | SA-320 L7          |          |                    |
| Bolt Allowable Stress At Temperature        | Sb                 | 172.00   | N./mm <sup>2</sup> |
| Bolt Allowable Stress At Ambient            | Sa                 | 172.00   | N./mm <sup>2</sup> |
| Diameter of Bolt Circle                     | C                  | 1945.000 | mm.                |
| Nominal Bolt Diameter                       | a                  | 36.0000  | mm.                |
| Type of Threads                             | TEMA Metric        |          |                    |
| Number of Bolts                             | 56                 |          |                    |
| Flange Face Outside Diameter                | Fod                | 1791.000 | mm.                |
| Flange Face Inside Diameter                 | Fid                | 1750.000 | mm.                |
| Flange Facing Sketch                        | Sketch 1a          |          |                    |
| Gasket Outside Diameter                     | Go                 | 1788.000 | mm.                |
| Gasket Inside Diameter                      | Gi                 | 1779.000 | mm.                |
| Gasket Factor                               | m                  | 3.7500   |                    |
| Gasket Design Seating Stress                | y                  | 62.06    | N./mm <sup>2</sup> |
| Gasket Thickness                            | tg                 | 4.5000   | mm.                |
| Raised face depth                           | Fd                 | 0.0000   | mm.                |



**ASME Code, Section VIII Division 1, 2025**

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.1 \cdot (1750.0/2 + 1.5)) / (138.0 \cdot 1.0 - 0.6 \cdot 25.1) + Ca$$



= 17.6210 mm.

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 44.5) / (876.5 + 0.6 \cdot 44.5)$$

$$= 67.986 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 46.0) / (875.0 + 0.6 \cdot 46.0)$$

$$= 70.325 \text{ bars}$$

|                                          |              |
|------------------------------------------|--------------|
| Corroded Flange Thickness, tc = T-ci     | 93.500 mm.   |
| Corroded Flange ID, Bcor = B+2*ci        | 1753.000 mm. |
| Corroded Large Hub, g1Cor = g1-ci        | 48.500 mm.   |
| Corroded Small Hub, g0Cor = go-ci        | 44.500 mm.   |
| Code R Dimension, R = ((C-Bcor)/2)-g1cor | 47.500 mm.   |

|                                           |              |
|-------------------------------------------|--------------|
| Gasket Contact Width, N = (Go - Gi)/2     | 4.500 mm.    |
| Basic Gasket Width, bo = N/2              | 2.250 mm.    |
| Effective Gasket Width, b = bo            | 2.250 mm.    |
| Gasket Reaction Diameter, G = (Go + Gi)/2 | 1783.500 mm. |

#### **Basic Flange and Bolt Loads:**

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 1783.5^2 \cdot 25.103$$

$$= 639519.188 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 2.25 \cdot \pi \cdot 1783.5 \cdot 3.75 \cdot 25.1$$

$$= 24203.922 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 1753.0^2 \cdot 25.1027$$

$$= 617833.000 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 639519 - 617833$$

$$= 21686.162 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max( W_{o,1} + W_{o,2} + H_p, 0 )$$

$$= \max( 639519 + 24204 + 0, 0 )$$

$$= 663723.125 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 62.06 \cdot 2.25 \cdot \pi \cdot 1783.5 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 79772.867 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max( ( W_o + F_a + 4 M_e / G ) / S_b, W_g / S_a )$$

$$= \max( ( 663723 + 0 + 4 \cdot 0 / 1783.5 ) / 172, 79773 / 172 )$$

$$= 378.438 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 36.0 + 6 \cdot 93.5 / (3.75 + 0.5)$$

$$= 204.000 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin( \pi / n )$$

$$= 1945.0 \cdot \sin( \pi / 56 )$$

$$= 109.057 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a+t)}, 1)$$

$$= \max(\sqrt{109.057/(2 \cdot 36.0 + 93.5)}, 1)$$

$$= 1.0000$$

#### Bolting Information for TEMA Metric Thread Series (Non Mandatory):

|                                         | Minimum | Actual  | Maximum |
|-----------------------------------------|---------|---------|---------|
| Bolt Area:                              | 378.438 | 413.288 |         |
| Radial Distance between Hub and Bolts:  | 47.250  | 47.500  |         |
| Radial Distance between Bolts and Edge: | 37.500  | 37.500  |         |
| Circ. Spacing between the Bolts:        | 82.600  | 109.057 | 204.000 |

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 413.288 \cdot 172.0 / (62.06 \cdot \pi(1788.0 + 1779.0))$$

$$= 10.222 \text{ mm. [Note: Exceeds actual gasket width, 4.500]}$$

#### Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm  
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 413.2884$$

$$= 724845.00 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 663723 - 639519$$

$$= 24203.93 \text{ Kg}$$

#### Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1945.0 - 1783.5) / 2$$

$$= 80.7500 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1945.0 - 1753.0) / 2 + 80.75)$$

$$= 88.3750 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1945.0 - 1753.0 - 48.5) / 2$$

$$= 71.7501 \text{ mm.}$$

#### Summary of Moments for Internal Pressure: (Kg-m.)

| Loading              | Force   | Distance | Bolt Corr | Moment |
|----------------------|---------|----------|-----------|--------|
| End Pressure, Md     | 617833. | 71.7501  | 1.0000    | 44330. |
| Face Pressure, Mt    | 21686.  | 88.3750  | 1.0000    | 1917.  |
| Gasket Load, Mg      | 24204.  | 80.7500  | 1.0000    | 1955.  |
| Gasket Seating, Matm | 724845. | 80.7500  | 1.0000    | 58532. |

Total Moment for Operation, Mo 48201. Kg-m.  
 Total Moment for Gasket seating, Mg 58532. Kg-m.

Effective Hub Length,  $h_o = \sqrt{Bcor \cdot goCor}$  279.300 mm.  
 Hub Ratio,  $h/h_o = HL / H_o$  0.251  
 Thickness Ratio,  $g1/g_o = (g1Cor/goCor)$  1.090

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.911  
 Factor V per 4.16.5 0.503



Factor f per 4.16.5

1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 2020.0/1753.0$$

$$= 1.152$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.152^2 \cdot \log_{10}(1.152))) / (1.152^2 - 1)) / (1.152 - 1)$$

$$= 13.749$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.152^2(1 + 8.55246 \cdot \log_{10}(1.152)) - 1) / ((1.0472 + 1.9448 \cdot 1.152^2) \cdot (1.152 - 1))$$

$$= 1.858$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.152^2(1 + 8.55246 \cdot \log_{10}(1.152)) - 1) / (1.36136(1.152^2 - 1) \cdot (1.152 - 1))$$

$$= 15.109$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.152^2 + 1) / (1.152^2 - 1)$$

$$= 7.101$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 15.109 \cdot 44.5^2 \cdot 279.3 / 0.503$$

$$= 16602632.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.911/279.3$$

$$= 0.003 \text{ 1/mm.}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (93.5 \cdot 0.003 + 1) / 1.858 + 93.5^3/16602632$$

$$= 0.752$$

## Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 48201 / 1753.0) / (0.7517 \cdot 48.5^2)$$

$$= 152.51 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 58532 / 1753.0) / (0.7517 \cdot 48.5^2)$$

$$= 185.20 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.4066 \cdot 48201 / 1753.0) / (0.7517 \cdot 93.5^2)$$

$$= 57.72 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.4066 \cdot 58532 / 1753.0) / (0.7517 \cdot 93.5^2)$$

$$= 70.09 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (13.7493 \cdot 48201 / (93.5^2 \cdot 1753.0)) - 7.1009 \cdot 58$$

$$= 14.22 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (13.7493 \cdot 58532 / (93.5^2 \cdot 1753.0)) - 7.1009 \cdot 70$$

$$= 17.27 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (153 + \max(58, 14)) / 2$$

$$= 105.11 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRa, STa)) / 2$$

$$= (185 + \max(70, 17)) / 2$$

$$= 127.64 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 663723 / 413.2884$$

$$= 157.49 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (79773 / 413.2884)$$

$$= 18.93 \text{ N./mm}^2$$

#### Flange Stress Analysis Results: N./mm<sup>2</sup>

|                   | Actual | Operating Allowed | Gasket Seating Actual | Allowed |
|-------------------|--------|-------------------|-----------------------|---------|
| Longitudinal Hub  | 152.51 | 207.00            | 185.20                | 207.00  |
| Radial Flange     | 57.72  | 138.00            | 70.09                 | 138.00  |
| Tangential Flange | 14.22  | 138.00            | 17.27                 | 138.00  |
| Maximum Average   | 105.11 | 138.00            | 127.64                | 138.00  |
| Bolting           | 157.49 | 172.00            | 18.93                 | 172.00  |

Minimum Required Flange Thickness 87.147 mm.

Estimated M.A.W.P. ( Operating ) 27 bars

Estimated Finished Weight of Flange at given Thk. 735.7 kg.

Estimated Unfinished Weight of Forging at given Thk 1022.4 kg.

#### Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 58532.4 / 1.0 \cdot 9806.637 \cdot 0.503 / (0.726 \cdot 202713 \cdot 44.5^2 \cdot 279.3 \cdot 0.3)$$

$$= 0.616 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 48201.5 / 1.0 \cdot 9806.637 \cdot 0.503 / (0.726 \cdot 199054 \cdot 44.5^2 \cdot 279.3 \cdot 0.3)$$

$$= 0.517 \text{ (should be } \leq 1)$$

#### Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 58532.4 / 1.0 \cdot 9806.637 \cdot 0.503 / (0.752 \cdot 202713 \cdot 44.5^2 \cdot 279.3 \cdot 0.3)$$

$$= 0.595 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 48201.5 / 1.0 \cdot 9806.637 \cdot 0.503 / (0.752 \cdot 199054 \cdot 44.5^2 \cdot 279.3 \cdot 0.3)$$

$$= 0.499 \text{ (should be } \leq 1)$$

#### Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

**Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:**

**Flange Stress Analysis Results: N./mm<sup>2</sup>**

|                   | Actual | Operating<br>Allowed | Gasket Seating<br>Actual      Allowed |
|-------------------|--------|----------------------|---------------------------------------|
| Longitudinal Hub  | 145.38 | 207.00               | 174.67      207.00                    |
| Radial Flange     | 56.66  | 138.00               | 68.07      138.00                     |
| Tangential Flange | 13.45  | 138.00               | 16.16      138.00                     |
| Maximum Average   | 101.02 | 138.00               | 121.37      138.00                    |
| Bolting           | 157.49 | 172.00               | 18.93      172.00                     |

Minimum Req'd Thickness, ambient, no Corrosion: 82.372 mm.

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### Internal Pressure Results Summary:

#### Element Thickness, Pressure, Diameter and Allowable Stress :

| From      | To | Int. Press<br>+ Liq. Hd<br>bars | Nominal<br>Thickness<br>mm. | Total Corr<br>Allowance<br>mm. | Element<br>Diameter<br>mm. | Allowable<br>Stress(SE)<br>N./mm <sup>2</sup> | Element# |
|-----------|----|---------------------------------|-----------------------------|--------------------------------|----------------------------|-----------------------------------------------|----------|
| CH. COVER |    | 191.11                          | 78                          | 1.5                            | 1750                       | 138                                           | 1        |
| CH. COVER |    | 191.1                           | 135                         | 1.5                            | 1750                       | 138                                           | 2        |
| SH. FLG.  |    | 25.103                          | 100                         | 1.5                            | 1750                       | 138                                           | 3        |
| SHELL-1   |    | 25.103                          | 46                          | 1.5                            | 1750                       | 138                                           | 4        |
| SHELL-2   |    | 25.103                          | 46                          | 1.5                            | 3250                       | 138                                           | 5        |
| K-SHELL   |    | 25.103                          | 35                          | 1.5                            | 3250                       | 138                                           | 6        |
| SH. COVER |    | 25.103                          | 41                          | 1.5                            | 3250                       | 138                                           | 7        |

#### Element Required Thickness and MAWP :

| From      | To | Design<br>Pressure<br>bars | M.A.W.P.<br>Corroded<br>bars | M.A.P.<br>New & Cold<br>bars | Minimum<br>Thickness<br>mm. | Required<br>Thickness<br>mm. | Element# |
|-----------|----|----------------------------|------------------------------|------------------------------|-----------------------------|------------------------------|----------|
| CH. COVER |    | 191                        | 200.029                      | 205.073                      | 66                          | 63.0493                      | 1        |
| CH. COVER |    | 191                        | 192.471                      | 194.86                       | 135                         | 133.888                      | 2        |
| SH. FLG.  |    | 25                         | 27.3117                      | 27.4144                      | 95                          | 87.1474                      | 3        |
| SHELL-1   |    | 25                         | 67.8836                      | 70.3248                      | 46                          | 17.6211                      | 4        |
| SHELL-2   |    | 25                         | 32.1293                      | 33.3379                      | 46                          | 36.0486                      | 5        |
| K-SHELL   |    | 25                         | 27.971                       | 29.3415                      | 35                          | 31.4157                      | 6        |
| SH. COVER |    | 25                         | 28.2943                      | 29.6568                      | 35                          | 31.1067                      | 7        |

#### Summary of Heat Exchanger Maximum Allowable Working Pressures :

##### Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

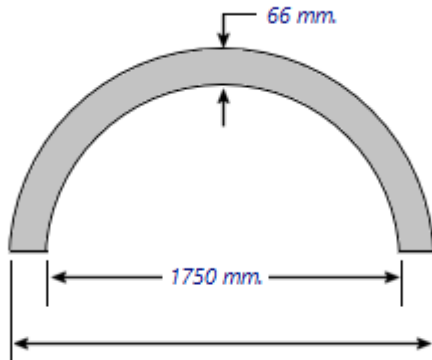
Shell Side MAWP = 27 bars  
 Shell Side MAPnc = 27 bars  
 Channel Side MAWP = 192 bars  
 Channel Side MAPnc = 195 bars

Elements Suitable for Design Internal Pressure.

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### Internal Pressure Calculation Results:

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#### Internal Pressure Results for: CH. COVER

| Spherical Head Design Information:    |    |                           |                         |        |                          |
|---------------------------------------|----|---------------------------|-------------------------|--------|--------------------------|
| Design Pressure                       | P  | 191.112 bars              | Int. Design Temperature |        | 85 °C                    |
| Material                              |    | SA-516 70                 | Ext. Design Temperature |        | 85 °C                    |
| Length                                |    | 0.000 mm.                 | External Pressure Chart |        | CS-2                     |
|                                       |    |                           | UNS Number              |        | K02700                   |
| Int. Corr. All.                       | c  | 1.5000 mm.                | Allowable Stress (ope)  | S      | 138.0 N./mm <sup>2</sup> |
| Ext. Corr. All.                       | ce | 0.0000 mm.                | Allowable Stress (amb)  | Sa     | 138.0 N./mm <sup>2</sup> |
| Inside Diameter                       | Di | 1750.000 mm.              | Actual Stress           |        | 131.8 N./mm <sup>2</sup> |
| Spec. Min. Thk t <sub>e</sub> , or sa |    | 66.0000 mm.               | Spec. Nominal Thk.      |        | 78.000 mm.               |
| Surface Area                          |    | 57064.5 cm <sup>2</sup>   | SG of Contents          |        | 0.655                    |
| Element Volume                        |    | 1403332.4 cm <sup>3</sup> | Weight of Contents      |        | 918.6 Kg                 |
| Empty Weight                          |    | 3175.1 Kg                 | Operating Weight        |        | 4093.7 Kg                |
|                                       |    |                           | Comp. All. Ope. Stress  |        | 147.3 N./mm <sup>2</sup> |
|                                       |    |                           |                         |        |                          |
| Radiography Information:              |    |                           |                         |        |                          |
| User Defined                          |    |                           | User Defined            |        |                          |
| Circ. Joint Efficiency                |    | 100 %                     | Long. Joint Eff.        | E or v | 100 %                    |

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 21.875 mm.  
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.938 mm.

#### Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (191.112 \cdot 876.5) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 191.112)$$

$$= 61.5494 + 1.5000 = 63.0493 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.112 bars

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 64.5) / (876.5 + 0.2 \cdot 64.5)$$

$$= 200 - 0 = 200 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 66.0) / (875.0 + 0.2 \cdot 66.0)$$

$$= 205 \text{ bars}$$



Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned} &= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t) \\ &= (191.112 \cdot (876.5 + 0.2 \cdot 64.5)) / (2 \cdot 1.0 \cdot 64.5) \\ &= 131.772 \text{ N./mm}^2 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57  $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$  6.400 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

#### MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 21.875 mm.

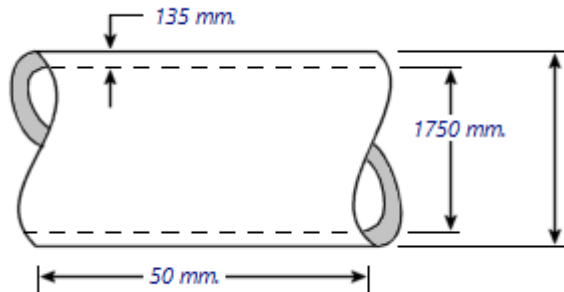
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.938 mm.

Material treatment options: +N+IT

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### Internal Pressure Calculation Results:

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#### Internal Pressure Results for: CH. COVER

| Cylindrical Shell Design Information:             |    |                          |                         |    |                          |
|---------------------------------------------------|----|--------------------------|-------------------------|----|--------------------------|
| Design Pressure                                   | P  | 191.103 bars             | Int. Design Temperature |    | 85 °C                    |
| Material                                          |    | SA-516 70                | Ext. Design Temperature |    | 85 °C                    |
| Length                                            |    | 50.000 mm.               | External Pressure Chart |    | CS-2                     |
|                                                   |    |                          | UNS Number              |    | K02700                   |
| Int. Corr. All.                                   | c  | 1.5000 mm.               | Allowable Stress (ope)  | S  | 138.0 N./mm <sup>2</sup> |
| Ext. Corr. All.                                   | ce | 0.0000 mm.               | Allowable Stress (amb)  | Sa | 138.0 N./mm <sup>2</sup> |
| Inside Diameter                                   | Di | 1750.000 mm.             | Actual Stress           |    | 136.9 N./mm <sup>2</sup> |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 135.0000 mm.             | Spec. Nominal Thk.      |    | 135.000 mm.              |
| Surface Area                                      |    | 3173.0 cm <sup>2</sup>   | SG of Contents          |    | 0.655                    |
| Element Volume                                    |    | 120285.6 cm <sup>3</sup> | Weight of Contents      |    | 75.5 Kg                  |
| Empty Weight                                      |    | 309.8 Kg                 | Operating Weight        |    | 385.3 Kg                 |
|                                                   |    |                          | Comp. All. Ope. Stress  |    | 147.3 N./mm <sup>2</sup> |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (191.103 \cdot 876.5) / (138.0 \cdot 1.0 - 0.6 \cdot 191.103)$$

$$= 132.3880 + 1.5000 = 133.8880 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.103 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 133.5) / (876.5 + 0.6 \cdot 133.5)$$

$$= 193 - 0 = 192 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 135.0) / (875.0 + 0.6 \cdot 135.0)$$

$$= 195 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (191.103 \cdot (876.5 + 0.6 \cdot 133.5)) / (1.0 \cdot 133.5)$$

$$= 136.944 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 7.162 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

### Minimum Design Metal Temperature Results:

**Warning:**

*The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.*

**Note:**

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

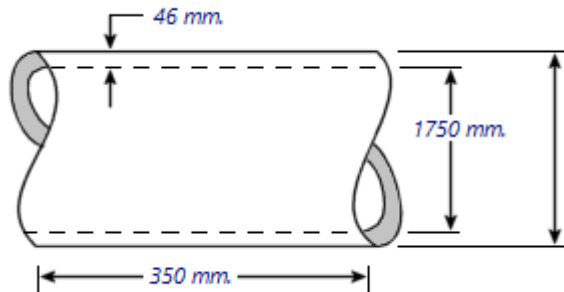
| MDMT Design Information: |             |                        |              |
|--------------------------|-------------|------------------------|--------------|
| MDMT Curve               | B           | Press. at MDMT         | 192.471 bars |
| UCS-66(b) red.           | No          | Min. Des. Mtl. Temp.   | -35 °C       |
| Governing Thk.           | 135.000 mm. | Basic MDMT             | -17 °C       |
| Stress/Thk. Ratio        | -1.000      | Temperature Difference | 0 °C         |
| UCS-66(c) red.           | No          | Computed Min. Temp.    | -17 °C       |
|                          |             | Joint Efficiency E*    | 100 %        |

**Material treatment options: +N**

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### Internal Pressure Calculation Results:

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#### Internal Pressure Results for: SHELL-1

| Cylindrical Shell Design Information:             |    |                          |                         |    |                          |
|---------------------------------------------------|----|--------------------------|-------------------------|----|--------------------------|
| Design Pressure                                   | P  | 25.103 bars              | Int. Design Temperature |    | 85 °C                    |
| Material                                          |    | SA-516 70                | Ext. Design Temperature |    | 85 °C                    |
| Length                                            |    | 350.000 mm.              | External Pressure Chart |    | CS-2                     |
|                                                   |    |                          | UNS Number              |    | K02700                   |
| Int. Corr. All.                                   | c  | 1.5000 mm.               | Allowable Stress (ope)  | S  | 138.0 N./mm <sup>2</sup> |
| Ext. Corr. All.                                   | ce | 0.0000 mm.               | Allowable Stress (amb)  | Sa | 138.0 N./mm <sup>2</sup> |
| Inside Diameter                                   | Di | 1750.000 mm.             | Actual Stress           |    | 51.0 N./mm <sup>2</sup>  |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 46.0000 mm.              | Spec. Nominal Thk.      |    | 46.000 mm.               |
| Surface Area                                      |    | 20253.8 cm <sup>2</sup>  | SG of Contents          |    | 0.655                    |
| Element Volume                                    |    | 428350.7 cm <sup>3</sup> | Weight of Contents      |    | 280.4 Kg                 |
| Empty Weight                                      |    | 704.1 Kg                 | Operating Weight        |    | 984.5 Kg                 |
|                                                   |    |                          | Comp. All. Ope. Stress  |    | 145.7 N./mm <sup>2</sup> |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Material treatment options: +Q

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.103 \cdot 876.5) / (138.0 \cdot 1.0 - 0.6 \cdot 25.103)$$

$$= 16.1211 + 1.5000 = 17.6211 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.103 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 44.5) / (876.5 + 0.6 \cdot 44.5)$$

$$= 68 - 0 = 68 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 46.0) / (875.0 + 0.6 \cdot 46.0)$$

$$= 70 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (25.103 \cdot (876.5 + 0.6 \cdot 44.5)) / (1.0 \cdot 44.5)$$

$$= 50.953 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 2.561 %

#### Minimum Design Metal Temperature Results:

Govrn. thk, tg = 46.0, tr = 17.354, c = 1.5 mm., E\* = 1.0  
 Thickness Ratio =  $tr * E^* / (tg - c) = 0.39$ , Temp. Reduction = 53 °C

Min Metal Temp. w/o impact per UCS-66, Curve B      15 °C  
 Min Metal Temp. at Required thickness (UCS 66.1)      -38 °C

**Note:**  
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

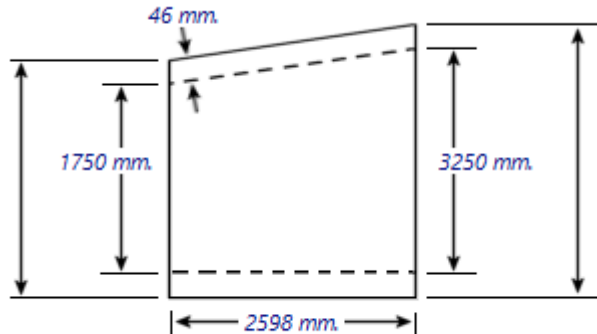
| MDMT Design Information: |            |                        |             |
|--------------------------|------------|------------------------|-------------|
| MDMT Curve               | B          | Press. at MDMT         | 27.000 bars |
| UCS-66(b) red.           | Yes        | Min. Des. Mtl. Temp.   | -35 °C      |
| Governing Thk.           | 46.000 mm. | Basic MDMT             | 15 °C       |
| Stress/Thk. Ratio        | 0.390      | Temperature Difference | 53 °C       |
| UCS-66(c) red.           | No         | Computed Min. Temp.    | -38 °C      |
|                          |            | Joint Efficiency E*    | 100 %       |

**Material treatment options: +Q**

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: SHELL-2

| Conical Head/Section Design Information: |           |                            |                         |                          |                          |
|------------------------------------------|-----------|----------------------------|-------------------------|--------------------------|--------------------------|
| Design Pressure                          | P         | 25.103 bars                | Int. Design Temperature | 85 °C                    |                          |
| Material                                 |           | SA-516 70                  | Ext. Design Temperature | 85 °C                    |                          |
| Length                                   |           | 2598.000 mm.               | External Pressure Chart | CS-2                     |                          |
|                                          |           |                            | UNS Number              | K02700                   |                          |
| Int. Corr. All.                          | c         | 1.5000 mm.                 | Allowable Stress (ope)  | S                        | 138.0 N./mm <sup>2</sup> |
| Ext. Corr. All.                          | ce        | 0.0000 mm.                 | Allowable Stress (amb)  | Sa                       | 138.0 N./mm <sup>2</sup> |
| Inside Diameter                          | Di        | 3250.000 mm.               | Actual Stress           | 107.5 N./mm <sup>2</sup> |                          |
| Spec. Min. Thk t <sub>e</sub> , or sa    |           | 46.0000 mm.                | Spec. Nominal Thk.      | 46.000 mm.               |                          |
| Surface Area                             |           | 221403.3 cm <sup>2</sup>   | SG of Contents          | 0.655                    |                          |
| Element Volume                           |           | 10067387.0 cm <sup>3</sup> | Weight of Contents      | 3213.6 Kg <sub>f</sub>   |                          |
| Empty Weight                             |           | 8578.6 Kg <sub>f</sub>     | Operating Weight        | 11792.2 Kg <sub>f</sub>  |                          |
|                                          |           |                            | Comp. All. Ope. Stress  | 147.0 N./mm <sup>2</sup> |                          |
|                                          |           |                            | To End Diameter         | 1750.000 mm.             |                          |
| Line of Support                          | Both Ends |                            |                         |                          |                          |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Material treatment options: +Q+IT

Required Thickness due to Internal Pressure [tr]:  

$$= (P \cdot D) / (2 \cdot \cos(a) \cdot (S \cdot E - 0.6 \cdot P))$$
 per Appendix 1-4 (e)  

$$= (25.103 \cdot 3253.4641) / (2 \cdot 0.866 \cdot (138.0 \cdot 1.0 - 0.6 \cdot 25.103))$$
  

$$= 34.5486 + 1.5000 = 36.0486 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.103 bars  

$$= (2 \cdot S \cdot E \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a))$$
 per App 1-4(e)  

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 44.5 \cdot 0.866) / (3253.464 + 1.2 \cdot 44.5 \cdot 0.866)$$
  

$$= 32 - 0 = 32 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a))$$
 per App 1-4(e)  

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 46.0 \cdot 0.866) / (3250.0 + 1.2 \cdot 46.0 \cdot 0.866)$$
  

$$= 33 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (D + 1.2 \cdot t \cdot \cos(a))) / (2 \cdot E \cdot \cos(a))$$
  

$$= (25.103 \cdot (3253.4641 + 1.2 \cdot 44.5 \cdot 0.866)) / (2 \cdot 1.0 \cdot 44.5 \cdot 0.866)$$
  

$$= 107.474 \text{ N./mm}^2$$



% Elongation per Table UG-79-1 ( $50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$ ) 2.551 %

#### Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +Q+IT

#### EXTERNAL PRESSURE RESULTS for: SHELL-2

##### Cone From 50 to 60 Ext. Chart: CS-2 at 85 °C

Elastic Modulus from Chart: CS-2 at 85 °C: 199943392 KPa.

| Results for Maximum Allowable Ext. Pressure |           |         |       | MAEP   |           |          |
|---------------------------------------------|-----------|---------|-------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t  | L/D    | Factor A  | Factor B |
| 38.538                                      | 3356.23   | 2014.97 | 87.09 | 0.6004 | 0.0029065 | 111.34   |

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 111.3395) / (3 \cdot 87.0886) = 17.0452 \text{ bars}$$

Note:

The cone thickness used in the calculation has been modified per UG-33(f),  $t_e = t \cdot \cos(\alpha)$ .

| Results for Required Thickness |           |         |        | Tca    |           |          |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 8.708                          | 3356.23   | 2014.97 | 385.43 | 0.6004 | 0.0002991 | 29.90    |

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 29.8986) / (3 \cdot 385.4296) = 1.0342 \text{ bars}$$

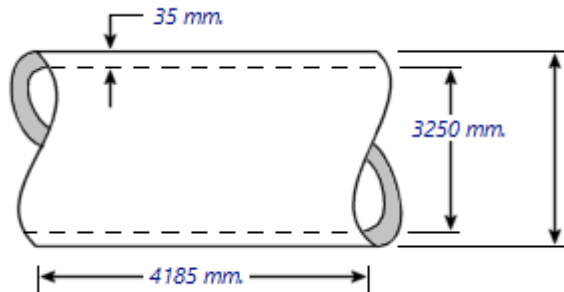
Note:

The cone thickness used in the calculation has been modified per UG-33(f),  $t_e = t \cdot \cos(\alpha)$ .

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



### Internal Pressure Results for: K-SHELL

| Cylindrical Shell Design Information:             |    |                            |                         |    |                          |
|---------------------------------------------------|----|----------------------------|-------------------------|----|--------------------------|
| Design Pressure                                   | P  | 25.103 bars                | Int. Design Temperature |    | 85 °C                    |
| Material                                          |    | SA-516 70                  | Ext. Design Temperature |    | 85 °C                    |
| Length                                            |    | 4185.000 mm.               | External Pressure Chart |    | CS-2                     |
|                                                   |    |                            | UNS Number              |    | K02700                   |
| Int. Corr. All.                                   | c  | 1.5000 mm.                 | Allowable Stress (ope)  | S  | 138.0 N./mm <sup>2</sup> |
| Ext. Corr. All.                                   | ce | 0.0000 mm.                 | Allowable Stress (amb)  | Sa | 138.0 N./mm <sup>2</sup> |
| Inside Diameter                                   | Di | 3250.000 mm.               | Actual Stress           |    | 123.4 N./mm <sup>2</sup> |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 35.0000 mm.                | Spec. Nominal Thk.      |    | 35.000 mm.               |
| Surface Area                                      |    | 436499.2 cm <sup>2</sup>   | SG of Contents          |    | 0.655                    |
| Element Volume                                    |    | 30900208.0 cm <sup>3</sup> | Weight of Contents      |    | 20227.3 Kg               |
| Empty Weight                                      |    | 11716.0 Kg                 | Operating Weight        |    | 31943.3 Kg               |
|                                                   |    |                            | Comp. All. Ope. Stress  |    | 130.4 N./mm <sup>2</sup> |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

### Material treatment options: +Q+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.103 \cdot 1626.4999) / (138.0 \cdot 1.0 - 0.6 \cdot 25.103)$$

$$= 29.9157 + 1.5000 = 31.4157 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.103 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 33.5) / (1626.4999 + 0.6 \cdot 33.5)$$

$$= 28 - 0 = 28 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 35.0) / (1625.0 + 0.6 \cdot 35.0)$$

$$= 29 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (25.103 \cdot (1626.4999 + 0.6 \cdot 33.5)) / (1.0 \cdot 33.5)$$

$$= 123.393 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 1.065 %

### Minimum Design Metal Temperature Results:



Note: This Element/Detail was specified as being Impact Tested.

Note:  
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +Q+IT

# EXTERNAL PRESSURE RESULTS for: K-SHELL

## Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 85 °C

Elastic Modulus from Chart: CS-2 at 85 °C: 199943392 KPa.

| Results for Maximum Allowable Ext. Pressure |           |         |       |        | MAEP      |          |
|---------------------------------------------|-----------|---------|-------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t  | L/D    | Factor A  | Factor B |
| 33.500                                      | 3320.00   | 4506.63 | 99.10 | 1.3574 | 0.0009834 | 84.78    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*84.7829)/(3*99.1045) = 11.4059 \text{ bars}$$

| Results for Required Thickness |           |         |        |        | Tca       |          |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.085                         | 3320.00   | 4506.63 | 274.72 | 1.3574 | 0.0002132 | 21.31    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*21.3109)/(3*274.7204) = 1.0342 \text{ bars}$$

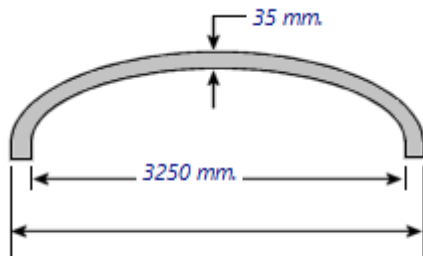
| Results for Maximum Stiffened Length |           |           |       |         | Slen      |          |
|--------------------------------------|-----------|-----------|-------|---------|-----------|----------|
| Tca                                  | Outer Dia | Slen      | Do/t  | L/D     | Factor A  | Factor B |
| 33.500                               | 3320.00   | 105418.47 | 99.10 | 31.7526 | 0.0001124 | 11.24    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*11.2363)/(3*99.1045) = 1.5116 \text{ bars}$$

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: SH. COVER

| Elliptical Head Design Information:               |    |                           |  |                         |    |                          |  |
|---------------------------------------------------|----|---------------------------|--|-------------------------|----|--------------------------|--|
| Design Pressure                                   | P  | 25.103 bars               |  | Int. Design Temperature |    | 85 °C                    |  |
| Material                                          |    | SA-516 70                 |  | Ext. Design Temperature |    | 85 °C                    |  |
| Length                                            |    | 50.800 mm.                |  | External Pressure Chart |    | CS-2                     |  |
|                                                   |    |                           |  | UNS Number              |    | K02700                   |  |
| Int. Corr. All.                                   | c  | 1.5000 mm.                |  | Allowable Stress (ope)  | S  | 138.0 N./mm <sup>2</sup> |  |
| Ext. Corr. All.                                   | ce | 0.0000 mm.                |  | Allowable Stress (amb)  | Sa | 138.0 N./mm <sup>2</sup> |  |
| Inside Diameter                                   | Di | 3250.000 mm.              |  | Actual Stress           |    | 122.0 N./mm <sup>2</sup> |  |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 35.0000 mm.               |  | Spec. Nominal Thk.      |    | 41.000 mm.               |  |
| Surface Area                                      |    | 125664.1 cm <sup>2</sup>  |  | SG of Contents          |    | 0.655                    |  |
| Element Volume                                    |    | 4915847.0 cm <sup>3</sup> |  | Weight of Contents      |    | 1571.8 Kg <sub>f</sub>   |  |
| Empty Weight                                      |    | 4286.1 Kg <sub>f</sub>    |  | Operating Weight        |    | 5857.9 Kg <sub>f</sub>   |  |
|                                                   |    |                           |  | Comp. All. Ope. Stress  |    | 130.4 N./mm <sup>2</sup> |  |
|                                                   |    |                           |  | Aspect Ratio            |    | 2.00                     |  |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 40.625 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 20.312 mm.

#### Material treatment options: +Q+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (25.103 \cdot 3253.0 \cdot 0.999) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 25.103)$$

$$= 29.6067 + 1.5000 = 31.1067 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.103 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 33.5) / (0.999 \cdot 3253.0 + 0.2 \cdot 33.5)$$

$$= 28 - 0 = 28 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 35.0) / (1.0 \cdot 3250.0 + 0.2 \cdot 35.0)$$

$$= 30 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (25.103 \cdot (0.999 \cdot 3253.0 + 0.2 \cdot 33.5)) / (2 \cdot 1.0 \cdot 33.5)$$

= 121.988 N./mm<sup>2</sup>

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (25.103 \cdot 1626.4999) / (138.0 \cdot 1.0 - 0.6 \cdot 25.103) + 1.5$$

$$= 31.416 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.103 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 39.5) / (1626.4999 + 0.6 \cdot 39.5)$$

$$= 33 - 0 = 33 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (3253.0 / (2 \cdot 814.0)))^2 / 6$$

$$= 0.998772$$

Percent Elong. per UCS-79, VIII-1-01-57 (75 \* t<sub>nom</sub> / R<sub>f</sub>) \* (1 - R<sub>f</sub> / R<sub>o</sub>) 5.366 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

**MDMT Calculations in the Knuckle Portion:**

Note: This Element/Detail was specified as being Impact Tested.

**MDMT Calculations in the Head Straight Flange:**

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

**Tolerance for Formed Heads per UG-81(a):**

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 40.625 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 20.312 mm.

**Material treatment options: +Q+IT**

**EXTERNAL PRESSURE RESULTS for: SH. COVER**

**Elliptical Head From 70 to 80 Ext. Chart: CS-2 at 85 °C**

Elastic Modulus from Chart: CS-2 at 85 °C: 199943392 KPa.

| Results for Maximum Allowable Ext. Pressure |           |       |           | MAEP     |
|---------------------------------------------|-----------|-------|-----------|----------|
| Tca                                         | Outer Dia | Do/t  | Factor A  | Factor B |
| 33.500                                      | 3320.00   | 99.10 | 0.0014014 | 95.00    |

$$MAEP = B / (K_0 \cdot Do / t) = 95.0 / (0.9 \cdot 99.1045) = 10.6503 \text{ bars}$$

| Results for Required Thickness |           |        |           | Tca      |
|--------------------------------|-----------|--------|-----------|----------|
| Tca                            | Outer Dia | Do/t   | Factor A  | Factor B |
| 8.596                          | 3320.00   | 386.22 | 0.0003596 | 35.95    |

$$MAEP = B / (K_0 \cdot Do / t) = 35.9533 / (0.9 \cdot 386.217) = 1.0343 \text{ bars}$$

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### Hydrostatic Test Pressure Results:

#### Exchanger Shell Side Hydrostatic Test Pressures:

|                 |                               |             |
|-----------------|-------------------------------|-------------|
| UG99(b)         | = 1.30 * M.A.W.P. * Sa/S      | 35.099 bars |
| UG99b(b)(35)    | = 1.30 * Design Pres * Sa/S   | 32.500 bars |
| UG99(c)         | = 1.30 * M.A.P. - Head(Hyd)   | 35.319 bars |
| UG100           | = 1.10 * M.A.W.P. * Sa/S      | 29.699 bars |
| PED             | = max(1.43*DP, 1.25*DP*ratio) | 35.750 bars |
| PED&UG99(b)(35) | = max(1.43*DP, 1.3*DP*ratio)  | 35.750 bars |
| App 27-4        | = M.A.W.P.                    | 27.000 bars |

#### Exchanger Channel Side Hydrostatic Test Pressures:

|                 |                               |              |
|-----------------|-------------------------------|--------------|
| UG99(b)         | = 1.30 * M.A.W.P. * Sa/S      | 250.212 bars |
| UG99b(b)(35)    | = 1.30 * Design Pres * Sa/S   | 248.300 bars |
| UG99(c)         | = 1.30 * M.A.P. - Head(Hyd)   | 252.999 bars |
| UG100           | = 1.10 * M.A.W.P. * Sa/S      | 211.718 bars |
| PED             | = max(1.43*DP, 1.25*DP*ratio) | 273.130 bars |
| PED&UG99(b)(35) | = max(1.43*DP, 1.3*DP*ratio)  | 273.130 bars |
| App 27-4        | = M.A.W.P.                    | 192.471 bars |

#### Horizontal Test performed per: UG-99b

*Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.*

#### Stresses on Elements due to Test Pressure (N./mm<sup>2</sup> & bars):

| From To   | Stress | Allowable | Ratio | Pressure |
|-----------|--------|-----------|-------|----------|
| CH. COVER | 172.6  | 234.0     | 0.738 | 250.38   |
| CH. COVER | 179.4  | 234.0     | 0.767 | 250.38   |
| SHELL-1   | 71.6   | 234.0     | 0.306 | 35.27    |
| SHELL-2   | 151.6  | 234.0     | 0.648 | 35.42    |
| K-SHELL   | 174.1  | 234.0     | 0.744 | 35.42    |
| SH. COVER | 172.1  | 234.0     | 0.736 | 35.42    |

#### Stress ratios for Nozzle and Pad Materials (N./mm<sup>2</sup>):

| Description     | Pad/Nozzle | Ambient | Operating | Ratio |
|-----------------|------------|---------|-----------|-------|
| CH. IN NOZ A1   | Nozzle     | 138.00  | 138.00    | 1.000 |
| CH. OUT NOZ B1  | Nozzle     | 138.00  | 138.00    | 1.000 |
| MANHOLE M1      | Nozzle     | 138.00  | 138.00    | 1.000 |
| LIQ OUT NOZ B3  | Nozzle     | 138.00  | 138.00    | 1.000 |
| SS IN NOZ. A2   | Nozzle     | 138.00  | 138.00    | 1.000 |
| SS OUT NOZ. B2  | Nozzle     | 138.00  | 138.00    | 1.000 |
| VENT NOZ V1     | Nozzle     | 138.00  | 138.00    | 1.000 |
| PR. IN P1       | Nozzle     | 138.00  | 138.00    | 1.000 |
| PR. IN NOZ. P2  | Nozzle     | 138.00  | 138.00    | 1.000 |
| SAFETY VALVE S1 | Nozzle     | 138.00  | 138.00    | 1.000 |
| LEVEL GUAGE L1  | Nozzle     | 138.00  | 138.00    | 1.000 |
| LEVEL GUAGE L2  | Nozzle     | 138.00  | 138.00    | 1.000 |
| LEVEL GUAGE L3  | Nozzle     | 138.00  | 138.00    | 1.000 |
| LEVEL GUAGE L4  | Nozzle     | 138.00  | 138.00    | 1.000 |
| TEMP IN. T1     | Nozzle     | 138.00  | 138.00    | 1.000 |

Minimum 1.000

#### Stress ratios for Pressurized Vessel Elements (N./mm<sup>2</sup>):

| Description | Ambient | Operating | Ratio |
|-------------|---------|-----------|-------|
| CH. COVER   | 138.00  | 138.00    | 1.000 |
| CH. COVER   | 138.00  | 138.00    | 1.000 |
| SH. FLG.    | 138.00  | 138.00    | 1.000 |



|           |        |        |       |
|-----------|--------|--------|-------|
| SHELL-1   | 138.00 | 138.00 | 1.000 |
| SHELL-2   | 138.00 | 138.00 | 1.000 |
| K-SHELL   | 138.00 | 138.00 | 1.000 |
| SH. COVER | 138.00 | 138.00 | 1.000 |

Minimum 1.000

**Stress ratios for Exchanger Materials (N./mm<sup>2</sup>):**

| Description        | Ambient | Operating | Ratio |
|--------------------|---------|-----------|-------|
| Tube Material      | 118.00  | 118.00    | 1.000 |
| Tubesheet Material | 138.00  | 138.00    | 1.000 |

Minimum 1.000

**Hoop Stress in Nozzle Wall during Pressure Test (N./mm<sup>2</sup>):**

| Description     | Ambient | Operating | Ratio |
|-----------------|---------|-----------|-------|
| CH. IN NOZ A1   | 147.17  | 179.40    | 0.820 |
| CH. OUT NOZ B1  | 146.93  | 179.40    | 0.819 |
| MANHOLE M1      | 146.93  | 179.40    | 0.819 |
| LIQ OUT NOZ B3  | 14.94   | 179.40    | 0.083 |
| SS IN NOZ. A2   | 34.17   | 179.40    | 0.190 |
| SS OUT NOZ. B2  | 20.82   | 179.40    | 0.116 |
| VENT NOZ V1     | 13.34   | 179.40    | 0.074 |
| PR. IN P1       | 13.34   | 179.40    | 0.074 |
| PR. IN NOZ. P2  | 13.34   | 179.40    | 0.074 |
| SAFETY VALVE S1 | 13.34   | 179.40    | 0.074 |
| LEVEL GUAGE L1  | 13.34   | 179.40    | 0.074 |
| LEVEL GUAGE L2  | 13.34   | 179.40    | 0.074 |
| LEVEL GUAGE L3  | 13.34   | 179.40    | 0.074 |
| LEVEL GUAGE L4  | 13.34   | 179.40    | 0.074 |
| TEMP IN. T1     | 13.34   | 179.40    | 0.074 |

Elements Suitable for Test Pressure.

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**Input Echo, Tubesheet Number 1, Description: U-TS**

**Shell Data:**

**Main Shell Description: SHELL-1**

|                                         |        |              |
|-----------------------------------------|--------|--------------|
| Shell Test Pressure                     |        | 35.10 bars   |
| Shell Thickness                         | ts     | 46.0000 mm.  |
| Shell Internal Corrosion Allowance      | cas    | 0.0000 mm.   |
| Shell External Corrosion Allowance      | casext | 0.0000 mm.   |
| Inside Diameter of Shell                | Ds     | 1750.000 mm. |
| Shell Temperature for Internal Pressure | Ts     | 21.11 °C     |
| Shell Material                          |        | SA-516 70    |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                       |    |                           |
|---------------------------------------|----|---------------------------|
| Shell Material UNS Number             |    | K02700                    |
| Shell Allowable Stress at Temperature | Ss | 138.00 N./mm <sup>2</sup> |
| Shell Allowable Stress at Ambient     |    | 138.00 N./mm <sup>2</sup> |

**Channel Data:**

**Channel Description: CH. COVER**

|                                      |        |              |
|--------------------------------------|--------|--------------|
| Channel Type:                        |        | Cylinder     |
| Channel Test Pressure                |        | 250.21 bars  |
| Channel Thickness                    | tc     | 135.0000 mm. |
| Channel Internal Corrosion Allowance | cac    | 0.0000 mm.   |
| Channel External Corrosion Allowance | cacext | 0.0000 mm.   |
| Inside Diameter of Channel           | Dc     | 1750.000 mm. |
| Channel Design Temperature           | TEMPC  | 21.11 °C     |
| Channel Flange Matl                  |        | SA-516 70    |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                         |    |                           |
|-----------------------------------------|----|---------------------------|
| Channel Material UNS Number             |    | K02700                    |
| Channel Allowable Stress at Temperature | Sc | 138.00 N./mm <sup>2</sup> |
| Channel Allowable Stress at Ambient     |    | 138.00 N./mm <sup>2</sup> |

**Tube Data:**

|                                                  |                   |                           |
|--------------------------------------------------|-------------------|---------------------------|
| Number of Tube Holes                             | Nt                | 2332                      |
| Tube Wall Thickness                              | et                | 2.7700 mm.                |
| Tube Outside Diameter                            | D                 | 25.4000 mm.               |
| Total Straight Tube Length                       | Lt                | 5400.00 mm.               |
| Straight Tube Length (bet. inner tubsht faces) L |                   | 4960.00 mm.               |
| Design Temperature of the Tubes                  |                   | 21.11 °C                  |
| Tube Material                                    |                   | SA-334 6                  |
| Tube Material UNS Number                         |                   | K03006                    |
| Is this a Welded Tube                            |                   | No                        |
| Tube Material Specification used                 | Smls. & wld. tube |                           |
| Tube Allowable Stress at Temperature             |                   | 118.00 N./mm <sup>2</sup> |
| Tube Allowable Stress At Ambient                 |                   | 118.00 N./mm <sup>2</sup> |
| Tube Yield Stress At design Temperature          | Syt               | 240.00 N./mm <sup>2</sup> |
| Tube Pitch (Center to Center Spacing)            | P                 | 31.7500 mm.               |
| Tube Layout Pattern                              |                   | Triangular                |
| Fillet Weld Leg                                  | af                | 2.7700 mm.                |
| Groove Weld Leg                                  | ag                | 2.7700 mm.                |
| Tube-Tubesheet Joint Weld Type                   |                   | Full Strength             |
| Method for Tube-Tubesheet Jt. Allowable          |                   | UW-20.3                   |
| Is the Tube Inset                                |                   | No                        |
| Radius to Outermost Tube Hole Center             | ro                | 845.800 mm.               |
| Largest Center-to-Center Tube Distance           | Ul                | 76.2000 mm.               |
| Length of Expanded Portion of Tube               | ltx               | 0.0000 mm.                |
| Tube-side pass partition groove depth            | hg                | 6.0000 mm.                |

**Tubesheet Data:**

**Tubesheet TYPE: U-tube, Integral with Channel, Conf. e**

|                                    |   |            |
|------------------------------------|---|------------|
| Tubesheet Design Metal Temperature | T | 21.11 °C   |
| Tubesheet Material                 |   | SA-350 LF2 |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                           |      |                           |
|-------------------------------------------|------|---------------------------|
| Tubesheet Material UNS Number             |      | K03011                    |
| Tubesheet Allowable Stress at Temperature | S    | 138.00 N./mm <sup>2</sup> |
| Tubesheet Allowable Stress at Ambient     | Tt   | 138.00 N./mm <sup>2</sup> |
| Thickness of Tubesheet                    | h    | 440.0000 mm.              |
| Tubesheet Corr. Allowance (Shell side)    | Cats | 0.0000 mm.                |
| Tubesheet Corr. Allowance (Channel side)  | Catc | 0.0000 mm.                |
| Tubesheet Outside Diameter                | A    | 2030.000 mm.              |

**Additional Data for Stepped Tubesheets:**

|                                                 |     |             |
|-------------------------------------------------|-----|-------------|
| Is the Tubesheet Stepped?                       |     | Yes         |
| Is the Tubesheet Flat on Tubeside?              |     | Yes         |
| Is the Tubesheet Flat on Shellside?             |     | No          |
| Step 1 Diameter on the Shellside                | ds1 | 1632.00 mm. |
| Step 1 Depth on the Shellside                   | hs1 | 6.00 mm.    |
| Step 2 Diameter on the Shellside                | ds2 | 1597.00 mm. |
| Step 2 Depth on the Shellside                   | hs2 | 6.00 mm.    |
| Calculated Tubesheet Diameter per Div 2, 4.18.4 |     | 2030.00 mm. |

Note: Tubesheet diameter is now: 2030.000 mm. Div 2, 4.18.4.

|                                      |    |                             |
|--------------------------------------|----|-----------------------------|
| Area of the Untubed Lanes            | AL | 1308.3 cm <sup>2</sup>      |
| Junction Stress Reduction option     |    | Perform Plastic Calculation |
| Perform Differential Pressure Design |    | No                          |
| Run Multiple Load Cases              |    | Yes                         |

**Additional Data for Gasketed Tubesheets:**

|                                |     |                          |
|--------------------------------|-----|--------------------------|
| Tubesheet Gasket on which Side |     | Shell                    |
| Flange Outside Diameter        | A   | 2030.000 mm.             |
| Flange Inside Diameter         | B   | 1750.000 mm.             |
| Flange Face Outside Diameter   | Fod | 1791.000 mm.             |
| Flange Face Inside Diameter    | Fid | 1750.000 mm.             |
| Gasket Outside Diameter        | Go  | 1788.000 mm.             |
| Gasket Inside Diameter         | Gi  | 1779.000 mm.             |
| Small end Hub thk.             | g0  | 46.0000 mm.              |
| Large end Hub thk.             | g1  | 50.0000 mm.              |
| Gasket Factor,                 | m   | 3.75                     |
| Gasket Design Seating Stress   | y   | 62.06 N./mm <sup>2</sup> |
| Flange Facing Sketch           |     | Code Sketch 1a           |
| Gasket Thickness               | tg  | 4.5000 mm.               |
| Full face Gasket Flange Option |     | Program Selects          |

**Bolting Information:**

|                                       |    |                           |
|---------------------------------------|----|---------------------------|
| Diameter of Bolt Circle               | C  | 1945.000 mm.              |
| Nominal Bolt Diameter                 | dB | 36.0000 mm.               |
| Thread Series                         |    | TEMA Metric Thread        |
| Number of Bolts                       | n  | 56                        |
| Bolt Material                         |    | SA-320 L7                 |
| Bolt Allowable Stress At Temperature  | Sb | 172.00 N./mm <sup>2</sup> |
| Bolt Allowable Stress At Ambient      | Sa | 172.00 N./mm <sup>2</sup> |
| Weld between Flange and Shell/Channel |    | 0.0000 mm.                |

Note: The full bolt load Sa\*Ab, was used for all load cases.

|                                            |    |              |
|--------------------------------------------|----|--------------|
| Tubesheet Integral with                    |    | Channel      |
| Tubesheet Extended as Flange               |    | Yes          |
| Thickness of Extended Portion of Tubesheet | Tf | 406.0000 mm. |
| Is Bolt Load Transferred to the Tubesheet  |    | Yes          |

A schematic diagram of a vertical pipe system. The total height of the pipe is labeled as 2030mm. The distance from the top to the first horizontal branch is 135mm. The distance between the first and second horizontal branches is 1945mm. The distance between the second and third horizontal branches is 1750mm. The pressure in the main vertical pipe is 250.2bars. The pressure in the horizontal branches is 35.1bars. The diameter of the horizontal branches is 440mm. The diagram also shows a vertical pipe section labeled G<sub>s</sub> on the right side.

| TCA                                                            | ODCA  | SLN     | D/T  | L/D     | Factor A  | B      |
|----------------------------------------------------------------|-------|---------|------|---------|-----------|--------|
| 2.7700                                                         | 25.40 | 5805.80 | 9.17 | 50.0000 | 0.0130823 | 122.73 |
| $EMAWP = (2.167 / (D/T) - 0.0833) * B = 187.7951 \text{ bars}$ |       |         |      |         |           |        |

Results for Req'd Thickness for Ext. Pressure (Tca):

| TCA    | ODCA  | SLEN    | D/T   | L/D     | Factor A  | B     |
|--------|-------|---------|-------|---------|-----------|-------|
| 0.7764 | 25.40 | 5805.80 | 32.72 | 50.0000 | 0.0010277 | 86.14 |

EMAWP =  $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 86.1358) / (3 \cdot 32.7161) = 35.1023$  bars

**Summary of Tube Required Thickness Results:**

|                                                   |             |
|---------------------------------------------------|-------------|
| Total Required Thickness including Corrosion all. | 2.4826 mm.  |
| Allowable Internal Pressure at Corroded thickness | 281.95 bars |
| Required Internal Design Pressure                 | 250.21 bars |
| Allowable External Pressure at Corroded thickness | 187.80 bars |
| Required External Design Pressure                 | 35.10 bars  |
| Required Thickness due to Shell Side pressure     | 0.7764 mm.  |

**Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)**

**Intermediate Calculations For Tubesheets Extended As Flanges:**

**ASME Code, Section VIII Division 1, 2025**

|                           |                             |              |
|---------------------------|-----------------------------|--------------|
| Gasket Contact Width,     | $N = (G_{oc} - G_{ic}) / 2$ | 4.500 mm.    |
| Basic Gasket Width,       | $b_0 = N / 2$               | 2.250 mm.    |
| Effective Gasket Width,   | $b = b_0$                   | 2.250 mm.    |
| Gasket Reaction Diameter, | $G = (G_o + G_i) / 2$       | 1783.500 mm. |

**Warning:** Available bolt area, 413.288 cm<sup>2</sup> for the flange is < required 529.144 cm<sup>2</sup>

**Note that for the pressure test case, ASME VIII-2 paragraph 4.18.13 (c)(9) does not require that the bolt area requirement be met.**

**Results for ASME U-tube Tubesheet Calculations for Configuration e:**

**Per Edition 2025, Original Thickness :**

**ASME Section VIII-2 Part 4.18.7.4 Step 1:**

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 845.8 + 25.4$$

$$= 1717.000 \text{ mm.}$$

Basic ligament efficiency for shear [ $\mu$ ]:

$$= (p - dt) / p$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.200$$

**ASME Section VIII-2 Part 4.18.7.4 Step 2:**

Ratio [ $\rho_s$ ]:

$$= G_s / D_o \text{ (Configurations d, e, f)}$$

$$= 1783.5 / 1717.0$$

$$= 1.0387$$

Ratio [ $\rho_c$ ]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1750.0 / 1717.0$$

$$= 1.0192$$

**ASME Section VIII-2 Part 4.18.7.4 Step 3:**

Ratio [ $\rho$ ]:

$$= l_{tx} / h$$

$$= 0 / 440.0$$

$$= 0.000 \text{ ( must be } 0 \leq \rho \leq 1 \text{ )}$$

Effective tube hole Diameter [ $d^*$ ]:

$$= \max( dt - 2tt( E_t T / E ) \cdot ( S_t T / S ) \cdot ( \rho ); dt - 2tt )$$

$$= \max( 25.4 - 2 \cdot 2.77 ( 0.20270E+09 / 2.0270E+09 ) \cdot$$

$$= (117/138) \cdot (0); 25.4 - 2 \cdot 2.77$$

$$= 25.4000 \text{ mm.}$$

Effective tube pitch [p\*]:

$$= p / \sqrt{(1 - 4 \cdot \min(A; 4 \cdot Do \cdot p)) / (\pi \cdot Do^2)}$$

$$= 31.75 / \sqrt{(1 - 4 \cdot \min(1308.35; 4 \cdot 1717.0 \cdot 31.75)) / (\pi \cdot 1717.0^2)}$$

$$= 32.6869 \text{ mm.}$$

Effective ligament efficiency for bending [ $\mu^*$ ]:

$$= (p^* - d^*) / p^*$$

$$= (32.6869 - 25.4) / 32.6869$$

$$= 0.2229$$

$$h/p = 13.858267; \mu^* = 0.222931; E^* = 37074484 \text{ KPa.}$$

$$E^*/E = 0.182902; \nu^* = 0.427388$$

Note: As h/p (13.858) is > 2, data values for h/p = 2 were used.

#### ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

$\beta_s$ ,  $k_s$ ,  $\lambda_s$ ,  $\delta_s$  and  $\omega_s$  are all = 0

Channel coefficient [ $\beta_c$ ]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1750.0 + 135.0)135.0}$$

$$= 0.0036 \text{ 1/mm.}$$

Channel coefficient [ $k_c$ ]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.004 \cdot 2013223 \cdot 135.0^3 / (6(1 - 0.3^2))$$

$$= 3269140736.0000 \text{ bars} \cdot \text{mm.}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [ $\lambda_c$ ]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1750.0 + 135.0 / \cos(0)) \cdot 0.0036 / 440.0^3 \cdot (1.0 + 440.0 \cdot 0.004 \cdot (1.0 + 1.0) / 2 + (440.0 \cdot 0.004)^2 / 2 \cdot 1.0)$$

$$= 1667870.4 \text{ bars}$$

Channel coefficient [ $\delta_c$ ]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1750.0^2 / (4 \cdot 0.20132E+09 \cdot 135.0) \cdot (1 - 1750.0 \cdot 0.3 / (2(1750.0 + 135.0)))$$

$$= 0.24246E-01 \text{ mm./N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ $\omega_c$ ]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.019 \cdot 0.32691E+1 [0.024246(1.0 + 0.004 \cdot 1.0 \cdot 440.0) - 0(1.0/440.0 + 1.0 \cdot 0.004/2)]$$

$$= 75275.6797 \text{ mm.}^2$$

#### ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / Do$$

$$= 2030.0 / 1717.0$$

$$= 1.1823$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 37074484 \cdot (1667870 + 0.20270E+09 \cdot \ln(1.18))$$

$$= 3.1003$$

#### ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:  

$$= Do^2/16 \cdot [ (ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt ]$$
  

$$= 1717.0^2/16 \cdot [ (1.039 - 1) \cdot (1.039^2 + 1) \cdot 35.099 -$$
  

$$(1.019 - 1) \cdot (1.019^2 + 1) \cdot 250.212 ]$$
  

$$= -1285803.7500 \text{ bars} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M\*]:  

$$= Mts + (wc \cdot Pt) + W* \cdot (C - Gs) / (2 \cdot \pi \cdot Do)$$
  

$$= -1285803 + (18834884) + 724845 \cdot (1945.0 - 1783.5) / (2 \cdot \pi \cdot 1717.0)$$
  

$$= 18613152.0000 \text{ bars} \cdot \text{mm}^2$$

#### ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:  

$$= (M*) - Do^2/32 \cdot F(Ps - Pt) / (1 + F)$$
  

$$= (18613152) - 1717.0^2/32 \cdot 3.1 (35.1 - 250.21) / (1 + 3.1)$$
  

$$= 19524044.0000 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:  

$$= Mp + Do^2/64 \cdot (3 + \nu*) \cdot (Ps - Pt)$$
  

$$= 19524044 + 1717.0^2/64 \cdot (3 + 0.427) \cdot (35.1 - 250.21)$$
  

$$= -14437689.0000 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:  

$$= \max(|Mp|; |Mo|)$$
  

$$= \max(|19524044|; |-0.1E+08|)$$
  

$$= 19524044.0000 \text{ bars} \cdot \text{mm}^2$$

#### ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:  

$$= 6 \cdot M / (\mu* (h - hg')^2)$$
  

$$= 6 \cdot 19524044 / (0.2229 (440.0 - 6.0)^2)$$
  

$$= 278.9950 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [ $\sigma_{All}$ ]:  

$$= 2 \cdot \beta T \cdot Sy$$
  

$$= 2 \cdot 0.95 \cdot 250.0$$
  

$$= 474.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [ $\sigma$ ]:  

$$= 6 \cdot M / (\mu* (h - hg')^2)$$
  

$$= 6 \cdot 19156508 / (0.2229 (335.4719 - 6.0)^2)$$
  

$$= 474.9913 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:  

$$= h + Cats + Catc$$
  

$$= 335.4719 + 0 + 0$$
  

$$= 335.4719 \text{ mm.}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:  

$$= \max(HreqB; HreqS)$$
  

$$= \max(335.4719; 243.0109)$$
  

$$= 335.4719 \text{ mm.}$$

#### ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [|Ps - Pt|]:  

$$= |Ps - Pt|$$
  

$$= |35.1 - 250.21|$$
  

$$= 215.113 \text{ bars}$$

Shear Stress check [ $\tau_{limit}$ ]:  

$$= 4 \cdot \mu \cdot h/Do \cdot 0.8 \cdot \beta T \cdot Sy$$
  

$$= 4 \cdot 0.2 \cdot 440.0/1717.0 \cdot 0.8 \cdot 0.95 \cdot 249.995$$
  

$$= 38.951 \text{ N./mm}^2$$

|Ps - Pt| is  $\leq$  the Shear Stress limit ( $\tau_{limit}$ ). Therefore, the required thickness

for shear is for information only.

Minimum required thickness for Shear [HreqS]:  
 $= 1/(4 \cdot \mu) \cdot (Do / (0.8 \cdot \beta T \cdot Sy)) \cdot |Ps - Pt| + Cats + Catc$   
 $= 1/(4 \cdot 0.2) \cdot (1717.0/189.997) \cdot |35.1 - 250.21| + 0$   
 $= 243.0109 \text{ mm.}$

#### ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 135.000 mm.  
 Min. Channel len. adjacent to the tubesheet is: 874.900mm.

Tube Metal Area [At]:  
 $= \pi (do - t)t$   
 $= \pi (25.4 - 2.77)2.77$   
 $= 1.969 \text{ cm}^2$

#### Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:  
 $= \pi \cdot t (do - t)S$   
 $= \pi \cdot 2.77 (25.4 - 2.77)227.996$   
 $= 4578.387 \text{ Kgf}$

Groove Weld Strength [Fg]:  
 $= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag)Sw \text{ (but not } > Ft)$   
 $= 0.85 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77)228.0$   
 $= 4578.3872 \text{ Kgf}$

Weld Strength Factor [fw]:  
 $= S / Sw \text{ where } Sw = \min(S \text{ and } St)$   
 $= 227.996/227.996$   
 $= 1.000$

Design Strength Ratio [fd]:  
 $= 1 \text{ for full strength welds}$

Fillet Weld Strength [Ff]:  
 $= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af)Sw \text{ (but not } > Ft)$   
 $= 0.55 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77)228.0$   
 $= 3032.8518 \text{ Kgf}$

Minimum Length of Weld [ar]:  
 $= 2 \cdot \sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t)fw \cdot fd) - 0.75 \cdot do}$   
 $= 2 \cdot \sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77)1.0 \cdot 1.0) - 0.75 \cdot 25.4}$   
 $= 3.372 \text{ mm.}$

Minimum Required Fillet Weld Leg [afr]: 1.6859 mm.  
 Minimum Required Groove Weld Leg [agr]: 1.6859 mm.

Tube-Tubesheet Jt allowable, 4578.39 is  $\geq$  tube strength 4578.39 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

#### Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

| Stress Description     | Actual       | Allowable                | Pass/Fail |
|------------------------|--------------|--------------------------|-----------|
| Tubesheet bend. stress | 279.0 $\leq$ | 475.0 N./mm <sup>2</sup> | Ok        |
| Tubesheet shear stress | 104.9 $\leq$ | 190.0 N./mm <sup>2</sup> | Ok        |

#### Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

| Thickness (mm.)                  | Required | Actual  | P/F |
|----------------------------------|----------|---------|-----|
| Tubesheet Thickness :            | 335.472  | 440.000 | Ok  |
| Tube-Tubesheet Fillet Weld Leg : | 1.686    | 2.770   | Ok  |

Tube-Tubesheet Groove Weld Leg : 1.686 2.770 Ok

### U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

#### Results for 3 Load Cases:

| Case# | --Reqd. Thk. + CA<br>Tbsht Extnsn | ----       | Tubesheet<br>Bend | Stresses<br>Allwd | Shear<br>Allwd | Case<br>Type | Pass/<br>Fail |
|-------|-----------------------------------|------------|-------------------|-------------------|----------------|--------------|---------------|
| D1uc  | 355.426                           | 57.416     | 308               | 475               | 122            | Ps+Pt D1     | Ok            |
| D2uc  | 138.600                           | ...        | 50                | 475               | 17             | Ps+Pt D2     | Ok            |
| D3uc  | 335.472                           | ...        | 279               | 475               | 105            | Ps+Pt D3     | Ok            |
| Max:  | 355.4261                          | 57.416 mm. |                   | 0.649             | 0.642          | (Str. Ratio) |               |

#### Load Case Definitions:

[Ps & Pt]:  
 Shell-side and Tube-side Design or Operating Pressures  
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the  
 Load Case Tables

[c]:  
 With or Without Corrosion Allowance

[D1, D2, D3]:  
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:  
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

#### Shell and Channel Stress Summary:

| Case# | -- Integral Cyl. Req'd. Thk.<br>Shell Channel | Shell<br>Stress | Stress<br>Allwd | Channel<br>Stress | Stress<br>Allwd | Pass<br>Fail |
|-------|-----------------------------------------------|-----------------|-----------------|-------------------|-----------------|--------------|
| D1uc  | ...                                           | ...             | ...             | ...               | 370             | Ok           |
| D2uc  | ...                                           | ...             | ...             | ...               | 370             | Ok           |
| D3uc  | ...                                           | ...             | ...             | ...               | 370             | Ok           |
| Max   | ...                                           | ...             | ...             | ...               |                 |              |

Note:  
 Because there was no net pressure differential across the tubesheet due to  
 the specified pressures psd,min and ptd,min; load case D4 was not processed as  
 the resulting tubesheet bending stress would be zero.

#### Summary of Thickness Comparisons for 3 Load Cases:

| Thickness (mm.)                         | Required | Actual  | P/F |
|-----------------------------------------|----------|---------|-----|
| Tubesheet thickness :                   | 355.426  | 440.000 | Ok  |
| Tubesheet thickness flanged extension : | 57.416   | 406.000 | Ok  |
| Tube thickness :                        | 2.483    | 2.770   | Ok  |
| Tube-tubesheet fillet weld leg :        | 1.686    | 2.770   | Ok  |
| Tube-tubesheet groove weld leg :        | 1.686    | 2.770   | Ok  |

Given Shell Thickness: 46.0000 mm.  
 Given Channel Thickness: 135.0000 mm.

Min Channel length of thk, (135.000) adj. to tubesheet: 874.900 mm.

Note: This is a full strength Tube to Tubesheet Joint.



## External Pressure Calculation Results Summary:

### ASME Code, Section VIII Division 1, 2025

#### External Pressure Calculations:

| From | To | Section Length<br>mm. | Outside Diameter<br>mm. | Corroded Thickness<br>mm. | Factor A   | Factor B<br>N./mm <sup>2</sup> |
|------|----|-----------------------|-------------------------|---------------------------|------------|--------------------------------|
| 10   | 20 | No Calc               | 1882                    | 64.5                      | 0.008568   | 122.731                        |
| 20   | 30 | 341.667               | 2020                    | 133.5                     | 0.096839   | 122.731                        |
| 30   | 40 | No Calc               | ...                     | 93.5                      | No Calc    | No Calc                        |
| 40   | 50 | 350                   | 1842                    | 44.5                      | 0.039641   | 122.731                        |
| 50   | 60 | 2014.97               | 3356.23                 | 44.5                      | 0.0029065  | 111.34                         |
| 60   | 70 | 4506.63               | 3320                    | 33.5                      | 0.00098338 | 84.7829                        |
| 70   | 80 | No Calc               | 3320                    | 33.5                      | 0.0014014  | 94.9996                        |

#### External Pressure Calculations:

| From | To | External Actual T.<br>mm. | External Required T.<br>mm. | External Design Pressure<br>bars | External M.A.W.P.<br>bars |
|------|----|---------------------------|-----------------------------|----------------------------------|---------------------------|
| 10   | 20 | 66                        | No Calc                     | ...                              | 84.12                     |
| 20   | 30 | 135                       | No Calc                     | ...                              | 108.143                   |
| 30   | 40 | 95                        | 87.1474                     | ...                              | No Calc                   |
| 40   | 50 | 46                        | No Calc                     | ...                              | 39.531                    |
| 50   | 60 | 46                        | 11.5549                     | 1.03419                          | 17.0452                   |
| 60   | 70 | 35                        | 13.585                      | 1.03419                          | 11.4059                   |
| 70   | 80 | 35                        | 10.0962                     | 1.03419                          | 10.6503                   |

Minimum

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#### External Pressure Calculations:

| From | To | Actual Length<br>Bet. Stiffeners<br>mm. | Allowable Length<br>Bet. Stiffeners<br>mm. | Ring Inertia<br>Required<br>cm <sup>4</sup> | Ring Inertia<br>Available<br>cm <sup>4</sup> |
|------|----|-----------------------------------------|--------------------------------------------|---------------------------------------------|----------------------------------------------|
| 10   | 20 | No Calc                                 | No Calc                                    | No Calc                                     | No Calc                                      |
| 20   | 30 | 341.667                                 | No Calc                                    | No Calc                                     | No Calc                                      |
| 30   | 40 | No Calc                                 | No Calc                                    | No Calc                                     | No Calc                                      |
| 40   | 50 | 350                                     | No Calc                                    | No Calc                                     | No Calc                                      |
| 50   | 60 | 2014.97                                 | 2014.97                                    | No Calc                                     | No Calc                                      |
| 60   | 70 | 4506.63                                 | 105418                                     | No Calc                                     | No Calc                                      |
| 70   | 80 | No Calc                                 | No Calc                                    | No Calc                                     | No Calc                                      |

Elements Suitable for External Pressure.

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### Element and Detail Weights:

| From  | To | Element<br>Metal Wgt.<br>kg. | Element<br>ID Volume<br>cm <sup>3</sup> | Corroded<br>Metal Wgt.<br>kg. | Corroded<br>ID Volume<br>cm <sup>3</sup> | Extra due<br>Misc %<br>kg. |
|-------|----|------------------------------|-----------------------------------------|-------------------------------|------------------------------------------|----------------------------|
| 10    | 20 | 3175.12                      | 1403332                                 | 3119.1                        | 1410561                                  | 317.512                    |
| 20    | 30 | 309.81                       | 120286                                  | 306.611                       | 120698                                   | 30.981                     |
| 30    | 40 | 766.718                      | 409797                                  | 755.843                       | 410375                                   | 76.6718                    |
| 40    | 50 | 704.064                      | 428351                                  | 681.674                       | 431240                                   | 70.4064                    |
| 50    | 60 | 8578.6                       | 10067387                                | 8304.5                        | 10102760                                 | 857.86                     |
| 60    | 70 | 11716                        | 30900208                                | 11219                         | 30964342                                 | 1171.6                     |
| 70    | 80 | 4286.07                      | 4915847                                 | 4129.27                       | 4929082                                  | 428.607                    |
| Total |    | 29536                        | 48245208.00                             | 28515                         | 48369060.00                              | 2953                       |



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

### Weight of Details:

| From | Type | Weight of<br>Detail<br>kg. | X Offset,<br>Dtl. Cent.<br>mm. | Y Offset,<br>Dtl. Cent.<br>mm. | Z Offset<br>Dtl. Cent.<br>mm. | Description     |
|------|------|----------------------------|--------------------------------|--------------------------------|-------------------------------|-----------------|
| 10   | Liqd | 918.622                    | -291.667                       | ...                            | ...                           | Liquid 10       |
| 10   | Insl | 18.7032                    | -437.5                         | ...                            | ...                           | Ins: 20         |
| 10   | Nozl | 961.737                    | -636.887                       | 600                            | ...                           | CH. IN NOZ A1   |
| 10   | Nozl | 685.558                    | -526.402                       | -350                           | 605                           | CH. OUT NOZ B1  |
| 10   | Nozl | 2786.33                    | -495.606                       | -399.999                       | -600                          | MANHOLE M1      |
| 10   | Wght | 850                        | -495.606                       | ...                            | ...                           | PP PLATE        |
| 20   | Liqd | 75.4722                    | 25                             | 75                             | ...                           | Liquid 20       |
| 20   | Insl | 0.92031                    | 25                             | ...                            | ...                           | Ins: 20         |
| 30   | Liqd | 257.124                    | 82.5                           | 75                             | ...                           | Liquid 30       |
| 30   | Insl | 3.49817                    | 117.5                          | ...                            | ...                           | Ins: 20         |
| 40   | Sadl | 415.721                    | 175                            | 1096.5                         | ...                           | FIXED SADDLE    |
| 40   | Liqd | 280.399                    | 175                            | 75                             | ...                           | Liquid 40       |
| 40   | Insl | 5.89348                    | 175                            | ...                            | ...                           | Ins: 20         |
| 50   | Liqd | 3213.59                    | 995.409                        | 450                            | ...                           | Liquid 50       |
| 50   | Insl | 69.826                     | 1299                           | ...                            | ...                           | Ins: 20         |
| 50   | Nozl | 29.5637                    | 250                            | -1658.32                       | ...                           | LIQ OUT NOZ B3  |
| 60   | Sadl | 1161.64                    | 400                            | 1852                           | ...                           | SLIDING SADDLE- |
| 60   | Sadl | 1161.64                    | 3785                           | 1852                           | ...                           | SLIDING SADDLE  |
| 60   | Liqd | 20227.3                    | 2092.5                         | 825                            | ...                           | Liquid 60       |
| 60   | Insl | 124.943                    | 2092.5                         | ...                            | ...                           | Ins: 20         |
| 60   | Nozl | 188.67                     | 3750                           | 1746.44                        | ...                           | SS IN NOZ. A2   |
| 60   | Nozl | 882.53                     | 1700                           | 1929.8                         | ...                           | SS OUT NOZ. B2  |
| 60   | Nozl | 8.6364                     | 400                            | 1646.42                        | ...                           | VENT NOZ V1     |
| 60   | Nozl | 12.3915                    | 150                            | 1646.42                        | ...                           | PR. IN P1       |
| 60   | Nozl | 12.3915                    | 3250                           | 1646.42                        | ...                           | PR. IN NOZ. P2  |
| 60   | Nozl | 12.3915                    | 3000                           | 1646.42                        | ...                           | SAFETY VALVE S1 |
| 60   | Nozl | 12.3915                    | 800                            | 835.143                        | 1418.89                       | LEVEL GUAGE L1  |
| 60   | Nozl | 12.3915                    | 800                            | -835.143                       | 1418.89                       | LEVEL GUAGE L2  |
| 60   | Nozl | 12.3915                    | 1500                           | 835.143                        | 1418.89                       | LEVEL GUAGE L3  |
| 60   | Nozl | 12.3915                    | 1500                           | -835.143                       | 1418.89                       | LEVEL GUAGE L4  |
| 60   | Wght | 900                        | 1700                           | ...                            | ...                           | VAPOR GET. DEV  |
| 60   | Wght | 14550                      | ...                            | ...                            | ...                           | Datsheet weight |
| 70   | Liqd | 1571.83                    | 321.633                        | 825                            | ...                           | Liquid 70       |
| 70   | Insl | 45.708                     | 431.65                         | ...                            | ...                           | Ins: 20         |
| 70   | Nozl | 12.3915                    | 784.051                        | -700                           | ...                           | TEMP IN. T1     |
| 20   | FTsh | 7708.05                    | 270                            | ...                            | ...                           | U-TS            |
| 20   | Tube | 25583.4                    | 2750                           | ...                            | ...                           | Tubes           |
| 30   | Baf1 | 282.339                    | 1070                           | ...                            | ...                           | Baffles         |
| 30   | Baf1 | 282.339                    | 2140                           | ...                            | ...                           | Baffles         |
| 30   | Baf1 | 282.339                    | 3210                           | ...                            | ...                           | Baffles         |
| 30   | Baf1 | 282.339                    | 4280                           | ...                            | ...                           | Baffles         |
| 30   | Baf1 | 283.082                    | 5350                           | ...                            | ...                           | Baffles         |

### Total Weight of Each Detail Type:

|                      |         |
|----------------------|---------|
| Saddles              | 2739.0  |
| Liquid               | 26544.3 |
| Insulation           | 269.5   |
| Nozzles              | 5642.2  |
| Weights              | 16300.0 |
| Exchanger Components | 34703.9 |
| Liquid in Tubes      | 3210.7  |

Sum of the Detail Weights 89409.6 kg.

### Weight Summation Results: (kg.)

|               | Fabricated | Shop Test | Shipping | Erected | Empty   | Operating |
|---------------|------------|-----------|----------|---------|---------|-----------|
| Main Elements | 32490.0    | 32490.0   | 32490.0  | 32490.0 | 32490.0 | 32490.0   |
| Saddles       | 2739.0     | 2739.0    | 2739.0   | 2739.0  | 2739.0  | 2739.0    |
| Nozzles       | 5642.2     | 5642.2    | 5642.2   | 5642.2  | 5642.2  | 5642.2    |
| Wld Weights   | 1750.0     | 1750.0    | 1750.0   | 1750.0  | 1750.0  | 1750.0    |
| Exchanger     | 34703.9    | 34703.9   | 34703.9  | 34703.9 | 34703.9 | 34703.9   |
| Insulation    | ...        | ...       | 269.5    | 269.5   | 269.5   | 269.5     |
| Ope Weights   | ...        | ...       | ...      | ...     | ...     | 14550.0   |
| Ope. Liquid   | ...        | ...       | ...      | ...     | ...     | 26544.3   |
| Tube Ope Lqd  | ...        | ...       | ...      | ...     | ...     | 3210.7    |
| Test Liquid   | ...        | 48215.8   | ...      | ...     | ...     | ...       |
| Tube Tst Lqd  | ...        | 4901.8    | ...      | ...     | ...     | ...       |
| Totals        | 77325.1    | 130442.7  | 77594.6  | 77594.6 | 77594.6 | 121899.6  |

### Field Installation Options:

#### Miscellaneous Weight Percent: 10.0 %

*Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.*

### Weight Summary:

|                 |                                                     |              |
|-----------------|-----------------------------------------------------|--------------|
| Fabricated Wt.  | - Bare weight without removable internals           | 77325.1 kg.  |
| Shop Test Wt.   | - Fabricated weight + water ( full )                | 130442.7 kg. |
| Shipping Wt.    | - Fab. weight + removable intls.+ shipping app.     | 77594.6 kg.  |
| Erected Wt.     | - Fab. wt + or - loose items (trays,platforms etc.) | 77594.6 kg.  |
| Ope. Wt. no Liq | - Fab. weight + internals. + details + weights      | 77594.6 kg.  |
| Operating Wt.   | - Empty weight + operating liq. uncorroded          | 121899.6 kg. |
| Oper. Wt. + CA  | - Corr wt. + operating liquid                       | 120777.2 kg. |

### Exchanger Tube Data

|                              |                           |
|------------------------------|---------------------------|
| Volume of Exchanger tubes :  | 4904829.5 cm <sup>3</sup> |
| Weight of Ope Liq in tubes : | 3210.7 kg.                |
| Weight of Water in tubes :   | 4901.8 kg.                |

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

### Surface Areas of Elements:

| From | To | Outside Surface Area cm <sup>2</sup> | Inside Surface Area cm <sup>2</sup> |
|------|----|--------------------------------------|-------------------------------------|
| 10   | 20 | 57064.5                              | 48105.6                             |
| 20   | 30 | 3173.01                              | 2748.89                             |
| 30   | 40 | 18408.9                              | 9346.24                             |
| 40   | 50 | 20253.8                              | 19242.3                             |



|    |    |        |        |
|----|----|--------|--------|
| 50 | 60 | 221403 | 212379 |
| 60 | 70 | 436499 | 427296 |
| 70 | 80 | 125664 | 119683 |

|       |          |                          |
|-------|----------|--------------------------|
| Total | 882466.9 | 838800.4 cm <sup>2</sup> |
| Total | 949.9    | 902.9 ft <sup>2</sup>    |

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### Nozzle Flange MAWP Results: (bars & °C)

| Nozzle Description | Flange Rating |         | Des. Temp | Class | Grade/Group | Equiv. Press | Max Pressure |     |     |
|--------------------|---------------|---------|-----------|-------|-------------|--------------|--------------|-----|-----|
|                    | Ope.          | Ambient |           |       |             |              | UG-44        | 50% | DNV |
| MANHOLE M1         | 238.12        | 255.10  | 85        | 1500  | GR 1.1      | ...          | ...          | ... | ... |
| LIQ OUT NOZ B3     | 47.63         | 51.00   | 85        | 300   | GR 1.1      | ...          | ...          | ... | ... |
| SS IN NOZ. A2      | 47.63         | 51.00   | 85        | 300   | GR 1.1      | ...          | ...          | ... | ... |
| SS OUT NOZ. B2     | 47.63         | 51.00   | 85        | 300   | GR 1.1      | ...          | ...          | ... | ... |
| VENT NOZ V1        | 47.63         | 51.00   | 85        | 300   | GR 1.1      | ...          | ...          | ... | ... |
| PR. IN P1          | 47.63         | 51.00   | 85        | 300   | GR 1.1      | ...          | ...          | ... | ... |
| PR. IN NOZ. P2     | 47.63         | 51.00   | 85        | 300   | GR 1.1      | ...          | ...          | ... | ... |
| SAFETY VALVE S     | 47.63         | 51.00   | 85        | 300   | GR 1.1      | ...          | ...          | ... | ... |
| LEVEL GUAGE L1     | 47.63         | 51.00   | 85        | 300   | GR 1.1      | ...          | ...          | ... | ... |
| LEVEL GUAGE L2     | 47.63         | 51.00   | 85        | 300   | GR 1.1      | ...          | ...          | ... | ... |
| LEVEL GUAGE L3     | 47.63         | 51.00   | 85        | 300   | GR 1.1      | ...          | ...          | ... | ... |
| LEVEL GUAGE L4     | 47.63         | 51.00   | 85        | 300   | GR 1.1      | ...          | ...          | ... | ... |
| TEMP IN. T1        | 47.63         | 51.00   | 85        | 300   | GR 1.1      | ...          | ...          | ... | ... |

#### Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 47.631 bars  
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 51.000 bars

#### Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide ]: 238.123 bars  
 Lowest Flange Pressure Rating was (Amb)[TubeSide ]: 255.100 bars

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

#### Warning:

*There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.*

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## Wind Load Calculation Results:

### Input Values:

|                                         |                       |
|-----------------------------------------|-----------------------|
| Wind Design Code                        | ASCE/SEI 7-22         |
| Wind Load Reduction Scale Factor        | 1.100                 |
| Basic Wind Speed                        | [V] 180 Km/hr         |
| Surface Roughness Category              | D: Flat, unobstructed |
| Type of Surface                         | 0                     |
| Base Elevation                          | 300 mm.               |
| Percent Wind for Hydrotest              | 33.0                  |
| Using User defined Wind Press. Vs Elev. | N                     |
| Height of Hill or Escarpment H or Hh    | 0 mm.                 |
| Distance Upwind of Crest Lh             | 0 mm.                 |
| Distance from Crest to the Vessel x     | 0 mm.                 |
| Type of Terrain ( Hill, Escarpment )    | Flat                  |
| Damping Factor (Beta) for Wind (Ope)    | 0.0100                |
| Damping Factor (Beta) for Wind (Empty)  | 0.0000                |
| Damping Factor (Beta) for Wind (Filled) | 0.0000                |
| Site Elevation above Sea Level          | 0 mm.                 |

### Wind Analysis Results

*The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.*

Static Gust-Effect Factor, Operating Case [G]:  
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$   
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.194 \cdot 0.98) / (1 + 1.7 \cdot 3.4 \cdot 0.194)))$   
 $= \min(0.85, 0.915)$   
 $= 0.850$

|                                         |           |
|-----------------------------------------|-----------|
| Natural Frequency of Vessel (Operating) | 33.000 Hz |
| Natural Frequency of Vessel (Empty)     | 33.000 Hz |
| Natural Frequency of Vessel (Test)      | 33.000 Hz |

|                                    |            |
|------------------------------------|------------|
| User Entered Importance Factor is  | 1.000      |
| Force Coefficient                  | [Cf] 0.734 |
| Structure Height to Diameter ratio | 3.026      |

*This is classified as a rigid structure. Static analysis performed.*

### Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:  
 Exposure Category: D from Table 26.11-1  
 $\alpha = 11.5$ ;  $Z_g = 589788$  mm.

Effective Height [z]:  
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$   
 $= 2130.0 + 300.0 = 2430.0$  mm.  
 $= 7.972$  ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:  
 Because  $z$  (7.972 ft.) < 15 ft.  
 $= 2.41(15 / Z_g)^{2/\alpha}$   
 $= 2.41(15 / 1935.0)^{2/11.5}$   
 $= 1.035$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:  
 $= 1$ , per Table 26.6-1

As there is No Hill Present: [Kzt]:  
 $K_1 = 0$ ,  $K_2 = 0$ ,  $K_3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf } [161.849 \text{ Kgs/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.734 \cdot 17.283$$

$$= 357.3 \text{ lbs. } [162.1 \text{ Kg}]$$

| Element   | Hgt (z)<br>mm. | K1    | K2    | K3    | Kz    | Kzt   | qz<br>Kgs/m <sup>2</sup> |
|-----------|----------------|-------|-------|-------|-------|-------|--------------------------|
| CH. COVER | 2430.0         | 0.000 | 0.000 | 0.000 | 1.035 | 1.000 | 161.849                  |
| CH. COVER | 2430.0         | 0.000 | 0.000 | 0.000 | 1.035 | 1.000 | 161.849                  |
| SH. FLG.  | 2430.0         | 0.000 | 0.000 | 0.000 | 1.035 | 1.000 | 161.849                  |
| SHELL-1   | 2430.0         | 0.000 | 0.000 | 0.000 | 1.035 | 1.000 | 161.849                  |
| SHELL-2   | 2430.0         | 0.000 | 0.000 | 0.000 | 1.035 | 1.000 | 161.849                  |
| K-SHELL   | 2430.0         | 0.000 | 0.000 | 0.000 | 1.035 | 1.000 | 161.849                  |
| SH. COVER | 2430.0         | 0.000 | 0.000 | 0.000 | 1.035 | 1.000 | 161.849                  |

#### Wind Loads on Masses/Equipment/Piping

| ID            | Wind Area<br>cm <sup>2</sup> | Elevation<br>mm. | Pressure<br>Kgs/m <sup>2</sup> | Force<br>Kgf |
|---------------|------------------------------|------------------|--------------------------------|--------------|
| PP PLATE      | 0.00                         | 2430.00          | 161.85                         | 0.00         |
| VAPOR GET. DE | 0.00                         | 2430.00          | 161.85                         | 0.00         |
| Datsheet weig | 0.00                         | 2430.00          | 161.85                         | 0.00         |

#### **Wind Load Calculation:**

| From | To | Wind Height<br>mm. | Wind Diameter<br>mm. | Wind Area<br>cm <sup>2</sup> | Wind Pressure<br>Kgs/m <sup>2</sup> | Element<br>Wind Load<br>Kgf |
|------|----|--------------------|----------------------|------------------------------|-------------------------------------|-----------------------------|
| 10   | 20 | 2430               | 2426.4               | 16056.2                      | 161.849                             | 178.29                      |
| 20   | 30 | 2430               | 2592                 | 1296                         | 161.849                             | 14.391                      |
| 30   | 40 | 2430               | 2268                 | 3742.2                       | 161.849                             | 41.554                      |
| 40   | 50 | 2430               | 2378.4               | 8324.4                       | 161.849                             | 92.4355                     |
| 50   | 60 | 2430               | 3278.4               | 85172.8                      | 161.849                             | 945.773                     |
| 60   | 70 | 2430               | 4152                 | 173761                       | 161.849                             | 1929.47                     |
| 70   | 80 | 2430               | 4152                 | 27042.1                      | 161.849                             | 300.28                      |

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

*Be sure the wind speed is in accordance with the specified wind design code.*

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## Earthquake Load Calculation Results:

### Earthquake Load Calculation:

#### Input Values:

|                                  |          |               |     |
|----------------------------------|----------|---------------|-----|
| Seismic Design Code              |          | ASCE/SEI 7-22 |     |
| Importance Factor                |          | 1.500         |     |
| Force Modification Factor R      |          | 3.000         |     |
| Site Class                       |          | D             |     |
| Component Elevation Ratio        | z/h      | 0.000         |     |
| Amplification Factor             | Ap       | 0.000         |     |
| Force Factor                     |          | 0.000         |     |
| Consider Vertical Acceleration   |          | Yes           |     |
| Minimum Acceleration Multiplier  |          | 0.000         |     |
| User Value of Sds (used if > 0 ) |          | 0.000         |     |
| User Value of Sd1 (used if > 0 ) |          | 0.000         |     |
| Component resonance duct. fact.  | [CAR]    | 0.000         |     |
| Component strength factor        | [Rpo]    | 0.000         |     |
| Overstrength factor              | [Omega0] | 0.000         |     |
| Response Acceleration Factor     | [SMS]    | 0.326         |     |
| Response Acceleration Factor     | [SM1]    | 0.139         |     |
| Building Importance Factor       | [Ie]     | 1.000         |     |
| Site Latitude                    |          | 0.000         | deg |
| Site Longitude                   |          | 0.000         | deg |

#### Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:  
 =  $C_t \cdot h_{nx}$  where  $C_t = 0.020$ ,  $x = 0.75$  and  $h_n$  = Structural Height (ft.)  
 =  $0.020(12.31960.75)$   
 = 0.132 seconds

The Coefficient  $C_u$  from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:  
 = ( 1/Natural Frequency ) = ( 1/33.0 )  
 = 0.030

Check the Value of T which is the smaller of  $C_u \cdot T_a$  and T:  
 = Minimum Value of (  $1.7 \cdot 0.132$ , 0.03 ) per 12.8.2  
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:  
 =  $0.3 \cdot Sds \cdot W \cdot I$   
 =  $0.3 \cdot 0.217 \cdot 118038 \cdot 1.5$   
 = 11544.137 Kgf

Vertical load per 12.4-4, [Ev]:  
 =  $0.2 \cdot Sds \cdot W$   
 =  $0.2 \cdot 0.217 \cdot 118038$   
 = 5130.7 Kgf

### Earthquake Load Calculation:

| From | To | Earthquake Height mm. | Earthquake Weight Kgf | Element Ope Load Kgf |
|------|----|-----------------------|-----------------------|----------------------|
| 10   | 20 | 875                   | 11803.8               | 859.67               |
| 20   | 30 | 875                   | 11803.8               | 859.67               |
| 30   | 40 | 875                   | 11803.8               | 859.67               |
| 40   | 50 | 875                   | 11803.8               | 859.67               |
| 40   | 50 | 875                   | 11803.8               | 859.67               |
| 50   | 60 | 875                   | 11803.8               | 859.67               |



|    |    |      |         |         |
|----|----|------|---------|---------|
| 60 | 0  | 1625 | 11803.8 | 1596.53 |
| 60 | 70 | 1625 | 11803.8 | 1596.53 |
| 60 | 70 | 1625 | 11803.8 | 1596.53 |
| 70 | 80 | 1625 | 11803.8 | 1596.53 |

The Earthquake Weight does not include the weight of the saddles.

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**Shop/Field Installation Options :**

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

|                                                    |              |
|----------------------------------------------------|--------------|
| Center of Gravity of the Saddles                   | 4961.522 mm. |
| Center of Gravity of the Liquid                    | 5071.097 mm. |
| Center of Gravity of the Insulation                | 4631.588 mm. |
| Center of Gravity of the Nozzles                   | 770.129 mm.  |
| Center of Gravity of the Added Weights (Operating) | 3487.399 mm. |
| Center of Gravity of the Added Weights (Empty)     | 2488.848 mm. |
| Center of Gravity of the Tubesheet(s)              | 270.000 mm.  |
| Center of Gravity of the Tubes                     | 2750.000 mm. |
| Center of Gravity of the Baffles                   | 3705.625 mm. |
|                                                    |              |
| Center of Gravity of Bare Shell New and Cold       | 4152.879 mm. |
| Center of Gravity of Bare Shell Corroded           | 4133.338 mm. |
|                                                    |              |
| Vessel CG in the Operating Condition               | 3533.683 mm. |
| Vessel CG in the Fabricated (Shop/Empty) Condition | 3043.191 mm. |
| Vessel CG in the Test Condition                    | 3753.831 mm. |

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**ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle**

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

**ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle**

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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**ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle**

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

**ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle**

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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### ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

There are more than two saddles present in this model. The load on the saddle supports has been computed as the total weight divided by the number of saddles. Ensure that this is an appropriate assumption for this vessel model. It may be necessary to manually enter the bending moments at the saddle through the Saddle Input dialog. Also note that the dimension Ls is taken to be the distance between the adjoining saddles (first two or last two).

Note:

Wear Pad Width (260.00) is less than  $1.56 \cdot \sqrt{R_m \cdot t}$   
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

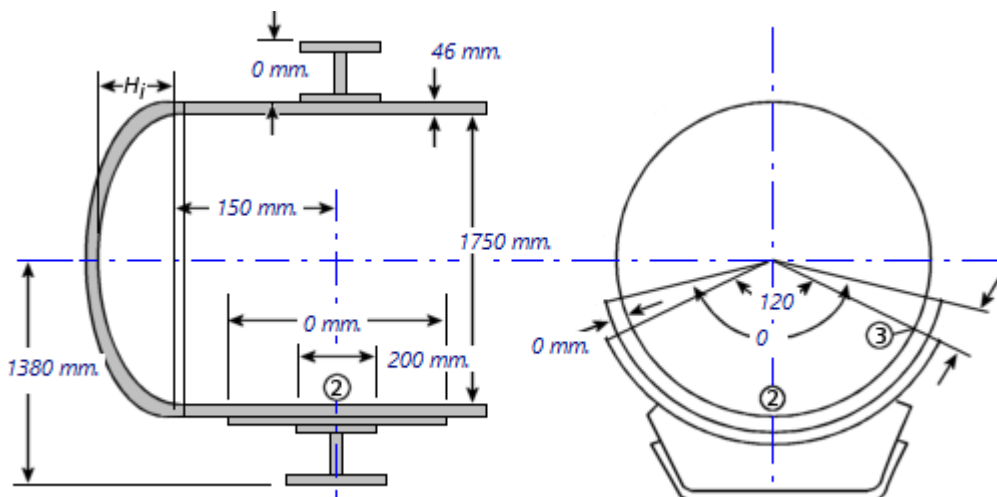
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{898.75 \cdot 44.5}, 2 \cdot 150.0)$$

$$= 300.0000 \text{ mm.}$$

#### Input and Calculated Values:

|                                               |                |         |                    |
|-----------------------------------------------|----------------|---------|--------------------|
| Vessel Mean Radius                            | Rm             | 898.75  | mm.                |
| Shell Thickness used in this Case             | t              | 46.000  | mm.                |
| Stiffened Vessel Length per 4.15.6            | L              | 7348.80 | mm.                |
| Distance from Saddle to Vessel tangent        | a1 or a        | 150.00  | mm.                |
| Saddle Width                                  | b1 or b        | 200.00  | mm.                |
| Saddle Bearing Angle                          | delta or theta | 120.00  | degrees            |
| Shell Allowable Stress used in Calculation    |                | 138.00  | N./mm <sup>2</sup> |
| Head Allowable Stress used in Calculation     |                | 138.00  | N./mm <sup>2</sup> |
| Circumferential Efficiency in Plane of Saddle |                | 1.00    |                    |
| Circumferential Efficiency at Mid-Span        |                | 1.00    |                    |
| Distance from Saddle Base to Centerline       | B              | 1380.00 | mm.                |
| Coefficient of Friction                       | mu             | 0.60    |                    |



#### Results for Horizontal Vessel Analysis:

|                                |          |      |
|--------------------------------|----------|------|
| Saddle Force Q, Operating Case | 50951.27 | Kgf  |
| Pressure used in Analysis      | P 25.047 | bars |

| Horizontal Vessel Analysis Results: |            | Actual<br>N./mm <sup>2</sup> | Allowable<br>N./mm <sup>2</sup> |
|-------------------------------------|------------|------------------------------|---------------------------------|
| Long. Stress at Top                 | of Midspan | 23.36                        | 138.00                          |
| Long. Stress at Bottom              | of Midspan | 43.84                        | 138.00                          |
| Long. Stress at Top                 | of Saddles | 25.06                        | 138.00                          |

|                                   |       |        |
|-----------------------------------|-------|--------|
| Long. Stress at Bottom of Saddles | 25.52 | 138.00 |
| Tangential Shear in Shell         | 10.99 | 110.40 |
| Tangential Shear in Head          | 5.23  | 110.40 |
| Circ. Stress at Horn of Saddle    | 10.48 | 172.50 |
| Circ. Compressive Stress in Shell | 1.67  | 138.00 |

#### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:  
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$   
 $= 3.0(3502.2/3 + 0) \cdot 1380.0/1610.0$   
 $= 3001.9 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:  
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$   
 $= \max(790.98, 0) \cdot 1380.0/3173.0$   
 $= 344.0 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:  
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$   
 $= \max(11544.14, 0) \cdot 1380.0/3173.0$   
 $= 5020.8 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:  
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$   
 $= 3.0(11544/3 + 0) \cdot 1380.0/1610.0$   
 $= 9895.0 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:  
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$   
 $= 41056 + \max(344, 3002, 5021, 9895)$   
 $= 50951.3 \text{ Kgf}$

Longitudinal Wind Force [Fl]:  
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$   
 $= 1.1 \cdot 1587.169(0.0 + \pi/4(1.982 \cdot 1.2)^2)$   
 $= 7756.729 \text{ N}$

#### Summary of Loads at the Base of this Saddle:

|                                         |          |     |
|-----------------------------------------|----------|-----|
| Vertical Load (including saddle weight) | 51367.00 | Kgf |
| Transverse Shear Load                   | 3848.05  | Kgf |
| Longitudinal Shear Load                 | 11544.14 | Kgf |

Note: Wear Plate is Welded to the Shell,  $k = 0.1$

Saddle Dimension [E]:  
 $= \text{Baseplate Length}$   
 $= 1610.000 \text{ mm.}$

#### The Computed K values from Table 4.15.1:

|             |              |              |             |
|-------------|--------------|--------------|-------------|
| K1 = 0.1066 | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011 |
| K5 = 0.7603 | K6 = 0.0529  | K7 = 0.0132  | K8 = 0.3405 |
| K9 = 0.2711 | K10 = 0.0581 | K1* = 0.1923 |             |

Note: Dimension a is less than  $Rm/2$ .

Moment per Equation 4.15.1 [M1]:  
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$   
 $= -50951 \cdot 150.0 [1 - (1 - 150.0/7348.8 + (898.75^2 - 0.0^2)/(2 \cdot 150.0 \cdot 7348.8))/(1 + (4 \cdot 0.0)/(3 \cdot 7348.8))]$   
 $= 2644.2 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:  
 $= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$   
 $= 50951 \cdot 7349/4((1 + 2(899^2 - 0^2)/(7349^2))/(1 + (4 \cdot 0)/(3 \cdot 7349)) - 4 \cdot 150/7349)$   
 $= 88766.9 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:  
 $= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$   
 $= 25.047 \cdot 898.75 / (2 \cdot 33.5) - 88766.9 / (\pi \cdot 898.8^2 \cdot 33.5)$   
 $= 23.36 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:  
 $= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$   
 $= 25.047 \cdot 898.75 / (2 \cdot 33.5) + 88766.9 / (\pi \cdot 898.8^2 \cdot 33.5)$   
 $= 43.84 \text{ N./mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.6) [ $\sigma_3$ ]:  
 $= P \cdot Rm / (2t) - M1 / (\pi \cdot Rm^2 \cdot t)$   
 $= 25.047 \cdot 898.75 / (2 \cdot 44.5) - 2644.2 / (\pi \cdot 898.8^2 \cdot 44.5)$   
 $= 25.06 \text{ N./mm}^2$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [ $\sigma_4$ ]:  
 $= P \cdot Rm / (2t) + M1 / (\pi \cdot Rm^2 \cdot t)$   
 $= 25.047 \cdot 898.75 / (2 \cdot 44.5) + 2644.2 / (\pi \cdot 898.8^2 \cdot 44.5)$   
 $= 25.52 \text{ N./mm}^2$

Maximum Shear Force in the Saddle (4.15.3) [T]:  
 $= Q(L-2a) / (L + (4 \cdot hm / 3))$   
 $= 50951(7348.8 - 2 \cdot 150.0) / (7348.8 + (4 \cdot 0.0 / 3))$   
 $= 48871.3 \text{ Kg}$

Shear Stress in the shell no rings, stiffened (4.15.13) [ $\tau_3$ ]:  
 $= K3 \cdot Q / (Rm \cdot t)$   
 $= 0.8799 \cdot 50951 / (898.75 \cdot 44.5)$   
 $= 10.99 \text{ N./mm}^2$

Shear Stress in the head, shell stiffened (4.15.14) [ $\tau_3^*$ ]:  
 $= K3 \cdot Q / (Rm \cdot th)$   
 $= 0.8799 \cdot 50951 / (898.75 \cdot 93.5)$   
 $= 5.23 \text{ N./mm}^2$

Decay Length (4.15.20) [ $x_1, x_2$ ]:  
 $= 0.78 \cdot \sqrt{(Rm \cdot t)}$   
 $= 0.78 \cdot \sqrt{(898.75 \cdot 44.5)}$   
 $= 155.989 \text{ mm.}$

Circumferential Stress in shell, no rings (4.15.21) [ $\sigma_6$ ]:  
 $= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$   
 $= -0.7603 \cdot 50951 \cdot 0.1 / (44.5(200.0 + 155.99 + 155.99))$   
 $= -1.67 \text{ N./mm}^2$

Circ. Comp. Stress at Horn of Saddle,  $L \geq 8Rm$  (4.15.22) [ $\sigma_7$ ]:  
 $= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$   
 $= -50951 / (4 \cdot 44.5(200.0 + 155.989 + 155.989)) -$   
 $3 \cdot 0.0132 \cdot 50951 / (2 \cdot 44.5^2)$   
 $= -10.48 \text{ N./mm}^2$

Effective reinforcing plate width (4.15.24) [B1]:  
 $= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$   
 $= \min(200.0 + 1.56 \cdot \sqrt{(898.75 \cdot 44.5)}, 2 \cdot 150.0)$   
 $= 300.00 \text{ mm.}$

#### Results for Vessel Ribs, Web and Base:

|                                      |        |           |                    |
|--------------------------------------|--------|-----------|--------------------|
| Baseplate Length                     | Bplen  | 1610.0000 | mm.                |
| Baseplate Thickness                  | Bpthk  | 28.0000   | mm.                |
| Baseplate Width                      | Bpwid  | 240.0000  | mm.                |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 4         |                    |
| Rib Thickness                        | Ribtk  | 16.0000   | mm.                |
| Web Thickness                        | Webtk  | 16.0000   | mm.                |
| Web Location                         | Webloc | Side      |                    |
| Saddle Design Temperature            |        | 21.1      | °C                 |
| Saddle Yield Stress                  | Sy     | 242.0     | N./mm <sup>2</sup> |
| Height of Web at Center              | Hw,c   | 306.0     | mm.                |
| Friction Coefficient                 | mu     | 0.600     |                    |

#### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

**Inertia of Shell [ShellInertia]:**

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(876.5 \cdot 44.5)} + 260.0)^3) / 12$$

$$= 417.174 \text{ cm}^4$$

**Resolved Inertia for the Shell [I,shell]:**

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 417.174 + 252.801 (136.678 - 22.25)^2$$

$$= 33518.469 \text{ cm}^4$$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>(2)</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|------------------------------------------|
| Shell     | 568.1    | 44.5     | 22.2     | 252.8                | 562482.1               | 0.335E+05                                |
| Wearplate | 260.0    | 16.0     | 52.5     | 41.6                 | 218400.0               | 0.296E+04                                |
| Web       | 16.0     | 415.0    | 268.0    | 66.4                 | 1779520.8              | 0.210E+05                                |
| BasePlate | 240.0    | 28.0     | 489.5    | 67.2                 | 3289440.5              | 0.837E+05                                |
| Totals    | ...      | ...      | ...      | 428.0                | 5849843.0              | 0.141E+06                                |

**Distance to Centroid [C1]:**

$$= AY / A$$

$$= 5849843 / 428.001$$

$$= 136.678 \text{ mm.}$$

**Angle [β]:**

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

**Saddle Splitting Coefficient [K1]:**

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

**Saddle Splitting Force [Fh]:**

$$= K1 \cdot Q$$

$$= 0.204 \cdot 50951.273$$

$$= 10369.6895 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 5.8044 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 145.1982 \text{ N./mm}^2$$

**Saddle Splitting Dimension [d]:**

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1380.0 - 876.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 655.140 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 6793.7358 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 6.4512 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 161.3313 \text{ N./mm}^2$$

**Minimum Thickness of Baseplate per Moss:**

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(50951 + 416) \cdot 240.0) / (2 \cdot 1610.0 \cdot 161.331)}$$

$$= 26.423 \text{ mm.}$$

**Calculation of Axial Load, Intermediate Values and Compressive Stress:**

**Web Length Dimension [Web Length]:**

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (875.0 + 46.0 + 16.0)$$

$$= 1622.932 \text{ mm.}$$

**Distance between Ribs [e]:**

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 1622.9315 / (4 - 1)$$

$$= 540.977 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 540.9772 \cdot 240.0 / 2$$

$$= 649.173 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= ( Q + SaddleWt ) / ( BasePlateLength \cdot BasePlateWidth )$$

$$= 51366.996 / ( 1610.0 \cdot 240.0 )$$

$$= 13.294 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 649.2 \cdot 13.29$$

$$= 8629.930 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 29.44 + 43.278$$

$$= 72.718 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 8629.9 / 72.7182$$

$$= 11.638 \text{ N./mm}^2$$

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|        | B     | D     | Y     | A    | AY       | Io        |
|--------|-------|-------|-------|------|----------|-----------|
| Rib    | 16.0  | 184.0 | 108.0 | 29.4 | 317951.9 | 0.187E+04 |
| Web    | 270.5 | 16.0  | 8.0   | 43.3 | 34622.5  | 719.      |
| Totals | ...   | ...   | ...   | 72.7 | 352574.5 | 0.259E+04 |

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 200.0 - 16.0$$

$$= 184.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 352574.5 / 72.718$$

$$= 48.485 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max( ytot, Saddle \text{ Width} - ytot )$$

$$= \max( 48.485, 200.0 - 48.485 )$$

$$= 151.515 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ ( Total \text{ Inertia} / Total \text{ Area} ) }$$

$$= \sqrt{ ( 2591.9 / 72.718 ) }$$

$$= 59.702 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= Saddle \text{ Height} - \cos( saddle \text{ bearing angle} / 2 ) ( radius + shlthk + wpdthk ) - bpthk$$

$$= 1380.0 - \cos( 120.0 / 2 ) ( 875.0 + 46.0 + 16.0 ) - 28.0$$

$$= 883.500 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin( saddle \text{ bearing angle} / 2 ) ( radius + shlthk + wpdthk )$$

$$= 2 \cdot \sin( 120.0 / 2 ) ( 875.0 + 46.0 + 16.0 )$$

$$= 1622.932 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= Baseplate \text{ length} - clearance$$

$$= 1610.0 - 50.8$$

$$= 1559.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (1622.932 - 1559.2) / 2$$

$$= 31.866 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{883.5^2 + 31.866^2}$$

$$= 884.075 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19904\text{E}+09 / 242.0)}$$

$$= 127.422$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 884.075 / 59.702$$

$$= 14.808$$

Bending Moment [Rm]:

$$= Fl / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 11544.1 / (2 \cdot 1610.0) \cdot 540.977 \cdot 884.07 / 2$$

$$= 857.338 \text{ Kg-m.}$$

Compressive Allowable,  $KL/r < Cc$  (  $14.808 < 127.4221$  ) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (14.81)^2 / (2 \cdot 127.42^2)) 242 /$$

$$(5/3 + 3 \cdot (14.81) / (8 \cdot 127.42) - (14.81^3) / (8 \cdot 127.42^3))$$

$$= 140.6 \text{ N./mm}^2$$

**AISC Unity Check of Outside Ribs ( must be  $\leq 1$  )**

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 11.64 / 140.56 + (857.34 \cdot 151.515 / 25919526) / 161.33$$

$$= 0.387$$

**Check of Inside Ribs:**

Inertia of Saddle, Inner Ribs - Axial Direction

|        | B     | D     | Y     | A     | AY       | Io        |
|--------|-------|-------|-------|-------|----------|-----------|
| Rib    | 16.0  | 184.0 | 108.0 | 29.4  | 317951.9 | 0.247E+04 |
| Web    | 541.0 | 16.0  | 8.0   | 86.6  | 69245.1  | 576.      |
| Totals | ...   | ...   | ...   | 116.0 | 387197.0 | 0.305E+04 |

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 387197.031 / 115.996$$

$$= 33.380 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}( \text{Saddle Width} - ytot ))$$

$$= \max(33.38, \text{abs}(200.0 - 33.38))$$

$$= 166.620 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2} - Bpthk$$

$$= 1380.0 - \sqrt{(937.0 + 16.0)^2 - (540.977/2)^2} - 28.0$$

$$= 454.891 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3045.9 / 115.996)}$$

$$= 51.243 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 454.891 / 51.243$$

$$= 8.877$$

Unit Force [Force,u]:  
 $= FI / ( 2 \cdot \text{Baseplate Length} )$   
 $= 11544.137 / ( 2 \cdot 1610.0 )$   
 $= 3.585 \text{ Kg/mm.}$

Moment at base of inner Rib [Mbase,c]:  
 $= \text{Unit Force} \cdot e \cdot L$   
 $= 3.585 \cdot 540.977 \cdot 454.891$   
 $= 882.268 \text{ Kg-m.}$

Bending Stress due to Transverse Force and Weight Load [ $\sigma_B$ ,base,c]:  
 $= \text{Bending Moment} / \text{Section Modulus}$   
 $= 882.268 / 182803.797$   
 $= 47.330 \text{ N./mm}^2$

Compressive Allowable,  $KL/r < C_c$  (  $8.8771 < 127.4221$  ) per AISC E2-1 [Sca]:  
 $= ( 1 - (KL/r)^2 / (2 \cdot C_c^2) ) F_y / ( 5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3) )$   
 $= ( 1 - ( 8.88 )^2 / ( 2 \cdot 127.42^2 ) ) 242 /$   
 $( 5/3 + 3 \cdot (8.88) / (8 \cdot 127.42) - ( 8.88^3 ) / (8 \cdot 127.42^3) )$   
 $= 142.6 \text{ N./mm}^2$

**AISC Unity Check of Inside Ribs ( must be  $\leq 1$  )**  
 $= S_c / S_{ca} + ( M_{base,c} \cdot C1/I ) / S_{ba}$   
 $= 14.3 / 142.61 + ( 882.27 \cdot 166.62 / 3045.872 ) / 161.33$   
 $= 0.394$

#### **Input Data for Base Plate Bolting Calculations:**

|                                            |         |                           |
|--------------------------------------------|---------|---------------------------|
| Total Number of Bolts per BasePlate        | Nbolts  | 3                         |
| Total Number of Bolts in Tension/Baseplate | Nbt     | 2                         |
| Bolt Design Temperature                    |         | 21.1 °C                   |
| Bolt Material Specification                |         | SA-193 B7                 |
| Bolt Allowable Stress                      | Stba    | 150.00 N./mm <sup>2</sup> |
| Bolt Corrosion Allowance                   | Bca     | 0.0 mm.                   |
| Distance from Bolts to Edge                | Edgedis | 85.0 mm.                  |
| Nominal Bolt Diameter                      | Bnd     | 30.0000 mm.               |
| Thread Series                              | Series  | TEMA Metric               |
| BasePlate Allowable Stress                 | S       | 138.00 N./mm <sup>2</sup> |
| Area Available in a Single Bolt            | BltArea | 5.0297 cm <sup>2</sup>    |
| Saddle Load QO (Weight)                    | QO      | 41472.0 Kg                |
| Saddle Load QL (Wind/Seismic contribution) | QL      | 5020.8 Kg                 |
| Maximum Transverse Force                   | Ft      | 3848.0 Kg                 |
| Maximum Longitudinal Force                 | Fl      | 11544.1 Kg                |
| Saddle Bolted to Steel Foundation          |         | No                        |

Shear Stress in a Single Bolt, Longitudinal Direction [ $\tau_b$ ,l]:  
 $= FI / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$   
 $= 11544.1 / ( 5.03 \cdot 3 )$   
 $= 75.0 \text{ N./mm}^2. \text{ Must be less than } 100.0 \text{ N./mm}^2.$

Shear Stress in a Single Bolt, Transverse Direction [ $\tau_b$ ,t]:  
 $= Ft / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$   
 $= 3848.0 / ( 5.03 \cdot 3 )$   
 $= 25.0 \text{ N./mm}^2. \text{ Must be less than } 100.0 \text{ N./mm}^2.$

#### **Bolt Area Calculation per Dennis R. Moss**

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:  
 $= 0.0$  ( QO > QL --> No Uplift in Longitudinal direction )

Bolt Area due to Shear Load [Bltarears]:  
 $= FI / ( \text{BoltShearAllowable} \cdot \text{Nbolts} )$   
 $= 11544.14 / ( 100.0 \cdot 3.0 )$   
 $= 3.7737 \text{ cm}^2$

#### **Bolt Area due to Transverse Load:**

Moment on Baseplate Due to Transverse Load [Rmom]:  
 $= B \cdot Ft + \text{Sum of X Moments}$   
 $= 1380.0 \cdot 3848.05 + 0.0$

= 5310.41 Kg-m.

Eccentricity (e):

= Rmom / QO

= 5310.41/41472.02

= 128.05 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0.0, 3.7737, 0.0]

= 3.7737 cm<sup>2</sup>

#### MDMT Calculations for: FIXED SADDLE, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 16.0, c = 1.5 mm., E\* = 1.0

Thickness Ratio = tr \* E\*/(tg - c) = 0.39, Temp. Reduction = 53 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C

Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -31 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

#### ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (480.00) is less than  $1.56 \cdot \sqrt{rm \cdot t}$

and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

= min( b +  $1.56 \cdot \sqrt{Rm \cdot t}$  ), 2a )

= min( 350.0 +  $1.56 \cdot \sqrt{1643.25 \cdot 33.5}$  ), 2 • 400.0 )

= 716.0149 mm.

#### Input and Calculated Values:

|                                               |                |         |                    |
|-----------------------------------------------|----------------|---------|--------------------|
| Vessel Mean Radius                            | Rm             | 1643.25 | mm.                |
| Shell Thickness used in this Case             | t              | 35.000  | mm.                |
| Stiffened Vessel Length per 4.15.6            | L              | 7348.80 | mm.                |
| Distance from Saddle to Vessel tangent        | a1 or a        | 400.00  | mm.                |
| Saddle Width                                  | b1 or b        | 350.00  | mm.                |
| Saddle Bearing Angle                          | delta or theta | 120.00  | degrees            |
| Inside or Mean Depth of Head                  | Hi or hm       | 830.75  | mm.                |
| Shell Allowable Stress used in Calculation    |                | 138.00  | N./mm <sup>2</sup> |
| Head Allowable Stress used in Calculation     |                | 138.00  | N./mm <sup>2</sup> |
| Circumferential Efficiency in Plane of Saddle |                | 1.00    |                    |
| Circumferential Efficiency at Mid-Span        |                | 1.00    |                    |
| Distance from Saddle Base to Centerline       | B              | 2130.00 | mm.                |
| Coefficient of Friction                       | mu             | 0.10    |                    |

#### Results for Horizontal Vessel Analysis:

|                                |   |          |      |
|--------------------------------|---|----------|------|
| Saddle Force Q, Operating Case |   | 49526.75 | Kgf  |
| Pressure used in Analysis      | P | 25.000   | bars |

| Horizontal Vessel Analysis Results: | Actual<br>N./mm <sup>2</sup> | Allowable<br>N./mm <sup>2</sup> |
|-------------------------------------|------------------------------|---------------------------------|
| Long. Stress at Top of Midspan      | 59.07                        | 138.00                          |
| Long. Stress at Bottom of Midspan   | 63.57                        | 138.00                          |
| Long. Stress at Top of Saddles      | 61.24                        | 138.00                          |
| Long. Stress at Bottom of Saddles   | 61.40                        | 138.00                          |

|                                   |        |        |
|-----------------------------------|--------|--------|
| Tangential Shear in Shell         | 7.76   | 110.40 |
| Tangential Shear in Head          | 7.76   | 110.40 |
| Circ. Stress at Horn of Saddle    | 20.41  | 172.50 |
| Addl. Stress in Head as Stiffener | 122.37 | 172.50 |
| Circ. Compressive Stress in Shell | 1.54   | 138.00 |

#### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:  
 $= Ftr( Ft/Num\ of\ Saddles + Z\ Force\ Load ) \cdot B / E$   
 $= 3.0( 3502.2/3 + 0 ) \cdot 2130.0/2902.9172$   
 $= 2569.7\ Kg$

Longitudinal Saddle Reaction Force [Fwl]:  
 $= \max( Fl,wind; Sum\ of\ X\ Forces ) \cdot B / Ls$   
 $= \max( 2410.51, 0 ) \cdot 2130.0/3385.0$   
 $= 1516.8\ Kg$

Saddle Reaction Force due to Earthquake Fl [Fsl]:  
 $= \max( Fl,eq; Sum\ of\ X\ Forces ) \cdot B / Ls$   
 $= \max( 11544.14, 0 ) \cdot 2130.0/3385.0$   
 $= 7264.1\ Kg$

Saddle Reaction Force due to Earthquake Ft [Fst]:  
 $= Ftr( Ft/Num\ of\ Saddles + Z\ Force\ Load ) \cdot B / E$   
 $= 3.0( 11544/3 + 0 ) \cdot 2130.0/2902.9172$   
 $= 8470.4\ Kg$

Load Combination Results for Q + Wind or Seismic [Q]:  
 $= Saddle\ Load + \max( Fwl, Fwt, Fsl, Fst )$   
 $= 41056 + \max( 1517, 2570, 7264, 8470 )$   
 $= 49526.7\ Kg$

Longitudinal Wind Force [Fl]:  
 $= WindScalar \cdot WindPress( Platform\ Area + ( \pi/4( OD \cdot WindDiaMult )^2 ) )$   
 $= 1.1 \cdot 1587.169( 0.0 + \pi/4( 3.46 \cdot 1.2 )^2 )$   
 $= 23638.697\ N$

#### Summary of Loads at the Base of this Saddle:

|                                         |          |     |
|-----------------------------------------|----------|-----|
| Vertical Load (including saddle weight) | 50688.39 | Kgf |
| Transverse Shear Load                   | 3848.05  | Kgf |
| Longitudinal Shear Load                 | 11544.14 | Kgf |

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:  
 $= \min( 2( ShellID/2 + t + WearPadThickness )\sin(\theta/2), 2 \cdot Rm )$   
 $= \min( 2( 3250.0/2 + 35.0 + 16.0 )\sin( 120.0/2 ), 2 \cdot 1643.25 )$   
 $= 2902.917\ mm$

#### The Computed K values from Table 4.15.1:

|             |              |              |             |
|-------------|--------------|--------------|-------------|
| K1 = 0.1066 | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011 |
| K5 = 0.7603 | K6 = 0.0529  | K7 = 0.0132  | K8 = 0.3405 |
| K9 = 0.2711 | K10 = 0.0581 | K1* = 0.1923 |             |

Note: Dimension a is less than Rm/2.

Moment per Equation 4.15.1 [M1]:  
 $= -Q \cdot a [ 1 - ( 1 - a/L + (Rm^2 - hm^2)/(2a \cdot L) ) / ( 1 + (4hm)/3L ) ]$   
 $= -49527 \cdot 400.0 [ 1 - ( 1 - 400.0/7348.8 + (1643.25^2 - 830.75^2)/(2 \cdot 400.0 \cdot 7348.8) ) / ( 1 + (4 \cdot 830.75)/(3 \cdot 7348.8) ) ]$   
 $= 2354.4\ Kg \cdot m$

Moment per Equation 4.15.2 [M2]:  
 $= Q \cdot L/4( (1 + 2(Rm^2 - hm^2)/(L^2)) / ( 1 + (4hm^2)/(3L) ) - 4a/L )$   
 $= 49527 \cdot 7349/4( (1 + 2(1643^2 - 831^2)/(7349^2)) / ( 1 + (4 \cdot 831)/(3 \cdot 7349) ) - 4 \cdot 400/7349 )$   
 $= 65149.1\ Kg \cdot m$

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:  
 $= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$   
 $= 25.0 \cdot 1643.25 / (2 \cdot 33.5) - 65149.1 / (\pi \cdot 1643.2^2 \cdot 33.5)$   
 $= 59.07 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:  
 $= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$   
 $= 25.0 \cdot 1643.25 / (2 \cdot 33.5) + 65149.1 / (\pi \cdot 1643.2^2 \cdot 33.5)$   
 $= 63.57 \text{ N./mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.6) [ $\sigma_3$ ]:  
 $= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$   
 $= 25.0 \cdot 1643.25 / (2 \cdot 33.5) - 2354.4 / (\pi \cdot 1643.2^2 \cdot 33.5)$   
 $= 61.24 \text{ N./mm}^2$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [ $\sigma_4$ ]:  
 $= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$   
 $= 25.0 \cdot 1643.25 / (2 \cdot 33.5) + 2354.4 / (\pi \cdot 1643.2^2 \cdot 33.5)$   
 $= 61.40 \text{ N./mm}^2$

Maximum Shear Force in the Saddle (4.15.3) [T]:  
 $= Q(L-2a) / (L + (4 \cdot h_m / 3))$   
 $= 49527(7348.8 - 2 \cdot 400.0) / (7348.8 + (4 \cdot 830.75 / 3))$   
 $= 38354.2 \text{ Kgf}$

Shear Stress in the shell no rings, stiffened (4.15.13) [ $\tau_3$ ]:  
 $= K_3 \cdot Q / (R_m \cdot t)$   
 $= 0.8799 \cdot 49527 / (1643.25 \cdot 33.5)$   
 $= 7.76 \text{ N./mm}^2$

Shear Stress in the head, shell stiffened (4.15.14) [ $\tau_3^*$ ]:  
 $= K_3 \cdot Q / (R_m \cdot t_h)$   
 $= 0.8799 \cdot 49527 / (1643.25 \cdot 33.5)$   
 $= 7.76 \text{ N./mm}^2$

Membrane stress in the head as a stiffener (4.15.16) [ $\sigma_5$ ]:  
 $= K_4 \cdot Q / (R_m \cdot t_h) + ((P \cdot R_i) / (2 \cdot t_h)) \cdot (R_i / h_m)$   
 $= 0.4011 \cdot 49527 / (1643.25 \cdot 33.5) +$   
 $((25.0 \cdot 1626.5) / (2 \cdot 33.5)) \cdot (1626.5 / 830.75)$   
 $= 122.37 \text{ N./mm}^2$

Decay Length (4.15.20) [ $x_1, x_2$ ]:  
 $= 0.78 \cdot \sqrt{R_m \cdot t}$   
 $= 0.78 \cdot \sqrt{1643.25 \cdot 33.5}$   
 $= 183.007 \text{ mm.}$

Circumferential Stress in shell, no rings (4.15.21) [ $\sigma_6$ ]:  
 $= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$   
 $= -0.7603 \cdot 49527 \cdot 0.1 / (33.5(350.0 + 183.01 + 183.01))$   
 $= -1.54 \text{ N./mm}^2$

Circ. Comp. Stress at Horn of Saddle,  $L < 8R_m$  (4.15.23) [ $\sigma_7^*$ ]:  
 $= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$   
 $= -49527 / (4 \cdot 33.5(350.0 + 183.007 + 183.007)) -$   
 $12 \cdot 0.013 \cdot 49527 \cdot 1643.25 / (7348.799 \cdot 33.5^2)$   
 $= -20.41 \text{ N./mm}^2$

Effective reinforcing plate width (4.15.24) [B1]:  
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$   
 $= \min(350.0 + 1.56 \cdot \sqrt{1643.25 \cdot 33.5}, 2 \cdot 400.0)$   
 $= 716.01 \text{ mm.}$

#### Results for Vessel Ribs, Web and Base:

|                                      |       |           |     |
|--------------------------------------|-------|-----------|-----|
| Baseplate Length                     | Bplen | 3010.0000 | mm. |
| Baseplate Thickness                  | Bpthk | 28.0000   | mm. |
| Baseplate Width                      | Bpwid | 390.0000  | mm. |
| Number of Ribs ( inc. outside ribs ) | Nribs | 5         |     |
| Rib Thickness                        | Ribtk | 16.0000   | mm. |
| Web Thickness                        | Webtk | 16.0000   | mm. |

|                           |        |       |                    |
|---------------------------|--------|-------|--------------------|
| Web Location              | Webloc | Side  |                    |
| Saddle Design Temperature |        | 21.1  | °C                 |
| Saddle Yield Stress       | Sy     | 242.0 | N./mm <sup>2</sup> |
| Height of Web at Center   | Hw,c   | 306.0 | mm.                |
| Friction Coefficient      | mu     | 0.100 |                    |

#### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:  
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$   
 $= ((1.56 \cdot \sqrt{1626.5 \cdot 33.5} + 480.0)^3) / 12$   
 $= 264.466 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:  
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$   
 $= 264.466 + 282.789 (147.629 - 16.75)^2$   
 $= 48704.332 \text{ cm}^4$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>(2)</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|------------------------------------------|
| Shell     | 844.1    | 33.5     | 16.8     | 282.8                | 473670.8               | 0.487E+05                                |
| Wearplate | 480.0    | 16.0     | 41.5     | 76.8                 | 318720.0               | 0.867E+04                                |
| Web       | 16.0     | 426.0    | 262.5    | 68.2                 | 1789200.0              | 0.193E+05                                |
| BasePlate | 390.0    | 28.0     | 489.5    | 109.2                | 5345340.0              | 0.128E+06                                |
| Totals    | ...      | ...      | ...      | 536.9                | 7926930.5              | 0.204E+06                                |

Distance to Centroid [C1]:  
 $= AY / A$   
 $= 7926931 / 536.949$   
 $= 147.629 \text{ mm.}$

Angle [β]:  
 $= 180 - \text{Saddle Angle} / 2$   
 $= 180 - 120.0 / 2$   
 $= 120.0$

Saddle Splitting Coefficient [K1]:  
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$   
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$   
 $= 0.2035$

Saddle Splitting Force [Fh]:  
 $= K1 \cdot Q$   
 $= 0.204 \cdot 49526.746$   
 $= 10079.7666 \text{ Kg}$

Tension Stress, St = ( Fh/As ) = 3.8893 N./mm<sup>2</sup>  
 Allowed Stress, Sa = 0.6 \* Yield Str = 145.1984 N./mm<sup>2</sup>

Saddle Splitting Dimension [d]:  
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$   
 $= 2130.0 - 1626.5 \cdot \sin(120.0/2) / 1.0472$   
 $= 784.895 \text{ mm.}$

Bending Moment, M = Fh \* d = 7911.7178 Kg-m.

Bending Stress, Sb = ( M \* C1 / I ) = 5.6046 N./mm<sup>2</sup>  
 Allowed Stress, Sa = 2/3 \* Yield Str = 161.3315 N./mm<sup>2</sup>

#### Minimum Thickness of Baseplate per Moss:

$= \sqrt[3]{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$   
 $= \sqrt[3]{(3(49527 + 1162) \cdot 390.0) / (2 \cdot 3010.0 \cdot 161.332)}$   
 $= 24.471 \text{ mm.}$

#### Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [ Web Length ]:  
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) ( \text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk} )$   
 $= 2 \cdot \cos(90 - 120.0/2) ( 1625.0 + 35.0 + 16.0 )$

= 2902.917 mm.

Distance between Ribs [e]:  
 = Web Length / ( Nribs - 1 )  
 = 2902.9172 / ( 5 - 1 )  
 = 725.729 mm.

Baseplate Pressure Area [Ap]:  
 = e • Bpwid / 2  
 = 725.7293 • 390.0 / 2  
 = 1415.172 cm<sup>2</sup>

Bearing Pressure [Bp]:  
 = ( Q + SaddleWt ) / ( BasePlateLength • BasePlateWidth )  
 = 50688.387 / ( 3010.0 • 390.0 )  
 = 4.318 Kg/cm<sup>2</sup>

Axial Load [P]:  
 = Ap • Bp  
 = 1415.2 • 4.32  
 = 6110.640 Kg

Area of the Rib and Web [Ar]:  
 = Rib Area + Web Area  
 = 53.44 + 58.058  
 = 111.498 cm<sup>2</sup>

Compressive Stress [Sc]:  
 = P / Ar  
 = 6110.6 / 111.4983  
 = 5.375 N./mm<sup>2</sup>

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|        | B     | D     | Y     | A     | AY        | Io        |
|--------|-------|-------|-------|-------|-----------|-----------|
| Rib    | 16.0  | 334.0 | 183.0 | 53.4  | 977951.9  | 0.941E+04 |
| Web    | 362.9 | 16.0  | 8.0   | 58.1  | 46446.7   | 0.410E+04 |
| Totals | ...   | ...   | ...   | 111.5 | 1024398.6 | 0.135E+05 |

Rib dimension [D]:  
 = Saddle Width - Web Thickness  
 = 350.0 - 16.0  
 = 334.000 mm.

Distance to Centroid from Datum [ytot]:  
 = AY / A  
 = 1024399 / 111.498  
 = 91.876 mm.

Distance to Centroid [C1]:  
 = max( ytot, Saddle Width - ytot )  
 = max( 91.876, 350.0 - 91.876 )  
 = 258.124 mm.

Radius of Gyration [r]:  
 =  $\sqrt{\text{Total Inertia} / \text{Total Area}}$   
 =  $\sqrt{13502.3 / 111.498}$   
 = 110.045 mm.

Length of Outer Rib [L]:  
 = Saddle Height - cos( theta/2 )( radius + shlthk + wpdthk ) - bpthk  
 = 2130.0 - cos( 120.0/2 )( 1625.0 + 35.0 + 16.0 ) - 28.0  
 = 1264.000 mm.

Intermediate Term [Cc]:  
 =  $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$   
 =  $\sqrt{2 \cdot \pi^2 \cdot 0.19904\text{E}+09 / 242.0}$

$$= 127.422$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1264.0/110.045$$

$$= 11.486$$

Bending Moment [Rm]:

$$= FI / ( 2 \cdot B_{plen} ) \cdot e \cdot L / 2$$

$$= 11544.1 / ( 2 \cdot 3010.0 ) \cdot 725.729 \cdot 1264.0 / 2$$

$$= 879.560 \text{ Kg-m.}$$

Compressive Allowable,  $KL/r < C_c$  ( 11.4862 < 127.4221 ) per AISC E2-1 [Sca]:

$$= ( 1 - (KL/r)^2 / (2 \cdot C_c^2) ) F_y / ( 5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3) )$$

$$= ( 1 - ( 11.49 )^2 / ( 2 \cdot 127.42^2 ) ) 242 / ( 5/3 + 3 \cdot ( 11.49 ) / ( 8 \cdot 127.42 ) - ( 11.49^3 ) / ( 8 \cdot 127.42^3 ) )$$

$$= 141.7 \text{ N./mm}^2$$

**AISC Unity Check of Outside Ribs ( must be  $\leq 1$  )**

$$= S_c / S_{ca} + ( R_m \cdot C_1 / I ) / S_{ba}$$

$$= 5.37 / 141.74 + ( 879.56 \cdot 258.124 / 13502E+09 ) / 161.33$$

$$= 0.140$$

**Check of Inside Ribs:**

Inertia of Saddle, Inner Ribs - Axial Direction

|        | B     | D     | Y     | A     | AY        | Io        |
|--------|-------|-------|-------|-------|-----------|-----------|
| Rib    | 16.0  | 334.0 | 183.0 | 53.4  | 977951.9  | 0.126E+05 |
| Web    | 725.7 | 16.0  | 8.0   | 116.1 | 92893.3   | 0.356E+04 |
| Totals | ...   | ...   | ...   | 169.6 | 1070845.2 | 0.162E+05 |

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1070845 / 169.557$$

$$= 63.156 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max( y_{tot}, \text{abs}( \text{Saddle Width} - y_{tot} ) )$$

$$= \max( 63.156, \text{abs}( 350.0 - 63.156 ) )$$

$$= 286.844 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 2130.0 - \cos( 1676.0/2 ) ( 28.0 + 0.0 + 0.0 ) - 0.0$$

$$= 426.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ ( \text{Total Inertia} / \text{Total Area} ) }$$

$$= \sqrt{ ( 16200.6 / 169.557 ) }$$

$$= 97.748 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 426.0 / 97.748$$

$$= 4.358$$

Unit Force [Force,u]:

$$= FI / ( 2 \cdot \text{Baseplate Length} )$$

$$= 11544.137 / ( 2 \cdot 3010.0 )$$

$$= 1.918 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.918 \cdot 725.729 \cdot 426.0$$

$$= 592.868 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [ $\sigma_B$ ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 592.868 / 564786$$

= 10.294 N./mm<sup>2</sup>

Compressive Allowable,  $KL/r < C_c$  ( 4.3581 < 127.4221 ) per AISC E2-1 [Sca]:

$$= ( 1 - (KL/r)^2 / (2 \cdot C_c^2) ) F_y / ( 5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3) )$$

$$= ( 1 - ( 4.36 )^2 / ( 2 \cdot 127.42^2 ) ) 242 / ( 5/3 + 3 \cdot ( 4.36 ) / ( 8 \cdot 127.42 ) - ( 4.36^3 ) / ( 8 \cdot 127.42^3 ) )$$

$$= 144 \text{ N./mm}^2$$

#### AISC Unity Check of Inside Ribs ( must be $\leq 1$ )

$$= S_c / S_{ca} + ( M_{base,c} \cdot C_1 / I ) / S_{ba}$$

$$= 6.97 / 144.01 + ( 592.87 \cdot 286.844 / 16200.564 ) / 161.33$$

$$= 0.112$$

#### Input Data for Base Plate Bolting Calculations:

|                                            |         |                           |
|--------------------------------------------|---------|---------------------------|
| Total Number of Bolts per BasePlate        | Nbolts  | 4                         |
| Total Number of Bolts in Tension/Baseplate | Nbt     | 2                         |
| Bolt Design Temperature                    |         | 21.1 °C                   |
| Bolt Material Specification                |         | SA-193 B7                 |
| Bolt Allowable Stress                      | Stba    | 150.00 N./mm <sup>2</sup> |
| Bolt Corrosion Allowance                   | Bca     | 1.5 mm.                   |
| Distance from Bolts to Edge                | Edgedis | 330.0 mm.                 |
| Nominal Bolt Diameter                      | Bnd     | 30.0000 mm.               |
| Thread Series                              | Series  | TEMA Metric               |
| BasePlate Allowable Stress                 | S       | 138.00 N./mm <sup>2</sup> |
| Area Available in a Single Bolt            | BltArea | 3.9088 cm <sup>2</sup>    |
| Saddle Load Q0 (Weight)                    | Q0      | 42217.9 Kg                |
| Saddle Load QL (Wind/Seismic contribution) | QL      | 7264.1 Kg                 |
| Maximum Transverse Force                   | Ft      | 3848.0 Kg                 |
| Maximum Longitudinal Force                 | Fl      | 11544.1 Kg                |
| Saddle Bolted to Steel Foundation          |         | No                        |

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= Fl / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$$

$$= 11544.1 / ( 3.909 \cdot 4 )$$

$$= 72.4 \text{ N./mm}^2. \text{ Must be less than } 100.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$$

$$= 3848.0 / ( 3.909 \cdot 4 )$$

$$= 24.1 \text{ N./mm}^2. \text{ Must be less than } 100.0 \text{ N./mm}^2.$$

#### Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ ( } Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction )}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / ( \text{BoltShearAllowable} \cdot \text{Nbolts} )$$

$$= 11544.14 / ( 100.0 \cdot 4.0 )$$

$$= 2.8303 \text{ cm}^2$$

#### Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2130.0 \cdot 3848.05 + 0.0$$

$$= 8196.50 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / Q_0$$

$$= 8196.5 / 42217.94$$

$$= 194.14 \text{ mm.} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 2.8303, 0.0]$$

= 2.8303 cm<sup>2</sup>

**MDMT Calculations for: SLIDING SADDLE, Saddle to Wearplate Junction, Curve: B**

Govrn. thk, tg = 16.0, c = 1.5 mm.,  $E^* = 1.0$   
Thickness Ratio =  $tr * E^* / (tg - c) = 0.961$ , Temp. Reduction = 2 °C  
Attachment governing, Attachment stress = Shell stress per Code

|                                                      |        |
|------------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B       | -14 °C |
| Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit | -31 °C |
| Min Metal Temp. at Required thickness (UCS 66.1)     | -33 °C |

*Review notes about nozzle loadings and supports in the Warnings report.*

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### ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

There are more than two saddles present in this model. The load on the saddle supports has been computed as the total weight divided by the number of saddles. Ensure that this is an appropriate assumption for this vessel model. It may be necessary to manually enter the bending moments at the saddle through the Saddle Input dialog. Also note that the dimension Ls is taken to be the distance between the adjoining saddles (first two or last two).

Note:

Wear Pad Width (260.00) is less than  $1.56 \cdot \sqrt{R_m \cdot t}$   
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

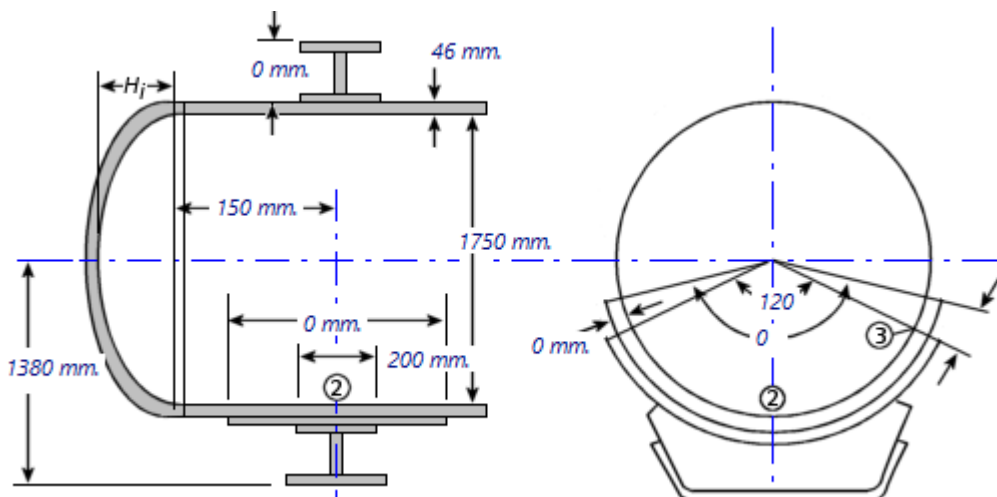
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{898.75 \cdot 44.5}, 2 \cdot 150.0)$$

$$= 300.0000 \text{ mm.}$$

#### Input and Calculated Values:

|                                               |                |         |                    |
|-----------------------------------------------|----------------|---------|--------------------|
| Vessel Mean Radius                            | Rm             | 898.75  | mm.                |
| Shell Thickness used in this Case             | t              | 46.000  | mm.                |
| Stiffened Vessel Length per 4.15.6            | L              | 7348.80 | mm.                |
| Distance from Saddle to Vessel tangent        | a1 or a        | 150.00  | mm.                |
| Saddle Width                                  | b1 or b        | 200.00  | mm.                |
| Saddle Bearing Angle                          | delta or theta | 120.00  | degrees            |
| Shell Allowable Stress used in Calculation    |                | 234.00  | N./mm <sup>2</sup> |
| Head Allowable Stress used in Calculation     |                | 234.00  | N./mm <sup>2</sup> |
| Circumferential Efficiency in Plane of Saddle |                | 1.00    |                    |
| Circumferential Efficiency at Mid-Span        |                | 1.00    |                    |
| Distance from Saddle Base to Centerline       | B              | 1380.00 | mm.                |
| Coefficient of Friction                       | mu             | 0.60    |                    |



#### Results for Horizontal Vessel Analysis:

|                                           |   |          |      |
|-------------------------------------------|---|----------|------|
| Saddle Force Q, Test Case, no Ext. Forces |   | 43558.52 | Kgf  |
| Pressure used in Analysis                 | P | 35.187   | bars |

| Horizontal Vessel Analysis Results: |            | Actual<br>N./mm <sup>2</sup> | Allowable<br>N./mm <sup>2</sup> |
|-------------------------------------|------------|------------------------------|---------------------------------|
| Long. Stress at Top                 | of Midspan | 38.45                        | 234.00                          |
| Long. Stress at Bottom              | of Midspan | 55.96                        | 234.00                          |
| Long. Stress at Top                 | of Saddles | 35.34                        | 234.00                          |

|                                   |       |        |
|-----------------------------------|-------|--------|
| Long. Stress at Bottom of Saddles | 35.73 | 234.00 |
| Tangential Shear in Shell         | 9.40  | 138.58 |
| Tangential Shear in Head          | 4.47  | 122.36 |
| Circ. Stress at Horn of Saddle    | 8.96  | 350.99 |
| Circ. Compressive Stress in Shell | 1.43  | 234.00 |

#### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:  
 $= Ftr( Ft/Num\ of\ Saddles + Z\ Force\ Load ) \cdot B / E$   
 $= 3.0( 1155.7/3 + 0 ) \cdot 1380.0/1610.0$   
 $= 990.6\ Kgf$

Longitudinal Saddle Reaction Force [Fwl]:  
 $= \max( Fl,wind; Sum\ of\ X\ Forces ) \cdot B / Ls$   
 $= \max( 261.02, 0 ) \cdot 1380.0/3173.0$   
 $= 113.5\ Kgf$

Load Combination Results for Q + Wind or Seismic [Q]:  
 $= Saddle\ Load + \max( Fwl, Fwt, Fsl, Fst )$   
 $= 42568 + \max( 114, 991, 0, 0 )$   
 $= 43558.5\ Kgf$

Longitudinal Wind Force [Fl]:  
 $= WindScalar \cdot WindPress( Platform\ Area + ( \pi/4( OD \cdot WindDiaMult )^2 ) )$   
 $= 1.1 \cdot 1587.169( 0.0 + \pi/4( 1.982 \cdot 1.2 )^2 )$   
 $= 7756.729\ N$

#### Summary of Loads at the Base of this Saddle:

|                                         |          |     |
|-----------------------------------------|----------|-----|
| Vertical Load (including saddle weight) | 43974.24 | Kgf |
| Transverse Shear Load                   | 385.24   | Kgf |
| Longitudinal Shear Load                 | 261.02   | Kgf |

Note: Wear Plate is Welded to the Shell,  $k = 0.1$

Saddle Dimension [E]:  
 $= Baseplate\ Length$   
 $= 1610.000\ mm.$

#### The Computed K values from Table 4.15.1:

|             |              |              |             |
|-------------|--------------|--------------|-------------|
| K1 = 0.1066 | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011 |
| K5 = 0.7603 | K6 = 0.0529  | K7 = 0.0132  | K8 = 0.3405 |
| K9 = 0.2711 | K10 = 0.0581 | K1* = 0.1923 |             |

Note: Dimension a is less than  $Rm/2$ .

Moment per Equation 4.15.1 [M1]:  
 $= -Q \cdot a [ 1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L) ]$   
 $= -43559 \cdot 150.0 [ 1 - (1 - 150.0/7348.8 + (898.75^2 - 0.0^2)/(2 \cdot 150.0 \cdot 7348.8))/(1 + (4 \cdot 0.0)/(3 \cdot 7348.8)) ]$   
 $= 2260.6\ Kg-m.$

Moment per Equation 4.15.2 [M2]:  
 $= Q \cdot L/4 \{ (1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L \}$   
 $= 43559 \cdot 7349/4 \{ (1 + 2(899^2 - 0^2)/(7349^2))/(1 + (4 \cdot 0)/(3 \cdot 7349)) - 4 \cdot 150/7349 \}$   
 $= 75887.3\ Kg-m.$

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:  
 $= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$   
 $= 35.187 \cdot 898.75/(2 \cdot 33.5) - 75887.3/(\pi \cdot 898.8^2 \cdot 33.5)$   
 $= 38.45\ N./mm^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:  
 $= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$   
 $= 35.187 \cdot 898.75/(2 \cdot 33.5) + 75887.3/(\pi \cdot 898.8^2 \cdot 33.5)$   
 $= 55.96\ N./mm^2$

Longitudinal Stress at Top of Shell at Support (4.15.6) [ $\sigma_3$ ]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 35.187 \cdot 898.75 / (2 \cdot 44.5) - 2260.6 / (\pi \cdot 898.8^2 \cdot 44.5)$$

$$= 35.34 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [ $\sigma_4$ ]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 35.187 \cdot 898.75 / (2 \cdot 44.5) + 2260.6 / (\pi \cdot 898.8^2 \cdot 44.5)$$

$$= 35.73 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 43559(7348.8 - 2 \cdot 150.0) / (7348.8 + (4 \cdot 0.0 / 3))$$

$$= 41780.3 \text{ Kgf}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [ $\tau_3$ ]:

$$= K_3 \cdot Q / (R_m \cdot t)$$

$$= 0.8799 \cdot 43559 / (898.75 \cdot 44.5)$$

$$= 9.40 \text{ N./mm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [ $\tau_{3*}$ ]:

$$= K_3 \cdot Q / (R_m \cdot t_h)$$

$$= 0.8799 \cdot 43559 / (898.75 \cdot 93.5)$$

$$= 4.47 \text{ N./mm}^2$$

Decay Length (4.15.20) [ $x_1, x_2$ ]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(898.75 \cdot 44.5)}$$

$$= 155.989 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [ $\sigma_6$ ]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 43559 \cdot 0.1 / (44.5(200.0 + 155.99 + 155.99))$$

$$= -1.43 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle,  $L \geq 8R_m$  (4.15.22) [ $\sigma_7$ ]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -43559 / (4 \cdot 44.5(200.0 + 155.989 + 155.989)) - 3 \cdot 0.0132 \cdot 43559 / (2 \cdot 44.5^2)$$

$$= -8.96 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{(898.75 \cdot 44.5)}, 2 \cdot 150.0)$$

$$= 300.00 \text{ mm.}$$

#### Results for Vessel Ribs, Web and Base:

|                                      |        |           |                    |
|--------------------------------------|--------|-----------|--------------------|
| Baseplate Length                     | Bplen  | 1610.0000 | mm.                |
| Baseplate Thickness                  | Bpthk  | 28.0000   | mm.                |
| Baseplate Width                      | Bpwid  | 240.0000  | mm.                |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 4         |                    |
| Rib Thickness                        | Ribtk  | 16.0000   | mm.                |
| Web Thickness                        | Webtk  | 16.0000   | mm.                |
| Web Location                         | Webloc | Side      |                    |
| Saddle Yield Stress                  | Sy     | 260.0     | N./mm <sup>2</sup> |
| Height of Web at Center              | Hw,c   | 306.0     | mm.                |
| Friction Coefficient                 | mu     | 0.600     |                    |

#### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(876.5 \cdot 44.5)} + 260.0)^3) / 12$$

$$= 417.174 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 417.174 + 252.801(136.678 - 22.25)^2$$

$$= 33518.469 \text{ cm}^4$$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>(2)</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|------------------------------------------|
| Shell     | 568.1    | 44.5     | 22.2     | 252.8                | 562482.1               | 0.335E+05                                |
| Wearplate | 260.0    | 16.0     | 52.5     | 41.6                 | 218400.0               | 0.296E+04                                |
| Web       | 16.0     | 415.0    | 268.0    | 66.4                 | 1779520.8              | 0.210E+05                                |
| BasePlate | 240.0    | 28.0     | 489.5    | 67.2                 | 3289440.5              | 0.837E+05                                |
| Totals    | ...      | ...      | ...      | 428.0                | 5849843.0              | 0.141E+06                                |

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5849843/428.001$$

$$= 136.678 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 43558.52$$

$$= 8865.1035 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 4.9622 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1380.0 - 876.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 655.140 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 5808.0020 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 5.5152 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

**Minimum Thickness of Baseplate per Moss:**

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(43559 + 416)240.0 / (2 \cdot 1610.0 \cdot 173.33)}$$

$$= 23.587 \text{ mm.}$$

**Calculation of Axial Load, Intermediate Values and Compressive Stress:**

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (875.0 + 46.0 + 16.0)$$

$$= 1622.932 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 1622.9315 / (4 - 1)$$

$$= 540.977 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 540.9772 \cdot 240.0/2$$

$$= 649.173 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 43974.242 / (1610.0 \cdot 240.0)$$

$$= 11.380 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 649.2 \cdot 11.38$$

$$= 7387.908 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 29.44 + 43.278$$

$$= 72.718 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 7387.9 / 72.7182$$

$$= 9.963 \text{ N./mm}^2$$

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|        | B     | D     | Y     | A    | AY       | Io        |
|--------|-------|-------|-------|------|----------|-----------|
| Rib    | 16.0  | 184.0 | 108.0 | 29.4 | 317951.9 | 0.187E+04 |
| Web    | 270.5 | 16.0  | 8.0   | 43.3 | 34622.5  | 719.      |
| Totals | ...   | ...   | ...   | 72.7 | 352574.5 | 0.259E+04 |

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200.0 - 16.0$$

$$= 184.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 352574.5 / 72.718$$

$$= 48.485 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max( y_{tot}, \text{Saddle Width} - y_{tot} )$$

$$= \max( 48.485, 200.0 - 48.485 )$$

$$= 151.515 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 2591.9 / 72.718 }$$

$$= 59.702 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos( \text{saddle bearing angle} / 2 ) ( \text{radius} + \text{shlthk} + \text{wpdthk} ) - \text{bpthk}$$

$$= 1380.0 - \cos( 120.0 / 2 ) ( 875.0 + 46.0 + 16.0 ) - 28.0$$

$$= 883.500 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin( \text{saddle bearing angle} / 2 ) ( \text{radius} + \text{shlthk} + \text{wpdthk} )$$

$$= 2 \cdot \sin( 120.0 / 2 ) ( 875.0 + 46.0 + 16.0 )$$

$$= 1622.932 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 1610.0 - 50.8$$

$$= 1559.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= ( L_w - B_d ) / 2$$

$$= ( 1622.932 - 1559.2 ) / 2$$

$$= 31.866 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{ L_v^2 + D_d^2 }$$

$$= \sqrt{ 883.5^2 + 31.866^2 }$$

$$= 884.075 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19904\text{E}+09 / 260.0)}$$

$$= 122.933$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 884.075 / 59.702$$

$$= 14.808$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 261.0 / (2 \cdot 1610.0) \cdot 540.977 \cdot 884.07 / 2$$

$$= 19.385 \text{ Kg-m.}$$

Compressive Allowable,  $KL/r < C_c$  (  $14.808 < 122.9326$  ) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (14.81)^2 / (2 \cdot 122.93^2)) 260 /$$

$$(5/3 + 3 \cdot (14.81) / (8 \cdot 122.93) - (14.81)^3 / (8 \cdot 122.93^3))$$

$$= 150.8 \text{ N./mm}^2$$

**AISC Unity Check of Outside Ribs ( must be  $\leq 1$  )**

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 9.96 / 150.8 + (19.39 \cdot 151.515 / 25919526) / 173.33$$

$$= 0.072$$

#### Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

|        | B     | D     | Y     | A     | AY       | Io        |
|--------|-------|-------|-------|-------|----------|-----------|
| Rib    | 16.0  | 184.0 | 108.0 | 29.4  | 317951.9 | 0.247E+04 |
| Web    | 541.0 | 16.0  | 8.0   | 86.6  | 69245.1  | 576.      |
| Totals | ...   | ...   | ...   | 116.0 | 387197.0 | 0.305E+04 |

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 387197.031 / 115.996$$

$$= 33.380 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}( \text{Saddle Width} - y_{tot} ))$$

$$= \max(33.38, \text{abs}(200.0 - 33.38))$$

$$= 166.620 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{((Ro + W_{pdthk})^2 - (\text{Pitch}/2)^2)} - B_{pthk}$$

$$= 1380.0 - \sqrt{(937.0 + 16.0)^2 - (540.977/2)^2} - 28.0$$

$$= 454.891 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3045.9 / 115.996)}$$

$$= 51.243 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 454.891 / 51.243$$

$$= 8.877$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 261.023 / (2 \cdot 1610.0)$$

$$= 0.081 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.081 \cdot 540.977 \cdot 454.891$$

$$= 19.949 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [ $\sigma_{B,base,c}$ ]:

= Bending Moment / Section Modulus  
 = 19.949/182803.797  
 = 1.070 N./mm<sup>2</sup>

Compressive Allowable,  $KL/r < C_c$  ( 8.8771 < 122.9326 ) per AISC E2-1 [Sca]:  
 =  $(1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$   
 =  $(1 - (8.88)^2 / (2 \cdot 122.93^2)) 260 /$   
 $(5/3 + 3 \cdot (8.88) / (8 \cdot 122.93) - (8.88^3) / (8 \cdot 122.93^3))$   
 = 153.1 N./mm<sup>2</sup>

#### AISC Unity Check of Inside Ribs ( must be $\leq 1$ )

=  $Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$   
 =  $12.25/153.11 + (19.95 \cdot 166.62/3045.872) / 173.33$   
 = 0.086

#### Input Data for Base Plate Bolting Calculations:

|                                            |         |                           |
|--------------------------------------------|---------|---------------------------|
| Total Number of Bolts per BasePlate        | Nbolts  | 3                         |
| Total Number of Bolts in Tension/Baseplate | Nbt     | 2                         |
| Bolt Material Specification                |         | SA-193 B7                 |
| Bolt Allowable Stress                      | Stba    | 172.00 N./mm <sup>2</sup> |
| Bolt Corrosion Allowance                   | Bca     | 0.0 mm.                   |
| Distance from Bolts to Edge                | Edgedis | 85.0 mm.                  |
| Nominal Bolt Diameter                      | Bnd     | 30.0000 mm.               |
| Thread Series                              | Series  | TEMA Metric               |
| BasePlate Allowable Stress                 | S       | 138.00 N./mm <sup>2</sup> |
| Area Available in a Single Bolt            | BltArea | 5.0297 cm <sup>2</sup>    |
| Saddle Load QO (Weight)                    | QO      | 42983.6 Kg                |
| Saddle Load QL (Wind/Seismic contribution) | QL      | 113.5 Kg                  |
| Maximum Transverse Force                   | Ft      | 385.2 Kg                  |
| Maximum Longitudinal Force                 | Fl      | 261.0 Kg                  |
| Saddle Bolted to Steel Foundation          |         | No                        |

Shear Stress in a Single Bolt, Longitudinal Direction [ $\tau_b, l$ ]:  
 =  $Fl / (Bolt Area \cdot Number of Bolts)$   
 =  $261.0 / (5.03 \cdot 3)$   
 = 1.7 N./mm<sup>2</sup>. Must be less than 103.2 N./mm<sup>2</sup>.

Shear Stress in a Single Bolt, Transverse Direction [ $\tau_b, t$ ]:  
 =  $Ft / (Bolt Area \cdot Number of Bolts)$   
 =  $385.2 / (5.03 \cdot 3)$   
 = 2.5 N./mm<sup>2</sup>. Must be less than 103.2 N./mm<sup>2</sup>.

#### Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:  
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:  
 =  $Fl / (BoltShearAllowable \cdot Nbolts)$   
 =  $261.02 / (103.2 \cdot 3.0)$   
 = 0.0827 cm<sup>2</sup>

#### Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:  
 =  $B \cdot Ft + \text{Sum of X Moments}$   
 =  $1380.0 \cdot 385.24 + 0.0$   
 = 531.64 Kg-m.

Eccentricity (e):  
 =  $Rmom / QO$   
 =  $531.64 / 42983.62$   
 = 12.37 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:  
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:  
 = max[Bltarearl, Bltarears, Bltareart]

= max[0.0, 0.0827, 0.0]  
 = 0.0827 cm<sup>2</sup>

### ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

#### Note:

Wear Pad Width (480.00) is less than  $1.56 \cdot \sqrt{rm \cdot t}$   
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

= min( b +  $1.56 \cdot \sqrt{Rm \cdot t}$ , 2a )  
 = min( 350.0 +  $1.56 \cdot \sqrt{1643.25 \cdot 33.5}$ , 2 • 400.0 )  
 = 716.0149 mm.

#### Input and Calculated Values:

|                                               |                |         |                    |
|-----------------------------------------------|----------------|---------|--------------------|
| Vessel Mean Radius                            | Rm             | 1643.25 | mm.                |
| Shell Thickness used in this Case             | t              | 35.000  | mm.                |
| Stiffened Vessel Length per 4.15.6            | L              | 7348.80 | mm.                |
| Distance from Saddle to Vessel tangent        | a1 or a        | 400.00  | mm.                |
| Saddle Width                                  | b1 or b        | 350.00  | mm.                |
| Saddle Bearing Angle                          | delta or theta | 120.00  | degrees            |
| Inside or Mean Depth of Head                  | Hi or hm       | 830.75  | mm.                |
| Shell Allowable Stress used in Calculation    |                | 234.00  | N./mm <sup>2</sup> |
| Head Allowable Stress used in Calculation     |                | 234.00  | N./mm <sup>2</sup> |
| Circumferential Efficiency in Plane of Saddle |                | 1.00    |                    |
| Circumferential Efficiency at Mid-Span        |                | 1.00    |                    |
| Distance from Saddle Base to Centerline       | B              | 2130.00 | mm.                |
| Coefficient of Friction                       | mu             | 0.10    |                    |

#### Results for Horizontal Vessel Analysis:

|                                           |   |          |      |
|-------------------------------------------|---|----------|------|
| Saddle Force Q, Test Case, no Ext. Forces |   | 43415.91 | Kgf  |
| Pressure used in Analysis                 | P | 35.260   | bars |

| Horizontal Vessel Analysis Results: | Actual<br>N./mm <sup>2</sup> | Allowable<br>N./mm <sup>2</sup> |
|-------------------------------------|------------------------------|---------------------------------|
| Long. Stress at Top of Midspan      | 84.51                        | 234.00                          |
| Long. Stress at Bottom of Midspan   | 88.46                        | 234.00                          |
| Long. Stress at Top of Saddles      | 86.41                        | 234.00                          |
| Long. Stress at Bottom of Saddles   | 86.56                        | 234.00                          |
| Tangential Shear in Shell           | 6.81                         | 138.58                          |
| Tangential Shear in Head            | 6.81                         | 128.98                          |
| Circ. Stress at Horn of Saddle      | 17.89                        | 350.99                          |
| Addl. Stress in Head as Stiffener   | 170.70                       | 234.00                          |
| Circ. Compressive Stress in Shell   | 1.35                         | 234.00                          |

#### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr( Ft/Num of Saddles + Z Force Load ) • B / E  
 = 3.0( 1155.7/3 + 0 ) • 2130.0/2902.9172  
 = 848.0 Kgf

Longitudinal Saddle Reaction Force [Fwl]:

= max( Fl,wind; Sum of X Forces ) • B / Ls  
 = max( 795.47, 0 ) • 2130.0/3385.0  
 = 500.5 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max( Fwl, Fwt, Fsl, Fst )  
 = 42568 + max( 501, 848, 0, 0 )

= 43415.9 Kgf

Longitudinal Wind Force [FI]:  
 = WindScalar • WindPress( Platform Area + (  $\pi/4$ ( OD • WindDiaMult )<sup>2</sup> ) )  
 = 1.1 • 1587.169( 0.0 +  $\pi/4$ ( 3.46 • 1.2 )<sup>2</sup> )  
 = 23638.697 N

**Summary of Loads at the Base of this Saddle:**

|                                         |          |     |
|-----------------------------------------|----------|-----|
| Vertical Load (including saddle weight) | 44577.55 | Kgf |
| Transverse Shear Load                   | 385.24   | Kgf |
| Longitudinal Shear Load                 | 795.47   | Kgf |

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:  
 = min( 2( ShellID/2 + t + WearPadThickness )sin( $\theta/2$ ), 2•Rm )  
 = min( 2( 3250.0/2 + 35.0 + 16.0 )sin( 120.0/2 ), 2•1643.25 )  
 = 2902.917 mm.

**The Computed K values from Table 4.15.1:**

|             |              |              |             |
|-------------|--------------|--------------|-------------|
| K1 = 0.1066 | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011 |
| K5 = 0.7603 | K6 = 0.0529  | K7 = 0.0132  | K8 = 0.3405 |
| K9 = 0.2711 | K10 = 0.0581 | K1* = 0.1923 |             |

Note: Dimension a is less than Rm/2.

Moment per Equation 4.15.1 [M1]:  
 =  $-Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$   
 =  $-43416 \cdot 400.0 [1 - (1 - 400.0/7348.8 + (1643.25^2 - 830.75^2)/(2 \cdot 400.0 \cdot 7348.8))/(1 + (4 \cdot 830.75)/(3 \cdot 7348.8))]$   
 = 2063.9 Kg-m.

Moment per Equation 4.15.2 [M2]:  
 =  $Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$   
 =  $43416 \cdot 7349/4((1 + 2(1643^2 - 831^2)/(7349^2))/(1 + (4 \cdot 831)/(3 \cdot 7349)) - 4 \cdot 400/7349)$   
 = 57110.7 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:  
 =  $P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$   
 =  $35.26 \cdot 1643.25/(2 \cdot 33.5) - 57110.7/(\pi \cdot 1643.2^2 \cdot 33.5)$   
 = 84.51 N./mm<sup>2</sup>

Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:  
 =  $P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$   
 =  $35.26 \cdot 1643.25/(2 \cdot 33.5) + 57110.7/(\pi \cdot 1643.2^2 \cdot 33.5)$   
 = 88.46 N./mm<sup>2</sup>

Longitudinal Stress at Top of Shell at Support (4.15.6) [ $\sigma_3$ ]:  
 =  $P \cdot Rm/(2t) - M1/(\pi \cdot Rm^2 \cdot t)$   
 =  $35.26 \cdot 1643.25/(2 \cdot 33.5) - 2063.9/(\pi \cdot 1643.2^2 \cdot 33.5)$   
 = 86.41 N./mm<sup>2</sup>

Longitudinal Stress at bottom of Shell at Support (4.15.7) [ $\sigma_4$ ]:  
 =  $P \cdot Rm/(2t) + M1/(\pi \cdot Rm^2 \cdot t)$   
 =  $35.26 \cdot 1643.25/(2 \cdot 33.5) + 2063.9/(\pi \cdot 1643.2^2 \cdot 33.5)$   
 = 86.56 N./mm<sup>2</sup>

Maximum Shear Force in the Saddle (4.15.3) [T]:  
 =  $Q(L - 2a)/(L + (4 \cdot hm/3))$   
 =  $43416(7348.8 - 2 \cdot 400.0)/(7348.8 + (4 \cdot 830.75/3))$   
 = 33621.9 Kgf

Shear Stress in the shell no rings, stiffened (4.15.13) [ $\tau_3$ ]:  
 =  $K3 \cdot Q / (Rm \cdot t)$   
 =  $0.8799 \cdot 43416 / (1643.25 \cdot 33.5)$   
 = 6.81 N./mm<sup>2</sup>

Shear Stress in the head, shell stiffened (4.15.14) [ $\tau_{3*}$ ]:  
 =  $K3 \cdot Q / (Rm \cdot th)$

$$= 0.8799 \cdot 43416 / (1643.25 \cdot 33.5)$$

$$= 6.81 \text{ N./mm}^2$$

Membrane stress in the head as a stiffener (4.15.16) [ $\sigma_5$ ]:

$$= K_4 \cdot Q / (R_m \cdot t_h) + ((P \cdot R_i) / (2 \cdot t_h)) \cdot (R_i / h_m)$$

$$= 0.4011 \cdot 43416 / (1643.25 \cdot 33.5) + ((35.26 \cdot 1626.5) / (2 \cdot 33.5)) \cdot (1626.5 / 830.75)$$

$$= 170.70 \text{ N./mm}^2$$

Decay Length (4.15.20) [ $x_1, x_2$ ]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1643.25 \cdot 33.5)}$$

$$= 183.007 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [ $\sigma_6$ ]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 43416 \cdot 0.1 / (33.5(350.0 + 183.01 + 183.01))$$

$$= -1.35 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle,  $L < 8R_m$  (4.15.23) [ $\sigma_7^*$ ]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -43416 / (4 \cdot 33.5(350.0 + 183.007 + 183.007)) - 12 \cdot 0.013 \cdot 43416 \cdot 1643.25 / (7348.799 \cdot 33.5^2)$$

$$= -17.89 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(350.0 + 1.56 \cdot \sqrt{(1643.25 \cdot 33.5)}, 2 \cdot 400.0)$$

$$= 716.01 \text{ mm.}$$

#### Results for Vessel Ribs, Web and Base:

|                                      |        |           |                    |
|--------------------------------------|--------|-----------|--------------------|
| Baseplate Length                     | Bplen  | 3010.0000 | mm.                |
| Baseplate Thickness                  | Bpthk  | 28.0000   | mm.                |
| Baseplate Width                      | Bpwid  | 390.0000  | mm.                |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 5         |                    |
| Rib Thickness                        | Ribtk  | 16.0000   | mm.                |
| Web Thickness                        | Webtk  | 16.0000   | mm.                |
| Web Location                         | Webloc | Side      |                    |
| Saddle Yield Stress                  | Sy     | 260.0     | N./mm <sup>2</sup> |
| Height of Web at Center              | Hw,c   | 306.0     | mm.                |
| Friction Coefficient                 | mu     | 0.100     |                    |

#### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1626.5 \cdot 33.5)} + 480.0)^3) / 12$$

$$= 264.466 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 264.466 + 282.789(147.629 - 16.75)^2$$

$$= 48704.332 \text{ cm}^4$$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>2</sup> (2)<br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|--------------------------------------------|
| Shell     | 844.1    | 33.5     | 16.8     | 282.8                | 473670.8               | 0.487E+05                                  |
| Wearplate | 480.0    | 16.0     | 41.5     | 76.8                 | 318720.0               | 0.867E+04                                  |
| Web       | 16.0     | 426.0    | 262.5    | 68.2                 | 1789200.0              | 0.193E+05                                  |
| BasePlate | 390.0    | 28.0     | 489.5    | 109.2                | 5345340.0              | 0.128E+06                                  |
| Totals    | ...      | ...      | ...      | 536.9                | 7926930.5              | 0.204E+06                                  |

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7926931 / 536.949$$

$$= 147.629 \text{ mm.}$$

Angle [ $\beta$ ]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120.0/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.204 \cdot 43415.906 \\
 &= 8836.0781 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh/As) = 3.4094 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 2130.0 - 1626.5 \cdot \sin(120.0/2) / 1.0472 \\
 &= 784.895 \text{ mm.}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 6935.5327 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 4.9130 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2
 \end{aligned}$$

#### Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{(3(43416 + 1162) \cdot 390.0) / (2 \cdot 3010.0 \cdot 173.33)} \\
 &= 22.140 \text{ mm.}
 \end{aligned}$$

#### Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120.0/2) (1625.0 + 35.0 + 16.0) \\
 &= 2902.917 \text{ mm.}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (N_{\text{ribs}} - 1) \\
 &= 2902.9172 / (5 - 1) \\
 &= 725.729 \text{ mm.}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 725.7293 \cdot 390.0/2 \\
 &= 1415.172 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 44577.547 / (3010.0 \cdot 390.0) \\
 &= 3.797 \text{ Kgf/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap \cdot Bp \\
 &= 1415.2 \cdot 3.8 \\
 &= 5373.958 \text{ Kgf}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 53.44 + 58.058 \\
 &= 111.498 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P/Ar \\
 &= 5374.0/111.4983 \\
 &= 4.727 \text{ N./mm}^2
 \end{aligned}$$

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|        | B     | D     | Y     | A     | AY        | Io        |
|--------|-------|-------|-------|-------|-----------|-----------|
| Rib    | 16.0  | 334.0 | 183.0 | 53.4  | 977951.9  | 0.941E+04 |
| Web    | 362.9 | 16.0  | 8.0   | 58.1  | 46446.7   | 0.410E+04 |
| Totals | ...   | ...   | ...   | 111.5 | 1024398.6 | 0.135E+05 |

Rib dimension [D]:  
 = Saddle Width - Web Thickness  
 = 350.0 - 16.0  
 = 334.000 mm.

Distance to Centroid from Datum [ytot]:  
 = AY / A  
 = 1024399/111.498  
 = 91.876 mm.

Distance to Centroid [C1]:  
 = max( ytot, Saddle Width - ytot )  
 = max( 91.876, 350.0 - 91.876 )  
 = 258.124 mm.

Radius of Gyration [r]:  
 =  $\sqrt{\text{Total Inertia} / \text{Total Area}}$   
 =  $\sqrt{13502.3/111.498}$   
 = 110.045 mm.

Length of Outer Rib [L]:  
 = Saddle Height - cos( theta/2 )( radius + shlthk + wpdthk ) - bpthk  
 = 2130.0 - cos( 120.0/2 )( 1625.0 + 35.0 + 16.0 ) - 28.0  
 = 1264.000 mm.

Intermediate Term [Cc]:  
 =  $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$   
 =  $\sqrt{2 \cdot \pi^2 \cdot 0.19904\text{E}+09/260.0}$   
 = 122.933

Slenderness ratio [KL/r]:  
 = KL/r  
 = 1 • 1264.0/110.045  
 = 11.486

Bending Moment [Rm]:  
 = FI / ( 2 • Bplen ) • e • L / 2  
 = 795.5/ ( 2 • 3010.0 ) • 725.729 • 1264.0/2  
 = 60.608 Kg-m.

Compressive Allowable, KL/r < Cc ( 11.4862 < 122.9326 ) per AISC E2-1 [Sca]:  
 =  $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$   
 =  $(1 - (11.49)^2 / (2 \cdot 122.93^2)) 260 / (5/3 + 3 \cdot (11.49) / (8 \cdot 122.93) - (11.49^3) / (8 \cdot 122.93^3))$   
 = 152.1 N./mm<sup>2</sup>

**AISC Unity Check of Outside Ribs ( must be ≤ 1 )**  
 = Sc/Sca + ( Rm • C1 / I )/Sba  
 = 4.73/152.13 + ( 60.61 • 258.124/.13502E+09 )/173.33  
 = 0.038

**Check of Inside Ribs:**

Inertia of Saddle, Inner Ribs - Axial Direction

|        | B     | D     | Y     | A     | AY        | Io        |
|--------|-------|-------|-------|-------|-----------|-----------|
| Rib    | 16.0  | 334.0 | 183.0 | 53.4  | 977951.9  | 0.126E+05 |
| Web    | 725.7 | 16.0  | 8.0   | 116.1 | 92893.3   | 0.356E+04 |
| Totals | ...   | ...   | ...   | 169.6 | 1070845.2 | 0.162E+05 |

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1070845/169.557$$

$$= 63.156 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max( y_{tot}, \text{abs}( \text{Saddle Width} - y_{tot} ) )$$

$$= \max( 63.156, \text{abs}( 350.0 - 63.156 ) )$$

$$= 286.844 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 2130.0 - \cos( 1676.0/2 )( 28.0 + 0.0 + 0.0 ) - 0.0$$

$$= 426.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{16200.6/169.557}$$

$$= 97.748 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 426.0/97.748$$

$$= 4.358$$

Unit Force [Force,u]:

$$= FI / ( 2 \cdot \text{Baseplate Length} )$$

$$= 795.469/( 2 \cdot 3010.0 )$$

$$= 0.132 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.132 \cdot 725.729 \cdot 426.0$$

$$= 40.853 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [ $\sigma_B$ ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 40.853/564786$$

$$= 0.709 \text{ N./mm}^2$$

Compressive Allowable,  $KL/r < C_c$  ( 4.3581 < 122.9326 ) per AISC E2-1 [Sca]:

$$= ( 1 - (KL/r)^2 / (2 \cdot C_c^2) ) F_y / ( 5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3) )$$

$$= ( 1 - (4.36)^2 / (2 \cdot 122.93^2) ) 260 / ( 5/3 + 3 \cdot (4.36) / (8 \cdot 122.93) - (4.36^3) / (8 \cdot 122.93^3) )$$

$$= 154.7 \text{ N./mm}^2$$

#### AISC Unity Check of Inside Ribs ( must be $\leq 1$ )

$$= Sc/Sca + ( M_{base,c} \cdot C_1/I ) / S_{ba}$$

$$= 6.13/154.67 + ( 40.85 \cdot 286.844/16200.564 ) / 173.33$$

$$= 0.044$$

#### Input Data for Base Plate Bolting Calculations:

|                                            |         |                           |
|--------------------------------------------|---------|---------------------------|
| Total Number of Bolts per BasePlate        | Nbolts  | 4                         |
| Total Number of Bolts in Tension/Baseplate | Nbt     | 2                         |
| Bolt Material Specification                |         | SA-193 B7                 |
| Bolt Allowable Stress                      | Stba    | 172.00 N./mm <sup>2</sup> |
| Bolt Corrosion Allowance                   | Bca     | 1.5 mm.                   |
| Distance from Bolts to Edge                | Edgedis | 330.0 mm.                 |
| Nominal Bolt Diameter                      | Bnd     | 30.0000 mm.               |
| Thread Series                              | Series  | TEMA Metric               |
| BasePlate Allowable Stress                 | S       | 138.00 N./mm <sup>2</sup> |
| Area Available in a Single Bolt            | BltArea | 3.9088 cm <sup>2</sup>    |
| Saddle Load Q0 (Weight)                    | Q0      | 43729.5 Kg                |
| Saddle Load QL (Wind/Seismic contribution) | QL      | 500.5 Kg                  |
| Maximum Transverse Force                   | Ft      | 385.2 Kg                  |
| Maximum Longitudinal Force                 | Fl      | 795.5 Kg                  |
| Saddle Bolted to Steel Foundation          |         | No                        |

Shear Stress in a Single Bolt, Longitudinal Direction [ $\tau_b$ ,l]:

$$= FI / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$$



=  $795.5 / (3.909 \cdot 4)$   
= 5.0 N./mm<sup>2</sup>. Must be less than 103.2 N./mm<sup>2</sup>.

Shear Stress in a Single Bolt, Transverse Direction [ $\tau_b, t$ ]:  
=  $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$   
=  $385.2 / (3.909 \cdot 4)$   
= 2.4 N./mm<sup>2</sup>. Must be less than 103.2 N./mm<sup>2</sup>.

#### **Bolt Area Calculation per Dennis R. Moss**

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:  
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:  
=  $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$   
=  $795.47 / (103.2 \cdot 4.0)$   
= 0.1890 cm<sup>2</sup>

#### **Bolt Area due to Transverse Load:**

Moment on Baseplate Due to Transverse Load [Rmom]:  
=  $B \cdot F_t + \text{Sum of X Moments}$   
=  $2130.0 \cdot 385.24 + 0.0$   
= 820.58 Kg-m.

Eccentricity (e):  
=  $R_{mom} / QO$   
=  $820.58 / 43729.54$   
= 18.76 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:  
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:  
=  $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$   
=  $\max[0.0, 0.189, 0.0]$   
= 0.1890 cm<sup>2</sup>

*Review notes about nozzle loadings and supports in the Warnings report.*

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### ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

There are more than two saddles present in this model. The load on the saddle supports has been computed as the total weight divided by the number of saddles. Ensure that this is an appropriate assumption for this vessel model. It may be necessary to manually enter the bending moments at the saddle through the Saddle Input dialog. Also note that the dimension Ls is taken to be the distance between the adjoining saddles (first two or last two).

Note:

Wear Pad Width (260.00) is less than  $1.56 \cdot \sqrt{r_m \cdot t}$   
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

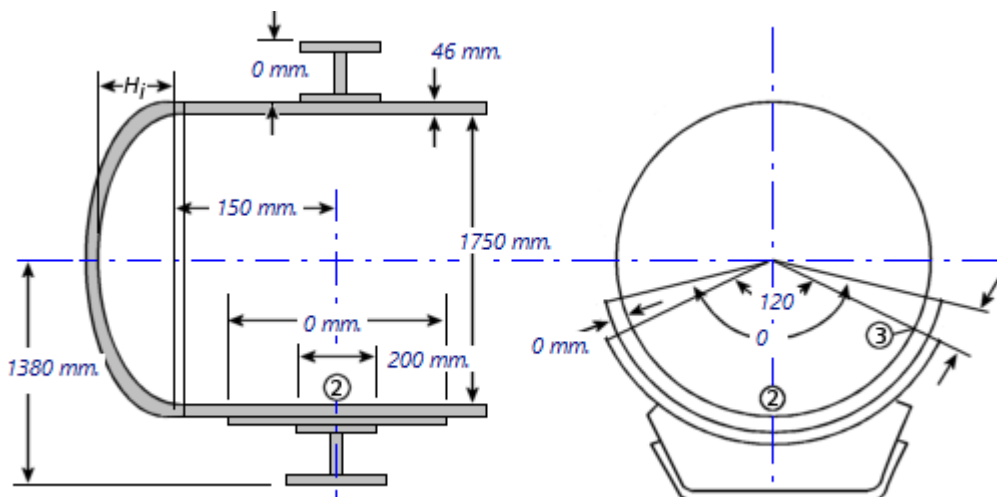
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{898.75 \cdot 44.5}, 2 \cdot 150.0)$$

$$= 300.0000 \text{ mm.}$$

#### Input and Calculated Values:

|                                               |                |         |                    |
|-----------------------------------------------|----------------|---------|--------------------|
| Vessel Mean Radius                            | Rm             | 898.75  | mm.                |
| Shell Thickness used in this Case             | t              | 46.000  | mm.                |
| Stiffened Vessel Length per 4.15.6            | L              | 7348.80 | mm.                |
| Distance from Saddle to Vessel tangent        | a1 or a        | 150.00  | mm.                |
| Saddle Width                                  | b1 or b        | 200.00  | mm.                |
| Saddle Bearing Angle                          | delta or theta | 120.00  | degrees            |
| Shell Allowable Stress used in Calculation    |                | 138.00  | N./mm <sup>2</sup> |
| Head Allowable Stress used in Calculation     |                | 138.00  | N./mm <sup>2</sup> |
| Circumferential Efficiency in Plane of Saddle |                | 1.00    |                    |
| Circumferential Efficiency at Mid-Span        |                | 1.00    |                    |
| Distance from Saddle Base to Centerline       | B              | 1380.00 | mm.                |
| Coefficient of Friction                       | mu             | 0.60    |                    |



#### Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case 50951.27 Kgf  
 Pressure used in Analysis P -0.000 bars

| Horizontal Vessel Analysis Results: |            | Actual<br>N./mm <sup>2</sup> | Allowable<br>N./mm <sup>2</sup> |
|-------------------------------------|------------|------------------------------|---------------------------------|
| Long. Stress at Top                 | of Midspan | -10.24                       | -235.80                         |
| Long. Stress at Top                 | of Midspan | 10.24                        | 138.00                          |
| Long. Stress at Bottom              | of Midspan | 10.24                        | 138.00                          |

|                                   |       |         |
|-----------------------------------|-------|---------|
| Long. Stress at Top of Saddles    | -0.23 | -235.80 |
| Long. Stress at Top of Saddles    | 0.23  | 138.00  |
| Long. Stress at Bottom of Saddles | 0.23  | 138.00  |
| Tangential Shear in Shell         | 10.99 | 110.40  |
| Tangential Shear in Head          | 5.23  | 110.40  |
| Circ. Stress at Horn of Saddle    | 10.48 | 172.50  |
| Circ. Compressive Stress in Shell | 1.67  | 138.00  |

#### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(3502.2/3 + 0) \cdot 1380.0/1610.0$$

$$= 3001.9 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(790.98, 0) \cdot 1380.0/3173.0$$

$$= 344.0 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(11544.14, 0) \cdot 1380.0/3173.0$$

$$= 5020.8 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(11544/3 + 0) \cdot 1380.0/1610.0$$

$$= 9895.0 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 41056 + \max(344, 3002, 5021, 9895)$$

$$= 50951.3 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169(0.0 + \pi/4(1.982 \cdot 1.2)^2)$$

$$= 7756.729 \text{ N}$$

#### Summary of Loads at the Base of this Saddle:

|                                         |          |     |
|-----------------------------------------|----------|-----|
| Vertical Load (including saddle weight) | 51367.00 | Kgf |
| Transverse Shear Load                   | 3848.05  | Kgf |
| Longitudinal Shear Load                 | 11544.14 | Kgf |

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1610.000 \text{ mm.}$$

#### The Computed K values from Table 4.15.1:

|             |              |              |             |
|-------------|--------------|--------------|-------------|
| K1 = 0.1066 | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011 |
| K5 = 0.7603 | K6 = 0.0529  | K7 = 0.0132  | K8 = 0.3405 |
| K9 = 0.2711 | K10 = 0.0581 | K1* = 0.1923 |             |

Note: Dimension a is less than Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/(3L))]$$

$$= -50951 \cdot 150.0 [1 - (1 - 150.0/7348.8 + (898.75^2 - 0.0^2)/(2 \cdot 150.0 \cdot 7348.8))/(1 + (4 \cdot 0.0)/(3 \cdot 7348.8))]$$

$$= 2644.2 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm)/(3L)) - 4a/L)$$

$$= 50951 \cdot 7349/4((1 + 2(899^2 - 0^2)/(7349^2))/(1 + (4 \cdot 0)/(3 \cdot 7349)) - 4 \cdot 150/7349)$$

= 88766.9 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= -0.0 \cdot 898.75 / (2 \cdot 33.5) - 88766.9 / (\pi \cdot 898.8^2 \cdot 33.5)$$

$$= -10.24 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0.0 \cdot 898.75 / (2 \cdot 33.5) + 88766.9 / (\pi \cdot 898.8^2 \cdot 33.5)$$

$$= 10.24 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [ $\sigma_3$ ]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0.0 \cdot 898.75 / (2 \cdot 44.5) - 2644.2 / (\pi \cdot 898.8^2 \cdot 44.5)$$

$$= -0.23 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [ $\sigma_4$ ]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 t)$$

$$= -0.0 \cdot 898.75 / (2 \cdot 44.5) + 2644.2 / (\pi \cdot 898.8^2 \cdot 44.5)$$

$$= 0.23 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 50951(7348.8 - 2 \cdot 150.0) / (7348.8 + (4 \cdot 0.0 / 3))$$

$$= 48871.3 \text{ Kgf}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [ $\tau_3$ ]:

$$= K_3 \cdot Q / (R_m \cdot t)$$

$$= 0.8799 \cdot 50951 / (898.75 \cdot 44.5)$$

$$= 10.99 \text{ N./mm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [ $\tau_{3*}$ ]:

$$= K_3 \cdot Q / (R_m \cdot t_h)$$

$$= 0.8799 \cdot 50951 / (898.75 \cdot 93.5)$$

$$= 5.23 \text{ N./mm}^2$$

Decay Length (4.15.20) [ $x_1, x_2$ ]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(898.75 \cdot 44.5)}$$

$$= 155.989 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [ $\sigma_6$ ]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 50951 \cdot 0.1 / (44.5(200.0 + 155.99 + 155.99))$$

$$= -1.67 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle,  $L \geq 8R_m$  (4.15.22) [ $\sigma_7$ ]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -50951 / (4 \cdot 44.5(200.0 + 155.989 + 155.989)) -$$

$$3 \cdot 0.0132 \cdot 50951 / (2 \cdot 44.5^2)$$

$$= -10.48 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{(898.75 \cdot 44.5)}, 2 \cdot 150.0)$$

$$= 300.00 \text{ mm.}$$

#### Results for Vessel Ribs, Web and Base:

|                                      |        |           |                    |
|--------------------------------------|--------|-----------|--------------------|
| Baseplate Length                     | Bplen  | 1610.0000 | mm.                |
| Baseplate Thickness                  | Bpthk  | 28.0000   | mm.                |
| Baseplate Width                      | Bpwid  | 240.0000  | mm.                |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 4         |                    |
| Rib Thickness                        | Ribtk  | 16.0000   | mm.                |
| Web Thickness                        | Webtk  | 16.0000   | mm.                |
| Web Location                         | Webloc | Side      |                    |
| Saddle Design Temperature            |        | 21.1      | °C                 |
| Saddle Yield Stress                  | Sy     | 242.0     | N./mm <sup>2</sup> |
| Height of Web at Center              | Hw,c   | 306.0     | mm.                |
| Friction Coefficient                 | mu     | 0.600     |                    |

### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:  

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(876.5 \cdot 44.5)} + 260.0) 44.5^3) / 12$$

$$= 417.174 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:  

$$= \text{ShellInertia} + A_{\text{shell}} (C1 - Y)^2$$

$$= 417.174 + 252.801 (136.678 - 22.25)^2$$

$$= 33518.469 \text{ cm}^4$$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>2</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|----------------------------------------|
| Shell     | 568.1    | 44.5     | 22.2     | 252.8                | 562482.1               | 0.335E+05                              |
| Wearplate | 260.0    | 16.0     | 52.5     | 41.6                 | 218400.0               | 0.296E+04                              |
| Web       | 16.0     | 415.0    | 268.0    | 66.4                 | 1779520.8              | 0.210E+05                              |
| BasePlate | 240.0    | 28.0     | 489.5    | 67.2                 | 3289440.5              | 0.837E+05                              |
| Totals    | ...      | ...      | ...      | 428.0                | 5849843.0              | 0.141E+06                              |

Distance to Centroid [C1]:  

$$= AY / A$$

$$= 5849843 / 428.001$$

$$= 136.678 \text{ mm.}$$

Angle [β]:  

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:  

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:  

$$= K1 \cdot Q$$

$$= 0.204 \cdot 50951.273$$

$$= 10369.6895 \text{ Kg}$$

Tension Stress,  $St = (Fh / As) = 5.8044 \text{ N./mm}^2$   
 Allowed Stress,  $Sa = 0.6 \cdot \text{Yield Str} = 145.1982 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:  

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1380.0 - 876.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 655.140 \text{ mm.}$$

Bending Moment,  $M = Fh \cdot d = 6793.7358 \text{ Kg-m.}$

Bending Stress,  $Sb = (M \cdot C1 / I) = 6.4512 \text{ N./mm}^2$   
 Allowed Stress,  $Sa = 2/3 \cdot \text{Yield Str} = 161.3313 \text{ N./mm}^2$

**Minimum Thickness of Baseplate per Moss:**  

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(50951 + 416)240.0 / (2 \cdot 1610.0 \cdot 161.331)}$$

$$= 26.423 \text{ mm.}$$

### Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:  

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (875.0 + 46.0 + 16.0)$$

$$= 1622.932 \text{ mm.}$$

Distance between Ribs [e]:  

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 1622.9315 / (4 - 1)$$

= 540.977 mm.

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2  
 = 540.9772 • 240.0/2  
 = 649.173 cm<sup>2</sup>

Bearing Pressure [Bp]:

= ( Q + SaddleWt ) / ( BasePlateLength • BasePlateWidth )  
 = 51366.996/( 1610.0 • 240.0 )  
 = 13.294 Kgf/cm<sup>2</sup>

Axial Load [P]:

= Ap • Bp  
 = 649.2 • 13.29  
 = 8629.930 Kgf

Area of the Rib and Web [Ar]:

= Rib Area + Web Area  
 = 29.44 + 43.278  
 = 72.718 cm<sup>2</sup>

Compressive Stress [Sc]:

= P/Ar  
 = 8629.9/72.7182  
 = 11.638 N./mm<sup>2</sup>

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|        | B     | D     | Y     | A    | AY       | Io        |
|--------|-------|-------|-------|------|----------|-----------|
| Rib    | 16.0  | 184.0 | 108.0 | 29.4 | 317951.9 | 0.187E+04 |
| Web    | 270.5 | 16.0  | 8.0   | 43.3 | 34622.5  | 719.      |
| Totals | ...   | ...   | ...   | 72.7 | 352574.5 | 0.259E+04 |

Rib dimension [D]:

= Saddle Width - Web Thickness  
 = 200.0 - 16.0  
 = 184.000 mm.

Distance to Centroid from Datum [ytot]:

= AY / A  
 = 352574.5/72.718  
 = 48.485 mm.

Distance to Centroid [C1]:

= max( ytot, Saddle Width - ytot )  
 = max( 48.485, 200.0 - 48.485 )  
 = 151.515 mm.

Radius of Gyration [r]:

=  $\sqrt{\text{Total Inertia} / \text{Total Area}}$   
 =  $\sqrt{2591.9/72.718}$   
 = 59.702 mm.

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos( saddle bearing angle / 2 )( radius + shlthk + wpdthk ) - bpthk  
 = 1380.0 - cos( 120.0/2 )( 875.0 + 46.0 + 16.0 ) - 28.0  
 = 883.500 mm.

Length of Outer Rib, step 2, [Lw]:

= 2 • sin( saddle bearing angle / 2 )( radius + shlthk + wpdthk )  
 = 2 • sin( 120.0/2 )( 875.0 + 46.0 + 16.0 )  
 = 1622.932 mm.

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance  
 = 1610.0 - 50.8

= 1559.200 mm.

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (1622.932 - 1559.2) / 2$$

$$= 31.866 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{883.5^2 + 31.866^2}$$

$$= 884.075 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19904\text{E}+09 / 242.0)}$$

$$= 127.422$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 884.075 / 59.702$$

$$= 14.808$$

Bending Moment [Rm]:

$$= Fl / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 11544.1 / (2 \cdot 1610.0) \cdot 540.977 \cdot 884.07 / 2$$

$$= 857.338 \text{ Kg-m.}$$

Compressive Allowable,  $KL/r < Cc$  (  $14.808 < 127.4221$  ) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (14.81)^2 / (2 \cdot 127.42^2)) 242 /$$

$$(5/3 + 3 \cdot (14.81) / (8 \cdot 127.42) - (14.81^3) / (8 \cdot 127.42^3))$$

$$= 140.6 \text{ N./mm}^2$$

**AISC Unity Check of Outside Ribs ( must be  $\leq 1$  )**

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 11.64 / 140.56 + (857.34 \cdot 151.515 / 25919526) / 161.33$$

$$= 0.387$$

**Check of Inside Ribs:**

Inertia of Saddle, Inner Ribs - Axial Direction

|        | B     | D     | Y     | A     | AY       | Io        |
|--------|-------|-------|-------|-------|----------|-----------|
| Rib    | 16.0  | 184.0 | 108.0 | 29.4  | 317951.9 | 0.247E+04 |
| Web    | 541.0 | 16.0  | 8.0   | 86.6  | 69245.1  | 576.      |
| Totals | ...   | ...   | ...   | 116.0 | 387197.0 | 0.305E+04 |

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 387197.031 / 115.996$$

$$= 33.380 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}( \text{Saddle Width} - ytot ))$$

$$= \max(33.38, \text{abs}(200.0 - 33.38))$$

$$= 166.620 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2} - Bpthk$$

$$= 1380.0 - \sqrt{(937.0 + 16.0)^2 - (540.977/2)^2} - 28.0$$

$$= 454.891 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{3045.9 / 115.996}$$

$$= 51.243 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 454.891 / 51.243$$

= 8.877

Unit Force [Force,u]:  
 =  $F_l / (2 \cdot \text{Baseplate Length})$   
 =  $11544.137 / (2 \cdot 1610.0)$   
 = 3.585 Kg/mm.

Moment at base of inner Rib [Mbase,c]:  
 = Unit Force  $\cdot e \cdot L$   
 =  $3.585 \cdot 540.977 \cdot 454.891$   
 = 882.268 Kg-m.

Bending Stress due to Transverse Force and Weight Load [ $\sigma_B$ ,base,c]:  
 = Bending Moment / Section Modulus  
 =  $882.268 / 182803.797$   
 = 47.330 N./mm<sup>2</sup>

Compressive Allowable,  $KL/r < C_c$  ( 8.8771 < 127.4221 ) per AISC E2-1 [Sca]:  
 =  $(1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$   
 =  $(1 - (8.88)^2 / (2 \cdot 127.42^2)) 242 / (5/3 + 3 \cdot (8.88) / (8 \cdot 127.42) - (8.88)^3 / (8 \cdot 127.42^3))$   
 = 142.6 N./mm<sup>2</sup>

**AISC Unity Check of Inside Ribs ( must be  $\leq 1$  )**  
 =  $S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$   
 =  $14.3 / 142.61 + (882.27 \cdot 166.62 / 3045.872) / 161.33$   
 = 0.394

#### ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:  
 Wear Pad Width (480.00) is less than  $1.56 \cdot \sqrt{r_m \cdot t}$   
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:  
 =  $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$   
 =  $\min(350.0 + 1.56 \cdot \sqrt{1643.25 \cdot 33.5}, 2 \cdot 400.0)$   
 = 716.0149 mm.

#### Input and Calculated Values:

|                                               |                |         |                    |
|-----------------------------------------------|----------------|---------|--------------------|
| Vessel Mean Radius                            | Rm             | 1643.25 | mm.                |
| Shell Thickness used in this Case             | t              | 35.000  | mm.                |
| Stiffened Vessel Length per 4.15.6            | L              | 7348.80 | mm.                |
| Distance from Saddle to Vessel tangent        | a1 or a        | 400.00  | mm.                |
| Saddle Width                                  | b1 or b        | 350.00  | mm.                |
| Saddle Bearing Angle                          | delta or theta | 120.00  | degrees            |
| Inside or Mean Depth of Head                  | Hi or hm       | 830.75  | mm.                |
| Shell Allowable Stress used in Calculation    |                | 138.00  | N./mm <sup>2</sup> |
| Head Allowable Stress used in Calculation     |                | 138.00  | N./mm <sup>2</sup> |
| Circumferential Efficiency in Plane of Saddle |                | 1.00    |                    |
| Circumferential Efficiency at Mid-Span        |                | 1.00    |                    |
| Distance from Saddle Base to Centerline       | B              | 2130.00 | mm.                |
| Coefficient of Friction                       | mu             | 0.10    |                    |

#### Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case 49526.75 Kg  
 Pressure used in Analysis P -1.034 bars

| Horizontal Vessel Analysis Results: |       | Actual             | Allowable          |
|-------------------------------------|-------|--------------------|--------------------|
|                                     |       | N./mm <sup>2</sup> | N./mm <sup>2</sup> |
| Long. Stress at Top of Midspan      | -4.78 | -217.80            |                    |
| Long. Stress at Top of Midspan      | 4.78  | 138.00             |                    |
| Long. Stress at Bottom of Midspan   | -0.29 | -217.80            |                    |

|                                   |        |         |
|-----------------------------------|--------|---------|
| Long. Stress at Bottom of Midspan | 0.29   | 138.00  |
| Long. Stress at Top of Saddles    | -2.62  | -217.80 |
| Long. Stress at Top of Saddles    | 2.62   | 138.00  |
| Long. Stress at Bottom of Saddles | -2.46  | -217.80 |
| Long. Stress at Bottom of Saddles | 2.46   | 138.00  |
| Tangential Shear in Shell         | 7.76   | 110.40  |
| Tangential Shear in Head          | 7.76   | 110.40  |
| Circ. Stress at Horn of Saddle    | 20.41  | 172.50  |
| Addl. Stress in Head as Stiffener | 122.37 | 172.50  |
| Circ. Compressive Stress in Shell | 1.54   | 138.00  |

#### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:  
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$   
 $= 3.0 (3502.2/3 + 0) \cdot 2130.0/2902.9172$   
 $= 2569.7 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:  
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$   
 $= \max(2410.51, 0) \cdot 2130.0/3385.0$   
 $= 1516.8 \text{ Kg}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:  
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$   
 $= \max(11544.14, 0) \cdot 2130.0/3385.0$   
 $= 7264.1 \text{ Kg}$

Saddle Reaction Force due to Earthquake Ft [Fst]:  
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$   
 $= 3.0 (11544/3 + 0) \cdot 2130.0/2902.9172$   
 $= 8470.4 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:  
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$   
 $= 41056 + \max(1517, 2570, 7264, 8470)$   
 $= 49526.7 \text{ Kg}$

Longitudinal Wind Force [Fl]:  
 $= \text{WindScalar} \cdot \text{WindPress} ( \text{Platform Area} + ( \pi/4 ( \text{OD} \cdot \text{WindDiaMult} )^2 ) )$   
 $= 1.1 \cdot 1587.169 ( 0.0 + \pi/4 ( 3.46 \cdot 1.2 )^2 )$   
 $= 23638.697 \text{ N}$

#### Summary of Loads at the Base of this Saddle:

|                                         |          |     |
|-----------------------------------------|----------|-----|
| Vertical Load (including saddle weight) | 50688.39 | Kgf |
| Transverse Shear Load                   | 3848.05  | Kgf |
| Longitudinal Shear Load                 | 11544.14 | Kgf |

Note: Wear Plate is Welded to the Shell,  $k = 0.1$

Saddle Dimension [E]:  
 $= \min( 2 ( \text{ShellID}/2 + t + \text{WearPadThickness} ) \sin(\theta/2), 2 \cdot R_m )$   
 $= \min( 2 ( 3250.0/2 + 35.0 + 16.0 ) \sin( 120.0/2 ), 2 \cdot 1643.25 )$   
 $= 2902.917 \text{ mm.}$

#### The Computed K values from Table 4.15.1:

|             |              |              |             |
|-------------|--------------|--------------|-------------|
| K1 = 0.1066 | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011 |
| K5 = 0.7603 | K6 = 0.0529  | K7 = 0.0132  | K8 = 0.3405 |
| K9 = 0.2711 | K10 = 0.0581 | K1* = 0.1923 |             |

Note: Dimension a is less than  $R_m/2$ .

Moment per Equation 4.15.1 [M1]:  
 $= -Q \cdot a [ 1 - ( 1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L) ) / ( 1 + (4h_m)/3L ) ]$   
 $= -49527 \cdot 400.0 [ 1 - ( 1 - 400.0/7348.8 + (1643.25^2 - 830.75^2)/(2 \cdot 400.0 \cdot 7348.8) ) / ( 1 + (4 \cdot 830.75)/(3 \cdot 7348.8) ) ]$   
 $= 2354.4 \text{ Kg-m.}$



Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L)$$

$$= 49527 \cdot 7349 / 4 \cdot ((1 + 2(1643^2 - 831^2) / (7349^2)) / (1 + (4 \cdot 831) / (3 \cdot 7349)) - 4 \cdot 400 / 7349)$$

$$= 65149.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -1.034 \cdot 1643.25 / (2 \cdot 33.5) - 65149.1 / (\pi \cdot 1643.2^2 \cdot 33.5)$$

$$= -4.78 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -1.034 \cdot 1643.25 / (2 \cdot 33.5) + 65149.1 / (\pi \cdot 1643.2^2 \cdot 33.5)$$

$$= -0.29 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [ $\sigma_3$ ]:

$$= P \cdot Rm / (2t) - M1 / (\pi \cdot Rm^2 \cdot t)$$

$$= -1.034 \cdot 1643.25 / (2 \cdot 33.5) - 2354.4 / (\pi \cdot 1643.2^2 \cdot 33.5)$$

$$= -2.62 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [ $\sigma_4$ ]:

$$= P \cdot Rm / (2t) + M1 / (\pi \cdot Rm^2 \cdot t)$$

$$= -1.034 \cdot 1643.25 / (2 \cdot 33.5) + 2354.4 / (\pi \cdot 1643.2^2 \cdot 33.5)$$

$$= -2.46 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 49527(7348.8 - 2 \cdot 400.0) / (7348.8 + (4 \cdot 830.75 / 3))$$

$$= 38354.2 \text{ Kgf}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [ $\tau_3$ ]:

$$= K3 \cdot Q / (Rm \cdot t)$$

$$= 0.8799 \cdot 49527 / (1643.25 \cdot 33.5)$$

$$= 7.76 \text{ N./mm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [ $\tau_3^*$ ]:

$$= K3 \cdot Q / (Rm \cdot th)$$

$$= 0.8799 \cdot 49527 / (1643.25 \cdot 33.5)$$

$$= 7.76 \text{ N./mm}^2$$

Membrane stress in the head as a stiffener (4.15.16) [ $\sigma_5$ ]:

$$= K4 \cdot Q / (Rm \cdot th) + ((P \cdot Ri) / (2 \cdot th)) \cdot (Ri / hm)$$

$$= 0.4011 \cdot 49527 / (1643.25 \cdot 33.5) + ((25.0 \cdot 1626.5) / (2 \cdot 33.5)) \cdot (1626.5 / 830.75)$$

$$= 122.37 \text{ N./mm}^2$$

Decay Length (4.15.20) [ $x_1, x_2$ ]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1643.25 \cdot 33.5)}$$

$$= 183.007 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [ $\sigma_6$ ]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 49527 \cdot 0.1 / (33.5(350.0 + 183.01 + 183.01))$$

$$= -1.54 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle,  $L < 8Rm$  (4.15.23) [ $\sigma_7^*$ ]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2)$$

$$= -49527 / (4 \cdot 33.5(350.0 + 183.007 + 183.007)) - 12 \cdot 0.013 \cdot 49527 \cdot 1643.25 / (7348.799 \cdot 33.5^2)$$

$$= -20.41 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(350.0 + 1.56 \cdot \sqrt{(1643.25 \cdot 33.5)}, 2 \cdot 400.0)$$

$$= 716.01 \text{ mm.}$$

**Results for Vessel Ribs, Web and Base:**



|                                      |        |           |                    |
|--------------------------------------|--------|-----------|--------------------|
| Baseplate Length                     | Bplen  | 3010.0000 | mm.                |
| Baseplate Thickness                  | Bpthk  | 28.0000   | mm.                |
| Baseplate Width                      | Bpwid  | 390.0000  | mm.                |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 5         |                    |
| Rib Thickness                        | Ribtk  | 16.0000   | mm.                |
| Web Thickness                        | Webtk  | 16.0000   | mm.                |
| Web Location                         | Webloc | Side      |                    |
| Saddle Design Temperature            |        | 21.1      | °C                 |
| Saddle Yield Stress                  | Sy     | 242.0     | N./mm <sup>2</sup> |
| Height of Web at Center              | Hw,c   | 306.0     | mm.                |
| Friction Coefficient                 | mu     | 0.100     |                    |

#### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:  
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$   
 $= ((1.56 \cdot \sqrt{1626.5 \cdot 33.5} + 480.0)^3) / 12$   
 $= 264.466 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:  
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$   
 $= 264.466 + 282.789 (147.629 - 16.75)^2$   
 $= 48704.332 \text{ cm}^4$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>(2)</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|------------------------------------------|
| Shell     | 844.1    | 33.5     | 16.8     | 282.8                | 473670.8               | 0.487E+05                                |
| Wearplate | 480.0    | 16.0     | 41.5     | 76.8                 | 318720.0               | 0.867E+04                                |
| Web       | 16.0     | 426.0    | 262.5    | 68.2                 | 1789200.0              | 0.193E+05                                |
| BasePlate | 390.0    | 28.0     | 489.5    | 109.2                | 5345340.0              | 0.128E+06                                |
| Totals    | ...      | ...      | ...      | 536.9                | 7926930.5              | 0.204E+06                                |

Distance to Centroid [C1]:  
 $= AY / A$   
 $= 7926931 / 536.949$   
 $= 147.629 \text{ mm.}$

Angle [β]:  
 $= 180 - \text{Saddle Angle} / 2$   
 $= 180 - 120.0 / 2$   
 $= 120.0$

Saddle Splitting Coefficient [K1]:  
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$   
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$   
 $= 0.2035$

Saddle Splitting Force [Fh]:  
 $= K1 \cdot Q$   
 $= 0.204 \cdot 49526.746$   
 $= 10079.7666 \text{ Kg}$

Tension Stress, St = ( Fh/As ) = 3.8893 N./mm<sup>2</sup>  
 Allowed Stress, Sa = 0.6 \* Yield Str = 145.1984 N./mm<sup>2</sup>

Saddle Splitting Dimension [d]:  
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$   
 $= 2130.0 - 1626.5 \cdot \sin(120.0/2) / 1.0472$   
 $= 784.895 \text{ mm.}$

Bending Moment, M = Fh \* d = 7911.7178 Kg-m.

Bending Stress, Sb = ( M \* C1 / I ) = 5.6046 N./mm<sup>2</sup>  
 Allowed Stress, Sa = 2/3 \* Yield Str = 161.3315 N./mm<sup>2</sup>

#### Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$   
 $= \sqrt{3(49527 + 1162)390.0 / (2 \cdot 3010.0 \cdot 161.332)}$   
 $= 24.471 \text{ mm.}$



### Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [ Web Length ]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1625.0 + 35.0 + 16.0)$$

$$= 2902.917 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 2902.9172 / (5 - 1)$$

$$= 725.729 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 725.7293 \cdot 390.0/2$$

$$= 1415.172 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 50688.387 / (3010.0 \cdot 390.0)$$

$$= 4.318 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 1415.2 \cdot 4.32$$

$$= 6110.640 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 53.44 + 58.058$$

$$= 111.498 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/\text{Ar}$$

$$= 6110.6/111.4983$$

$$= 5.375 \text{ N./mm}^2$$

### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|        | B     | D     | Y     | A     | AY        | Io        |
|--------|-------|-------|-------|-------|-----------|-----------|
| Rib    | 16.0  | 334.0 | 183.0 | 53.4  | 977951.9  | 0.941E+04 |
| Web    | 362.9 | 16.0  | 8.0   | 58.1  | 46446.7   | 0.410E+04 |
| Totals | ...   | ...   | ...   | 111.5 | 1024398.6 | 0.135E+05 |

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 350.0 - 16.0$$

$$= 334.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 1024399/111.498$$

$$= 91.876 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(91.876, 350.0 - 91.876)$$

$$= 258.124 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(13502.3/111.498)}$$

$$= 110.045 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$



$$= 2130.0 - \cos(120.0/2) (1625.0 + 35.0 + 16.0) - 28.0$$

$$= 1264.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19904\text{E}+09 / 242.0)}$$

$$= 127.422$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1264.0 / 110.045$$

$$= 11.486$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 11544.1 / (2 \cdot 3010.0) \cdot 725.729 \cdot 1264.0 / 2$$

$$= 879.560 \text{ Kg-m.}$$

Compressive Allowable,  $KL/r < C_c$  (  $11.4862 < 127.4221$  ) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.49)^2 / (2 \cdot 127.42^2)) 242 /$$

$$(5/3 + 3 \cdot (11.49) / (8 \cdot 127.42) - (11.49)^3 / (8 \cdot 127.42^3))$$

$$= 141.7 \text{ N./mm}^2$$

**AISC Unity Check of Outside Ribs ( must be  $\leq 1$  )**

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 5.37 / 141.74 + (879.56 \cdot 258.124 / 1.3502\text{E}+09) / 161.33$$

$$= 0.140$$

**Check of Inside Ribs:**

Inertia of Saddle, Inner Ribs - Axial Direction

|        | B     | D     | Y     | A     | AY        | Io        |
|--------|-------|-------|-------|-------|-----------|-----------|
| Rib    | 16.0  | 334.0 | 183.0 | 53.4  | 977951.9  | 0.126E+05 |
| Web    | 725.7 | 16.0  | 8.0   | 116.1 | 92893.3   | 0.356E+04 |
| Totals | ...   | ...   | ...   | 169.6 | 1070845.2 | 0.162E+05 |

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1070845 / 169.557$$

$$= 63.156 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{\text{tot}}, \text{abs}(\text{Saddle Width} - y_{\text{tot}}))$$

$$= \max(63.156, \text{abs}(350.0 - 63.156))$$

$$= 286.844 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{\text{pthk}}$$

$$= 2130.0 - \cos(1676.0/2) (28.0 + 0.0 + 0.0) - 0.0$$

$$= 426.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(16200.6 / 169.557)}$$

$$= 97.748 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 426.0 / 97.748$$

$$= 4.358$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 11544.137 / (2 \cdot 3010.0)$$

$$= 1.918 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$



$$= 1.918 \cdot 725.729 \cdot 426.0$$
$$= 592.868 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [ $\sigma_{B,base,c}$ ]:

$$= \text{Bending Moment} / \text{Section Modulus}$$
$$= 592.868 / 564786$$
$$= 10.294 \text{ N./mm}^2$$

Compressive Allowable,  $KL/r < C_c$  (  $4.3581 < 127.4221$  ) per AISC E2-1 [Sca]:

$$= ( 1 - (KL/r)^2 / (2 \cdot C_c^2) ) F_y / ( 5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3) )$$
$$= ( 1 - ( 4.36 )^2 / ( 2 \cdot 127.42^2 ) ) 242 /$$
$$( 5/3 + 3 \cdot ( 4.36 ) / ( 8 \cdot 127.42 ) - ( 4.36^3 ) / ( 8 \cdot 127.42^3 ) )$$
$$= 144 \text{ N./mm}^2$$

**AISC Unity Check of Inside Ribs ( must be  $\leq 1$  )**

$$= S_c / S_{ca} + ( M_{base,c} \cdot C_1 / I ) / S_{ba}$$
$$= 6.97 / 144.01 + ( 592.87 \cdot 286.844 / 16200.564 ) / 161.33$$
$$= 0.112$$

*Review notes about nozzle loadings and supports in the Warnings report.*

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## Conical Reinforcement Calculations, ASME VIII Div. 1, App. 1

### Input Values:

#### SHELL-2

|                                     |        |      |
|-------------------------------------|--------|------|
| Design Internal Pressure, Small End | 25.103 | bars |
| Design Internal Pressure, Large End | 25.103 | bars |
| Temperature for Internal Pressure   | 85.0   | °C   |
| Design External Pressure            | 1.034  | bars |
| Temperature for External Pressure   | 85.0   | °C   |

#### CONE Properties:

|                                          |       |           |                    |
|------------------------------------------|-------|-----------|--------------------|
| Cone Material                            |       | SA-516 70 | +Q+IT              |
| Cone Allowable Stress at Temperature     | Ss    | 138.00    | N./mm <sup>2</sup> |
| Cone Allowable Stress At Ambient         |       | 138.00    | N./mm <sup>2</sup> |
| Longitudinal Joint Efficiency of Cone    | Ec,E2 | 1.00      |                    |
| Circumferential Joint Efficiency of Cone |       | 1.00      |                    |
| Actual Thickness of Cone                 | Tc    | 46.0000   | mm.                |
| Corrosion Allowance for Cone             | Cac   | 1.5000    | mm.                |
| Diameter Basis for Cone and Cylinders    |       | ID        |                    |
| Diameter of Small End of Cone            | Ds    | 1750.000  | mm.                |
| Diameter of Large End of Cone            | Dl    | 3250.000  | mm.                |
| Half Apex Angle for Cone                 |       | 30.00     | degrees            |
| Axial Length of Cone                     | Lc    | 2598.000  | mm.                |

#### SMALL-END Properties:

|                                           |       |           |                    |
|-------------------------------------------|-------|-----------|--------------------|
| Attached Small End Material Specification |       | SA-516 70 | +Q                 |
| Operating Allowable Stress                | Ss    | 138.00    | N./mm <sup>2</sup> |
| Ambient Allowable Stress                  |       | 138.00    | N./mm <sup>2</sup> |
| Longitudinal Joint Efficiency             | Es,E1 | 1.00      |                    |
| Actual Thickness                          | Ts    | 46.0000   | mm.                |
| Corrosion Allowance                       | Cas   | 1.5000    | mm.                |
| Axial Length of Attached Element          | Ls    | 350.000   | mm.                |

#### LARGE-END Properties:

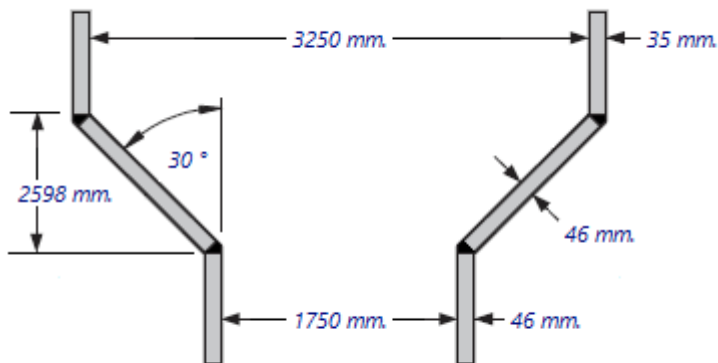
|                                           |       |           |                    |
|-------------------------------------------|-------|-----------|--------------------|
| Attached Large End Material Specification |       | SA-516 70 | +Q+IT              |
| Operating Allowable Stress                | Sl    | 138.00    | N./mm <sup>2</sup> |
| Ambient Allowable Stress                  |       | 138.00    | N./mm <sup>2</sup> |
| Longitudinal Joint Efficiency             | El,E1 | 1.00      |                    |
| Actual Thickness                          | Tl    | 35.0000   | mm.                |
| Corrosion Allowance                       | Cal   | 1.5000    | mm.                |
| Axial Length of Attached Element          | Ll    | 4506.633  | mm.                |

Type of Reinforcement at Large End of Cone: None

Type of Reinforcement at Small End of Cone: None

Force Arl & Ars calculation, cone -> flange No

Force Arl & Ars calculation, cone -> cylinder No



**Cone Geometry**

#### Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 85 °C

|                                            |                         |
|--------------------------------------------|-------------------------|
| Elastic Modulus of Cone Material           | 0.199E+09 KPa. at 85 °C |
| Elastic Modulus of Small Cylinder Material | 0.199E+09 KPa. at 85 °C |
| Elastic Modulus of Large Cylinder Material | 0.199E+09 KPa. at 85 °C |
| Elastic Modulus of Large End Reinforcement | 0.199E+09 KPa. at 85 °C |
| Elastic Modulus of Small End Reinforcement | 0.199E+09 KPa. at 85 °C |

Elastic Modulus Data from ASME Section II Part D at 85 °C

|                                            |                         |
|--------------------------------------------|-------------------------|
| Elastic Modulus of Cone Material           | 0.199E+09 KPa. at 85 °C |
| Elastic Modulus of Small Cylinder Material | 0.199E+09 KPa. at 85 °C |
| Elastic Modulus of Large Cylinder Material | 0.199E+09 KPa. at 85 °C |
| Elastic Modulus of Large End Reinforcement | 0.199E+09 KPa. at 85 °C |
| Elastic Modulus of Small End Reinforcement | 0.199E+09 KPa. at 85 °C |

Axial forces and moments are not computed for Horizontal geometries.  
 Compute them manually and enter the loads using the Force/Moment dialog.  
 Place the loads at the ends of the cone as needed.

Note: Both ends of the Cone are Lines of Support

|                                                   |             |
|---------------------------------------------------|-------------|
| Maximum Centroid Reinforcement Distance Large End | 58.9544 mm. |
| Maximum Centroid Reinforcement Distance Small End | 50.6116 mm. |

Note: No ring was found close enough to the large end to be considered.

Note: No ring was found close enough to the small end to be considered.

#### **Reinforcement Calculations for Cone / Large Cylinder:**

##### **Required Area of Reinforcement for Large End Under Internal Pressure**

Note: Large end is in compression, 1.0 will be used for E1 and E2.

Large end ratio of pressure to allowable stress [P/Ss E1]: 0.01819

Large End Maximum half apex angle w/o reinforcement [ $\delta$ ]:  
 $= 326.6 \cdot \sqrt{P / (Ss \cdot E1)}$  per 1-5 equation 1  
 $= 326.6 \cdot \sqrt{25.103 / (137.997 \cdot 1.0)}$   
 $= 44.051$  degrees

Large end actual half apex angle [alpha]: 30.000 degrees

**Because delta is greater than alpha, reinforcement calculations are not required**



for the large end under internal pressure.

### Required Area of Reinforcement for Large End Under External Pressure

Large end ratio of pressure to allowable stress [P/Ss E1]: 0.00075

Large End Maximum half apex angle w/o reinforcement [ $\delta$ ]:  
 $= 104 \cdot \sqrt{(P / (Ss \cdot E1))}$  per 1-8 equation 1  
 $= 104 \cdot \sqrt{(1.034 / (137.997 \cdot 1.0))}$   
 $= 2.847$  degrees

Large end actual half apex angle [alpha]: 30.000 degrees

Intermediate Value [k]:  
 $= \max(Y / (Srl \cdot Erl), 1)$   
 $= \max(0.27467E+11 / (138.0 \cdot 0.19904E+09), 1)$   
 $= 1.0000$

where [Y] is:  
 $= \text{Large End All. Stress} \cdot \text{Large End Elastic Modulus (Ext. temp.)}$   
 $= 138.0 \cdot 0.19904E+09$   
 $= 27467233280.0 \text{ (N./mm}^2\text{)}$

|                                                    |       |                    |
|----------------------------------------------------|-------|--------------------|
| Allowable Stress of Large End Material (Ext. Temp) | 138.0 | N./mm <sup>2</sup> |
| Allowable Stress of Cone Material (Ext. Temp)      | 138.0 | N./mm <sup>2</sup> |

Required Area of Reinforcement, Large End, External [Arl]:  
 $= (k \cdot QL \cdot RI \cdot \tan(\text{angle})) / (Ss \cdot E1) \cdot (1 - 1/4 \cdot ((P \cdot RI - QL) / QL) \cdot (\delta / \alpha))$   
 $= (1.0 \cdot 8.7534 \cdot 1660.0 \cdot 0.577 / (138 \cdot 1.0)) \cdot$   
 $(1 - 1/4 \cdot ((1.03 \cdot 1660.0 - 8.753) / 8.753) \cdot (2.847 / 30.0))$   
 $= 5.82 \text{ cm}^2$

Force per Length, Cone Large End External Pressure, compressive side [QL-]:  
 $= P_{ext}(RI/2) + f1$   
 $= 1.034(1660.0/2) + 0.0$   
 $= 8.753 \text{ Kg/mm.}$

Axial Load per Unit Circumference excluding Pressure [f1]:  
 $= F_{axial} / (\pi(DI + TI)) + \text{Moment} / (\pi(RI - TI/2)^2)$   
 $= +.0 / (\pi(3320.0 - 35.0)) + .0 / (\pi(1660.0 - 35.0/2)^2)$   
 $= 0.0 \text{ Kg/mm.}$

Force per Length, Cone Large End External Pressure, tensile side [QL+]:  
 $= P_{ext}(RI/2) + f1$   
 $= 1.034(1660.0/2) + 0.0$   
 $= 8.753 \text{ Kg/mm.}$

Axial Load per Unit Circumference excluding Pressure [f1]:  
 $= F_{axial} / (\pi(DI + TI)) - \text{Moment} / (\pi(RI - TI/2)^2)$   
 $= +.0 / (\pi(3320.0 - 35.0)) - 0.0 / (\pi(1660.0 - 35.0/2)^2)$   
 $= 0.0 \text{ Kg/mm.}$

Available Area of Reinforcement, Large End, External [Ael]:  
 $= 0.55[ \sqrt{(DI \cdot ts)(ts - t)} + \sqrt{(DI \cdot tc / \cos(\alpha))(tc - tr)} ]$  per 1-8 eqn. 3  
 $= 0.55[ \sqrt{(3320.0 \cdot 33.5)(33.5 - 12.085)} +$   
 $\sqrt{(3320.0 \cdot 44.5/0.866)(44.5 - 10.055)} ]$   
 $= 117.5282 \text{ cm}^2$

### Summary of Reinforcement Area, Large End, External Pressure:

|                                                |          |                 |
|------------------------------------------------|----------|-----------------|
| Area of reinforcement required per App. 1-8(1) | 5.8204   | cm <sup>2</sup> |
| Area of reinforcement in shell per App. 1-8(2) | 117.5282 | cm <sup>2</sup> |
| Area of reinforcement in stiffening ring       | 0.0000   | cm <sup>2</sup> |

### Intermediate Results, Large End, External Pressure:

Area Available in Cone, Shell, and Reinforcement [AT]:  
 $= LL \cdot ts/2 + Lc \cdot tc/2 + As$   
 $= 4506.633 \cdot 33.5/2 + 2701.06 \cdot 44.5/2 + 0.0$   
 $= 1355.847 \text{ cm}^2$

Determine the Meridian Length [M]:

$$= -RL \cdot \tan(\alpha) / 2 + LL / 2 + (RL^2 - RS^2) / (3 \cdot RL \cdot \tan(\alpha))$$

$$= -1660.0 \cdot \tan(30.0) / 2 + 4506.633 / 2 + (1660.0^2 - 921.0^2) / (3 \cdot 1660.0 \cdot \tan(30.0))$$

$$= 2437.498 \text{ mm.}$$

Force per Unit Length on the Cone/Shell Junction [FL]:

$$= P \cdot M + \tan(\alpha) ( \text{Force} / ( \pi \cdot D ) + \text{Moment} / ( \pi \cdot \text{Radius}^2 ) )$$

$$= 1.034 \cdot 2437.5 + \tan(30.0) ( 0.0 / ( \pi \cdot 3320.0 ) + 0.0 / ( \pi \cdot 1660.0^2 ) )$$

$$= 25.706 \text{ Kg/mm.}$$

Actual Buckling Stress associated with force FL, [B]:

$$= 0.75 ( FL \cdot DL / ATL )$$

$$= 0.75 ( 25.706 \cdot 3320.0 / 1355.847 )$$

$$= 4.630 \text{ N./mm}^2$$

Material Strain associated with this stress [A]: 0.000047

Required Moment of Inertia, Large End, External Pressure [I's]:

$$= A \cdot DI^2 \cdot Atl / 10.9$$

$$= 0.465E-04 \cdot 3320.0 \cdot 3320.0 \cdot 1355.85 / 10.9$$

$$= 6377910.00 \text{ mm.}^4$$

**Available Moment of Inertia, Large End, External Pressure:**

|     | Area    | Centroid | Ar*Ce     | Dist     | I        | Ar*Di <sup>2</sup> |
|-----|---------|----------|-----------|----------|----------|--------------------|
| Shl | 61.447  | 0.0000   | 0.000     | -32.0528 | 57.465   | 63129.355          |
| Con | 94.250  | -52.9497 | -4990.535 | 20.8969  | 1088.201 | 41157.297          |
| Sec | 0.000   | 16.7500  | 0.000     | -48.8028 | 0.000    | 0.000              |
| TOT | 155.697 |          | -4990.535 |          | 1145.667 | 104286.648         |

Centroid of Section -32.0528 Moment of Inertia 2188.53

**Summary of Large End Inertia Calculations**

|                                           |          |                 |
|-------------------------------------------|----------|-----------------|
| Available Moment of Inertia ( Large End ) | 2188.532 | cm <sup>4</sup> |
| Required Moment of Inertia ( Large End )  | 637.790  | cm <sup>4</sup> |

**Reinforcement Calculations for Cone / Small Cylinder:**

**Required Area of Reinforcement for Small End under Internal Pressure**

Small end ratio of pressure to allowable stress [P/Ss E1]: 0.01819

Small End Maximum half apex angle w/o reinforcement [δ]:

$$= 89 \cdot \sqrt{ ( P / ( Ss \cdot E1 ) ) } \text{ per 1-5 equation 6}$$

$$= 89 \cdot \sqrt{ ( 25.103 / ( 137.997 \cdot 1.0 ) ) }$$

$$= 12.004 \text{ degrees}$$

Small end actual half apex angle [alpha]: 30.000 degrees

Intermediate Value [k]:

$$= \max( Y / ( Sr \cdot Ers ), 1 )$$

$$= \max( 0.27467E+11 / ( 138.0 \cdot 0.19904E+09 ), 1 )$$

$$= 1.0000$$

where [Y] is:

$$= \text{Small End All. Stress} \cdot \text{Small End Elastic Modulus (Int. temp.)}$$

$$= 138.0 \cdot 0.19904E+09$$

$$= 27467335680.0 \text{ (N./mm}^2\text{)}^2$$

Decay Length, Cone Small End:

$$= 1.4 \cdot \sqrt{ ( Rs / ( ts - ca ) ) }$$

$$= 1.4 \cdot \sqrt{ ( 876.5 / ( 46.0 - 1.5 ) ) }$$

$$= 276.493 \text{ mm.}$$

Required Area of Reinforcement, Small End, Internal [Ars]:

$$= k \cdot QS \cdot Rs / ( Ss \cdot E1 ) \cdot ( 1 - \delta / \alpha ) \cdot \tan(\alpha) \text{ per 1-5 equation 7}$$

$$= 1.0 \cdot 112.1866 \cdot 876.5 / ( 138 \cdot 1.0 ) \cdot ( 1 - 12.0 / 30.0 ) \cdot 0.5774$$

$$= 24.2 \text{ cm}^2$$

Force per Length, Cone Small End, tensile side [QS+]:

$$= P ( Rs / 2 ) + f2$$

$$= 25.103(876.5/2) + 0.0$$

$$= 112.187 \text{ Kg/mm.}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= \text{Faxial}/(\pi(Ds + Ts)) + \text{Moment}/(\pi(Rs + Ts/2)^2)$$

$$= -0.0/(\pi(1753.0+46.0)) + 0.0/(\pi(876.5+46.0/2)^2)$$

$$= 0.0 \text{ Kg/mm.}$$

Force per Length, Cone Small End, compressive side [QS-]:

$$= P(Rs/2) + f2$$

$$= 25.103(876.5/2) + -0.0$$

$$= 112.187 \text{ Kg/mm.}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= \text{Faxial}/(\pi(Ds + Ts)) - \text{Moment}/(\pi(Rs + Ts/2)^2)$$

$$= -0.0/(\pi(1753.0+46.0)) - 0.0/(\pi(876.5+46.0/2)^2)$$

$$= -0.0 \text{ Kg/mm.}$$

Area of Reinforcement Available in Small End Shell [Aes]:

$$= 0.78[\sqrt{(Rs+ts)} \cdot (ts-t) + \sqrt{(Rs \cdot tc/\cos(\alpha))} \cdot (tc - tr)] \text{ per 1-5 eqn. 8}$$

$$= 0.78[\sqrt{(876.5 + 44.5)} \cdot (44.5 - 16.121) + \sqrt{(876.5 \cdot 44.5/0.866)} \cdot (44.5 - 18.615)]$$

$$= 86.5647 \text{ cm}^2$$

**Summary of Reinforcement Area, Small End, Internal Pressure:**

|                                                |         |                 |
|------------------------------------------------|---------|-----------------|
| Area of reinforcement required per App. 1-5(7) | 24.2015 | cm <sup>2</sup> |
| Area of reinforcement in shell per App. 1-5(8) | 86.5647 | cm <sup>2</sup> |
| Area of reinforcement in stiffening ring       | 0.0000  | cm <sup>2</sup> |

**Required Area of Reinforcement for Small End Under External Pressure**

Note: Small end is in compression, 1.0 will be used for E1 and E2.

|                                                    |       |                    |
|----------------------------------------------------|-------|--------------------|
| Allowable Stress of Small End Material (Ext. Temp) | 138.0 | N./mm <sup>2</sup> |
| Allowable Stress of Cone Material (Ext. Temp)      | 138.0 | N./mm <sup>2</sup> |

Intermediate Value [k]:

$$= \max(Y / (Srs \cdot Ers), 1)$$

$$= \max(0.27467E+11 / (138.0 \cdot 0.19904E+09), 1)$$

$$= 1.0000$$

where [Y] is:

$$= \text{Small End All. Stress} \cdot \text{Small End Elastic Modulus (Ext. temp.)}$$

$$= 138.0 \cdot 0.19904E+09$$

$$= 27467335680.0 \text{ (N./mm}^2\text{)}$$

Area of Reinforcement Required in Small End Shell [Ars]:

$$= k \cdot QS \cdot Rs \cdot \tan(\alpha) / (Ss \cdot E1)$$

$$= (1.0 \cdot 4.8565 \cdot 921.0 \cdot 0.5774) / (138 \cdot 1.0)$$

$$= 1.835 \text{ cm}^2$$

Force per Length, Cone Small End External Pressure, compressive side [QS-]:

$$= \text{Pext}(Rs/2) + f2$$

$$= 1.034(921.0/2) + 0.0$$

$$= 4.857 \text{ Kg/mm.}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= \text{Faxial}/(\pi(Ds + Ts)) + \text{Moment}/(\pi(Rs - Ts/2)^2)$$

$$= +0.0/(\pi(1842.0-46.0)) + 0.0/(\pi(921.0-46.0/2)^2)$$

$$= 0.0 \text{ Kg/mm.}$$

Force per Length, Cone Small End External Pressure, tensile side [QS+]:

$$= \text{Pext}(Rs/2) + f2$$

$$= 1.034(921.0/2) + 0.0$$

$$= 4.857 \text{ Kg/mm.}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= \text{Faxial}/(\pi(Ds + Ts)) - \text{Moment}/(\pi(Rs - Ts/2)^2)$$

$$= +0.0/(\pi(1842.0-46.0)) - 0.0/(\pi(921.0-46.0/2)^2)$$

$$= 0.0 \text{ Kg/mm.}$$

Area of Reinforcement Available in Small End Shell [Aes]:

$$= 0.55[ \sqrt{(Ds \cdot ts)(ts-t)} + \sqrt{(Ds \cdot tc / \cos(\alpha))(tc-tr)} ]$$

$$= 0.55[ \sqrt{(1842.0 \cdot 44.5)(44.5 - 85.647)} + \sqrt{(1842.0 \cdot 44.5 / 0.866)(44.5 - 7.145)} ]$$

$$= 63.2084 \text{ cm}^2$$

**Summary of Reinforcement Area, Small End, External Pressure:**

|                                                |         |                 |
|------------------------------------------------|---------|-----------------|
| Area of reinforcement required per App. 1-8(6) | 1.8352  | cm <sup>2</sup> |
| Area of reinforcement in shell per App. 1-8(7) | 63.2084 | cm <sup>2</sup> |
| Area of reinforcement in stiffening ring       | 0.0000  | cm <sup>2</sup> |

**Intermediate Results, Small End, External Pressure:**

|                                                   |          |                    |
|---------------------------------------------------|----------|--------------------|
| Area Available in Cone, Shell, and Reinforcement  | 678.86   | cm <sup>2</sup>    |
| Force per Unit Length on Shell / Cone Junction    | 10.95    | Kgf/mm.            |
| Actual Buckling Stress associated with this Force | 2.19     | N./mm <sup>2</sup> |
| Material Strain associated with this stress       | 0.000022 |                    |

Required Moment of Inertia, Small End, External Pressure [I's]:

$$= A \cdot Ds^2 \cdot Ats / 10.9$$

$$= 0.220E-04 \cdot 1842.0 \cdot 1842.0 \cdot 678.86 / 10.9$$

$$= 464174.53 \text{ mm}^4$$

**Available Moment of Inertia, Small End, External Pressure:**

|     | Area    | Centroid | Ar*Ce    | Dist     | I       | Ar*Di <sup>2</sup> |
|-----|---------|----------|----------|----------|---------|--------------------|
| Shl | 70.072  | 0.0000   | 0.000    | 24.3601  | 115.634 | 41582.078          |
| Con | 80.913  | 45.4566  | 3678.017 | -21.0965 | 735.330 | 36011.137          |
| Sec | 0.000   | 22.2500  | 0.000    | 2.1101   | 0.000   | 0.000              |
| TOT | 150.985 |          | 3678.017 |          | 850.964 | 77593.219          |

|                     |         |                   |         |
|---------------------|---------|-------------------|---------|
| Centroid of Section | 24.3601 | Moment of Inertia | 1626.90 |
|---------------------|---------|-------------------|---------|

**Summary of Small End Inertia Calculations**

|                                           |            |                 |
|-------------------------------------------|------------|-----------------|
| Available Moment of Inertia ( Small End ) | 0.1627E+04 | cm <sup>4</sup> |
| Required Moment of Inertia ( Small End )  | 0.4642E+02 | cm <sup>4</sup> |



*Because the 1/2 apex angle is less than 30 degrees, the following discontinuity stress evaluation results are for information only.*

The following calculations are only required per 1-5(g) and do include external loads due to wind or seismic. These discontinuity stresses are computed at the shell/cone junction and do not include effects of local stiffening from a junction ring.

**Results for Discontinuity Stresses per Bednar p. 236 2nd Edition:**

| Stress Type     | Stress  | Allowable | Location          |
|-----------------|---------|-----------|-------------------|
| Tensile Stress  | 99.37   | 483.99    | Small Cyl. Long.  |
| Compres. Stress | -48.67  | -483.99   | Small Cyl. Long.  |
| Membrane Stress | 91.44   | -207.00   | Small End Tang.   |
| Tensile Stress  | 103.29  | 483.99    | Cone Longitudinal |
| Compres. Stress | -44.75  | -483.99   | Cone Longitudinal |
| Compres Stress  | 99.29   | -207.00   | Cone Tangential   |
| Tensile Stress  | 350.41  | 483.99    | Large Cyl. Long.  |
| Compres. Stress | -227.27 | -483.99   | Large Cyl. Long.  |
| Membrane Stress | -2.61   | -207.00   | Large End Tang.   |
| Tensile Stress  | 217.21  | 483.99    | Cone Longitudinal |
| Compres. Stress | -110.17 | -483.99   | Cone Longitudinal |
| Compres Stress  | -18.71  | -207.00   | Cone Tangential   |

An asterisk (\*) denotes that this stress was not applicable for this combination of loads.

**Maximum Allowable Pressure Calculations for Cone to Shell Junction:**

| Pressure Case | Pressure bars | Reason for Failure at this Pressure |
|---------------|---------------|-------------------------------------|
|---------------|---------------|-------------------------------------|



|       |        |                                                    |
|-------|--------|----------------------------------------------------|
| MAWP  | 32.239 | Thickness due to internal pressure, Cone Large End |
| MAPnc | 33.344 | Thickness due to internal pressure, Cone Large End |

These pressures were determined by iteration.

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**Input, Nozzle Desc: CH. IN NOZ A1**

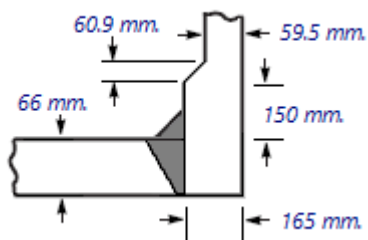
**From: 10**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 191.018   | bars               |
| Temperature for Internal Pressure             | Temp | 85        | °C                 |
| Parent Material                               |      | SA-516 70 | +N+IT              |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Hemispherical Head         | D    | 1750.00   | mm.                |
| Head Finished (Minimum) Thickness             | t    | 66.0000   | mm.                |
| Head Internal Corrosion Allowance             | c    | 1.5000    | mm.                |
| Head External Corrosion Allowance             | co   | 0.0000    | mm.                |
| Distance from Head Centerline                 | L1   | 600.0000  | mm.                |
| User Entered Minimum Design Metal Temperature |      | -35.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                 |      |            |                    |
|-------------------------------------------------|------|------------|--------------------|
| Material                                        |      | SA-350 LF2 | +IT                |
| Material UNS Number                             |      | K03011     |                    |
| Material Specification/Type                     |      | Forgings   |                    |
| Allowable Stress at Temperature                 | Sn   | 138.00     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 138.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside     |                    |
| Layout Angle                                    |      | 90.00      | deg                |
| Diameter                                        |      | 24.0000    | in.                |
| Size and Thickness Basis                        |      | Actual     |                    |
| Actual Thickness                                | tn   | 59.5400    | mm.                |
| Hub Height of Integral Nozzle                   | h    | 150.0000   | mm.                |
| Height of Beveled Transition                    | L'   | 60.8873    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 165.0000   | mm.                |
| Corrosion Allowance                             | can  | 1.5000     | mm.                |
| Joint Efficiency of Shell Seam at Nozzle        | E1   | 1.00       |                    |
| Joint Efficiency of Nozzle Neck                 | En   | 1.00       |                    |
| Outside Projection                              | ho   | 300.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 60.0000    | mm.                |
| Inside Projection                               | h    | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.                |

The Pressure Design option was Design Pressure + static head.



**Reinforcement CALCULATION, Description: CH. IN NOZ A1**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |        |     |
|--------------------------------------------|--------|-----|
| Actual Inside Diameter Used in Calculation | 24.000 | in. |
| Actual Thickness Used in Calculation       | 2.344  | in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]  
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$  per UG-27 (d)  
 $= 191.02 \cdot 876.5 / (2 \cdot 138 \cdot 1.0 - 0.2 \cdot 191.02)$   
 $= 61.5184 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 191.02 \cdot 306.3 / (138 \cdot 1.0 - 0.6 \cdot 191.02)$   
 $= 46.2416 \text{ mm.}$

#### Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$   
 $= 105.46 / 0.5773$   
 $= 182.6622 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

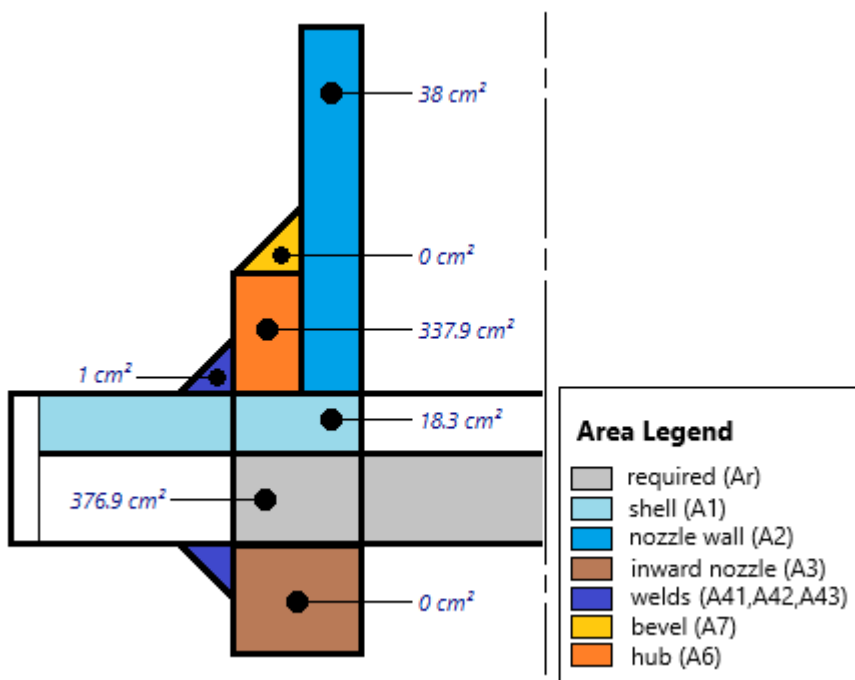
#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |           |     |
|-------------------------------------------------|------|-----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 1225.2000 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 612.6000  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 161.2500  | mm. |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$



**Results of Nozzle Reinforcement Area Calculations: (cm²)**

| AREA AVAILABLE, A1 to A5 |             | Design  | External | Mapnc |
|--------------------------|-------------|---------|----------|-------|
| Area Required            | Ar          | 376.862 | NA       | NA    |
| Area in Shell            | A1          | 18.265  | NA       | NA    |
| Area in Nozzle Wall      | A2          | 38.050  | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000   | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 1.000   | NA       | NA    |
| Area in Element          | A5          | 0.000   | NA       | NA    |
| Area in Hub + Bevel      | A6          | 337.916 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 395.232 | NA       | NA    |

**The Internal Pressure Case Governs the Analysis.**

Nozzle Angle Used in Area Calculations 90.00 degs.

**The area available without a pad is Sufficient.**

Area Required [A]:  

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (612.6 \cdot 61.5184 \cdot 1.0 + 2 \cdot 163.5 \cdot 61.5184 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 376.862 \text{ cm}^2$$

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:  

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 612.6(1.0 \cdot 64.5 - 1.0 \cdot 61.5184) - 2 \cdot 163.5$$

$$(1.0 \cdot 64.5 - 1.0 \cdot 61.5184) \cdot (1 - 1.0)$$

$$= 18.265 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:  

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 161.25)(58.04 - 46.24)1.0$$

$$= 38.050 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:  

$$= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(161.2, 300.0, 150.0)) \cdot (165.0 - 59.5) \cdot 1.0$$

$$= 337.916 \text{ cm}^2 \text{ Includes Bevel Area of } 21.536 \text{ cm}^2$$

**UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**

Wall Thickness for Internal/External pressures  $ta = 47.7416 \text{ mm.}$   
 Wall Thickness per UG16(b),  $tr16b = 3.0000 \text{ mm.}$   
 Wall Thickness, shell/head, internal pressure  $trb1 = 63.0184 \text{ mm.}$   
 Wall Thickness  $tb1 = \max(trb1, tr16b) = 63.0184 \text{ mm.}$   
 Wall Thickness  $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$   
 Wall Thickness per table UG-45  $tb3 = 9.8312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:  

$$= \min[ tb3, \max( tb1, tb2) ]$$

$$= \min[ 9.831, \max( 63.0184, 3.0 ) ]$$

$$= 9.8312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  

$$= \max( ta, tb )$$

$$= \max( 47.7416, 9.8312 )$$

$$= 47.7416 \text{ mm.}$$

Available Nozzle Neck Thickness = 59.5400 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**



|              |       |             |                          |        |
|--------------|-------|-------------|--------------------------|--------|
| Sustained :  | 66.7, | Allowable : | 138.0 N./mm <sup>2</sup> | Passed |
| Expansion :  | 0.0,  | Allowable : | 278.2 N./mm <sup>2</sup> | Passed |
| Occasional : | 46.0, | Allowable : | 183.5 N./mm <sup>2</sup> | Passed |
| Shear :      | 14.4, | Allowable : | 96.6 N./mm <sup>2</sup>  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

###### Note:

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

##### Weld Size Calculations, Description: CH. IN NOZ A1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

#### Results Per UW-16.1:

|             |                       |                       |
|-------------|-----------------------|-----------------------|
|             | Required Thickness    | Actual Thickness      |
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm. |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

##### Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (376.8617 - 18.2653 + 2 \cdot 163.5 \cdot 1.0 \cdot (1.0 \cdot 64.5 - 61.5184)) \cdot 138) \\
 &= 518319.00 \text{ Kgf}
 \end{aligned}$$

*For hub type nozzles, A2 includes the area of the hub.*

*Note: F is always set to 1.0 throughout the calculation.*

##### Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A4-(W_i - Can/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (375.9665 + 1. - 0.0 \cdot 1.0) \cdot 138 \\
 &= 530449.00 \text{ Kgf}
 \end{aligned}$$

##### Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (375.9665 + 0.0 + 1. + (210.915)) \cdot 138 \\
 &= 827238.44 \text{ Kgf}
 \end{aligned}$$

##### Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (375.9665 + 0.0 + 1. + 0.0 + (210.915)) \cdot 138 \\
 &= 827238.44 \text{ Kgf}
 \end{aligned}$$

#### Strength of Connection Elements for Failure Path Analysis

##### Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 939.6 \cdot 10.0 \cdot 0.49 \cdot 138 \\
 &= 101765. \text{ Kgf}
 \end{aligned}$$

##### Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 388.05) \cdot (165.0 - 1.5) \cdot 0.7 \cdot 138 \\
 &= 1963342. \text{ Kgf}
 \end{aligned}$$

##### Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 939.6 \cdot (60.0 - 0.0) \cdot 0.74 \cdot 138
 \end{aligned}$$



= 922122. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= ( \text{SONW} + \text{SNW} ) = ( 101765 + 1963342 ) = 2065107 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= ( \text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw} ) \\ = ( 101765 + 0 + 922122 + 0 ) = 1023887 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= ( \text{Sonw} + \text{Tngw} + \text{Sinw} ) \\ = ( 101765 + 922122 + 0 ) = 1023887 \text{ Kgf}$$

#### Summary of Failure Path Calculations:

Path 1-1 = 2065107 Kgf, must exceed W = 518319 Kgf or W1 = 530449 Kgf  
 Path 2-2 = 1023887 Kgf, must exceed W = 518319 Kgf or W2 = 827238 Kgf  
 Path 3-3 = 1023887 Kgf, must exceed W = 518319 Kgf or W3 = 827238 Kgf

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 195 bars

The Drop for this Nozzle is : 136.8178 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 502.8178 mm.

#### Input Echo, WRC107/537 Item 1, Description: CH. IN NOZ A1 :

|                                 |        |           |                    |
|---------------------------------|--------|-----------|--------------------|
| Diameter Basis for Vessel       | Vbasis | ID        |                    |
| Cylindrical or Spherical Vessel | Cylsph | Spherical |                    |
| Internal Corrosion Allowance    | Cas    | 1.5000    | mm.                |
| Vessel Diameter                 | Dv     | 1750.000  | mm.                |
| Vessel Thickness                | Tv     | 66.000    | mm.                |
| Design Temperature              | T1     | 85.0      | C                  |
| Vessel Material                 |        | SA-516 70 |                    |
| Vessel UNS Number               |        | K02700    |                    |
| Vessel Cold S.I. Allowable      | Smc    | 138.00    | N./mm <sup>2</sup> |
| Vessel Hot S.I. Allowable       | Smh    | 138.00    | N./mm <sup>2</sup> |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                  |        |            |                    |
|----------------------------------|--------|------------|--------------------|
| Attachment Type                  | Type   | Round      |                    |
| WRC107 Attachment Classification | Holsol | Hollow     |                    |
| Diameter Basis for Nozzle        | Nbasis | ID         |                    |
| Corrosion Allowance for Nozzle   | Can    | 1.5000     | mm.                |
| Nozzle Diameter                  | Dn     | 609.600    | mm.                |
| Nozzle Thickness                 | Tn     | 165.000    | mm.                |
| Nozzle Material                  |        | SA-350 LF2 |                    |
| Nozzle UNS Number                |        | K03011     |                    |
| Nozzle Cold S.I. Allowable       | SNmc   | 138.00     | N./mm <sup>2</sup> |
| Nozzle Hot S.I. Allowable        | SNmh   | 138.00     | N./mm <sup>2</sup> |
| Design Internal Pressure         | Dp     | 191.018    | bars               |
| Include Pressure Thrust          |        | No         |                    |

#### External Forces and Moments in WRC 107/537 Convention:

|                              |         |          |       |
|------------------------------|---------|----------|-------|
| Radial Load (SUS)            | P       | -19579.9 | Kgf   |
| Longitudinal Shear (SUS)     | (V1) V1 | 19579.9  | Kgf   |
| Circumferential Shear (SUS)  | (Vc) V2 | 14684.9  | Kgf   |
| Circumferential Moment (SUS) | (Mc) M1 | 30532.8  | Kg-m. |
| Longitudinal Moment (SUS)    | (Ml) M2 | 30532.8  | Kg-m. |
| Torsional Moment (SUS)       | Mt      | 35230.2  | Kg-m. |

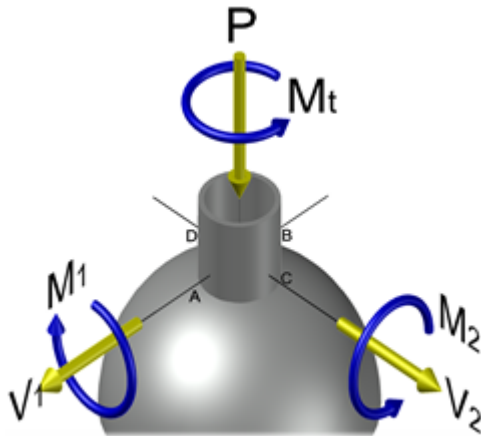
Use Interactive Control

No

WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No  
 Compute Pressure Stress per WRC-368 No  
 Local Loads applied at end of Nozzle/Attachment No

*WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".*



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean} (t - ca))}$   
 $= 939.6 + 2 \cdot 1.65 \cdot \sqrt{(908.75 (66.0 - 1.5))}$   
 $= 1738.543 \text{ mm.}$

WRC 107 Stress Calculation for SUSained loads:

|                        |         |          |       |
|------------------------|---------|----------|-------|
| Radial Load            | P       | -19579.9 | Kgf   |
| Circumferential Shear  | (VC) V2 | 14684.9  | Kgf   |
| Longitudinal Shear     | (VL) V1 | 19579.9  | Kgf   |
| Circumferential Moment | (MC) M1 | 30532.8  | Kg-m. |
| Longitudinal Moment    | (ML) M2 | 30532.8  | Kg-m. |
| Torsional Moment       | MT      | 35230.2  | Kg-m. |

Dimensionless Param: U = 1.94 TAU = 5.00 ( 2.37) RHO = 0.39

#### Dimensionless Loads for Spherical Shells at Attachment Junction:

| Curves read for 1979                   | Figure | Value   | Location  |
|----------------------------------------|--------|---------|-----------|
| $N(x) * T / P$                         | SP 1   | 0.01877 | (A,B,C,D) |
| $M(x) / P$                             | SP 1   | 0.01640 | (A,B,C,D) |
| $N(x) * T * \text{SQRT}(R_m * T) / MC$ | SM 1   | 0.01667 | (A,B,C,D) |
| $M(x) * \text{SQRT}(R_m * T) / MC$     | SM 1   | 0.01735 | (A,B,C,D) |
| $N(x) * T * \text{SQRT}(R_m * T) / ML$ | SM 1   | 0.01667 | (A,B,C,D) |
| $M(x) * \text{SQRT}(R_m * T) / ML$     | SM 1   | 0.01735 | (A,B,C,D) |
| $N(y) * T / P$                         | SP 1   | 0.02570 | (A,B,C,D) |
| $M(y) / P$                             | SP 1   | 0.00511 | (A,B,C,D) |
| $N(y) * T * \text{SQRT}(R_m * T) / MC$ | SM 1   | 0.01896 | (A,B,C,D) |
| $M(y) * \text{SQRT}(R_m * T) / MC$     | SM 1   | 0.00551 | (A,B,C,D) |
| $N(y) * T * \text{SQRT}(R_m * T) / ML$ | SM 1   | 0.01896 | (A,B,C,D) |
| $M(y) * \text{SQRT}(R_m * T) / ML$     | SM 1   | 0.00551 | (A,B,C,D) |

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

#### Stresses in the Vessel at the Attachment Junction (N./mm<sup>2</sup>)

| Type of | Stress Intensity Values at |
|---------|----------------------------|
|---------|----------------------------|



| Stress     | Load      | Au    | Al   | Bu   | Bl    | Cu    | Cl   | Du   | Dl    |
|------------|-----------|-------|------|------|-------|-------|------|------|-------|
| Rad.       | Memb. P   | 0.9   | 0.9  | 0.9  | 0.9   | 0.9   | 0.9  | 0.9  | 0.9   |
| Rad.       | Bend. P   | 4.5   | -4.5 | 4.5  | -4.5  | 4.5   | -4.5 | 4.5  | -4.5  |
| Rad.       | Memb. MC  | 0.0   | 0.0  | 0.0  | 0.0   | -5.0  | -5.0 | 5.0  | 5.0   |
| Rad.       | Bend. MC  | 0.0   | 0.0  | 0.0  | 0.0   | -30.9 | 30.9 | 30.9 | -30.9 |
| Rad.       | Memb. ML  | -5.0  | -5.0 | 5.0  | 5.0   | 0.0   | 0.0  | 0.0  | 0.0   |
| Rad.       | Bend. ML  | -30.9 | 30.9 | 30.9 | -30.9 | 0.0   | 0.0  | 0.0  | 0.0   |
| Tot. Rad.  | Str.      | -30.5 | 22.3 | 41.3 | -29.7 | -30.5 | 22.3 | 41.3 | -29.7 |
| Tang.      | Memb. P   | 1.2   | 1.2  | 1.2  | 1.2   | 1.2   | 1.2  | 1.2  | 1.2   |
| Tang.      | Bend. P   | 1.4   | -1.4 | 1.4  | -1.4  | 1.4   | -1.4 | 1.4  | -1.4  |
| Tang.      | Memb. MC  | 0.0   | 0.0  | 0.0  | 0.0   | -5.6  | -5.6 | 5.6  | 5.6   |
| Tang.      | Bend. MC  | 0.0   | 0.0  | 0.0  | 0.0   | -9.8  | 9.8  | 9.8  | -9.8  |
| Tang.      | Memb. ML  | -5.6  | -5.6 | 5.6  | 5.6   | 0.0   | 0.0  | 0.0  | 0.0   |
| Tang.      | Bend. ML  | -9.8  | 9.8  | 9.8  | -9.8  | 0.0   | 0.0  | 0.0  | 0.0   |
| Tot. Tang. | Str.      | -12.9 | 4.0  | 18.1 | -4.4  | -12.9 | 4.0  | 18.1 | -4.4  |
|            | Shear VC  | 1.5   | 1.5  | -1.5 | -1.5  | 0.0   | 0.0  | 0.0  | 0.0   |
|            | Shear VL  | 0.0   | 0.0  | 0.0  | 0.0   | -2.0  | -2.0 | 2.0  | 2.0   |
|            | Shear MT  | 3.9   | 3.9  | 3.9  | 3.9   | 3.9   | 3.9  | 3.9  | 3.9   |
| Tot.       | Shear     | 5.4   | 5.4  | 2.3  | 2.3   | 1.8   | 1.8  | 5.9  | 5.9   |
|            | Str. Int. | 32.0  | 23.8 | 41.5 | 29.9  | 30.7  | 22.5 | 42.7 | 31.0  |

#### WRC 107/537 Stress Summations:

##### Vessel Stress Summation at Attachment Junction (N./mm<sup>2</sup>)

| Type of Stress  | Load     | Stress Intensity Values at |       |       |       |       |       |       |       |
|-----------------|----------|----------------------------|-------|-------|-------|-------|-------|-------|-------|
|                 |          | Au                         | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Rad.            | Pm (SUS) | 130.2                      | 130.2 | 130.2 | 130.2 | 130.2 | 130.2 | 130.2 | 130.2 |
| Rad.            | Pl (SUS) | -4.1                       | -4.1  | 5.8   | 5.8   | -4.1  | -4.1  | 5.8   | 5.8   |
| Rad.            | Q (SUS)  | -26.4                      | 26.4  | 35.5  | -35.5 | -26.4 | 26.4  | 35.5  | -35.5 |
| Long.           | Pm (SUS) | 130.2                      | 130.2 | 130.2 | 130.2 | 130.2 | 130.2 | 130.2 | 130.2 |
| Long.           | Pl (SUS) | -4.5                       | -4.5  | 6.8   | 6.8   | -4.5  | -4.5  | 6.8   | 6.8   |
| Long.           | Q (SUS)  | -8.4                       | 8.4   | 11.2  | -11.2 | -8.4  | 8.4   | 11.2  | -11.2 |
| Shear           | Pm (SUS) | 0.0                        | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
| Shear           | Pl (SUS) | 1.5                        | 1.5   | -1.5  | -1.5  | -2.0  | -2.0  | 2.0   | 2.0   |
| Shear           | Q (SUS)  | 3.9                        | 3.9   | 3.9   | 3.9   | 3.9   | 3.9   | 3.9   | 3.9   |
| Pm (SUS)        |          | 130.2                      | 130.2 | 130.2 | 130.2 | 130.2 | 130.2 | 130.2 | 130.2 |
| Pm+Pl (SUS)     |          | 127.5                      | 127.5 | 138.2 | 138.2 | 128.0 | 128.0 | 138.6 | 138.6 |
| Pm+Pl+Q (Total) |          | 118.9                      | 154.0 | 171.8 | 126.0 | 117.6 | 152.7 | 173.0 | 127.1 |

##### Vessel Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 130.25    | 138.00         | Passed |
| Pm+Pl (SUS)         | 138.65    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 172.95    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

**Input, Nozzle Desc: CH. OUT NOZ B1**

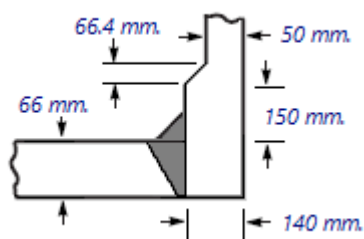
**From: 10**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 191.079   | bars               |
| Temperature for Internal Pressure             | Temp | 85        | °C                 |
| Parent Material                               |      | SA-516 70 | +N+IT              |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Hemispherical Head         | D    | 1750.00   | mm.                |
| Head Finished (Minimum) Thickness             | t    | 66.0000   | mm.                |
| Head Internal Corrosion Allowance             | c    | 1.5000    | mm.                |
| Head External Corrosion Allowance             | co   | 0.0000    | mm.                |
| Distance from Head Centerline                 | L1   | 698.9460  | mm.                |
| User Entered Minimum Design Metal Temperature |      | -35.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                 |      |            |                    |
|-------------------------------------------------|------|------------|--------------------|
| Material                                        |      | SA-350 LF2 | +IT                |
| Material UNS Number                             |      | K03011     |                    |
| Material Specification/Type                     |      | Forgings   |                    |
| Allowable Stress at Temperature                 | Sn   | 138.00     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 138.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside     |                    |
| Layout Angle                                    |      | -30.05     | deg                |
| Diameter                                        |      | 20.0000    | in.                |
| Size and Thickness Basis                        |      | Actual     |                    |
| Actual Thickness                                | tn   | 50.0000    | mm.                |
| Hub Height of Integral Nozzle                   | h    | 150.0000   | mm.                |
| Height of Beveled Transition                    | L'   | 66.3953    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 140.0000   | mm.                |
| Corrosion Allowance                             | can  | 1.5000     | mm.                |
| Joint Efficiency of Shell Seam at Nozzle        | E1   | 1.00       |                    |
| Joint Efficiency of Nozzle Neck                 | En   | 1.00       |                    |
| Outside Projection                              | ho   | 300.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 60.0000    | mm.                |
| Inside Projection                               | h    | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.                |

The Pressure Design option was Design Pressure + static head.



**Reinforcement CALCULATION, Description: CH. OUT NOZ B1**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |        |     |
|--------------------------------------------|--------|-----|
| Actual Inside Diameter Used in Calculation | 20.000 | in. |
| Actual Thickness Used in Calculation       | 1.969  | in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]  
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$  per UG-27 (d)  
 $= 191.08 \cdot 876.5 / (2 \cdot 138 \cdot 1.0 - 0.2 \cdot 191.08)$   
 $= 61.5383 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 191.08 \cdot 255.5 / (138 \cdot 1.0 - 0.6 \cdot 191.08)$   
 $= 38.5858 \text{ mm.}$

#### Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$   
 $= 90.0 / 0.5773$   
 $= 155.8846 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

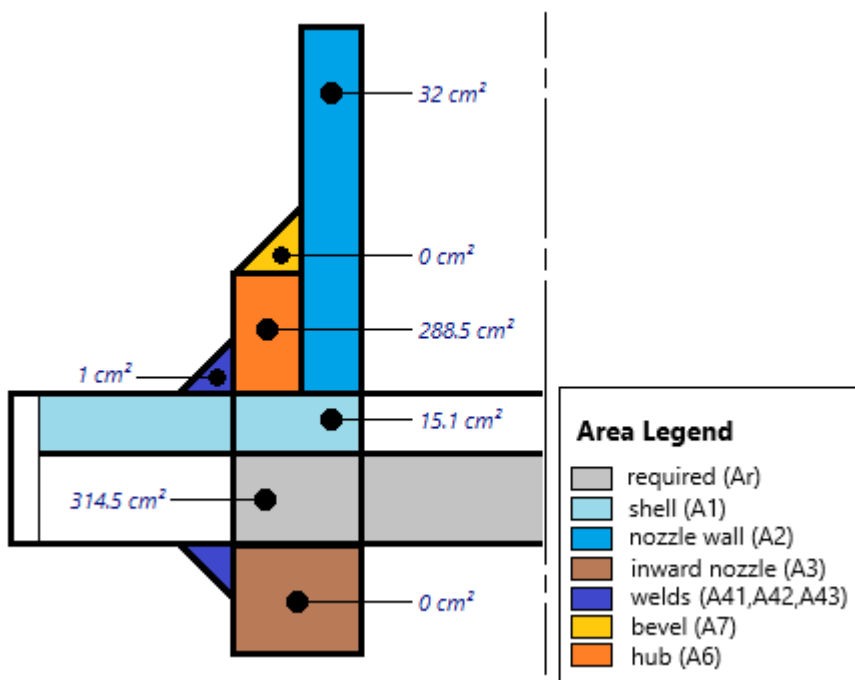
#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |           |     |
|-------------------------------------------------|------|-----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 1022.0000 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 511.0000  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 161.2500  | mm. |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$



**Results of Nozzle Reinforcement Area Calculations: (cm²)**

| AREA AVAILABLE, A1 to A5 |             | Design  | External | Mapnc |
|--------------------------|-------------|---------|----------|-------|
| Area Required            | Ar          | 314.461 | NA       | NA    |
| Area in Shell            | A1          | 15.134  | NA       | NA    |
| Area in Nozzle Wall      | A2          | 31.973  | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000   | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 1.000   | NA       | NA    |
| Area in Element          | A5          | 0.000   | NA       | NA    |
| Area in Hub + Bevel      | A6          | 288.534 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 336.642 | NA       | NA    |

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (511.0 \cdot 61.5383 \cdot 1.0 + 2 \cdot 138.5 \cdot 61.5383 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 314.461 \text{ cm}^2$

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 511.0(1.0 \cdot 64.5 - 1.0 \cdot 61.538) - 2 \cdot 138.5$   
 $(1.0 \cdot 64.5 - 1.0 \cdot 61.5383) \cdot (1 - 1.0)$   
 $= 15.134 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= (2 \cdot tnp)(tn - trn)fr2$   
 $= (2 \cdot 161.25)(48.5 - 38.59)1.0$   
 $= 31.973 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$   
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$   
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:  
 $= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$   
 $= (2 \cdot \min(161.2, 300.0, 150.0)) \cdot (140.0 - 50.0) \cdot 1.0$   
 $= 288.534 \text{ cm}^2$  Includes Bevel Area of 18.534 cm²

**UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**

Wall Thickness for Internal/External pressures  $ta = 40.0858 \text{ mm.}$   
 Wall Thickness per UG16(b),  $tr16b = 3.0000 \text{ mm.}$   
 Wall Thickness, shell/head, internal pressure  $trb1 = 63.0383 \text{ mm.}$   
 Wall Thickness  $tb1 = \max(trb1, tr16b) = 63.0383 \text{ mm.}$   
 Wall Thickness  $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$   
 Wall Thickness per table UG-45  $tb3 = 9.8312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2) ]$   
 $= \min[ 9.831, \max( 63.0383, 3.0 ) ]$   
 $= 9.8312 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 40.0858, 9.8312 )$   
 $= 40.0858 \text{ mm.}$

Available Nozzle Neck Thickness = 50.0000 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**



|              |       |             |                          |        |
|--------------|-------|-------------|--------------------------|--------|
| Sustained :  | 70.7, | Allowable : | 138.0 N./mm <sup>2</sup> | Passed |
| Expansion :  | 0.0,  | Allowable : | 274.3 N./mm <sup>2</sup> | Passed |
| Occasional : | 46.0, | Allowable : | 183.5 N./mm <sup>2</sup> | Passed |
| Shear :      | 17.2, | Allowable : | 96.6 N./mm <sup>2</sup>  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

###### Note:

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

##### Weld Size Calculations, Description: CH. OUT NOZ B1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

#### Results Per UW-16.1:

|             |                       |                       |
|-------------|-----------------------|-----------------------|
|             | Required Thickness    | Actual Thickness      |
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm. |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

##### Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (314.4608 - 15.1342 + 2 \cdot 138.5 \cdot 1.0 \cdot (1.0 \cdot 64.5 - 61.5383)) \cdot 138) \\
 &= 432742.06 \text{ Kgf}
 \end{aligned}$$

*For hub type nozzles, A2 includes the area of the hub.*

*Note: F is always set to 1.0 throughout the calculation.*

##### Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (320.5078 + 1. - 0.0 \cdot 1.0) \cdot 138 \\
 &= 452410.19 \text{ Kgf}
 \end{aligned}$$

##### Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (320.5078 + 0.0 + 1. + (178.665)) \cdot 138 \\
 &= 703818.94 \text{ Kgf}
 \end{aligned}$$

##### Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (320.5078 + 0.0 + 1. + 0.0 + (178.665)) \cdot 138 \\
 &= 703818.94 \text{ Kgf}
 \end{aligned}$$

#### Strength of Connection Elements for Failure Path Analysis

##### Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 788.0 \cdot 10.0 \cdot 0.49 \cdot 138 \\
 &= 85346. \text{ Kgf}
 \end{aligned}$$

##### Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 324.75) \cdot (140.0 - 1.5) \cdot 0.7 \cdot 138 \\
 &= 1391840. \text{ Kgf}
 \end{aligned}$$

##### Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 788.0 \cdot (60.0 - 0.0) \cdot 0.74 \cdot 138
 \end{aligned}$$

= 773342. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= ( \text{SONW} + \text{SNW} ) = ( 85346 + 1391840 ) = 1477186 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= ( \text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw} ) \\ = ( 85346 + 0 + 773342 + 0 ) = 858688 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= ( \text{Sonw} + \text{Tngw} + \text{Sinw} ) \\ = ( 85346 + 773342 + 0 ) = 858688 \text{ Kgf}$$

#### Summary of Failure Path Calculations:

Path 1-1 = 1477186 Kgf, must exceed W = 432742 Kgf or W1 = 452410 Kgf  
 Path 2-2 = 858688 Kgf, must exceed W = 432742 Kgf or W2 = 703818 Kgf  
 Path 3-3 = 858688 Kgf, must exceed W = 432742 Kgf or W3 = 703818 Kgf

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 196 bars

The Drop for this Nozzle is : 93.7261 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 459.7261 mm.

#### Input Echo, WRC107/537 Item 1, Description: CH. OUT NOZ B1:

|                                 |        |           |                    |
|---------------------------------|--------|-----------|--------------------|
| Diameter Basis for Vessel       | Vbasis | ID        |                    |
| Cylindrical or Spherical Vessel | Cylsph | Spherical |                    |
| Internal Corrosion Allowance    | Cas    | 1.5000    | mm.                |
| Vessel Diameter                 | Dv     | 1750.000  | mm.                |
| Vessel Thickness                | Tv     | 66.000    | mm.                |
| Design Temperature              | T1     | 85.0      | C                  |
| Vessel Material                 |        | SA-516 70 |                    |
| Vessel UNS Number               |        | K02700    |                    |
| Vessel Cold S.I. Allowable      | Smc    | 138.00    | N./mm <sup>2</sup> |
| Vessel Hot S.I. Allowable       | Smh    | 138.00    | N./mm <sup>2</sup> |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                  |        |            |                    |
|----------------------------------|--------|------------|--------------------|
| Attachment Type                  | Type   | Round      |                    |
| WRC107 Attachment Classification | Holsol | Hollow     |                    |
| Diameter Basis for Nozzle        | Nbasis | ID         |                    |
| Corrosion Allowance for Nozzle   | Can    | 1.5000     | mm.                |
| Nozzle Diameter                  | Dn     | 508.000    | mm.                |
| Nozzle Thickness                 | Tn     | 140.000    | mm.                |
| Nozzle Material                  |        | SA-350 LF2 |                    |
| Nozzle UNS Number                |        | K03011     |                    |
| Nozzle Cold S.I. Allowable       | SNmc   | 138.00     | N./mm <sup>2</sup> |
| Nozzle Hot S.I. Allowable        | SNmh   | 138.00     | N./mm <sup>2</sup> |
| Design Internal Pressure         | Dp     | 191.079    | bars               |
| Include Pressure Thrust          |        | No         |                    |

#### External Forces and Moments in WRC 107/537 Convention:

|                              |         |          |       |
|------------------------------|---------|----------|-------|
| Radial Load (SUS)            | P       | -16316.5 | Kgf   |
| Longitudinal Shear (SUS)     | (V1) V1 | 16316.5  | Kgf   |
| Circumferential Shear (SUS)  | (Vc) V2 | 12237.4  | Kgf   |
| Circumferential Moment (SUS) | (Mc) M1 | 21203.3  | Kg-m. |
| Longitudinal Moment (SUS)    | (Ml) M2 | 21203.3  | Kg-m. |
| Torsional Moment (SUS)       | Mt      | 24465.4  | Kg-m. |

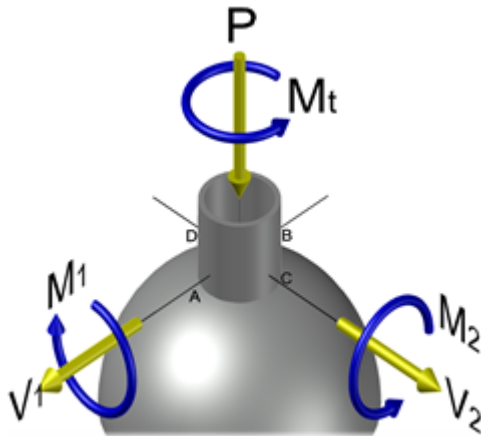
Use Interactive Control

No

WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No  
 Compute Pressure Stress per WRC-368 No  
 Local Loads applied at end of Nozzle/Attachment No

*WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".*



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$   
 $= 788.0 + 2 \cdot 1.65 \cdot \sqrt{908.75 (66.0 - 1.5)}$   
 $= 1586.943 \text{ mm.}$

WRC 107 Stress Calculation for SUSained loads:

|                        |         |          |       |
|------------------------|---------|----------|-------|
| Radial Load            | P       | -16316.5 | Kgf   |
| Circumferential Shear  | (VC) V2 | 12237.4  | Kgf   |
| Longitudinal Shear     | (VL) V1 | 16316.5  | Kgf   |
| Circumferential Moment | (MC) M1 | 21203.3  | Kg-m. |
| Longitudinal Moment    | (ML) M2 | 21203.3  | Kg-m. |
| Torsional Moment       | MT      | 24465.4  | Kg-m. |

Dimensionless Param: U = 1.63 TAU = 5.00 ( 2.34) RHO = 0.47

#### Dimensionless Loads for Spherical Shells at Attachment Junction:

| Curves read for 1979                   | Figure | Value   | Location  |
|----------------------------------------|--------|---------|-----------|
| $N(x) * T / P$                         | SP 1   | 0.02392 | (A,B,C,D) |
| $M(x) / P$                             | SP 1   | 0.02171 | (A,B,C,D) |
| $N(x) * T * \text{SQRT}(R_m * T) / MC$ | SM 1   | 0.02281 | (A,B,C,D) |
| $M(x) * \text{SQRT}(R_m * T) / MC$     | SM 1   | 0.02633 | (A,B,C,D) |
| $N(x) * T * \text{SQRT}(R_m * T) / ML$ | SM 1   | 0.02281 | (A,B,C,D) |
| $M(x) * \text{SQRT}(R_m * T) / ML$     | SM 1   | 0.02633 | (A,B,C,D) |
| $N(y) * T / P$                         | SP 1   | 0.03360 | (A,B,C,D) |
| $M(y) / P$                             | SP 1   | 0.00608 | (A,B,C,D) |
| $N(y) * T * \text{SQRT}(R_m * T) / MC$ | SM 1   | 0.02898 | (A,B,C,D) |
| $M(y) * \text{SQRT}(R_m * T) / MC$     | SM 1   | 0.00859 | (A,B,C,D) |
| $N(y) * T * \text{SQRT}(R_m * T) / ML$ | SM 1   | 0.02898 | (A,B,C,D) |
| $M(y) * \text{SQRT}(R_m * T) / ML$     | SM 1   | 0.00859 | (A,B,C,D) |

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

#### Stresses in the Vessel at the Attachment Junction (N./mm<sup>2</sup>)

| Type of | Stress Intensity Values at |
|---------|----------------------------|
|---------|----------------------------|



| Stress          | Load     | Au    | Al   | Bu   | Bl    | Cu    | Cl   | Du   | Dl    |
|-----------------|----------|-------|------|------|-------|-------|------|------|-------|
| Rad.            | Memb. P  | 0.9   | 0.9  | 0.9  | 0.9   | 0.9   | 0.9  | 0.9  | 0.9   |
| Rad.            | Bend. P  | 5.0   | -5.0 | 5.0  | -5.0  | 5.0   | -5.0 | 5.0  | -5.0  |
| Rad.            | Memb. MC | 0.0   | 0.0  | 0.0  | 0.0   | -4.7  | -4.7 | 4.7  | 4.7   |
| Rad.            | Bend. MC | 0.0   | 0.0  | 0.0  | 0.0   | -32.6 | 32.6 | 32.6 | -32.6 |
| Rad.            | Memb. ML | -4.7  | -4.7 | 4.7  | 4.7   | 0.0   | 0.0  | 0.0  | 0.0   |
| Rad.            | Bend. ML | -32.6 | 32.6 | 32.6 | -32.6 | 0.0   | 0.0  | 0.0  | 0.0   |
| Tot. Rad. Str.  |          | -31.4 | 23.8 | 43.3 | -32.0 | -31.4 | 23.8 | 43.3 | -32.0 |
| Tang.           | Memb. P  | 1.3   | 1.3  | 1.3  | 1.3   | 1.3   | 1.3  | 1.3  | 1.3   |
| Tang.           | Bend. P  | 1.4   | -1.4 | 1.4  | -1.4  | 1.4   | -1.4 | 1.4  | -1.4  |
| Tang.           | Memb. MC | 0.0   | 0.0  | 0.0  | 0.0   | -6.0  | -6.0 | 6.0  | 6.0   |
| Tang.           | Bend. MC | 0.0   | 0.0  | 0.0  | 0.0   | -10.6 | 10.6 | 10.6 | -10.6 |
| Tang.           | Memb. ML | -6.0  | -6.0 | 6.0  | 6.0   | 0.0   | 0.0  | 0.0  | 0.0   |
| Tang.           | Bend. ML | -10.6 | 10.6 | 10.6 | -10.6 | 0.0   | 0.0  | 0.0  | 0.0   |
| Tot. Tang. Str. |          | -13.9 | 4.5  | 19.3 | -4.8  | -13.9 | 4.5  | 19.3 | -4.8  |
|                 | Shear VC | 1.5   | 1.5  | -1.5 | -1.5  | 0.0   | 0.0  | 0.0  | 0.0   |
|                 | Shear VL | 0.0   | 0.0  | 0.0  | 0.0   | -2.0  | -2.0 | 2.0  | 2.0   |
|                 | Shear MT | 3.8   | 3.8  | 3.8  | 3.8   | 3.8   | 3.8  | 3.8  | 3.8   |
| Tot. Shear      |          | 5.3   | 5.3  | 2.3  | 2.3   | 1.8   | 1.8  | 5.8  | 5.8   |
| Str. Int.       |          | 32.9  | 25.2 | 43.5 | 32.2  | 31.6  | 24.0 | 44.6 | 33.2  |

#### WRC 107/537 Stress Summations:

##### Vessel Stress Summation at Attachment Junction (N./mm<sup>2</sup>)

| Type of Stress  | Load     | Stress Intensity Values at |       |       |       |       |       |       |       |
|-----------------|----------|----------------------------|-------|-------|-------|-------|-------|-------|-------|
|                 |          | Au                         | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Rad.            | Pm (SUS) | 130.3                      | 130.3 | 130.3 | 130.3 | 130.3 | 130.3 | 130.3 | 130.3 |
| Rad.            | Pl (SUS) | -3.8                       | -3.8  | 5.6   | 5.6   | -3.8  | -3.8  | 5.6   | 5.6   |
| Rad.            | Q (SUS)  | -27.6                      | 27.6  | 37.6  | -37.6 | -27.6 | 27.6  | 37.6  | -37.6 |
| Long.           | Pm (SUS) | 130.3                      | 130.3 | 130.3 | 130.3 | 130.3 | 130.3 | 130.3 | 130.3 |
| Long.           | Pl (SUS) | -4.7                       | -4.7  | 7.3   | 7.3   | -4.7  | -4.7  | 7.3   | 7.3   |
| Long.           | Q (SUS)  | -9.2                       | 9.2   | 12.0  | -12.0 | -9.2  | 9.2   | 12.0  | -12.0 |
| Shear           | Pm (SUS) | 0.0                        | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
| Shear           | Pl (SUS) | 1.5                        | 1.5   | -1.5  | -1.5  | -2.0  | -2.0  | 2.0   | 2.0   |
| Shear           | Q (SUS)  | 3.8                        | 3.8   | 3.8   | 3.8   | 3.8   | 3.8   | 3.8   | 3.8   |
| Pm (SUS)        |          | 130.3                      | 130.3 | 130.3 | 130.3 | 130.3 | 130.3 | 130.3 | 130.3 |
| Pm+Pl (SUS)     |          | 127.6                      | 127.6 | 138.5 | 138.5 | 128.1 | 128.1 | 138.9 | 138.9 |
| Pm+Pl+Q (Total) |          | 117.9                      | 155.5 | 173.8 | 125.7 | 116.6 | 154.3 | 174.9 | 126.7 |

##### Vessel Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 130.29    | 138.00         | Passed |
| Pm+Pl (SUS)         | 138.91    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 174.88    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

**Input, Nozzle Desc: MANHOLE M1**

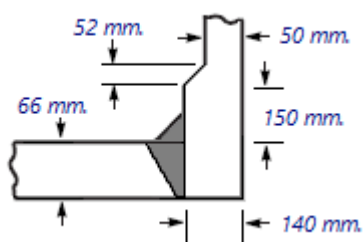
**From: 10**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 191.082   | bars               |
| Temperature for Internal Pressure             | Temp | 85        | °C                 |
| Parent Material                               |      | SA-516 70 | +N+IT              |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Hemispherical Head         | D    | 1750.00   | mm.                |
| Head Finished (Minimum) Thickness             | t    | 66.0000   | mm.                |
| Head Internal Corrosion Allowance             | c    | 1.5000    | mm.                |
| Head External Corrosion Allowance             | co   | 0.0000    | mm.                |
| Distance from Head Centerline                 | L1   | 721.1100  | mm.                |
| User Entered Minimum Design Metal Temperature |      | -35.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                  |     |            |                    |
|--------------------------------------------------|-----|------------|--------------------|
| Material                                         |     | SA-350 LF2 | +IT                |
| Material UNS Number                              |     | K03011     |                    |
| Material Specification/Type                      |     | Forgings   |                    |
| Allowable Stress at Temperature                  | Sn  | 138.00     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                      | Sna | 138.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)                |     | Inside     |                    |
| Layout Angle                                     |     | -146.31    | deg                |
| Diameter                                         |     | 20.0000    | in.                |
| Size and Thickness Basis                         |     | Actual     |                    |
| Actual Thickness                                 | tn  | 50.0000    | mm.                |
| Flange Material                                  |     | SA-350 LF2 | +IT                |
| Hub Height of Integral Nozzle                    | h   | 150.0000   | mm.                |
| Height of Beveled Transition                     | L'  | 51.9615    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp )  |     | 140.0000   | mm.                |
| Corrosion Allowance                              | can | 1.5000     | mm.                |
| Joint Efficiency of Shell Seam at Nozzle         | E1  | 1.00       |                    |
| Joint Efficiency of Nozzle Neck                  | En  | 1.00       |                    |
| Outside Projection                               | ho  | 300.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell       | Wo  | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel Wgnv |     | 60.0000    | mm.                |
| Inside Projection                                | h   | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell           | Wi  | 0.0000     | mm.                |
| Flange Class                                     |     | 1500       |                    |
| Flange Grade                                     |     | GR 1.1     |                    |

The Pressure Design option was Design Pressure + static head.



**Reinforcement CALCULATION, Description: MANHOLE M1**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 20.000 in.  
Actual Thickness Used in Calculation 1.969 in.

**Note:**

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]  
=  $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$  per UG-27 (d)  
=  $191.08 \cdot 876.5 / (2 \cdot 138 \cdot 1.0 - 0.2 \cdot 191.08)$   
= 61.5394 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
=  $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $191.08 \cdot 255.5 / (138 \cdot 1.0 - 0.6 \cdot 191.08)$   
= 38.5865 mm.

**Intermediate Hub Nozzle Calculations:**

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
= ( Hub Thickness - Neck Thickness ) / tan(30)  
= 90.0/0.5773  
= 155.8846 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

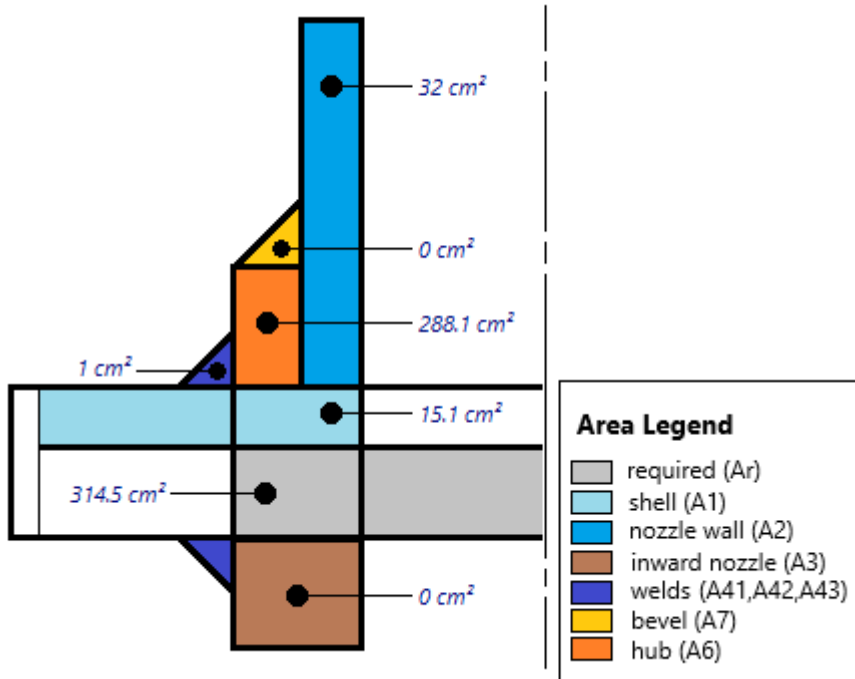
**UG-40, Limits of Reinforcement: [Internal Pressure]**

|                                                 |      |           |     |
|-------------------------------------------------|------|-----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 1022.0000 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 511.0000  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 161.2500  | mm. |

Weld Strength Reduction Factor [fr1]:  
= min( 1,  $S_n/S_v$  )  
= min( 1,  $138.0/138.0$  )  
= 1.000

Weld Strength Reduction Factor [fr2]:  
= min( 1,  $S_n/S_v$  )  
= min( 1,  $138.0/138.0$  )  
= 1.000

Weld Strength Reduction Factor [fr3]:  
= min( fr2, fr4 )  
= min( 1.0, 1.0 )  
= 1.000



#### Results of Nozzle Reinforcement Area Calculations: (cm²)

| AREA AVAILABLE, A1 to A5  |      | Design  | External | Mapnc |
|---------------------------|------|---------|----------|-------|
| Area Required             | Ar   | 314.466 | NA       | NA    |
| Area in Shell             | A1   | 15.129  | NA       | NA    |
| Area in Nozzle Wall       | A2   | 31.971  | NA       | NA    |
| Area in Inward Nozzle     | A3   | 0.000   | NA       | NA    |
| Area in Welds A41+A42+A43 |      | 1.000   | NA       | NA    |
| Area in Element           | A5   | 0.000   | NA       | NA    |
| Area in Hub + Bevel       | A6   | 288.058 | NA       | NA    |
| TOTAL AREA AVAILABLE      | Atot | 336.158 | NA       | NA    |

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (511.0 \cdot 61.5394 \cdot 1.0 + 2 \cdot 138.5 \cdot 61.5394 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 314.466 \text{ cm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 511.0(1.0 \cdot 64.5 - 1.0 \cdot 61.539) - 2 \cdot 138.5$   
 $(1.0 \cdot 64.5 - 1.0 \cdot 61.5394) \cdot (1 - 1.0)$   
 $= 15.129 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= (2 \cdot tlnp)(tn - trn)fr2$   
 $= (2 \cdot 161.25)(48.5 - 38.59)1.0$   
 $= 31.971 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$   
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$   
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:  

$$= (2 \cdot \min(T_{np}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(161.2, 300.0, 150.0)) \cdot (140.0 - 50.0) \cdot 1.0$$

$$= 288.058 \text{ cm}^2 \text{ Includes Bevel Area of } 18.058 \text{ cm}^2$$

**UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**

|                                                |                                                      |
|------------------------------------------------|------------------------------------------------------|
| Wall Thickness for Internal/External pressures | $t_a = 40.0865 \text{ mm.}$                          |
| Wall Thickness per UG16(b),                    | $tr_{16b} = 3.0000 \text{ mm.}$                      |
| Wall Thickness, shell/head, internal pressure  | $trb_1 = 63.0394 \text{ mm.}$                        |
| Wall Thickness                                 | $tb_1 = \max(trb_1, tr_{16b}) = 63.0394 \text{ mm.}$ |
| Wall Thickness                                 | $tb_2 = \max(trb_2, tr_{16b}) = 3.0000 \text{ mm.}$  |
| Wall Thickness per table UG-45                 | $tb_3 = 9.8312 \text{ mm.}$                          |

Determine Nozzle Thickness candidate [tb]:  

$$= \min[ tb_3, \max( tb_1, tb_2 ) ]$$

$$= \min[ 9.831, \max( 63.0394, 3.0 ) ]$$

$$= 9.8312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  

$$= \max( t_a, t_b )$$

$$= \max( 40.0865, 9.8312 )$$

$$= 40.0865 \text{ mm.}$$

Available Nozzle Neck Thickness = 50.0000 mm. -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**

**Nozzle Neck to Flange Weld (Impact tested) :**

*Note:*  
*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                    |        |
|----------------------------------------------------|--------|
| Impact Test Temperature provided per Specification | -46 °C |
|----------------------------------------------------|--------|

**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :**

*Note:*  
*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                    |        |
|----------------------------------------------------|--------|
| Impact Test Temperature provided per Specification | -46 °C |
|----------------------------------------------------|--------|

|                                                          |        |
|----------------------------------------------------------|--------|
| Gov. MDMT of the nozzle to shell joint welded assembly : | -46 °C |
|----------------------------------------------------------|--------|

**Flange MDMT including Temperature reduction per UCS-66.1:**

|                                                      |        |
|------------------------------------------------------|--------|
| MDMT of ASME B16.5/47 flange per Matl. Specification | -46 °C |
| Flange MDMT with Temp reduction per UCS-66(i)(2)     | -60 °C |

Where the Stress Reduction Ratio per UCS-66(i)(2) is :  

$$\text{Design Pressure/Ambient Rating} = 191.08/255.10 = 0.749$$

Weld Size Calculations, Description: MANHOLE M1

|                                           |           |             |
|-------------------------------------------|-----------|-------------|
| Intermediate Calc. for nozzle/shell Welds | $T_{min}$ | 19.0000 mm. |
|-------------------------------------------|-----------|-------------|

**Results Per UW-16.1:**

|             |                                |                                      |
|-------------|--------------------------------|--------------------------------------|
|             | Required Thickness             | Actual Thickness                     |
| Nozzle Weld | $6.0000 = \text{Min per Code}$ | $7.0700 = 0.7 \cdot W_o \text{ mm.}$ |

**Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:  

$$= \max( 0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - tr)) S_v )$$

$$= \max( 0, (314.4662 - 15.1288 + 2 \cdot 138.5 \cdot 1.0 \cdot (1.0 \cdot 64.5 - 61.5394) ) 138 )$$

$$= 432753.03 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

*Note: F is always set to 1.0 throughout the calculation.*



Weld Load [W1]:

$$\begin{aligned} &= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v \\ &= (320.0289 + 1. - 0.0 \cdot 1.0) \cdot 138 \\ &= 451736.38 \text{ Kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v \\ &= (320.0289 + 0.0 + 1. + (178.665)) \cdot 138 \\ &= 703145.19 \text{ Kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S \\ &= (320.0289 + 0.0 + 1. + 0.0 + (178.665)) \cdot 138 \\ &= 703145.19 \text{ Kgf} \end{aligned}$$

### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 788.0 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 85346. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 324.75) \cdot (140.0 - 1.5) \cdot 0.7 \cdot 138 \\ &= 1391840. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 788.0 \cdot (60.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 773342. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (85346 + 1391840) = 1477186 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (85346 + 0 + 773342 + 0) = 858688 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ &= (85346 + 773342 + 0) = 858688 \text{ Kgf} \end{aligned}$$

### Summary of Failure Path Calculations:

Path 1-1 = 1477186 Kgf, must exceed W = 432753 Kgf or W1 = 451736 Kgf  
Path 2-2 = 858688 Kgf, must exceed W = 432753 Kgf or W2 = 703145 Kgf  
Path 3-3 = 858688 Kgf, must exceed W = 432753 Kgf or W3 = 703145 Kgf

### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 196 bars

The Drop for this Nozzle is : 93.7261 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 459.7261 mm.

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**Input, Nozzle Desc: LIQ OUT NOZ B3**

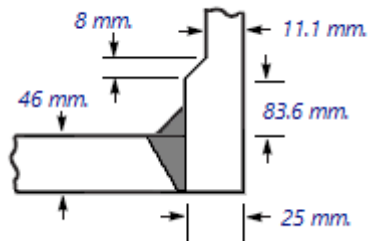
**From: 50**

|                                               |        |           |                    |
|-----------------------------------------------|--------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P      | 25.043    | bars               |
| Temperature for Internal Pressure             | Temp   | 85        | °C                 |
| Design External Pressure                      | Pext   | 1.03      | bars               |
| Temperature for External Pressure             | Tempex | 85        | °C                 |
| Parent Material                               |        | SA-516 70 | +Q+IT              |
| Parent Allowable Stress at Temperature        | Sv     | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva    | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cone at Nozzle Location    | D      | 1894.38   | mm.                |
| Equivalent Length of Conical Section          | Le     | 1871.7892 | mm.                |
| Cone Half Apex Angle                          | Alpha  | 30.00     | Degrees            |
| Shell Finished (Minimum) Thickness            | t      | 46.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c      | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co     | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |        | 1259.50   | mm.                |
| User Entered Minimum Design Metal Temperature |        | -35.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                 |      |            |                    |
|-------------------------------------------------|------|------------|--------------------|
| Material                                        |      | SA-350 LF2 | +IT                |
| Material UNS Number                             |      | K03011     |                    |
| Material Specification/Type                     |      | Forgings   |                    |
| Allowable Stress at Temperature                 | Sn   | 138.00     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 138.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside     |                    |
| Layout Angle                                    |      | -90.00     | deg                |
| Diameter                                        |      | 3.0000     | in.                |
| Size and Thickness Basis                        |      | Nominal    |                    |
| Nominal Thickness                               |      | 160        |                    |
| Flange Material                                 |      | SA-350 LF2 | +IT                |
| Hub Height of Integral Nozzle                   | h    | 83.6362    | mm.                |
| Height of Beveled Transition                    | L'   | 8.0106     | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 25.0000    | mm.                |
| Corrosion Allowance                             | can  | 1.5000     | mm.                |
| Joint Efficiency of Shell Seam at Nozzle        | E1   | 1.00       |                    |
| Joint Efficiency of Nozzle Neck                 | En   | 1.00       |                    |
| Outside Projection                              | ho   | 300.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 32.0000    | mm.                |
| Inside Projection                               | h    | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.                |
| Flange Class                                    |      | 300        |                    |
| Flange Grade                                    |      | GR 1.1     |                    |

The Pressure Design option was Design Pressure + static head.



### Reinforcement CALCULATION, Description: LIQ OUT NOZ B3

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 2.624 in. |
| Actual Thickness Used in Calculation       | 0.438 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Conical Transition,  $t_r$  [Int. Press]  
 $= P \cdot D / (2 \cdot \cos(a) \cdot (S_v \cdot E - 0.6 \cdot P))$  Appendix 1-4(e)  
 $= 25.043 \cdot 1897.846 / (2 \cdot \cos(30.0) \cdot (137.997 \cdot 1.0 - 0.6 \cdot 25.043))$   
 $= 20.1049 \text{ mm.}$

Reqd Cone thickness at Nozzle Location under External Pres. : 11.2778 mm.

Reqd thk per UG-37(a) of Nozzle Wall,  $t_{rn}$  [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 25.04 \cdot 34.82 / (138 \cdot 1.0 - 0.6 \cdot 25.04)$   
 $= 0.6390 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4727 mm.

### Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):  
 $= 2.5 \cdot \text{Corroded Hub Thickness}$   
 $= 2.5 \cdot 23.5$ , less than the hub height, use (e-2)  
 $= 58.7500 \text{ mm.}$

### UG-40, Limits of Reinforcement: [Internal Pressure]

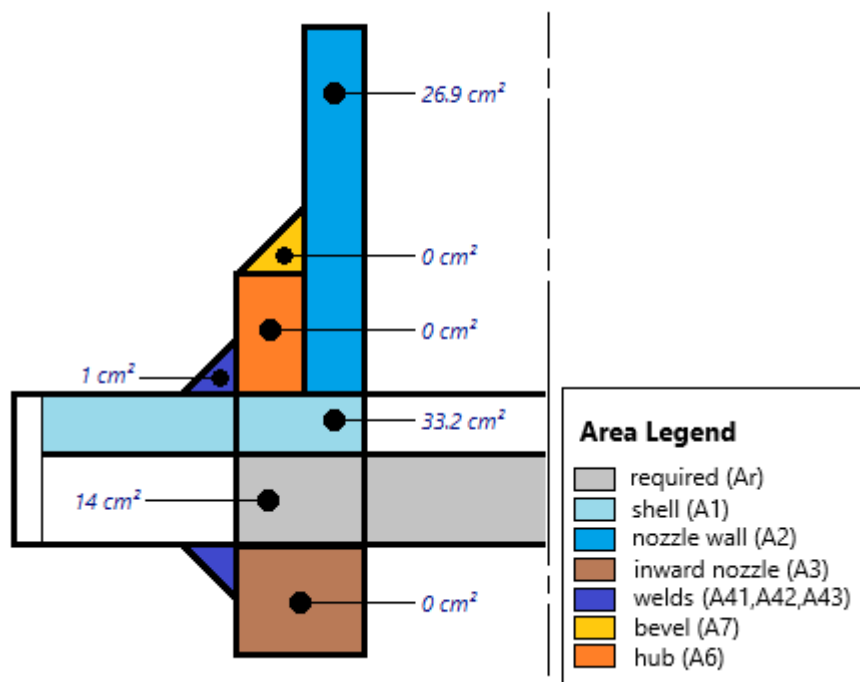
|                                                 |                 |              |
|-------------------------------------------------|-----------------|--------------|
| Parallel to Vessel Wall (Diameter Limit)        | D1              | 205.6496 mm. |
| Parallel to Vessel Wall                         | $R_n + t_n + t$ | 102.8248 mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | $T_{lnp}$       | 58.7500 mm.  |

Hub Height was  $\geq 2.5$  times Hub Thickness, using sketch UG-40 (e-2).

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n/S_v)$   
 $= \min(1, 138.0/138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n/S_v)$   
 $= \min(1, 138.0/138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$



#### Results of Nozzle Reinforcement Area Calculations: (cm²)

| AREA AVAILABLE, A1 to A5 |             | Design | External | Mapnc |
|--------------------------|-------------|--------|----------|-------|
| Area Required            | Ar          | 14.003 | 3.405    | NA    |
| Area in Shell            | A1          | 33.177 | 47.222   | NA    |
| Area in Nozzle Wall      | A2          | 26.862 | 27.057   | NA    |
| Area in Inward Nozzle    | A3          | 0.000  | 0.000    | NA    |
| Area in Welds            | A41+A42+A43 | 1.000  | 1.000    | NA    |
| Area in Element          | A5          | 0.000  | 0.000    | NA    |
| Area in Hub              | A6          | 0.000  | 0.000    | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 61.039 | 75.279   | NA    |

#### The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (69.6496 \cdot 20.1049 \cdot 1.0 + 2 \cdot 23.5 \cdot 20.1049 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 14.003 \text{ cm}^2$$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 136.0(1.0 \cdot 44.5 - 1.0 \cdot 20.105) - 2 \cdot 23.5$$

$$(1.0 \cdot 44.5 - 1.0 \cdot 20.1049) \cdot (1 - 1.0)$$

$$= 33.177 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:  

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 58.75)(23.5 - 0.64)1.0$$

$$= 26.862 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

*There are no hub area calculations because Figure UG-40 (e-2) was used.*

**UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**

Wall Thickness for Internal/External pressures  $t_a = 2.1390$  mm.  
 Wall Thickness per UG16(b),  $t_{r16b} = 3.0000$  mm.  
 Wall Thickness, shell/head, internal pressure  $t_{rb1} = 21.6049$  mm.  
 Wall Thickness  $t_{b1} = \max(t_{rb1}, t_{r16b}) = 21.6049$  mm.  
 Wall Thickness, shell/head, external pressure  $t_{rb2} = 2.3216$  mm.  
 Wall Thickness  $t_{b2} = \max(t_{rb2}, t_{r16b}) = 3.0000$  mm.  
 Wall Thickness per table UG-45  $t_{b3} = 6.3000$  mm.

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ t_{b3}, \max( t_{b1}, t_{b2} ) ]$   
 $= \min[ 6.3, \max( 21.6049, 3.0 ) ]$   
 $= 6.3000$  mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( t_a, t_b )$   
 $= \max( 2.139, 6.3 )$   
 $= 6.3000$  mm.

Available Nozzle Neck Thickness =  $0.875 \times 11.125 = 9.735$  mm. -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**

**Nozzle Neck to Flange Weld (Impact tested) :**

*Note:*  
*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification  $-46$  °C  
 Calculated Minimum Design Metal Temperature  $-104$  °C

**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :**

*Note:*  
*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification  $-46$  °C  
 Calculated Minimum Design Metal Temperature  $-104$  °C  
 Gov. MDMT of the nozzle to shell joint welded assembly :  $-104$  °C

**Flange MDMT including Temperature reduction per UCS-66.1:**

MDMT of ASME B16.5/47 flange per Matl. Specification  $-46$  °C  
 Flange MDMT with Temp reduction per UCS-66(i)(2)  $-79$  °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :  
 Design Pressure/Ambient Rating =  $25.04/51.00 = 0.491$

Weld Size Calculations, Description: LIQ OUT NOZ B3

Intermediate Calc. for nozzle/shell Welds  $T_{min} = 19.0000$  mm.

**Results Per UW-16.1:**

Required Thickness Actual Thickness  
 Nozzle Weld  $6.0000 = \text{Min per Code}$   $7.0700 = 0.7 \times W_o$  mm.

**Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:  
 $= \max( 0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) S_v )$   
 $= \max( 0, (14.003 - 33.1774 + 2 \cdot 23.5 \cdot 1.0 \cdot (1.0 \cdot 44.5 - 20.1049) ) 138 )$   
 $= \max( 0, -10847.28 )$  Kg

*For hub type nozzles, A2 includes the area of the hub.*

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot \text{Sv}$$
$$= (26.8617 + 1. - 0.0 \cdot 1.0) \cdot 138$$
$$= 39205.64 \text{ Kgf}$$

Weld Load [W2]:  
$$= (A2 + A3 + A4 + (2(\text{Hub Thickness}) \cdot t \cdot \text{fr}1)) \cdot \text{Sv}$$
$$= (26.8617 + 0.0 + 1. + (20.915)) \cdot 138$$
$$= 68636.21 \text{ Kgf}$$

Weld Load [W3]:  
$$= (A2 + A3 + A4 + (2(\text{Hub Thickness}) \cdot t \cdot \text{fr}1)) \cdot \text{Sv}$$
$$= (26.8617 + 0.0 + 1. + 0.0) \cdot 138$$
$$= 68636.21 \text{ Kgf}$$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
$$= (\pi/2) \cdot \text{Dlo} \cdot \text{Wo} \cdot 0.49 \cdot \text{Snw}$$
$$= (3.1416/2.0) \cdot 116.6496 \cdot 10.0 \cdot 0.49 \cdot 138$$
$$= 12634. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:  
$$= (\pi \cdot (\text{Dlr} + \text{Dlo}) / 4) \cdot (\text{Thk} - \text{Can}) \cdot 0.7 \cdot \text{Sn}$$
$$= (3.1416 \cdot 46.5748) \cdot (25.0 - 1.5) \cdot 0.7 \cdot 138$$
$$= 33870. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:  
$$= (\pi/2) \cdot \text{Dlo} \cdot \text{Wgnvi} \cdot 0.74 \cdot \text{Sng}$$
$$= (3.1416/2.0) \cdot 116.6496 \cdot (32.0 - 0.0) \cdot 0.74 \cdot 138$$
$$= 61056. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:  
$$= (\text{SONW} + \text{SNW}) = (12634 + 33870) = 46504 \text{ Kgf}$$

Failure Path [Path 2-2]:  
$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$
$$= (12634 + 0 + 61056 + 0) = 73690 \text{ Kgf}$$

Failure Path [Path 3-3]:  
$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$
$$= (12634 + 61056 + 0) = 73690 \text{ Kgf}$$

#### Summary of Failure Path Calculations:

|                  |                            |                   |
|------------------|----------------------------|-------------------|
| Path 1-1 = 46503 | Kgf, must exceed W = 0 Kgf | or W1 = 39205 Kgf |
| Path 2-2 = 73689 | Kgf, must exceed W = 0 Kgf | or W2 = 68636 Kgf |
| Path 3-3 = 73689 | Kgf, must exceed W = 0 Kgf | or W3 = 68636 Kgf |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 32 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 1.7975 mm.  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 347.7975 mm.

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**Input, Nozzle Desc: SS IN NOZ. A2**

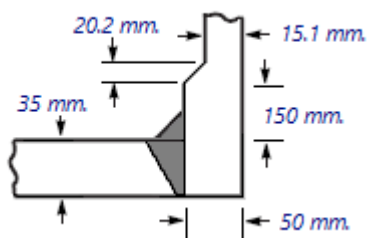
**From: 60**

|                                               |        |           |                    |
|-----------------------------------------------|--------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P      | 25.000    | bars               |
| Temperature for Internal Pressure             | Temp   | 85        | °C                 |
| Design External Pressure                      | Pext   | 1.03      | bars               |
| Temperature for External Pressure             | Tempex | 85        | °C                 |
| Parent Material                               |        | SA-516 70 | +Q+IT              |
| Parent Allowable Stress at Temperature        | Sv     | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva    | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D      | 3250.00   | mm.                |
| Design Length of Section                      | L      | 4506.6333 | mm.                |
| Shell Finished (Minimum) Thickness            | t      | 35.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c      | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co     | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |        | 7357.50   | mm.                |
| User Entered Minimum Design Metal Temperature |        | -35.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                 |      |            |                    |
|-------------------------------------------------|------|------------|--------------------|
| Material                                        |      | SA-350 LF2 | +IT                |
| Material UNS Number                             |      | K03011     |                    |
| Material Specification/Type                     |      | Forgings   |                    |
| Allowable Stress at Temperature                 | Sn   | 138.00     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 138.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside     |                    |
| Layout Angle                                    |      | 90.00      | deg                |
| Diameter                                        |      | 10.0000    | in.                |
| Size and Thickness Basis                        |      | Nominal    |                    |
| Nominal Thickness                               |      | 80         |                    |
| Flange Material                                 |      | SA-350 LF2 | +IT                |
| Hub Height of Integral Nozzle                   | h    | 150.0000   | mm.                |
| Height of Beveled Transition                    | L'   | 20.1567    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 50.0000    | mm.                |
| Corrosion Allowance                             | can  | 1.5000     | mm.                |
| Joint Efficiency of Shell Seam at Nozzle        | E1   | 1.00       |                    |
| Joint Efficiency of Nozzle Neck                 | En   | 1.00       |                    |
| Outside Projection                              | ho   | 300.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 30.0000    | mm.                |
| Inside Projection                               | h    | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.                |
| Flange Class                                    |      | 300        |                    |
| Flange Grade                                    |      | GR 1.1     |                    |

The Pressure Design option was Design Pressure + static head.





**Reinforcement CALCULATION, Description: SS IN NOZ. A2**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 9.562 in.  
Actual Thickness Used in Calculation 0.594 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $25.0 \cdot 1626.4999 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$   
= 29.7918 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
=  $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $25.0 \cdot 122.94 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$   
= 2.2518 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.9136 mm.

**Intermediate Hub Nozzle Calculations:**

Check to determine use of Sketch (e-1) or (e-2):  
=  $2.5 \cdot$  Corroded Hub Thickness  
=  $2.5 \cdot 48.5$ , less than the hub height, use (e-2)  
= 121.2500 mm.

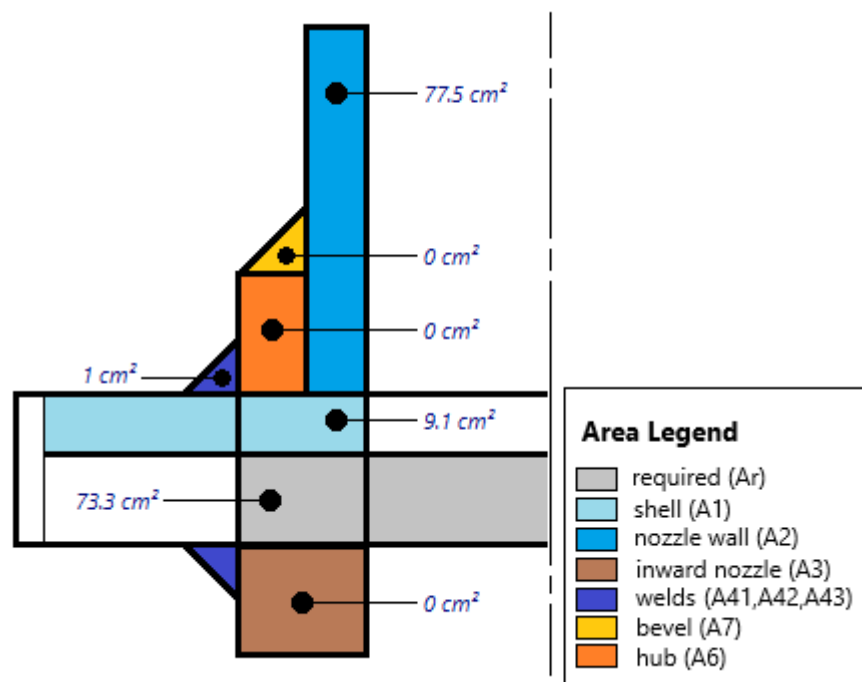
**UG-40, Limits of Reinforcement: [Internal Pressure]**

|                                                 |      |          |     |
|-------------------------------------------------|------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 491.7496 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 245.8748 | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 83.7500  | mm. |

Weld Strength Reduction Factor [fr1]:  
=  $\min(1, S_n / S_v)$   
=  $\min(1, 138.0 / 138.0)$   
= 1.000

Weld Strength Reduction Factor [fr2]:  
=  $\min(1, S_n / S_v)$   
=  $\min(1, 138.0 / 138.0)$   
= 1.000

Weld Strength Reduction Factor [fr3]:  
=  $\min(fr2, fr4)$   
=  $\min(1.0, 1.0)$   
= 1.000



#### Results of Nozzle Reinforcement Area Calculations: (cm²)

| AREA AVAILABLE, A1 to A5 |             | Design | External | Mapnc |
|--------------------------|-------------|--------|----------|-------|
| Area Required            | Ar          | 73.250 | 14.857   | NA    |
| Area in Shell            | A1          | 9.118  | 52.654   | NA    |
| Area in Nozzle Wall      | A2          | 77.466 | 79.707   | NA    |
| Area in Inward Nozzle    | A3          | 0.000  | 0.000    | NA    |
| Area in Welds            | A41+A42+A43 | 1.000  | 1.000    | NA    |
| Area in Element          | A5          | 0.000  | 0.000    | NA    |
| Area in Hub              | A6          | 0.000  | 0.000    | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 87.583 | 133.361  | NA    |

#### The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (245.8748 \cdot 29.7918 \cdot 1.0 + 2 \cdot 48.5 \cdot 29.7918 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 73.250 \text{ cm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 245.875 (1.0 \cdot 33.5 - 1.0 \cdot 29.792) - 2 \cdot 48.5$   
 $(1.0 \cdot 33.5 - 1.0 \cdot 29.7918) \cdot (1 - 1.0)$   
 $= 9.118 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= (2 \cdot tlnp) (tn - trn) fr2$   
 $= (2 \cdot 83.75) (48.5 - 2.25) 1.0$   
 $= 77.466 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$   
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$   
 $= 1.000 \text{ cm}^2$

*There are no hub area calculations because Figure UG-40 (e-2) was used.*

**UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**

Wall Thickness for Internal/External pressures  $t_a = 3.7518$  mm.  
 Wall Thickness per UG16(b),  $tr_{16b} = 3.0000$  mm.  
 Wall Thickness, shell/head, internal pressure  $trb_1 = 31.2918$  mm.  
 Wall Thickness  $tb_1 = \max(trb_1, tr_{16b}) = 31.2918$  mm.  
 Wall Thickness, shell/head, external pressure  $trb_2 = 2.7196$  mm.  
 Wall Thickness  $tb_2 = \max(trb_2, tr_{16b}) = 3.0000$  mm.  
 Wall Thickness per table UG-45  $tb_3 = 9.6026$  mm.

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb_3, \max( tb_1, tb_2 ) ]$   
 $= \min[ 9.603, \max( 31.2918, 3.0 ) ]$   
 $= 9.6026$  mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( t_a, t_b )$   
 $= \max( 3.7518, 9.6026 )$   
 $= 9.6026$  mm.

Available Nozzle Neck Thickness =  $0.875 * 15.088 = 13.202$  mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**

|            |   |       |           |   |                          |        |
|------------|---|-------|-----------|---|--------------------------|--------|
| Sustained  | : | 29.5, | Allowable | : | 138.0 N./mm <sup>2</sup> | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 315.5 N./mm <sup>2</sup> | Passed |
| Occasional | : | 10.7, | Allowable | : | 183.5 N./mm <sup>2</sup> | Passed |
| Shear      | : | 12.3, | Allowable | : | 96.6 N./mm <sup>2</sup>  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**

**Nozzle Neck to Flange Weld (Impact tested) :**

*Note:*  
*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification  $-46$  °C  
 Calculated Minimum Design Metal Temperature  $-104$  °C

**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :**

*Note:*  
*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification  $-46$  °C  
 Gov. MDMT of the nozzle to shell joint welded assembly :  $-46$  °C

**Flange MDMT including Temperature reduction per UCS-66.1:**

MDMT of ASME B16.5/47 flange per Matl. Specification  $-46$  °C  
 Flange MDMT with Temp reduction per UCS-66(i)(2)  $-79$  °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :  
 Design Pressure/Ambient Rating =  $25.00/51.00 = 0.490$

**Weld Size Calculations, Description: SS IN NOZ. A2**

Intermediate Calc. for nozzle/shell Welds  $T_{min} = 19.0000$  mm.

**Results Per UW-16.1:**

|             |                                |                          |
|-------------|--------------------------------|--------------------------|
|             | Required Thickness             | Actual Thickness         |
| Nozzle Weld | $6.0000 = \text{Min per Code}$ | $7.0700 = 0.7 * W_o$ mm. |



### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$   
 $= \max(0, (73.2505 - 9.1176 + 2 \cdot 48.5 \cdot 1.0 \cdot (1.0 \cdot 33.5 - 29.7918)) \cdot 138)$   
 $= 95306.23 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$   
 $= (77.4658 + 1. - 0.0 \cdot 1.0) \cdot 138$   
 $= 110413.27 \text{ Kgf}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2(\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v$   
 $= (77.4658 + 0.0 + 1. + (32.495)) \cdot 138$   
 $= 156138.67 \text{ Kgf}$

Weld Load [W3]:  
 $= (A2+A3+A4+(2 \cdot (\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v$   
 $= (77.4658 + 0.0 + 1. + 0.0) \cdot 138$   
 $= 156138.67 \text{ Kgf}$

### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$   
 $= (3.1416/2.0) \cdot 342.8748 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 37136. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$   
 $= (3.1416 \cdot 147.1874) \cdot (50.0 - 1.5) \cdot 0.7 \cdot 138$   
 $= 220904. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng}$   
 $= (3.1416/2.0) \cdot 342.8748 \cdot (30.0 - 0.0) \cdot 0.74 \cdot 138$   
 $= 168248. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

### Strength of Failure Paths:

Failure Path [Path 1-1]:  
 $= (\text{SONW} + \text{SNW}) = (37136 + 220904) = 258039 \text{ Kgf}$

Failure Path [Path 2-2]:  
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$   
 $= (37136 + 0 + 168248 + 0) = 205384 \text{ Kgf}$

Failure Path [Path 3-3]:  
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$   
 $= (37136 + 168248 + 0) = 205384 \text{ Kgf}$

### Summary of Failure Path Calculations:

|                                                  |                    |
|--------------------------------------------------|--------------------|
| Path 1-1 = 258039 Kgf, must exceed W = 95306 Kgf | or W1 = 110413 Kgf |
| Path 2-2 = 205384 Kgf, must exceed W = 95306 Kgf | or W2 = 156138 Kgf |
| Path 3-3 = 205384 Kgf, must exceed W = 95306 Kgf | or W3 = 156138 Kgf |

### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 27 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 9.0686 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 344.0686 mm.

**Input Echo, WRC297 Item 1, Description: SS IN NOZ. A2 :**

|                                      |           |                           |
|--------------------------------------|-----------|---------------------------|
| Diameter Basis for Cylindrical Shell | ID        |                           |
| Shell Internal Corrosion Allowance   | 1.5000    | mm.                       |
| Shell External Corrosion Allowance   | 0.0000    | mm.                       |
| Shell Diameter                       | 3250.000  | mm.                       |
| Shell Thickness                      | 35.0000   | mm.                       |
| Shell Stress Concentration Factor    | 1.000     |                           |
| Vessel Material                      | SA-516 70 |                           |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 N./mm <sup>2</sup> |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

|                                    |            |                           |
|------------------------------------|------------|---------------------------|
| Diameter Basis for Nozzle          | ID         |                           |
| Nozzle Corrosion Allowance         | 1.5000     | mm.                       |
| Nozzle Diameter                    | 242.875    | mm.                       |
| Nozzle Thickness                   | 50.0000    | mm.                       |
| Nozzle Stress Concentration Factor | 1.000      |                           |
| Nozzle Material                    | SA-350 LF2 |                           |
| Nozzle Cold S.I. Allowable         | SNmc       | 138.00 N./mm <sup>2</sup> |
| Nozzle Hot S.I. Allowable          | SNmh       | 138.00 N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUSstained loads.*

|                          |    |          |       |
|--------------------------|----|----------|-------|
| Design Internal Pressure | Dp | 25.00    | bars  |
| Radial Load              | P  | -1529.68 | Kgf   |
| Circumferential Shear    | Vc | 1147.26  | Kgf   |
| Longitudinal Shear       | Vl | 1529.68  | Kgf   |
| Circumferential Moment   | Mc | 993.91   | Kg-m. |
| Longitudinal Moment      | Ml | 993.91   | Kg-m. |
| Torsional Moment         | Mt | 1146.81  | Kg-m. |

|                                                 |     |
|-------------------------------------------------|-----|
| Include Axial Pressure Thrust                   | Yes |
| Include Pressure Stress Indices per Div. 2      | No  |
| Local Loads applied at end of Nozzle/Attachment | No  |

*Warning - The ratio, Dn/Tn (6.857) is < 10  
 check the Limitations of WRC bulletin 297.*

Note: The radial load includes the effect of the pressure thrust force (Pressure times Area). This may lead to overly conservative results in some cases.

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$   
 $= 342.875 + 2 \cdot 1.65 \cdot \sqrt{(1643.25(35.0 - 1.5))}$   
 $= 1117.137 \text{ mm.}$

WRC 297 Curve Access Parameters:

|                         |       |              |
|-------------------------|-------|--------------|
| Vessel Mean Diameter    | (D) = | 3286.500 mm. |
| Nozzle Outside Diameter | (d) = | 342.875 mm.  |
| Vessel Thickness used   | (T) = | 33.500 mm.   |
| Nozzle Thickness used   | (t) = | 48.500 mm.   |
| T / t                   | =     | 0.691        |
| d / t                   | =     | 10.000       |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.033$$

$$Nr/P = 0.111$$



Mr/P = 0.206  
 M0/P = 0.064  
 N0/P = 0.127  
 MrD/Mc = 0.296  
 NrDL/Mc = 0.112  
 M0d/Mc = 0.100  
 N0DL/Mc = 0.039  
 MrD/Ml = 0.193  
 NrDL/Ml = 0.096  
 M0D/Ml = 0.066  
 N0DL/Ml = 0.120

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 15                   | 15                  | 15                      | 15                     |
| Outplane Bending (P )                                            | 45                   | -45                 | 45                      | -45                    |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -3                   | -3                  | 3                       | 3                      |
| Outplane Bending (ML)                                            | -10                  | 10                  | 10                      | -10                    |
| Normal Pressure Stress                                           | 120                  | 122                 | 120                     | 122                    |
| Outplane Stress Summary                                          | 167                  | 99                  | 194                     | 85                     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 13                   | 13                  | 13                      | 13                     |
| Inplane Bending (P )                                               | 147                  | -147                | 147                     | -147                   |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -2                   | -2                  | 2                       | 2                      |
| Inplane Bending (ML)                                               | -29                  | 29                  | 29                      | -29                    |
| Inplane Pressure Stress                                            | 60                   | 60                  | 60                      | 60                     |
| Inplane Stress Summary                                             | 188                  | -47                 | 252                     | -100                   |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 1                    | 1                   | 1                       | 1                      |
| Shear Stress Summary                                                 | 2                    | 2                   | 1                       | 1                      |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Two * Max Shear Stress                     | 189                  | 146                 | 252                     | 185                    |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | 15                    | 15                   | 15                     | 15                    |
| Outplane Bending (P )                                                  | 45                    | -45                  | 45                     | -45                   |
| Outplane Membrane (Mc)                                                 | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (Mc)                                                  | -15                   | 15                   | 15                     | -15                   |
| Outplane Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |



|                        |    |    |    |    |
|------------------------|----|----|----|----|
| Outplane Bending (ML)  | 0  | 0  | 0  | 0  |
| Normal Pressure Stress | 60 | 60 | 60 | 60 |

|                         |     |    |     |    |
|-------------------------|-----|----|-----|----|
| Outplane Stress Summary | 104 | 43 | 137 | 15 |
|-------------------------|-----|----|-----|----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                              |              |             |               |              |
|----------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                        | Cu           | Cl          | Du            | Dl           |
| (Stresses parallel to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                         |     |      |     |      |
|-------------------------|-----|------|-----|------|
| Inplane Membrane (P )   | 13  | 13   | 13  | 13   |
| Inplane Bending (P )    | 147 | -147 | 147 | -147 |
| Inplane Membrane (Mc)   | -2  | -2   | 2   | 2    |
| Inplane Bending (Mc)    | -44 | 44   | 44  | -44  |
| Inplane Membrane (ML)   | 0   | 0    | 0   | 0    |
| Inplane Bending (ML)    | 0   | 0    | 0   | 0    |
| Inplane Pressure Stress | 120 | 122  | 120 | 122  |

|                        |     |    |     |     |
|------------------------|-----|----|-----|-----|
| Inplane Stress Summary | 232 | 30 | 328 | -53 |
|------------------------|-----|----|-----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                                |              |             |               |              |
|------------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                          | Cu           | Cl          | Du            | Dl           |
| (Shear stress normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Outplane Shear (Vc)  | 0 | 0 | 0 | 0 |
| Outplane Shear (Vl)  | 0 | 0 | 0 | 0 |
| Torsional Shear (Mt) | 1 | 1 | 1 | 1 |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 0 | 0 | 2 | 2 |
|----------------------|---|---|---|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                       |              |             |               |              |
|-----------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE | Cu           | Cl          | Du            | Dl           |
| (Stress Intensities)  | Left Outside | Left Inside | Right Outside | Right Inside |

|                        |     |    |     |    |
|------------------------|-----|----|-----|----|
| Two * Max Shear Stress | 232 | 43 | 328 | 68 |
|------------------------|-----|----|-----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                  |             |            |                |               |
|----------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE               | Au          | Al         | Bu             | Bl            |
| (Stresses in the hoop direction) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                      |    |    |    |    |
|----------------------|----|----|----|----|
| Hoop Membrane (P )   | 15 | 15 | 15 | 15 |
| Hoop Bending (P )    | 0  | 0  | 0  | 0  |
| Hoop Membrane (Mc)   | 0  | 0  | 0  | 0  |
| Hoop Bending (Mc)    | 0  | 0  | 0  | 0  |
| Hoop Membrane (ML)   | -3 | -3 | 3  | 3  |
| Hoop Bending (ML)    | 0  | 0  | 0  | 0  |
| Hoop Pressure Stress | 6  | 9  | 6  | 9  |

|                     |    |    |    |    |
|---------------------|----|----|----|----|
| Hoop Stress Summary | 18 | 21 | 24 | 27 |
|---------------------|----|----|----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                         |             |            |                |               |
|-----------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                      | Au          | Al         | Bu             | Bl            |
| (Stresses Normal to pipe cross-section) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                       |     |     |    |     |
|-----------------------|-----|-----|----|-----|
| Axial Membrane (P )   | 2   | 2   | 2  | 2   |
| Axial Bending (P )    | 51  | -51 | 51 | -51 |
| Axial Membrane (Mc)   | 0   | 0   | 0  | 0   |
| Axial Bending (Mc)    | 0   | 0   | 0  | 0   |
| Axial Membrane (ML)   | -3  | -3  | 3  | 3   |
| Axial Bending (ML)    | -10 | 10  | 10 | -10 |
| Axial Pressure Stress | 3   | 3   | 3  | 3   |

|                      |    |     |    |     |
|----------------------|----|-----|----|-----|
| Axial Stress Summary | 43 | -37 | 71 | -52 |
|----------------------|----|-----|----|-----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                    |    |    |    |    |
|--------------------|----|----|----|----|
| LONGITUDINAL PLANE | Au | Al | Bu | Bl |
|--------------------|----|----|----|----|



| (Shear stress)       | Outside | Inside | Outside | Inside |
|----------------------|---------|--------|---------|--------|
| Shear due to (Vc)    | 0       | 0      | 0       | 0      |
| Shear due to (Vl)    | 0       | 0      | 0       | 0      |
| Shear due to Torsion | 1       | 1      | 1       | 1      |
| Shear Stress Summary | 1       | 1      | 0       | 0      |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                     | 43            | 59           | 71            | 79           |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )                                           | 15                    | 15                   | 15                     | 15                    |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 6                     | 9                    | 6                      | 9                     |
| Hoop Stress Summary                                          | 20                    | 23                   | 22                     | 25                    |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Axial Membrane (P )                                                 | 2                     | 2                    | 2                      | 2                     |
| Axial Bending (P )                                                  | 51                    | -51                  | 51                     | -51                   |
| Axial Membrane (Mc)                                                 | -3                    | -3                   | 3                      | 3                     |
| Axial Bending (Mc)                                                  | -17                   | 17                   | 17                     | -17                   |
| Axial Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Axial Pressure Stress                                               | 3                     | 3                    | 3                      | 3                     |
| Axial Stress Summary                                                | 36                    | -31                  | 78                     | -59                   |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | 0             | 0            | 0             | 0            |
| Shear due to Torsion                    | 1             | 1            | 1             | 1            |
| Shear Stress Summary                    | 0             | 0            | 1             | 1            |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                        | 36            | 54           | 78            | 84           |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):**

| Type of<br>Stress Int.<br>Location | Stress Values at |     |     |     |     |     |     |     |
|------------------------------------|------------------|-----|-----|-----|-----|-----|-----|-----|
|                                    | Au               | Al  | Bu  | Bl  | Cu  | Cl  | Du  | Dl  |
| Circ. Pm (SUS)                     | 120              | 122 | 120 | 122 | 120 | 122 | 120 | 122 |
| Circ. Pl (SUS)                     | 12               | 12  | 18  | 18  | 11  | 11  | 15  | 15  |

|                 |       |       |       |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|-------|-------|-------|-------|
| Circ. Q (SUS)   | 35    | -35   | 55    | -55   | 103   | -103  | 191   | -191  |
| Long. Pm (SUS)  | 60    | 60    | 60    | 60    | 60    | 60    | 60    | 60    |
| Long. Pl (SUS)  | 11    | 11    | 15    | 15    | 15    | 15    | 15    | 15    |
| Long. Q (SUS)   | 118   | -118  | 176   | -176  | 30    | -30   | 60    | -60   |
| Shear Pm (SUS)  | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)  | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Q (SUS)   | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     |
| Pm (SUS)        | 120.0 | 122.0 | 120.0 | 122.0 | 120.0 | 122.0 | 120.0 | 122.0 |
| Pm+Pl (SUS)     | 132.0 | 134.0 | 138.0 | 140.0 | 131.0 | 133.0 | 135.0 | 137.0 |
| Pm+Pl+Q (Total) | 189.0 | 146.0 | 251.0 | 186.0 | 234.0 | 45.1  | 326.0 | 69.0  |

#### Vessel Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 122.00    | 138.00         | Passed |
| Pm+Pl (SUS)         | 140.00    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 326.01    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):

| Type of Stress Int. Location | Stress Values at |      |      |      |      |      |      |      |
|------------------------------|------------------|------|------|------|------|------|------|------|
|                              | Au               | Al   | Bu   | Bl   | Cu   | Cl   | Du   | Dl   |
| Circ. Pm (SUS)               | 6                | 9    | 6    | 9    | 6    | 9    | 6    | 9    |
| Circ. Pl (SUS)               | 12               | 12   | 18   | 18   | 15   | 15   | 15   | 15   |
| Circ. Q (SUS)                | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Long. Pm (SUS)               | 3                | 3    | 3    | 3    | 3    | 3    | 3    | 3    |
| Long. Pl (SUS)               | -1               | -1   | 5    | 5    | -1   | -1   | 5    | 5    |
| Long. Q (SUS)                | 41               | -41  | 61   | -61  | 34   | -34  | 68   | -68  |
| Shear Pm (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Q (SUS)                | 1                | 1    | 1    | 1    | 1    | 1    | 1    | 1    |
| Pm (SUS)                     | 6.0              | 9.0  | 6.0  | 9.0  | 6.0  | 9.0  | 6.0  | 9.0  |
| Pm+Pl (SUS)                  | 18.0             | 21.0 | 24.0 | 27.0 | 21.0 | 24.0 | 21.0 | 24.0 |
| Pm+Pl+Q (Total)              | 43.0             | 60.0 | 69.0 | 80.0 | 36.1 | 56.0 | 76.0 | 84.0 |

#### Nozzle Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 9.00      | 138.00         | Passed |
| Pm+Pl (SUS)         | 27.00     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 84.02     | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

**Input, Nozzle Desc: SS OUT NOZ. B2**

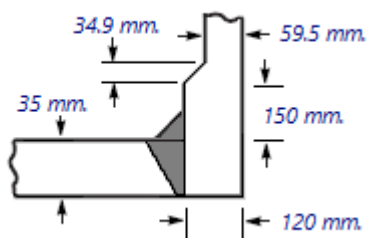
**From: 60**

|                                               |        |           |                    |
|-----------------------------------------------|--------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P      | 25.000    | bars               |
| Temperature for Internal Pressure             | Temp   | 85        | °C                 |
| Design External Pressure                      | Pext   | 1.03      | bars               |
| Temperature for External Pressure             | Tempex | 85        | °C                 |
| Parent Material                               |        | SA-516 70 | +Q+IT              |
| Parent Allowable Stress at Temperature        | Sv     | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva    | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D      | 3250.00   | mm.                |
| Design Length of Section                      | L      | 4506.6333 | mm.                |
| Shell Finished (Minimum) Thickness            | t      | 35.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c      | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co     | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |        | 5307.50   | mm.                |
| User Entered Minimum Design Metal Temperature |        | -35.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                 |      |            |                    |
|-------------------------------------------------|------|------------|--------------------|
| Material                                        |      | SA-350 LF2 | +IT                |
| Material UNS Number                             |      | K03011     |                    |
| Material Specification/Type                     |      | Forgings   |                    |
| Allowable Stress at Temperature                 | Sn   | 138.00     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 138.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside     |                    |
| Layout Angle                                    |      | 90.00      | deg                |
| Diameter                                        |      | 24.0000    | in.                |
| Size and Thickness Basis                        |      | Actual     |                    |
| Actual Thickness                                | tn   | 59.5400    | mm.                |
| Flange Material                                 |      | SA-350 LF2 | +IT                |
| Hub Height of Integral Nozzle                   | h    | 150.0000   | mm.                |
| Height of Beveled Transition                    | L'   | 34.9066    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 120.0000   | mm.                |
| Corrosion Allowance                             | can  | 1.5000     | mm.                |
| Joint Efficiency of Shell Seam at Nozzle        | E1   | 1.00       |                    |
| Joint Efficiency of Nozzle Neck                 | En   | 1.00       |                    |
| Outside Projection                              | ho   | 300.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 30.0000    | mm.                |
| Inside Projection                               | h    | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000     | mm.                |
| Flange Class                                    |      | 300        |                    |
| Flange Grade                                    |      | GR 1.1     |                    |

The Pressure Design option was Design Pressure + static head.



## Reinforcement CALCULATION, Description: SS OUT NOZ. B2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 24.000 in.  
 Actual Thickness Used in Calculation 2.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 25.0 \cdot 1626.4999 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$   
 $= 29.7918$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 25.0 \cdot 306.3 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$   
 $= 5.6103$  mm.

Required Nozzle thickness under External Pressure per UG-28 : 1.6144 mm.

### Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
 $= ( \text{Hub Thickness} - \text{Neck Thickness} ) / \tan(30)$   
 $= 60.46 / 0.5773$   
 $= 104.7198$  mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

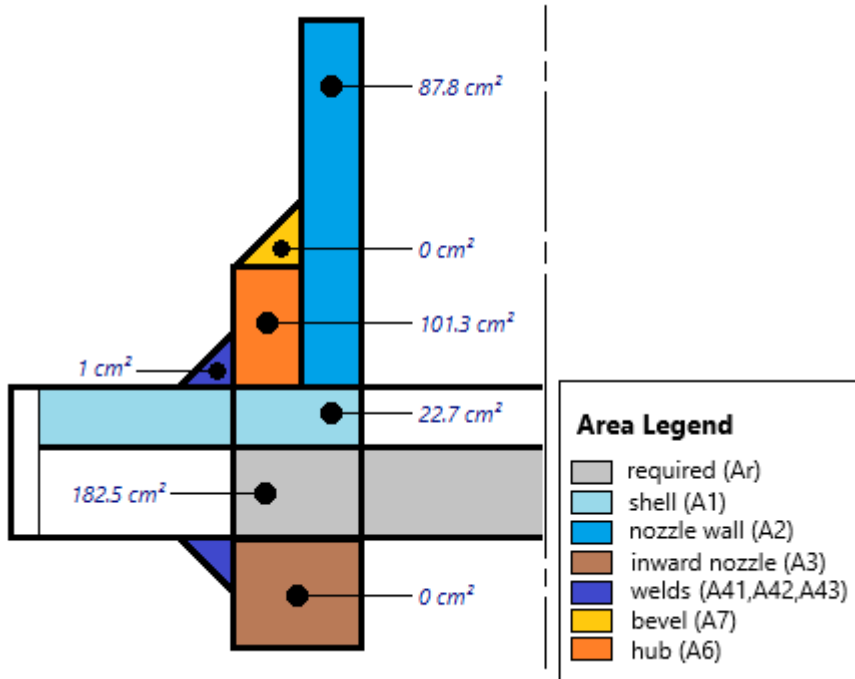
### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |           |     |
|-------------------------------------------------|------|-----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl   | 1225.2000 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 612.6000  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 83.7500   | mm. |

Weld Strength Reduction Factor [fr1]:  
 $= \min( 1, S_n / S_v )$   
 $= \min( 1, 138.0 / 138.0 )$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min( 1, S_n / S_v )$   
 $= \min( 1, 138.0 / 138.0 )$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min( fr2, fr4 )$   
 $= \min( 1.0, 1.0 )$   
 $= 1.000$



#### Results of Nozzle Reinforcement Area Calculations: (cm²)

| AREA AVAILABLE, A1 to A5 |             | Design  | External | Mapnc |
|--------------------------|-------------|---------|----------|-------|
| Area Required            | Ar          | 182.504 | 37.016   | NA    |
| Area in Shell            | A1          | 22.717  | 131.188  | NA    |
| Area in Nozzle Wall      | A2          | 87.820  | 94.513   | NA    |
| Area in Inward Nozzle    | A3          | 0.000   | 0.000    | NA    |
| Area in Welds            | A41+A42+A43 | 1.000   | 1.000    | NA    |
| Area in Element          | A5          | 0.000   | 0.000    | NA    |
| Area in Hub              | A6          | 101.271 | 101.271  | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 212.807 | 327.972  | NA    |

#### The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

#### The area available without a pad is Sufficient.

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (612.6 \cdot 29.7918 \cdot 1.0 + 2 \cdot 118.5 \cdot 29.7918 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 182.504 \text{ cm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 612.6(1.0 \cdot 33.5 - 1.0 \cdot 29.792) - 2 \cdot 118.5$   
 $(1.0 \cdot 33.5 - 1.0 \cdot 29.7918) \cdot (1 - 1.0)$   
 $= 22.717 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= (2 \cdot tlnp)(tn - trn)fr2$   
 $= (2 \cdot 83.75)(58.04 - 5.61)1.0$   
 $= 87.820 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$   
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$   
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:  

$$= (2 \cdot \min(T_{lp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot \text{fr2}$$

$$= (2 \cdot \min(83.8, 300.0, 150.0)) \cdot (120.0 - 59.5) \cdot 1.0$$

$$= 101.271 \text{ cm}^2$$

**UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**

Wall Thickness for Internal/External pressures  $t_a = 7.1103 \text{ mm.}$   
 Wall Thickness per UG16(b),  $\text{tr16b} = 3.0000 \text{ mm.}$   
 Wall Thickness, shell/head, internal pressure  $\text{trb1} = 31.2918 \text{ mm.}$   
 Wall Thickness  $\text{tb1} = \max(\text{trb1}, \text{tr16b}) = 31.2918 \text{ mm.}$   
 Wall Thickness, shell/head, external pressure  $\text{trb2} = 2.7196 \text{ mm.}$   
 Wall Thickness  $\text{tb2} = \max(\text{trb2}, \text{tr16b}) = 3.0000 \text{ mm.}$   
 Wall Thickness per table UG-45  $\text{tb3} = 9.8312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:  

$$= \min[\text{tb3}, \max(\text{tb1}, \text{tb2})]$$

$$= \min[9.831, \max(31.2918, 3.0)]$$

$$= 9.8312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  

$$= \max(t_a, t_b)$$

$$= \max(7.1103, 9.8312)$$

$$= 9.8312 \text{ mm.}$$

Available Nozzle Neck Thickness = 59.5400 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**

|            |   |      |           |   |                          |        |
|------------|---|------|-----------|---|--------------------------|--------|
| Sustained  | : | 9.9, | Allowable | : | 138.0 N./mm <sup>2</sup> | Passed |
| Expansion  | : | 0.0, | Allowable | : | 335.1 N./mm <sup>2</sup> | Passed |
| Occasional | : | 6.0, | Allowable | : | 183.5 N./mm <sup>2</sup> | Passed |
| Shear      | : | 2.7, | Allowable | : | 96.6 N./mm <sup>2</sup>  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**

**Nozzle Neck to Flange Weld (Impact tested) :**

*Note:*  
*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                    |         |
|----------------------------------------------------|---------|
| Impact Test Temperature provided per Specification | -46 °C  |
| Calculated Minimum Design Metal Temperature        | -104 °C |

**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :**

*Note:*  
*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |        |
|----------------------------------------------------------|--------|
| Impact Test Temperature provided per Specification       | -46 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -46 °C |

**Flange MDMT including Temperature reduction per UCS-66.1:**

|                                                      |        |
|------------------------------------------------------|--------|
| MDMT of ASME B16.5/47 flange per Matl. Specification | -46 °C |
| Flange MDMT with Temp reduction per UCS-66(i)(2)     | -79 °C |

Where the Stress Reduction Ratio per UCS-66(i)(2) is :  

$$\text{Design Pressure/Ambient Rating} = 25.00/51.00 = 0.490$$

Weld Size Calculations, Description: SS OUT NOZ. B2

|                                           |           |             |
|-------------------------------------------|-----------|-------------|
| Intermediate Calc. for nozzle/shell Welds | $T_{min}$ | 19.0000 mm. |
|-------------------------------------------|-----------|-------------|

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness      |
|-------------|-----------------------|-----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm. |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$   
 $= \max(0, (182.5045 - 22.7166 + 2 \cdot 118.5 \cdot 1.0 \cdot (1.0 \cdot 33.5 - 29.7918)) \cdot 138)$   
 $= 237212.56 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$   
 $= (189.0902 + 1. - 0.0 \cdot 1.0) \cdot 138$   
 $= 267485.81 \text{ Kgf}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$   
 $= (189.0902 + 0.0 + 1. + (79.395)) \cdot 138$   
 $= 379206.62 \text{ Kgf}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$   
 $= (189.0902 + 0.0 + 1. + 0.0 + (79.395)) \cdot 138$   
 $= 379206.62 \text{ Kgf}$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$   
 $= (3.1416/2.0) \cdot 849.6 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 92018. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$   
 $= (3.1416 \cdot 365.55) \cdot (120.0 - 1.5) \cdot 0.7 \cdot 138$   
 $= 1340465. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng}$   
 $= (3.1416/2.0) \cdot 849.6 \cdot (30.0 - 0.0) \cdot 0.74 \cdot 138$   
 $= 416898. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:  
 $= (\text{SONW} + \text{SNW}) = (92018 + 1340465) = 1432483 \text{ Kgf}$

Failure Path [Path 2-2]:  
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$   
 $= (92018 + 0 + 416898 + 0) = 508916 \text{ Kgf}$

Failure Path [Path 3-3]:  
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$   
 $= (92018 + 416898 + 0) = 508916 \text{ Kgf}$

#### Summary of Failure Path Calculations:

|                                                    |                    |
|----------------------------------------------------|--------------------|
| Path 1-1 = 1432483 Kgf, must exceed W = 237212 Kgf | or W1 = 267485 Kgf |
| Path 2-2 = 508915 Kgf, must exceed W = 237212 Kgf  | or W2 = 379206 Kgf |
| Path 3-3 = 508915 Kgf, must exceed W = 237212 Kgf  | or W3 = 379206 Kgf |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 27 bars



Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 56.5071 mm.  
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 391.5071 mm.

**Input Echo, WRC297 Item 1, Description: SS OUT NOZ. B2:**

|                                      |            |                    |
|--------------------------------------|------------|--------------------|
| Diameter Basis for Cylindrical Shell | ID         |                    |
| Shell Internal Corrosion Allowance   | 1.5000     | mm.                |
| Shell External Corrosion Allowance   | 0.0000     | mm.                |
| Shell Diameter                       | 3250.000   | mm.                |
| Shell Thickness                      | 35.0000    | mm.                |
| Shell Stress Concentration Factor    | 1.000      |                    |
| Vessel Material                      | SA-516 70  |                    |
| Vessel Cold S.I. Allowable           | Smc 138.00 | N./mm <sup>2</sup> |
| Vessel Hot S.I. Allowable            | Smh 138.00 | N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

|                                    |             |                    |
|------------------------------------|-------------|--------------------|
| Diameter Basis for Nozzle          | ID          |                    |
| Nozzle Corrosion Allowance         | 1.5000      | mm.                |
| Nozzle Diameter                    | 609.600     | mm.                |
| Nozzle Thickness                   | 120.0000    | mm.                |
| Nozzle Stress Concentration Factor | 1.000       |                    |
| Nozzle Material                    | SA-350 LF2  |                    |
| Nozzle Cold S.I. Allowable         | SNmc 138.00 | N./mm <sup>2</sup> |
| Nozzle Hot S.I. Allowable          | SNmh 138.00 | N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

**Note:**  
*External Forces and Moments in WRC 107/537 Convention:*  
*These loads are assumed to be SUSstained loads.*

|                          |    |          |       |
|--------------------------|----|----------|-------|
| Design Internal Pressure | Dp | 25.00    | bars  |
| Radial Load              | P  | -3671.23 | Kgf   |
| Circumferential Shear    | Vc | 2753.42  | Kgf   |
| Longitudinal Shear       | Vl | 3671.23  | Kgf   |
| Circumferential Moment   | Mc | 5724.90  | Kg-m. |
| Longitudinal Moment      | Ml | 5724.90  | Kg-m. |
| Torsional Moment         | Mt | 6605.65  | Kg-m. |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

*Warning - The ratio, Dn/Tn (7.080) is < 10*  
*check the limitations of WRC bulletin 297.*

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$   
 $= 849.6 + 2 \cdot 1.65 \cdot \sqrt{(1643.25(35.0 - 1.5))}$   
 $= 1623.862 \text{ mm.}$

WRC 297 Curve Access Parameters:

|                         |       |          |     |
|-------------------------|-------|----------|-----|
| Vessel Mean Diameter    | (D) = | 3286.500 | mm. |
| Nozzle Outside Diameter | (d) = | 849.600  | mm. |
| Vessel Thickness used   | (T) = | 33.500   | mm. |
| Nozzle Thickness used   | (t) = | 118.500  | mm. |
| T / t                   | =     | 0.283    |     |
| d / t                   | =     | 10.000   |     |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.561$$

$$Nr/P = 0.062$$



Mr/P = 0.120  
 M0/P = 0.036  
 N0/P = 0.074  
 MrD/Mc = 0.276  
 NrDL/Mc = 0.094  
 M0d/Mc = 0.083  
 N0DL/Mc = 0.083  
 MrD/Ml = 0.096  
 NrDL/Ml = 0.069  
 M0D/Ml = 0.028  
 N0DL/Ml = 0.094

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 2                    | 2                   | 2                       | 2                      |
| Outplane Bending (P )                                            | 6                    | -6                  | 6                       | -6                     |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -5                   | -5                  | 5                       | 5                      |
| Outplane Bending (ML)                                            | -9                   | 9                   | 9                       | -9                     |
| Normal Pressure Stress                                           | 120                  | 122                 | 120                     | 122                    |
| Outplane Stress Summary                                          | 114                  | 122                 | 144                     | 113                    |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 1                    | 1                   | 1                       | 1                      |
| Inplane Bending (P )                                               | 23                   | -23                 | 23                      | -23                    |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -4                   | -4                  | 4                       | 4                      |
| Inplane Bending (ML)                                               | -33                  | 33                  | 33                      | -33                    |
| Inplane Pressure Stress                                            | 60                   | 60                  | 60                      | 60                     |
| Inplane Stress Summary                                             | 47                   | 68                  | 122                     | 9                      |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 1                    | 1                   | 1                       | 1                      |
| Shear Stress Summary                                                 | 2                    | 2                   | 1                       | 1                      |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Two * Max Shear Stress                     | 114                  | 122                 | 144                     | 113                    |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | 2                     | 2                    | 2                      | 2                     |
| Outplane Bending (P )                                                  | 6                     | -6                   | 6                      | -6                    |
| Outplane Membrane (Mc)                                                 | -4                    | -4                   | 4                      | 4                     |
| Outplane Bending (Mc)                                                  | -29                   | 29                   | 29                     | -29                   |
| Outplane Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |



|                        |    |    |    |    |
|------------------------|----|----|----|----|
| Outplane Bending (ML)  | 0  | 0  | 0  | 0  |
| Normal Pressure Stress | 60 | 60 | 60 | 60 |

|                         |    |    |     |    |
|-------------------------|----|----|-----|----|
| Outplane Stress Summary | 35 | 80 | 103 | 30 |
|-------------------------|----|----|-----|----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                              |              |             |               |              |
|----------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                        | Cu           | Cl          | Du            | Dl           |
| (Stresses parallel to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                         |     |     |     |     |
|-------------------------|-----|-----|-----|-----|
| Inplane Membrane (P )   | 1   | 1   | 1   | 1   |
| Inplane Bending (P )    | 23  | -23 | 23  | -23 |
| Inplane Membrane (Mc)   | -5  | -5  | 5   | 5   |
| Inplane Bending (Mc)    | -97 | 97  | 97  | -97 |
| Inplane Membrane (ML)   | 0   | 0   | 0   | 0   |
| Inplane Bending (ML)    | 0   | 0   | 0   | 0   |
| Inplane Pressure Stress | 120 | 122 | 120 | 122 |

|                        |    |     |     |   |
|------------------------|----|-----|-----|---|
| Inplane Stress Summary | 42 | 193 | 248 | 9 |
|------------------------|----|-----|-----|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                                |              |             |               |              |
|------------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                          | Cu           | Cl          | Du            | Dl           |
| (Shear stress normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Outplane Shear (Vc)  | 0 | 0 | 0 | 0 |
| Outplane Shear (Vl)  | 0 | 0 | 0 | 0 |
| Torsional Shear (Mt) | 1 | 1 | 1 | 1 |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 0 | 0 | 2 | 2 |
|----------------------|---|---|---|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                       |              |             |               |              |
|-----------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE | Cu           | Cl          | Du            | Dl           |
| (Stress Intensities)  | Left Outside | Left Inside | Right Outside | Right Inside |

|                        |    |     |     |    |
|------------------------|----|-----|-----|----|
| Two * Max Shear Stress | 42 | 193 | 248 | 31 |
|------------------------|----|-----|-----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                  |             |            |                |               |
|----------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE               | Au          | Al         | Bu             | Bl            |
| (Stresses in the hoop direction) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                      |    |    |   |   |
|----------------------|----|----|---|---|
| Hoop Membrane (P )   | 2  | 2  | 2 | 2 |
| Hoop Bending (P )    | 0  | 0  | 0 | 0 |
| Hoop Membrane (Mc)   | 0  | 0  | 0 | 0 |
| Hoop Bending (Mc)    | 0  | 0  | 0 | 0 |
| Hoop Membrane (ML)   | -5 | -5 | 5 | 5 |
| Hoop Bending (ML)    | 0  | 0  | 0 | 0 |
| Hoop Pressure Stress | 6  | 9  | 6 | 9 |

|                     |   |   |    |    |
|---------------------|---|---|----|----|
| Hoop Stress Summary | 3 | 6 | 14 | 17 |
|---------------------|---|---|----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                         |             |            |                |               |
|-----------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                      | Au          | Al         | Bu             | Bl            |
| (Stresses Normal to pipe cross-section) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                       |    |    |   |    |
|-----------------------|----|----|---|----|
| Axial Membrane (P )   | 0  | 0  | 0 | 0  |
| Axial Bending (P )    | 1  | -1 | 1 | -1 |
| Axial Membrane (Mc)   | 0  | 0  | 0 | 0  |
| Axial Bending (Mc)    | 0  | 0  | 0 | 0  |
| Axial Membrane (ML)   | -1 | -1 | 1 | 1  |
| Axial Bending (ML)    | -1 | 1  | 1 | -1 |
| Axial Pressure Stress | 3  | 3  | 3 | 3  |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Axial Stress Summary | 1 | 2 | 7 | 1 |
|----------------------|---|---|---|---|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                    |    |    |    |    |
|--------------------|----|----|----|----|
| LONGITUDINAL PLANE | Au | Al | Bu | Bl |
|--------------------|----|----|----|----|



| (Shear stress)       | Outside | Inside | Outside | Inside |
|----------------------|---------|--------|---------|--------|
| Shear due to (Vc)    | 0       | 0      | 0       | 0      |
| Shear due to (Vl)    | 0       | 0      | 0       | 0      |
| Shear due to Torsion | 0       | 0      | 0       | 0      |
| Shear Stress Summary | 0       | 0      | 0       | 0      |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                     | 3             | 6            | 14            | 17           |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )                                           | 2                     | 2                    | 2                      | 2                     |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | -4                    | -4                   | 4                      | 4                     |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 6                     | 9                    | 6                      | 9                     |
| Hoop Stress Summary                                          | 3                     | 6                    | 13                     | 16                    |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Axial Membrane (P )                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (P )                                                  | 1                     | -1                   | 1                      | -1                    |
| Axial Membrane (Mc)                                                 | -1                    | -1                   | 1                      | 1                     |
| Axial Bending (Mc)                                                  | -6                    | 6                    | 6                      | -6                    |
| Axial Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Axial Pressure Stress                                               | 3                     | 3                    | 3                      | 3                     |
| Axial Stress Summary                                                | -2                    | 7                    | 12                     | -3                    |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | 0             | 0            | 0             | 0            |
| Shear due to Torsion                    | 0             | 0            | 0             | 0            |
| Shear Stress Summary                    | 0             | 0            | 0             | 0            |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                        | 6             | 7            | 14            | 19           |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):**

| Type of<br>Stress Int.<br>Location | Stress Values at |     |     |     |     |     |     |     |
|------------------------------------|------------------|-----|-----|-----|-----|-----|-----|-----|
|                                    | Au               | Al  | Bu  | Bl  | Cu  | Cl  | Du  | Dl  |
| Circ. Pm (SUS)                     | 120              | 122 | 120 | 122 | 120 | 122 | 120 | 122 |
| Circ. Pl (SUS)                     | -3               | -3  | 7   | 7   | -4  | -4  | 6   | 6   |

|                 |       |       |       |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|-------|-------|-------|-------|
| Circ. Q (SUS)   | -3    | 3     | 15    | -15   | -74   | 74    | 120   | -120  |
| Long. Pm (SUS)  | 60    | 60    | 60    | 60    | 60    | 60    | 60    | 60    |
| Long. P1 (SUS)  | -3    | -3    | 5     | 5     | -2    | -2    | 6     | 6     |
| Long. Q (SUS)   | -10   | 10    | 56    | -56   | -23   | 23    | 35    | -35   |
| Shear Pm (SUS)  | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear P1 (SUS)  | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Q (SUS)   | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     |
| Pm (SUS)        | 120.0 | 122.0 | 120.0 | 122.0 | 120.0 | 122.0 | 120.0 | 122.0 |
| Pm+P1 (SUS)     | 117.0 | 119.0 | 127.0 | 129.0 | 116.0 | 118.0 | 126.0 | 128.0 |
| Pm+P1+Q (Total) | 114.0 | 122.0 | 142.0 | 114.0 | 42.1  | 192.0 | 246.0 | 31.0  |

#### Vessel Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 122.00    | 138.00         | Passed |
| Pm+P1 (SUS)         | 129.00    | 207.00         | Passed |
| Pm+P1+Q (TOTAL)     | 246.01    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):

| Type of Stress Int. Location | Stress Values at |     |      |      |     |     |      |      |
|------------------------------|------------------|-----|------|------|-----|-----|------|------|
|                              | Au               | Al  | Bu   | Bl   | Cu  | Cl  | Du   | Dl   |
| Circ. Pm (SUS)               | 6                | 9   | 6    | 9    | 6   | 9   | 6    | 9    |
| Circ. P1 (SUS)               | -3               | -3  | 7    | 7    | -2  | -2  | 6    | 6    |
| Circ. Q (SUS)                | 0                | 0   | 0    | 0    | 0   | 0   | 0    | 0    |
| Long. Pm (SUS)               | 3                | 3   | 3    | 3    | 3   | 3   | 3    | 3    |
| Long. P1 (SUS)               | -1               | -1  | 1    | 1    | -1  | -1  | 1    | 1    |
| Long. Q (SUS)                | 0                | 0   | 2    | -2   | -5  | 5   | 7    | -7   |
| Shear Pm (SUS)               | 0                | 0   | 0    | 0    | 0   | 0   | 0    | 0    |
| Shear P1 (SUS)               | 0                | 0   | 0    | 0    | 0   | 0   | 0    | 0    |
| Shear Q (SUS)                | 0                | 0   | 0    | 0    | 0   | 0   | 0    | 0    |
| Pm (SUS)                     | 6.0              | 9.0 | 6.0  | 9.0  | 6.0 | 9.0 | 6.0  | 9.0  |
| Pm+P1 (SUS)                  | 3.0              | 6.0 | 13.0 | 16.0 | 4.0 | 7.0 | 12.0 | 15.0 |
| Pm+P1+Q (Total)              | 3.0              | 6.0 | 13.0 | 16.0 | 7.0 | 7.0 | 12.0 | 18.0 |

#### Nozzle Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 9.00      | 138.00         | Passed |
| Pm+P1 (SUS)         | 16.00     | 207.00         | Passed |
| Pm+P1+Q (TOTAL)     | 18.00     | 413.99         | Passed |

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 \* Smh.

**Input, Nozzle Desc: VENT NOZ V1**

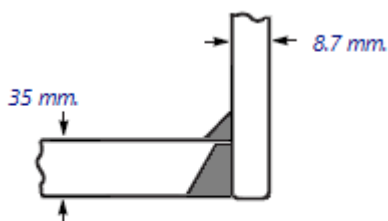
**From: 60**

|                                               |        |           |                    |
|-----------------------------------------------|--------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P      | 25.000    | bars               |
| Temperature for Internal Pressure             | Temp   | 85        | °C                 |
| Design External Pressure                      | Pext   | 1.03      | bars               |
| Temperature for External Pressure             | Tempex | 85        | °C                 |
| Parent Material                               |        | SA-516 70 | +Q+IT              |
| Parent Allowable Stress at Temperature        | Sv     | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva    | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D      | 3250.00   | mm.                |
| Design Length of Section                      | L      | 4506.6333 | mm.                |
| Shell Finished (Minimum) Thickness            | t      | 35.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c      | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co     | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |        | 4007.50   | mm.                |
| User Entered Minimum Design Metal Temperature |        | -35.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                  |     |                |                    |
|--------------------------------------------------|-----|----------------|--------------------|
| Material                                         |     | SA-350 LF2     | +IT                |
| Material UNS Number                              |     | K03011         |                    |
| Material Specification/Type                      |     | Forgings       |                    |
| Allowable Stress at Temperature                  | Sn  | 138.00         | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                      | Sna | 138.00         | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)                |     | Inside         |                    |
| Layout Angle                                     |     | 90.00          | deg                |
| Diameter                                         |     | 2.0000         | in.                |
| Size and Thickness Basis                         |     | Nominal        |                    |
| Nominal Thickness                                |     | 160            |                    |
| Flange Material                                  |     | SA-350 LF2     | +IT                |
| Flange Type                                      |     | Long Weld Neck |                    |
| Corrosion Allowance                              | can | 1.5000         | mm.                |
| Joint Efficiency of Shell Seam at Nozzle         | E1  | 1.00           |                    |
| Joint Efficiency of Nozzle Neck                  | En  | 1.00           |                    |
| Outside Projection                               | ho  | 300.0000       | mm.                |
| Weld leg size between Nozzle and Pad/Shell       | Wo  | 10.0000        | mm.                |
| Groove weld depth between Nozzle and Vessel Wgnv |     | 30.0000        | mm.                |
| Inside Projection                                | h   | 0.0000         | mm.                |
| Weld leg size, Inside Element to Shell           | Wi  | 0.0000         | mm.                |
| Flange Class                                     |     | 300            |                    |
| Flange Grade                                     |     | GR 1.1         |                    |

The Pressure Design option was Design Pressure + static head.



**Reinforcement CALCULATION, Description: VENT NOZ V1**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.687 in.  
Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $25.0 \cdot 1626.4999 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$   
= 29.7918 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
=  $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $25.0 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$   
= 0.4199 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |              |
|-------------------------------------------------|---------|--------------|
| Parallel to Vessel Wall (Diameter Limit)        | D1      | 127.3250 mm. |
| Parallel to Vessel Wall                         | Rn+tn+t | 63.6625 mm.  |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 18.0940 mm.  |

*Taking a UG-36(c)(3)(a) exemption for nozzle: VENT NOZ V1.  
This calculation is valid for nozzles that meet all the requirements of  
paragraph UG-36. Please check the Code carefully, especially for nozzles  
that are not isolated or do not meet Code spacing requirements. To force  
the computation of areas for small nozzles go to Tools->Configuration  
and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |               |
|------------------------------------------------|------------------------|---------------|
| Wall Thickness for Internal/External pressures | ta                     | = 1.9199 mm.  |
| Wall Thickness per UG16(b),                    | tr16b                  | = 3.0000 mm.  |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 31.2918 mm. |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 31.2918 mm. |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 2.7196 mm.  |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.0000 mm.  |
| Wall Thickness per table UG-45                 | tb3                    | = 4.9200 mm.  |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 4.92, max( 31.2918, 3.0 ) ]  
= 4.9200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 1.9199, 4.92 )  
= 4.9200 mm.

Available Nozzle Neck Thickness =  $0.875 \cdot 8.738 = 7.645$  mm. -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:  
This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |         |
|----------------------------------------------------------|---------|
| Impact Test Temperature provided per Specification       | -46 °C  |
| Calculated Minimum Design Metal Temperature              | -104 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -104 °C |

#### Weld Size Calculations, Description: VENT NOZ V1

|                                           |      |            |
|-------------------------------------------|------|------------|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 7.2376 mm. |
|-------------------------------------------|------|------------|

#### Results Per UW-16.1:

|                    |                  |
|--------------------|------------------|
| Required Thickness | Actual Thickness |
|--------------------|------------------|



Nozzle Weld       $5.0663 = 0.7 * t_{min}$ .       $7.0700 = 0.7 * W_o$  mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2799 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 335.2799 mm.

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**Input, Nozzle Desc: PR. IN P1**

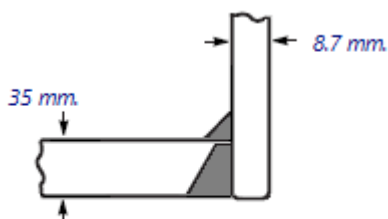
**From: 60**

|                                               |        |           |                    |
|-----------------------------------------------|--------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P      | 25.000    | bars               |
| Temperature for Internal Pressure             | Temp   | 85        | °C                 |
| Design External Pressure                      | Pext   | 1.03      | bars               |
| Temperature for External Pressure             | Tempex | 85        | °C                 |
| Parent Material                               |        | SA-516 70 | +Q+IT              |
| Parent Allowable Stress at Temperature        | Sv     | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva    | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D      | 3250.00   | mm.                |
| Design Length of Section                      | L      | 4506.6333 | mm.                |
| Shell Finished (Minimum) Thickness            | t      | 35.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c      | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co     | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |        | 3757.50   | mm.                |
| User Entered Minimum Design Metal Temperature |        | -35.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                  |     |                |                    |
|--------------------------------------------------|-----|----------------|--------------------|
| Material                                         |     | SA-350 LF2     | +IT                |
| Material UNS Number                              |     | K03011         |                    |
| Material Specification/Type                      |     | Forgings       |                    |
| Allowable Stress at Temperature                  | Sn  | 138.00         | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                      | Sna | 138.00         | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)                |     | Inside         |                    |
| Layout Angle                                     |     | 90.00          | deg                |
| Diameter                                         |     | 2.0000         | in.                |
| Size and Thickness Basis                         |     | Nominal        |                    |
| Nominal Thickness                                |     | 160            |                    |
| Flange Material                                  |     | SA-350 LF2     | +IT                |
| Flange Type                                      |     | Long Weld Neck |                    |
| Corrosion Allowance                              | can | 1.5000         | mm.                |
| Joint Efficiency of Shell Seam at Nozzle         | E1  | 1.00           |                    |
| Joint Efficiency of Nozzle Neck                  | En  | 1.00           |                    |
| Outside Projection                               | ho  | 300.0000       | mm.                |
| Weld leg size between Nozzle and Pad/Shell       | Wo  | 10.0000        | mm.                |
| Groove weld depth between Nozzle and Vessel Wgnv |     | 30.0000        | mm.                |
| Inside Projection                                | h   | 0.0000         | mm.                |
| Weld leg size, Inside Element to Shell           | Wi  | 0.0000         | mm.                |
| Flange Class                                     |     | 300            |                    |
| Flange Grade                                     |     | GR 1.1         |                    |

The Pressure Design option was Design Pressure + static head.



**Reinforcement CALCULATION, Description: PR. IN P1**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.687 in.  
 Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 25.0 \cdot 1626.4999 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$   
 $= 29.7918 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 25.0 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$   
 $= 0.4199 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |              |
|-------------------------------------------------|---------|--------------|
| Parallel to Vessel Wall (Diameter Limit)        | D1      | 127.3250 mm. |
| Parallel to Vessel Wall                         | Rn+tn+t | 63.6625 mm.  |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 18.0940 mm.  |

*Taking a UG-36(c)(3)(a) exemption for nozzle: PR. IN P1.  
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                      |
|------------------------------------------------|--------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 1.9199 mm.                      |
| Wall Thickness per UG16(b),                    | tr16b = 3.0000 mm.                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 31.2918 mm.                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 31.2918 mm. |
| Wall Thickness, shell/head, external pressure  | trb2 = 2.7196 mm.                    |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.0000 mm.  |
| Wall Thickness per table UG-45                 | tb3 = 4.9200 mm.                     |

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2 ) ]$   
 $= \min[ 4.92, \max( 31.2918, 3.0 ) ]$   
 $= 4.9200 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 1.9199, 4.92 )$   
 $= 4.9200 \text{ mm.}$

Available Nozzle Neck Thickness =  $0.875 \cdot 8.738 = 7.645 \text{ mm.} \rightarrow \text{OK}$

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:  
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |         |
|----------------------------------------------------------|---------|
| Impact Test Temperature provided per Specification       | -46 °C  |
| Calculated Minimum Design Metal Temperature              | -104 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -104 °C |

#### Weld Size Calculations, Description: PR. IN P1

|                                           |      |            |
|-------------------------------------------|------|------------|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 7.2376 mm. |
|-------------------------------------------|------|------------|

#### Results Per UW-16.1:

|                    |                  |
|--------------------|------------------|
| Required Thickness | Actual Thickness |
|--------------------|------------------|



Nozzle Weld       $5.0663 = 0.7 * t_{min}$ .     $7.0700 = 0.7 * W_o$  mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2799 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 335.2799 mm.

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**Input, Nozzle Desc: PR. IN NOZ. P2**

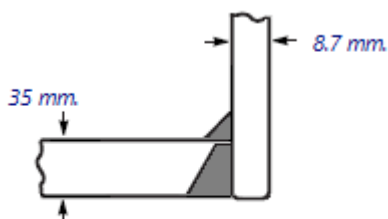
**From: 60**

|                                               |        |           |                    |
|-----------------------------------------------|--------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P      | 25.000    | bars               |
| Temperature for Internal Pressure             | Temp   | 85        | °C                 |
| Design External Pressure                      | Pext   | 1.03      | bars               |
| Temperature for External Pressure             | Tempex | 85        | °C                 |
| Parent Material                               |        | SA-516 70 | +Q+IT              |
| Parent Allowable Stress at Temperature        | Sv     | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva    | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D      | 3250.00   | mm.                |
| Design Length of Section                      | L      | 4506.6333 | mm.                |
| Shell Finished (Minimum) Thickness            | t      | 35.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c      | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co     | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |        | 6857.50   | mm.                |
| User Entered Minimum Design Metal Temperature |        | -35.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |                    |
|---------------------------------------------|------|----------------|--------------------|
| Material                                    |      | SA-350 LF2     | +IT                |
| Material UNS Number                         |      | K03011         |                    |
| Material Specification/Type                 |      | Forgings       |                    |
| Allowable Stress at Temperature             | Sn   | 138.00         | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                 | Sna  | 138.00         | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)           |      | Inside         |                    |
| Layout Angle                                |      | 90.00          | deg                |
| Diameter                                    |      | 2.0000         | in.                |
| Size and Thickness Basis                    |      | Nominal        |                    |
| Nominal Thickness                           |      | 160            |                    |
| Flange Material                             |      | SA-350 LF2     | +IT                |
| Flange Type                                 |      | Long Weld Neck |                    |
| Corrosion Allowance                         | can  | 1.5000         | mm.                |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |                    |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |                    |
| Outside Projection                          | ho   | 300.0000       | mm.                |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm.                |
| Groove weld depth between Nozzle and Vessel | Wgnv | 30.0000        | mm.                |
| Inside Projection                           | h    | 0.0000         | mm.                |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm.                |
| Flange Class                                |      | 300            |                    |
| Flange Grade                                |      | GR 1.1         |                    |

The Pressure Design option was Design Pressure + static head.



**Reinforcement CALCULATION, Description: PR. IN NOZ. P2**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.687 in.  
 Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 25.0 \cdot 1626.4999 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$   
 $= 29.7918$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 25.0 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$   
 $= 0.4199$  mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |     |
|-------------------------------------------------|---------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1      | 127.3250 | mm. |
| Parallel to Vessel Wall                         | Rn+tn+t | 63.6625  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 18.0940  | mm. |

*Taking a UG-36(c)(3)(a) exemption for nozzle: PR. IN NOZ. P2.  
 This calculation is valid for nozzles that meet all the requirements of  
 paragraph UG-36. Please check the Code carefully, especially for nozzles  
 that are not isolated or do not meet Code spacing requirements. To force  
 the computation of areas for small nozzles go to Tools->Configuration  
 and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                  |     |
|------------------------------------------------|----------------------------------|-----|
| Wall Thickness for Internal/External pressures | ta = 1.9199                      | mm. |
| Wall Thickness per UG16(b),                    | tr16b = 3.0000                   | mm. |
| Wall Thickness, shell/head, internal pressure  | trb1 = 31.2918                   | mm. |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 31.2918 | mm. |
| Wall Thickness, shell/head, external pressure  | trb2 = 2.7196                    | mm. |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.0000  | mm. |
| Wall Thickness per table UG-45                 | tb3 = 4.9200                     | mm. |

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2 ) ]$   
 $= \min[ 4.92, \max( 31.2918, 3.0 ) ]$   
 $= 4.9200$  mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 1.9199, 4.92 )$   
 $= 4.9200$  mm.

Available Nozzle Neck Thickness =  $0.875 \cdot 8.738 = 7.645$  mm. -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:  
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |         |
|----------------------------------------------------------|---------|
| Impact Test Temperature provided per Specification       | -46 °C  |
| Calculated Minimum Design Metal Temperature              | -104 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -104 °C |

##### Weld Size Calculations, Description: PR. IN NOZ. P2

|                                           |      |        |     |
|-------------------------------------------|------|--------|-----|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 7.2376 | mm. |
|-------------------------------------------|------|--------|-----|

#### Results Per UW-16.1:

|                    |                  |
|--------------------|------------------|
| Required Thickness | Actual Thickness |
|--------------------|------------------|



Nozzle Weld       $5.0663 = 0.7 * t_{min}$ .       $7.0700 = 0.7 * W_o$  mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2799 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 335.2799 mm.

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**Input, Nozzle Desc: SAFETY VALVE S1**

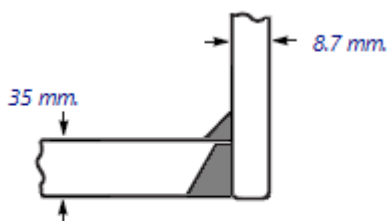
**From: 60**

|                                               |        |           |                    |
|-----------------------------------------------|--------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P      | 25.000    | bars               |
| Temperature for Internal Pressure             | Temp   | 85        | °C                 |
| Design External Pressure                      | Pext   | 1.03      | bars               |
| Temperature for External Pressure             | Tempex | 85        | °C                 |
| Parent Material                               |        | SA-516 70 | +Q+IT              |
| Parent Allowable Stress at Temperature        | Sv     | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva    | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D      | 3250.00   | mm.                |
| Design Length of Section                      | L      | 4506.6333 | mm.                |
| Shell Finished (Minimum) Thickness            | t      | 35.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c      | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co     | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |        | 6607.50   | mm.                |
| User Entered Minimum Design Metal Temperature |        | -35.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                  |     |                |                    |
|--------------------------------------------------|-----|----------------|--------------------|
| Material                                         |     | SA-350 LF2     | +IT                |
| Material UNS Number                              |     | K03011         |                    |
| Material Specification/Type                      |     | Forgings       |                    |
| Allowable Stress at Temperature                  | Sn  | 138.00         | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                      | Sna | 138.00         | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)                |     | Inside         |                    |
| Layout Angle                                     |     | 90.00          | deg                |
| Diameter                                         |     | 2.0000         | in.                |
| Size and Thickness Basis                         |     | Nominal        |                    |
| Nominal Thickness                                |     | 160            |                    |
| Flange Material                                  |     | SA-350 LF2     | +IT                |
| Flange Type                                      |     | Long Weld Neck |                    |
| Corrosion Allowance                              | can | 1.5000         | mm.                |
| Joint Efficiency of Shell Seam at Nozzle         | E1  | 1.00           |                    |
| Joint Efficiency of Nozzle Neck                  | En  | 1.00           |                    |
| Outside Projection                               | ho  | 300.0000       | mm.                |
| Weld leg size between Nozzle and Pad/Shell       | Wo  | 10.0000        | mm.                |
| Groove weld depth between Nozzle and Vessel Wgnv |     | 30.0000        | mm.                |
| Inside Projection                                | h   | 0.0000         | mm.                |
| Weld leg size, Inside Element to Shell           | Wi  | 0.0000         | mm.                |
| Flange Class                                     |     | 300            |                    |
| Flange Grade                                     |     | GR 1.1         |                    |

The Pressure Design option was Design Pressure + static head.



**Reinforcement CALCULATION, Description: SAFETY VALVE S1**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.687 in.  
 Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 25.0 \cdot 1626.4999 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$   
 $= 29.7918$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 25.0 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$   
 $= 0.4199$  mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |              |
|-------------------------------------------------|---------|--------------|
| Parallel to Vessel Wall (Diameter Limit)        | D1      | 127.3250 mm. |
| Parallel to Vessel Wall                         | Rn+tn+t | 63.6625 mm.  |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 18.0940 mm.  |

*Taking a UG-36(c)(3)(a) exemption for nozzle: SAFETY VALVE S1.  
 This calculation is valid for nozzles that meet all the requirements of  
 paragraph UG-36. Please check the Code carefully, especially for nozzles  
 that are not isolated or do not meet Code spacing requirements. To force  
 the computation of areas for small nozzles go to Tools->Configuration  
 and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                      |
|------------------------------------------------|--------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 1.9199 mm.                      |
| Wall Thickness per UG16(b),                    | tr16b = 3.0000 mm.                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 31.2918 mm.                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 31.2918 mm. |
| Wall Thickness, shell/head, external pressure  | trb2 = 2.7196 mm.                    |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.0000 mm.  |
| Wall Thickness per table UG-45                 | tb3 = 4.9200 mm.                     |

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2 ) ]$   
 $= \min[ 4.92, \max( 31.2918, 3.0 ) ]$   
 $= 4.9200$  mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 1.9199, 4.92 )$   
 $= 4.9200$  mm.

Available Nozzle Neck Thickness =  $0.875 \cdot 8.738 = 7.645$  mm. -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:  
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |         |
|----------------------------------------------------------|---------|
| Impact Test Temperature provided per Specification       | -46 °C  |
| Calculated Minimum Design Metal Temperature              | -104 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -104 °C |

#### Weld Size Calculations, Description: SAFETY VALVE S1

|                                           |      |            |
|-------------------------------------------|------|------------|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 7.2376 mm. |
|-------------------------------------------|------|------------|

#### Results Per UW-16.1:

|                    |                  |
|--------------------|------------------|
| Required Thickness | Actual Thickness |
|--------------------|------------------|



Nozzle Weld       $5.0663 = 0.7 * t_{min}$ .       $7.0700 = 0.7 * W_o$  mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2799 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 335.2799 mm.

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**Input, Nozzle Desc: LEVEL GUAGE L1**

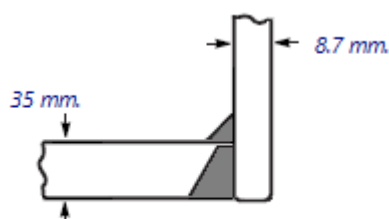
**From: 60**

|                                               |        |           |                    |
|-----------------------------------------------|--------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P      | 25.000    | bars               |
| Temperature for Internal Pressure             | Temp   | 85        | °C                 |
| Design External Pressure                      | Pext   | 1.03      | bars               |
| Temperature for External Pressure             | Tempex | 85        | °C                 |
| Parent Material                               |        | SA-516 70 | +Q+IT              |
| Parent Allowable Stress at Temperature        | Sv     | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva    | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D      | 3250.00   | mm.                |
| Design Length of Section                      | L      | 4506.6333 | mm.                |
| Shell Finished (Minimum) Thickness            | t      | 35.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c      | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co     | 0.0000    | mm.                |
| Distance from Cylinder/Cone Centerline        | L1     | 700.0000  | mm.                |
| Distance from Bottom/Left Tangent             |        | 4407.50   | mm.                |
| User Entered Minimum Design Metal Temperature |        | -35.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |                    |
|---------------------------------------------|------|----------------|--------------------|
| Material                                    |      | SA-350 LF2     | +IT                |
| Material UNS Number                         |      | K03011         |                    |
| Material Specification/Type                 |      | Forgings       |                    |
| Allowable Stress at Temperature             | Sn   | 138.00         | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                 | Sna  | 138.00         | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)           |      | Inside         |                    |
| Layout Angle                                |      | 30.48          | deg                |
| Diameter                                    |      | 2.0000         | in.                |
| Size and Thickness Basis                    |      | Nominal        |                    |
| Nominal Thickness                           |      | 160            |                    |
| Flange Material                             |      | SA-350 LF2     | +IT                |
| Flange Type                                 |      | Long Weld Neck |                    |
| Corrosion Allowance                         | can  | 1.5000         | mm.                |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |                    |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |                    |
| Outside Projection                          | ho   | 300.0000       | mm.                |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm.                |
| Groove weld depth between Nozzle and Vessel | Wgnv | 30.0000        | mm.                |
| Inside Projection                           | h    | 0.0000         | mm.                |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm.                |
| Flange Class                                |      | 300            |                    |
| Flange Grade                                |      | GR 1.1         |                    |

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.



### Reinforcement CALCULATION, Description: LEVEL GUAGE L1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 1.687 in. |
| Actual Thickness Used in Calculation       | 0.344 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $25.0 \cdot 1626.4999 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$   
= 29.7918 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
=  $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $25.0 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$   
= 0.4199 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |              |
|-------------------------------------------------|---------|--------------|
| Parallel to Vessel Wall (Diameter Limit)        | D1      | 132.1693 mm. |
| Parallel to Vessel Wall                         | Rn+tn+t | 66.0847 mm.  |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 18.0940 mm.  |

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL GUAGE L1.  
This calculation is valid for nozzles that meet all the requirements of  
paragraph UG-36. Please check the Code carefully, especially for nozzles  
that are not isolated or do not meet Code spacing requirements. To force  
the computation of areas for small nozzles go to Tools->Configuration  
and check the box to force the UG-37 small nozzle area calculation.*

### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

#### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:  
*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |         |
|----------------------------------------------------------|---------|
| Impact Test Temperature provided per Specification       | -46 °C  |
| Calculated Minimum Design Metal Temperature              | -104 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -104 °C |

### Weld Size Calculations, Description: LEVEL GUAGE L1

|                                           |      |            |
|-------------------------------------------|------|------------|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 7.2376 mm. |
|-------------------------------------------|------|------------|

### Results Per UW-16.1:

|             |                              |                              |
|-------------|------------------------------|------------------------------|
|             | Required Thickness           | Actual Thickness             |
| Nozzle Weld | $5.0663 = 0.7 \cdot t_{min}$ | $7.0700 = 0.7 \cdot W_o$ mm. |

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles  
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in plane parallel to the vessel axis.

### Reinforcement CALCULATION, Description: LEVEL GUAGE L1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 1.687 in. |
| Actual Thickness Used in Calculation       | 0.344 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.0 \cdot 1626.4999 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$$

$$= 29.7918 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.0 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$$

$$= 0.4199 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |     |
|-------------------------------------------------|---------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1      | 127.3250 | mm. |
| Parallel to Vessel Wall                         | Rn+tn+t | 63.6625  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 18.0940  | mm. |

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL GUAGE L1.*  
*This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |     |
|------------------------------------------------|------------------------|-----------|-----|
| Wall Thickness for Internal/External pressures | ta                     | = 1.9199  | mm. |
| Wall Thickness per UG16(b),                    | tr16b                  | = 3.0000  | mm. |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 31.2918 | mm. |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 31.2918 | mm. |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 2.7196  | mm. |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.0000  | mm. |
| Wall Thickness per table UG-45                 | tb3                    | = 4.9200  | mm. |

Determine Nozzle Thickness candidate [tb]:

$$= \min[ tb3, \max( tb1, tb2 ) ]$$

$$= \min[ 4.92, \max( 31.2918, 3.0 ) ]$$

$$= 4.9200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max( ta, tb )$$

$$= \max( 1.9199, 4.92 )$$

$$= 4.9200 \text{ mm.}$$

Available Nozzle Neck Thickness =  $0.875 \cdot 8.738 = 7.645 \text{ mm.} \rightarrow \text{OK}$

#### Weld Size Calculations, Description: LEVEL GUAGE L1

Intermediate Calc. for nozzle/shell Welds Tmin 7.2376 mm.

#### Results Per UW-16.1:

|             | Required Thickness           | Actual Thickness                     |
|-------------|------------------------------|--------------------------------------|
| Nozzle Weld | $5.0663 = 0.7 \cdot t_{min}$ | $7.0700 = 0.7 \cdot W_o \text{ mm.}$ |

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 14.7821 mm.  
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.4721 mm.

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**Input, Nozzle Desc: LEVEL GUAGE L2**

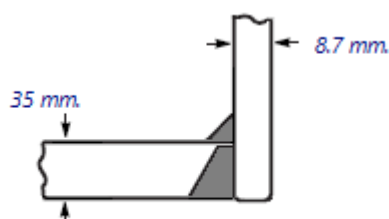
**From: 60**

|                                               |        |           |                    |
|-----------------------------------------------|--------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P      | 25.051    | bars               |
| Temperature for Internal Pressure             | Temp   | 85        | °C                 |
| Design External Pressure                      | Pext   | 1.03      | bars               |
| Temperature for External Pressure             | Tempex | 85        | °C                 |
| Parent Material                               |        | SA-516 70 | +Q+IT              |
| Parent Allowable Stress at Temperature        | Sv     | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva    | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D      | 3250.00   | mm.                |
| Design Length of Section                      | L      | 4506.6333 | mm.                |
| Shell Finished (Minimum) Thickness            | t      | 35.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c      | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co     | 0.0000    | mm.                |
| Distance from Cylinder/Cone Centerline        | L1     | 700.0000  | mm.                |
| Distance from Bottom/Left Tangent             |        | 4407.50   | mm.                |
| User Entered Minimum Design Metal Temperature |        | -35.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |                    |
|---------------------------------------------|------|----------------|--------------------|
| Material                                    |      | SA-350 LF2     | +IT                |
| Material UNS Number                         |      | K03011         |                    |
| Material Specification/Type                 |      | Forgings       |                    |
| Allowable Stress at Temperature             | Sn   | 138.00         | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                 | Sna  | 138.00         | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)           |      | Inside         |                    |
| Layout Angle                                |      | -30.48         | deg                |
| Diameter                                    |      | 2.0000         | in.                |
| Size and Thickness Basis                    |      | Nominal        |                    |
| Nominal Thickness                           |      | 160            |                    |
| Flange Material                             |      | SA-350 LF2     | +IT                |
| Flange Type                                 |      | Long Weld Neck |                    |
| Corrosion Allowance                         | can  | 1.5000         | mm.                |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |                    |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |                    |
| Outside Projection                          | ho   | 300.0000       | mm.                |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm.                |
| Groove weld depth between Nozzle and Vessel | Wgnv | 30.0000        | mm.                |
| Inside Projection                           | h    | 0.0000         | mm.                |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm.                |
| Flange Class                                |      | 300            |                    |
| Flange Grade                                |      | GR 1.1         |                    |

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.



### Reinforcement CALCULATION, Description: LEVEL GUAGE L2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 1.687 in. |
| Actual Thickness Used in Calculation       | 0.344 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $25.05 \cdot 1626.4999 / (138 \cdot 1.0 - 0.6 \cdot 25.05)$   
= 29.8536 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
=  $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $25.05 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.05)$   
= 0.4208 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |              |
|-------------------------------------------------|---------|--------------|
| Parallel to Vessel Wall (Diameter Limit)        | D1      | 132.1689 mm. |
| Parallel to Vessel Wall                         | Rn+tn+t | 66.0844 mm.  |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 18.0940 mm.  |

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL GUAGE L2.  
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

#### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:  
*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |         |
|----------------------------------------------------------|---------|
| Impact Test Temperature provided per Specification       | -46 °C  |
| Calculated Minimum Design Metal Temperature              | -104 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -104 °C |

### Weld Size Calculations, Description: LEVEL GUAGE L2

|                                           |      |            |
|-------------------------------------------|------|------------|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 7.2376 mm. |
|-------------------------------------------|------|------------|

### Results Per UW-16.1:

|             |                              |                              |
|-------------|------------------------------|------------------------------|
|             | Required Thickness           | Actual Thickness             |
| Nozzle Weld | $5.0663 = 0.7 \cdot t_{min}$ | $7.0700 = 0.7 \cdot W_o$ mm. |

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in plane parallel to the vessel axis.

### Reinforcement CALCULATION, Description: LEVEL GUAGE L2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 1.687 in. |
| Actual Thickness Used in Calculation       | 0.344 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.05 \cdot 1626.4999 / (138 \cdot 1.0 - 0.6 \cdot 25.05)$$

$$= 29.8536 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.05 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.05)$$

$$= 0.4208 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |     |
|-------------------------------------------------|---------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1      | 127.3250 | mm. |
| Parallel to Vessel Wall                         | Rn+tn+t | 63.6625  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 18.0940  | mm. |

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL GUAGE L2.*  
*This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |     |
|------------------------------------------------|------------------------|-----------|-----|
| Wall Thickness for Internal/External pressures | ta                     | = 1.9208  | mm. |
| Wall Thickness per UG16(b),                    | tr16b                  | = 3.0000  | mm. |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 31.3536 | mm. |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 31.3536 | mm. |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 2.7196  | mm. |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.0000  | mm. |
| Wall Thickness per table UG-45                 | tb3                    | = 4.9200  | mm. |

Determine Nozzle Thickness candidate [tb]:

$$= \min[ tb3, \max( tb1, tb2 ) ]$$

$$= \min[ 4.92, \max( 31.3536, 3.0 ) ]$$

$$= 4.9200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max( ta, tb )$$

$$= \max( 1.9208, 4.92 )$$

$$= 4.9200 \text{ mm.}$$

Available Nozzle Neck Thickness =  $0.875 \cdot 8.738 = 7.645 \text{ mm.} \rightarrow \text{OK}$

#### Weld Size Calculations, Description: LEVEL GUAGE L2

|                                           |      |        |     |
|-------------------------------------------|------|--------|-----|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 7.2376 | mm. |
|-------------------------------------------|------|--------|-----|

#### Results Per UW-16.1:

|             |                              |                                      |
|-------------|------------------------------|--------------------------------------|
|             | Required Thickness           | Actual Thickness                     |
| Nozzle Weld | $5.0663 = 0.7 \cdot t_{min}$ | $7.0700 = 0.7 \cdot W_o \text{ mm.}$ |

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 14.7821 mm.  
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.4721 mm.

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**Input, Nozzle Desc: LEVEL GUAGE L3**

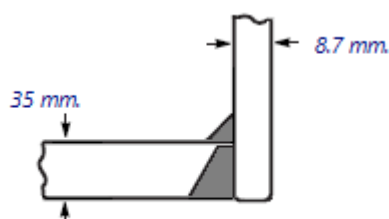
**From: 60**

|                                               |        |           |                    |
|-----------------------------------------------|--------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P      | 25.000    | bars               |
| Temperature for Internal Pressure             | Temp   | 85        | °C                 |
| Design External Pressure                      | Pext   | 1.03      | bars               |
| Temperature for External Pressure             | Tempex | 85        | °C                 |
| Parent Material                               |        | SA-516 70 | +Q+IT              |
| Parent Allowable Stress at Temperature        | Sv     | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva    | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D      | 3250.00   | mm.                |
| Design Length of Section                      | L      | 4506.6333 | mm.                |
| Shell Finished (Minimum) Thickness            | t      | 35.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c      | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co     | 0.0000    | mm.                |
| Distance from Cylinder/Cone Centerline        | L1     | 700.0000  | mm.                |
| Distance from Bottom/Left Tangent             |        | 5107.50   | mm.                |
| User Entered Minimum Design Metal Temperature |        | -35.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |                    |
|---------------------------------------------|------|----------------|--------------------|
| Material                                    |      | SA-350 LF2     | +IT                |
| Material UNS Number                         |      | K03011         |                    |
| Material Specification/Type                 |      | Forgings       |                    |
| Allowable Stress at Temperature             | Sn   | 138.00         | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                 | Sna  | 138.00         | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)           |      | Inside         |                    |
| Layout Angle                                |      | 30.48          | deg                |
| Diameter                                    |      | 2.0000         | in.                |
| Size and Thickness Basis                    |      | Nominal        |                    |
| Nominal Thickness                           |      | 160            |                    |
| Flange Material                             |      | SA-350 LF2     | +IT                |
| Flange Type                                 |      | Long Weld Neck |                    |
| Corrosion Allowance                         | can  | 1.5000         | mm.                |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |                    |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |                    |
| Outside Projection                          | ho   | 300.0000       | mm.                |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm.                |
| Groove weld depth between Nozzle and Vessel | Wgnv | 30.0000        | mm.                |
| Inside Projection                           | h    | 0.0000         | mm.                |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm.                |
| Flange Class                                |      | 300            |                    |
| Flange Grade                                |      | GR 1.1         |                    |

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

### Reinforcement CALCULATION, Description: LEVEL GUAGE L3

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 1.687 in. |
| Actual Thickness Used in Calculation       | 0.344 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 25.0 \cdot 1626.4999 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$   
 $= 29.7918$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 25.0 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$   
 $= 0.4199$  mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |              |
|-------------------------------------------------|---------|--------------|
| Parallel to Vessel Wall (Diameter Limit)        | D1      | 132.1693 mm. |
| Parallel to Vessel Wall                         | Rn+tn+t | 66.0847 mm.  |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 18.0940 mm.  |

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL GUAGE L3.  
 This calculation is valid for nozzles that meet all the requirements of  
 paragraph UG-36. Please check the Code carefully, especially for nozzles  
 that are not isolated or do not meet Code spacing requirements. To force  
 the computation of areas for small nozzles go to Tools->Configuration  
 and check the box to force the UG-37 small nozzle area calculation.*

### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

#### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:  
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |         |
|----------------------------------------------------------|---------|
| Impact Test Temperature provided per Specification       | -46 °C  |
| Calculated Minimum Design Metal Temperature              | -104 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -104 °C |

### Weld Size Calculations, Description: LEVEL GUAGE L3

|                                           |      |            |
|-------------------------------------------|------|------------|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 7.2376 mm. |
|-------------------------------------------|------|------------|

### Results Per UW-16.1:

|             |                              |                              |
|-------------|------------------------------|------------------------------|
|             | Required Thickness           | Actual Thickness             |
| Nozzle Weld | $5.0663 = 0.7 \cdot t_{min}$ | $7.0700 = 0.7 \cdot W_o$ mm. |

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles  
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in plane parallel to the vessel axis.

### Reinforcement CALCULATION, Description: LEVEL GUAGE L3

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 1.687 in. |
| Actual Thickness Used in Calculation       | 0.344 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.0 \cdot 1626.4999 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$$

$$= 29.7918 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.0 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$$

$$= 0.4199 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |     |
|-------------------------------------------------|---------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1      | 127.3250 | mm. |
| Parallel to Vessel Wall                         | Rn+tn+t | 63.6625  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 18.0940  | mm. |

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL GUAGE L3.*  
*This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |     |
|------------------------------------------------|------------------------|-----------|-----|
| Wall Thickness for Internal/External pressures | ta                     | = 1.9199  | mm. |
| Wall Thickness per UG16(b),                    | tr16b                  | = 3.0000  | mm. |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 31.2918 | mm. |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 31.2918 | mm. |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 2.7196  | mm. |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.0000  | mm. |
| Wall Thickness per table UG-45                 | tb3                    | = 4.9200  | mm. |

Determine Nozzle Thickness candidate [tb]:

$$= \min[ tb3, \max( tb1, tb2 ) ]$$

$$= \min[ 4.92, \max( 31.2918, 3.0 ) ]$$

$$= 4.9200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max( ta, tb )$$

$$= \max( 1.9199, 4.92 )$$

$$= 4.9200 \text{ mm.}$$

Available Nozzle Neck Thickness =  $0.875 \cdot 8.738 = 7.645 \text{ mm.} \rightarrow \text{OK}$

#### Weld Size Calculations, Description: LEVEL GUAGE L3

Intermediate Calc. for nozzle/shell Welds Tmin 7.2376 mm.

#### Results Per UW-16.1:

|             |                              |                                      |
|-------------|------------------------------|--------------------------------------|
|             | Required Thickness           | Actual Thickness                     |
| Nozzle Weld | $5.0663 = 0.7 \cdot t_{min}$ | $7.0700 = 0.7 \cdot W_o \text{ mm.}$ |

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 14.7821 mm.  
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.4721 mm.

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**Input, Nozzle Desc: LEVEL GUAGE L4**

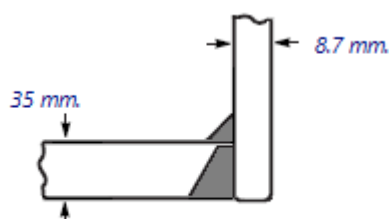
**From: 60**

|                                               |        |           |                    |
|-----------------------------------------------|--------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P      | 25.051    | bars               |
| Temperature for Internal Pressure             | Temp   | 85        | °C                 |
| Design External Pressure                      | Pext   | 1.03      | bars               |
| Temperature for External Pressure             | Tempex | 85        | °C                 |
| Parent Material                               |        | SA-516 70 | +Q+IT              |
| Parent Allowable Stress at Temperature        | Sv     | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva    | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D      | 3250.00   | mm.                |
| Design Length of Section                      | L      | 4506.6333 | mm.                |
| Shell Finished (Minimum) Thickness            | t      | 35.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c      | 1.5000    | mm.                |
| Shell External Corrosion Allowance            | co     | 0.0000    | mm.                |
| Distance from Cylinder/Cone Centerline        | L1     | 700.0000  | mm.                |
| Distance from Bottom/Left Tangent             |        | 5107.50   | mm.                |
| User Entered Minimum Design Metal Temperature |        | -35.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |                    |
|---------------------------------------------|------|----------------|--------------------|
| Material                                    |      | SA-350 LF2     | +IT                |
| Material UNS Number                         |      | K03011         |                    |
| Material Specification/Type                 |      | Forgings       |                    |
| Allowable Stress at Temperature             | Sn   | 138.00         | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                 | Sna  | 138.00         | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)           |      | Inside         |                    |
| Layout Angle                                |      | -30.48         | deg                |
| Diameter                                    |      | 2.0000         | in.                |
| Size and Thickness Basis                    |      | Nominal        |                    |
| Nominal Thickness                           |      | 160            |                    |
| Flange Material                             |      | SA-350 LF2     | +IT                |
| Flange Type                                 |      | Long Weld Neck |                    |
| Corrosion Allowance                         | can  | 1.5000         | mm.                |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |                    |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |                    |
| Outside Projection                          | ho   | 300.0000       | mm.                |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm.                |
| Groove weld depth between Nozzle and Vessel | Wgnv | 30.0000        | mm.                |
| Inside Projection                           | h    | 0.0000         | mm.                |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm.                |
| Flange Class                                |      | 300            |                    |
| Flange Grade                                |      | GR 1.1         |                    |

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.



### Reinforcement CALCULATION, Description: LEVEL GUAGE L4

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 1.687 in. |
| Actual Thickness Used in Calculation       | 0.344 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $25.05 \cdot 1626.4999 / (138 \cdot 1.0 - 0.6 \cdot 25.05)$   
= 29.8536 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
=  $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $25.05 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.05)$   
= 0.4208 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |              |
|-------------------------------------------------|---------|--------------|
| Parallel to Vessel Wall (Diameter Limit)        | D1      | 132.1689 mm. |
| Parallel to Vessel Wall                         | Rn+tn+t | 66.0844 mm.  |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 18.0940 mm.  |

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL GUAGE L4.  
This calculation is valid for nozzles that meet all the requirements of  
paragraph UG-36. Please check the Code carefully, especially for nozzles  
that are not isolated or do not meet Code spacing requirements. To force  
the computation of areas for small nozzles go to Tools->Configuration  
and check the box to force the UG-37 small nozzle area calculation.*

### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

#### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:  
*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |         |
|----------------------------------------------------------|---------|
| Impact Test Temperature provided per Specification       | -46 °C  |
| Calculated Minimum Design Metal Temperature              | -104 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -104 °C |

### Weld Size Calculations, Description: LEVEL GUAGE L4

|                                           |      |            |
|-------------------------------------------|------|------------|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 7.2376 mm. |
|-------------------------------------------|------|------------|

### Results Per UW-16.1:

|             |                              |                              |
|-------------|------------------------------|------------------------------|
|             | Required Thickness           | Actual Thickness             |
| Nozzle Weld | $5.0663 = 0.7 \cdot t_{min}$ | $7.0700 = 0.7 \cdot W_o$ mm. |

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles  
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in plane parallel to the vessel axis.

### Reinforcement CALCULATION, Description: LEVEL GUAGE L4

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 1.687 in. |
| Actual Thickness Used in Calculation       | 0.344 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.05 \cdot 1626.4999 / (138 \cdot 1.0 - 0.6 \cdot 25.05)$$

$$= 29.8536 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.05 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.05)$$

$$= 0.4208 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |     |
|-------------------------------------------------|---------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1      | 127.3250 | mm. |
| Parallel to Vessel Wall                         | Rn+tn+t | 63.6625  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 18.0940  | mm. |

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL GUAGE L4.*  
*This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |     |
|------------------------------------------------|------------------------|-----------|-----|
| Wall Thickness for Internal/External pressures | ta                     | = 1.9208  | mm. |
| Wall Thickness per UG16(b),                    | tr16b                  | = 3.0000  | mm. |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 31.3536 | mm. |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 31.3536 | mm. |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 2.7196  | mm. |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.0000  | mm. |
| Wall Thickness per table UG-45                 | tb3                    | = 4.9200  | mm. |

Determine Nozzle Thickness candidate [tb]:

$$= \min[ tb3, \max( tb1, tb2 ) ]$$

$$= \min[ 4.92, \max( 31.3536, 3.0 ) ]$$

$$= 4.9200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max( ta, tb )$$

$$= \max( 1.9208, 4.92 )$$

$$= 4.9200 \text{ mm.}$$

Available Nozzle Neck Thickness =  $0.875 \cdot 8.738 = 7.645 \text{ mm.} \rightarrow \text{OK}$

#### Weld Size Calculations, Description: LEVEL GUAGE L4

Intermediate Calc. for nozzle/shell Welds Tmin 7.2376 mm.

#### Results Per UW-16.1:

|             | Required Thickness           | Actual Thickness                     |
|-------------|------------------------------|--------------------------------------|
| Nozzle Weld | $5.0663 = 0.7 \cdot t_{min}$ | $7.0700 = 0.7 \cdot W_o \text{ mm.}$ |

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 14.7821 mm.  
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.4721 mm.

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**Input, Nozzle Desc: TEMP IN. T1**

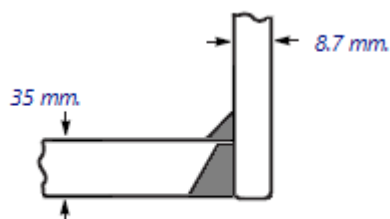
**From: 70**

|                                               |        |           |                    |
|-----------------------------------------------|--------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P      | 25.043    | bars               |
| Temperature for Internal Pressure             | Temp   | 85        | °C                 |
| Design External Pressure                      | Pext   | 1.03      | bars               |
| Temperature for External Pressure             | Tempex | 85        | °C                 |
| Parent Material                               |        | SA-516 70 | +Q+IT              |
| Parent Allowable Stress at Temperature        | Sv     | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva    | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Elliptical Head            | D      | 3250.00   | mm.                |
| Aspect Ratio of Elliptical Head               | Ar     | 2.00      |                    |
| Head Finished (Minimum) Thickness             | t      | 35.0000   | mm.                |
| Head Internal Corrosion Allowance             | c      | 1.5000    | mm.                |
| Head External Corrosion Allowance             | co     | 0.0000    | mm.                |
| Distance from Head Centerline                 | L1     | 700.0000  | mm.                |
| User Entered Minimum Design Metal Temperature |        | -35.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                  |     |                |                    |
|--------------------------------------------------|-----|----------------|--------------------|
| Material                                         |     | SA-350 LF2     | +IT                |
| Material UNS Number                              |     | K03011         |                    |
| Material Specification/Type                      |     | Forgings       |                    |
| Allowable Stress at Temperature                  | Sn  | 138.00         | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                      | Sna | 138.00         | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)                |     | Inside         |                    |
| Layout Angle                                     |     | 270.00         | deg                |
| Diameter                                         |     | 2.0000         | in.                |
| Size and Thickness Basis                         |     | Nominal        |                    |
| Nominal Thickness                                |     | 160            |                    |
| Flange Material                                  |     | SA-105         | +IT                |
| Flange Type                                      |     | Long Weld Neck |                    |
| Corrosion Allowance                              | can | 1.5000         | mm.                |
| Joint Efficiency of Shell Seam at Nozzle         | E1  | 1.00           |                    |
| Joint Efficiency of Nozzle Neck                  | En  | 1.00           |                    |
| Outside Projection                               | ho  | 300.0000       | mm.                |
| Weld leg size between Nozzle and Pad/Shell       | Wo  | 10.0000        | mm.                |
| Groove weld depth between Nozzle and Vessel Wgnv |     | 30.0000        | mm.                |
| Inside Projection                                | h   | 0.0000         | mm.                |
| Weld leg size, Inside Element to Shell           | Wi  | 0.0000         | mm.                |
| Flange Class                                     |     | 300            |                    |
| Flange Grade                                     |     | GR 1.1         |                    |

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: TEMP IN. T1**



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.687 in.  
 Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]  
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$  per Appendix 1-4(c)  
 $= 25.04 \cdot 3253.0 \cdot 0.8992 / (2 \cdot 138.0 \cdot 1.0 \cdot 0.2 \cdot 25.04)$   
 $= 26.5920$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 25.04 \cdot 22.92 / (138 \cdot 1.0 \cdot 0.6 \cdot 25.04)$   
 $= 0.4206$  mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |     |
|-------------------------------------------------|---------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl      | 128.6110 | mm. |
| Parallel to Vessel Wall                         | Rn+tn+t | 64.3055  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 18.0940  | mm. |

*Taking a UG-36(c)(3)(a) exemption for nozzle: TEMP IN. T1.  
 This calculation is valid for nozzles that meet all the requirements of  
 paragraph UG-36. Please check the Code carefully, especially for nozzles  
 that are not isolated or do not meet Code spacing requirements. To force  
 the computation of areas for small nozzles go to Tools->Configuration  
 and check the box to force the UG-37 small nozzle area calculation.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:  
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |         |
|----------------------------------------------------------|---------|
| Impact Test Temperature provided per Specification       | -46 °C  |
| Calculated Minimum Design Metal Temperature              | -104 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -104 °C |

##### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                        |        |
|--------------------------------------------------------|--------|
| Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) | -18 °C |
| Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)   | -48 °C |

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating =  $25.04 / 51.00 = 0.491$

##### Weld Size Calculations, Description: TEMP IN. T1

|                                           |      |        |     |
|-------------------------------------------|------|--------|-----|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 7.2376 | mm. |
|-------------------------------------------|------|--------|-----|

##### Results Per UW-16.1:

|             |                              |                              |
|-------------|------------------------------|------------------------------|
|             | Required Thickness           | Actual Thickness             |
| Nozzle Weld | $5.0663 = 0.7 \cdot t_{min}$ | $7.0700 = 0.7 \cdot W_o$ mm. |

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles  
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in the Latitudinal direction.

#### Reinforcement CALCULATION, Description: TEMP IN. T1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.687 in.  
 Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]  
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$  per Appendix 1-4(c)  
 $= 25.04 \cdot 3253.0 \cdot 0.8992 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 25.04)$   
 $= 26.5920$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 25.04 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.04)$   
 $= 0.4206$  mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |     |
|-------------------------------------------------|---------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl      | 127.3250 | mm. |
| Parallel to Vessel Wall                         | Rn+tn+t | 63.6625  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 18.0940  | mm. |

*Taking a UG-36(c)(3)(a) exemption for nozzle: TEMP IN. T1.  
 This calculation is valid for nozzles that meet all the requirements of  
 paragraph UG-36. Please check the Code carefully, especially for nozzles  
 that are not isolated or do not meet Code spacing requirements. To force  
 the computation of areas for small nozzles go to Tools->Configuration  
 and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |     |
|------------------------------------------------|------------------------|-----------|-----|
| Wall Thickness for Internal/External pressures | ta                     | = 1.9206  | mm. |
| Wall Thickness per UG16(b),                    | tr16b                  | = 3.0000  | mm. |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 31.0364 | mm. |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 31.0364 | mm. |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 2.7176  | mm. |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.0000  | mm. |
| Wall Thickness per table UG-45                 | tb3                    | = 4.9200  | mm. |

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2 ) ]$   
 $= \min[ 4.92, \max( 31.0364, 3.0 ) ]$   
 $= 4.9200$  mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 1.9206, 4.92 )$   
 $= 4.9200$  mm.

Available Nozzle Neck Thickness =  $0.875 \cdot 8.738 = 7.645$  mm. -> OK

#### Weld Size Calculations, Description: TEMP IN. T1

Intermediate Calc. for nozzle/shell Welds Tmin 7.2376 mm.

#### Results Per UW-16.1:

|             |                              |                              |
|-------------|------------------------------|------------------------------|
|             | Required Thickness           | Actual Thickness             |
| Nozzle Weld | $5.0663 = 0.7 \cdot t_{min}$ | $7.0700 = 0.7 \cdot W_o$ mm. |

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles  
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 7.5280 mm.  
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 343.5423 mm.

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**Input Echo, Tubesheet Number 1, Description: U-TS**

**Shell Data:**

**Main Shell Description: SHELL-1**

|                                         |          |              |
|-----------------------------------------|----------|--------------|
| Shell Maximum Design Pressure           | Psd,max  | 25.00 bars   |
| Shell Maximum Operating Pressure        | Psox,max | 25.00 bars   |
| Shell Minimum Operating Pressure        | Psox,min | 0.00 bars    |
| Shell Thickness                         | ts       | 46.0000 mm.  |
| Shell Internal Corrosion Allowance      | cas      | 1.5000 mm.   |
| Shell External Corrosion Allowance      | casext   | 0.0000 mm.   |
| Inside Diameter of Shell                | Ds       | 1750.000 mm. |
| Shell Temperature for Internal Pressure | Ts       | 85.00 °C     |
| Shell Material                          |          | SA-516 70    |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                       |    |                           |
|---------------------------------------|----|---------------------------|
| Shell Material UNS Number             |    | K02700                    |
| Shell Allowable Stress at Temperature | Ss | 138.00 N./mm <sup>2</sup> |
| Shell Allowable Stress at Ambient     |    | 138.00 N./mm <sup>2</sup> |

**Channel Data:**

**Channel Description: CH. COVER**

|                                      |          |              |
|--------------------------------------|----------|--------------|
| Channel Type:                        |          | Cylinder     |
| Channel Maximum Design Pressure      | Ptd,max  | 191.00 bars  |
| Channel Maximum Operating Pressure   | Ptox,max | 191.00 bars  |
| Channel Minimum Operating Pressure   | Ptox,min | 0.00 bars    |
| Channel Thickness                    | tc       | 135.0000 mm. |
| Channel Internal Corrosion Allowance | cac      | 1.5000 mm.   |
| Channel External Corrosion Allowance | cacext   | 0.0000 mm.   |
| Inside Diameter of Channel           | Dc       | 1750.000 mm. |
| Channel Design Temperature           | TEMPC    | 85.00 °C     |
| Channel Flange Matl                  |          | SA-516 70    |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                         |    |                           |
|-----------------------------------------|----|---------------------------|
| Channel Material UNS Number             |    | K02700                    |
| Channel Allowable Stress at Temperature | Sc | 138.00 N./mm <sup>2</sup> |
| Channel Allowable Stress at Ambient     |    | 138.00 N./mm <sup>2</sup> |

**Tube Data:**

|                                                  |                   |                           |
|--------------------------------------------------|-------------------|---------------------------|
| Number of Tube Holes                             | Nt                | 2332                      |
| Tube Wall Thickness                              | et                | 2.7700 mm.                |
| Tube Outside Diameter                            | D                 | 25.4000 mm.               |
| Total Straight Tube Length                       | Lt                | 5400.00 mm.               |
| Straight Tube Length (bet. inner tubsht faces) L |                   | 4960.00 mm.               |
| Design Temperature of the Tubes                  |                   | 85.00 °C                  |
| Tube Material                                    |                   | SA-334 6                  |
| Tube Material UNS Number                         |                   | K03006                    |
| Is this a Welded Tube                            |                   | No                        |
| Tube Material Specification used                 | Smls. & wld. tube |                           |
| Tube Allowable Stress at Temperature             |                   | 118.00 N./mm <sup>2</sup> |
| Tube Allowable Stress At Ambient                 |                   | 118.00 N./mm <sup>2</sup> |
| Tube Yield Stress At design Temperature          | Syt               | 223.00 N./mm <sup>2</sup> |
| Tube Pitch (Center to Center Spacing)            | P                 | 31.7500 mm.               |
| Tube Layout Pattern                              |                   | Triangular                |
| Fillet Weld Leg                                  | af                | 2.7700 mm.                |
| Groove Weld Leg                                  | ag                | 2.7700 mm.                |
| Tube-Tubesheet Joint Weld Type                   |                   | Full Strength             |
| Method for Tube-Tubesheet Jt. Allowable          |                   | UW-20.3                   |
| Is the Tube Inset                                |                   | No                        |
| Radius to Outermost Tube Hole Center             | ro                | 845.800 mm.               |
| Largest Center-to-Center Tube Distance           | Ul                | 76.2000 mm.               |
| Length of Expanded Portion of Tube               | ltx               | 0.0000 mm.                |
| Tube-side pass partition groove depth            | hg                | 6.0000 mm.                |

**Tubesheet Data:**

**Tubesheet TYPE: U-tube, Integral with Channel, Conf. e**

Tubesheet Design Metal Temperature T 85.00 °C  
 Tubesheet Material SA-350 LF2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                           |      |                           |
|-------------------------------------------|------|---------------------------|
| Tubesheet Material UNS Number             |      | K03011                    |
| Tubesheet Allowable Stress at Temperature | S    | 138.00 N./mm <sup>2</sup> |
| Tubesheet Allowable Stress at Ambient     | Tt   | 138.00 N./mm <sup>2</sup> |
| Thickness of Tubesheet                    | h    | 440.0000 mm.              |
| Tubesheet Corr. Allowance (Shell side)    | Cats | 1.5000 mm.                |
| Tubesheet Corr. Allowance (Channel side)  | Catc | 1.5000 mm.                |
| Tubesheet Outside Diameter                | A    | 2030.000 mm.              |

**Additional Data for Stepped Tubesheets:**

|                                                 |     |             |
|-------------------------------------------------|-----|-------------|
| Is the Tubesheet Stepped?                       |     | Yes         |
| Is the Tubesheet Flat on Tubeside?              |     | Yes         |
| Is the Tubesheet Flat on Shellside?             |     | No          |
| Step 1 Diameter on the Shellside                | ds1 | 1632.00 mm. |
| Step 1 Depth on the Shellside                   | hs1 | 6.00 mm.    |
| Step 2 Diameter on the Shellside                | ds2 | 1597.00 mm. |
| Step 2 Depth on the Shellside                   | hs2 | 6.00 mm.    |
| Calculated Tubesheet Diameter per Div 2, 4.18.4 |     | 2030.00 mm. |

Note: Tubesheet diameter is now: 2030.000 mm. Div 2, 4.18.4.

|                                      |    |                             |
|--------------------------------------|----|-----------------------------|
| Area of the Untubed Lanes            | AL | 1308.3 cm <sup>2</sup>      |
| Junction Stress Reduction option     |    | Perform Plastic Calculation |
| Perform Differential Pressure Design |    | No                          |
| Run Multiple Load Cases              |    | Yes                         |

**Additional Data for Gasketed Tubesheets:**

|                                |     |                          |
|--------------------------------|-----|--------------------------|
| Tubesheet Gasket on which Side |     | Shell                    |
| Flange Outside Diameter        | A   | 2030.000 mm.             |
| Flange Inside Diameter         | B   | 1750.000 mm.             |
| Flange Face Outside Diameter   | Fod | 1791.000 mm.             |
| Flange Face Inside Diameter    | Fid | 1750.000 mm.             |
| Gasket Outside Diameter        | Go  | 1788.000 mm.             |
| Gasket Inside Diameter         | Gi  | 1779.000 mm.             |
| Small end Hub thk.             | g0  | 46.0000 mm.              |
| Large end Hub thk.             | g1  | 50.0000 mm.              |
| Gasket Factor,                 | m   | 3.75                     |
| Gasket Design Seating Stress   | y   | 62.06 N./mm <sup>2</sup> |
| Flange Facing Sketch           |     | Code Sketch 1a           |
| Gasket Thickness               | tg  | 4.5000 mm.               |
| Full face Gasket Flange Option |     | Program Selects          |

**Bolting Information:**

|                                       |    |                           |
|---------------------------------------|----|---------------------------|
| Diameter of Bolt Circle               | C  | 1945.000 mm.              |
| Nominal Bolt Diameter                 | dB | 36.0000 mm.               |
| Thread Series                         |    | TEMA Metric Thread        |
| Number of Bolts                       | n  | 56                        |
| Bolt Material                         |    | SA-320 L7                 |
| Bolt Allowable Stress At Temperature  | Sb | 172.00 N./mm <sup>2</sup> |
| Bolt Allowable Stress At Ambient      | Sa | 172.00 N./mm <sup>2</sup> |
| Weld between Flange and Shell/Channel |    | 0.0000 mm.                |

Note: The full bolt load Sa\*Ab, was used for all load cases.

|                                            |    |              |
|--------------------------------------------|----|--------------|
| Tubesheet Integral with                    |    | Channel      |
| Tubesheet Extended as Flange               |    | Yes          |
| Thickness of Extended Portion of Tubesheet | Tf | 406.0000 mm. |
| Is Bolt Load Transferred to the Tubesheet  |    | Yes          |

| TCA    | ODCA  | SLen    | D/T  | L/D     | Factor A  | B      |
|--------|-------|---------|------|---------|-----------|--------|
| 2.7700 | 25.40 | 5805.80 | 9.17 | 50.0000 | 0.0130823 | 122.73 |

EMAWP = (2.167/(D/T)-0.0833)\*B = 187.7951 bars

Results for Req'd Thickness for Ext. Pressure (Tca):

| TCA                                                                   | ODCA  | SLEN    | D/T   | L/D     | Factor A  | B     |
|-----------------------------------------------------------------------|-------|---------|-------|---------|-----------|-------|
| 0.6538                                                                | 25.40 | 5805.80 | 38.85 | 50.0000 | 0.0007287 | 72.86 |
| EMAWP = (4*B)/(3*(D/T)) = ( 4 *72.855 )/( 3 *38.8524 ) = 25.0009 bars |       |         |       |         |           |       |

**Summary of Tube Required Thickness Results:**

|                                                   |             |
|---------------------------------------------------|-------------|
| Total Required Thickness including Corrosion all. | 1.9308 mm.  |
| Allowable Internal Pressure at Corroded thickness | 281.95 bars |
| Required Internal Design Pressure                 | 191.00 bars |
| Allowable External Pressure at Corroded thickness | 187.80 bars |
| Required External Design Pressure                 | 25.00 bars  |
| Required Thickness due to Shell Side pressure     | 0.6538 mm.  |

**Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)**

**Intermediate Calculations For Tubesheets Extended As Flanges:**

**ASME Code, Section VIII Division 1, 2025**

|                           |                   |              |
|---------------------------|-------------------|--------------|
| Gasket Contact Width,     | N = (Goc-Gic) / 2 | 4.500 mm.    |
| Basic Gasket Width,       | b0 = N / 2        | 2.250 mm.    |
| Effective Gasket Width,   | b = b0            | 2.250 mm.    |
| Gasket Reaction Diameter, | G = (Go+Gi) / 2   | 1783.500 mm. |

**Results for ASME U-tube Tubesheet Calculations for Configuration e:**

**Per Edition 2025, Original Thickness :**

**ASME Section VIII-2 Part 4.18.7.4 Step 1:**

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 845.8 + 25.4$$

$$= 1717.000 \text{ mm.}$$

Basic ligament efficiency for shear [ $\mu$ ]:

$$= (p - dt) / p$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.200$$

**ASME Section VIII-2 Part 4.18.7.4 Step 2:**

Ratio [ps]:

$$= G_s / D_o \text{ (Configurations d, e, f)}$$

$$= 1783.5 / 1717.0$$

$$= 1.0387$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1750.0 / 1717.0$$

$$= 1.0192$$

**ASME Section VIII-2 Part 4.18.7.4 Step 3:**

Ratio [p]:

$$= l_{tx} / h$$

$$= 0 / 440.0$$

$$= 0.000 \text{ ( must be } 0 \leq p \leq 1 \text{ )}$$

Effective tube hole Diameter [d\*]:

$$= \max( dt - 2t_t( E t / T ) \cdot ( S t / S ) \cdot ( p ); dt - 2t_t )$$

$$= \max( 25.4 - 2 \cdot 2.77 ( 0.19904E+09 / 0.19904E+09 ) \cdot ( 117 / 138 ) \cdot ( 0 ); 25.4 - 2 \cdot 2.77 )$$

$$= 25.4000 \text{ mm.}$$

Effective tube pitch [p\*]:

$$= p / \sqrt{ 1 - 4 \cdot \min( A_l; 4 \cdot D_o \cdot p ) / ( \pi \cdot D_o^2 ) }$$



$$= 31.75 / ((1 - 4 \cdot \min(1308.35; 4 \cdot 1717.0 \cdot 31.75) / (\pi \cdot 1717.0^2)))$$

$$= 32.6869 \text{ mm.}$$

Effective ligament efficiency for bending [ $\mu^*$ ]:

$$= (p^* - d^*) / p^*$$

$$= (32.6869 - 25.4) / 32.6869$$

$$= 0.2229$$

$h/p = 13.858267$ ;  $\mu^* = 0.222931$ ;  $E^* = 36405248 \text{ KPa}$ .

$$E^*/E = 0.182902$$
;  $\nu^* = 0.427388$

Note: As  $h/p$  (13.858) is  $> 2$ , data values for  $h/p = 2$  were used.

#### ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

$\beta_s$ ,  $k_s$ ,  $\lambda_s$ ,  $\delta_s$  and  $\omega_s$  are all = 0

Channel coefficient [ $\beta_c$ ]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{((D_c + t_c)t_c)}$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1750.0 + 135.0)135.0)$$

$$= 0.0036 \text{ 1/mm.}$$

Channel coefficient [ $k_c$ ]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.004 \cdot 1976633 \cdot 135.0^3 / (6(1 - 0.3^2))$$

$$= 3209725184.0000 \text{ bars} \cdot \text{mm.}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [ $\lambda_c$ ]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1750.0 + 135.0 / \cos(0)) \cdot 0.32097E+1 / 440.0^3 \cdot (1.0 + 440.0 \cdot 0.004 \cdot (1.0 + 1.0) / 2 + (440.0 \cdot 0.004)^2 / 2 \cdot 1.0)$$

$$= 1637557.4 \text{ bars}$$

Channel coefficient [ $\delta_c$ ]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1750.0^2 / (4 \cdot 0.19766E+09 \cdot 135.0) \cdot (1 - 1750.0 \cdot 0.3 / (2(1750.0 + 135.0)))$$

$$= 0.24695E-01 \text{ mm./N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ $\omega_c$ ]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.019 \cdot 0.32097E+1 [0.024695(1.0 + 0.004 \cdot 1.0 \cdot 440.0) - 0(1.0 / 440.0 + 1.0 \cdot 0.004 / 2)]$$

$$= 75275.6875 \text{ mm.}^2$$

#### ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2030.0 / 1717.0$$

$$= 1.1823$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 36405248 \cdot (1637557 + 0.19904E+09 \cdot \ln(1.18))$$

$$= 3.0999$$

#### ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures ( $P_s$ ,  $P_t$ ) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_t - 1) \cdot (p_t^2 + 1) \cdot P_t]$$

$$= 1717.0^2 / 16 \cdot [(1.039 - 1) \cdot (1.039^2 + 1) \cdot 25.0 -$$



$$\begin{aligned} & (1.019 - 1) \cdot (1.019^2 + 1) \cdot 191.0 ] \\ & = -1008125.7500 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M\*]  
$$\begin{aligned} & = Mts + ( \omega c \cdot Pt ) + W* \cdot ( C - Gs ) / ( 2 \cdot \pi \cdot Do ) \\ & = -1008125 + (14377656) + 724845 \cdot ( 1945.0 - 1783.5 ) / ( 2 \cdot \pi \cdot 1717.0 ) \\ & = 14433602.0000 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

#### ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:  
$$\begin{aligned} & = ( (M*) - Do^2/32 \cdot F( Ps - Pt ) ) / ( 1 + F ) \\ & = ( (14433602) - 1717.0^2/32 \cdot 3.1 ( 25.0 - 191.0 ) ) / ( 1 + 3.1 ) \\ & = 15083549.0000 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:  
$$\begin{aligned} & = Mp + Do^2/64 \cdot ( 3 + \nu* ) \cdot ( Ps - Pt ) \\ & = 15083549 + 1717.0^2/64 \cdot ( 3 + 0.427 ) \cdot ( 25.0 - 191.0 ) \\ & = -11124336.0000 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:  
$$\begin{aligned} & = \max( | Mp | ; | Mo | ) \\ & = \max( | 15083549 | ; | -0.1E+08 | ) \\ & = 15083549.0000 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

#### ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:  
$$\begin{aligned} & = 6 \cdot M / ( \mu* ( h - hg' )^2 ) \\ & = 6 \cdot 15083549 / ( 0.2229 ( 440.0 - 6.0 )^2 ) \\ & = 215.5411 \text{ N./mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [ $\sigma_{All}$ ]:  
$$\begin{aligned} & = 2 \cdot S \\ & = 2 \cdot 138.0 \\ & = 276.00 \text{ N./mm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [ $\sigma$ ]:  
$$\begin{aligned} & = 6 \cdot M / ( \mu* ( h - hg' )^2 ) \\ & = 6 \cdot 14896771 / ( 0.2229 ( 387.1816 - 6.0 )^2 ) \\ & = 275.9525 \text{ N./mm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:  
$$\begin{aligned} & = h + Cats + Catc \\ & = 387.1816 + 0 + 0 \\ & = 387.1816 \text{ mm.} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:  
$$\begin{aligned} & = \max( HreqB ; HreqS ) \\ & = \max( 387.1816 ; 322.7339 ) \\ & = 387.1816 \text{ mm.} \end{aligned}$$

#### ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [ | Ps - Pt | ]:  
$$\begin{aligned} & = | Ps - Pt | \\ & = | 25.0 - 191.0 | \\ & = 166.000 \text{ bars} \end{aligned}$$

Shear Stress check [ $\tau_{limit}$ ]:  
$$\begin{aligned} & = 4 \cdot \mu \cdot h / Do \cdot \min( 0.8 \cdot S ; 0.533 \cdot Sy ) \\ & = 4 \cdot 0.2 \cdot 440.0 / 1717.0 \cdot \min( 0.8 \cdot 138.0 ; 0.533 \cdot 229.6 ) \\ & = 22.633 \text{ N./mm}^2 \end{aligned}$$

| Ps - Pt | is  $\leq$  the Shear Stress limit ( $\tau_{limit}$ ). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:  
$$\begin{aligned} & = 1 / ( 4 \cdot \mu ) \cdot ( Do / \min( 0.8 \cdot S ; 0.533 \cdot Sy ) ) \cdot | Ps - Pt | + Cats + Catc \\ & = 1 / ( 4 \cdot 0.2 ) \cdot ( 1717.0 / 110.4 ) \cdot | 25.0 - 191.0 | + 0 \end{aligned}$$



= 322.7339 mm.

#### ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 135.000 mm.  
 Min. Channel len. adjacent to the tubesheet is: 874.900mm.

Axial Channel Membrane Stress [ $\sigma_c, m$ ]:  
 $= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$   
 $= 1750.0^2 / (4 \cdot 135.0 (1750.0 + 135.0) \cdot \cos(0)) \cdot 191.0$   
 $= 57.468 \text{ N./mm}^2$

Axial Channel Bending Stress [ $\sigma_c, b$ ]:  
 $= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$   
 $6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta +$   
 $h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$   
 $= 6 \cdot 72159144 / 135.0^2 \cdot [(0.024695 \cdot 1.0 \cdot 0.004 - 0 \cdot 1.0) \cdot 191.0 -$   
 $6((1 - 0.427) / 36405248) \cdot 1717.0 / 440.0^3 \cdot (1.0 +$   
 $0.004 \cdot 440.0 / 2 \cdot 1.0) \cdot (15083549 + 1717.0^2 / 32 \cdot (25.0 - 191.0))]$   
 $= 187.183 \text{ N./mm}^2$

Channel Membrane + Bending Stress [ $\sigma_c$ ]:  
 $= |\sigma_c, m| + |\sigma_c, b| \text{ (Should be } \leq 1.5 \cdot S_c)$   
 $= 244.652 \text{ N./mm}^2 \text{ (Should be } \leq 207.00)$

Step 11, The Cylinder-to-Tubesheet Juncture is Overstressed.  
 Performing a simplified Elastic-Plastic calculation  
 (option 3) to reduce the overstress condition.  
 Modify  $E_s$  and/or  $E_c$  and recompute from Step 4 onwards.

Modified Elasticity [ $E_c$ ]:  
 $= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)}$   
 $= 0.19766E+09 \cdot \sqrt{(1.5 \cdot 138.0 / 244.65)}$   
 $= 181816432.0 \text{ KPa.}$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:  
 $= 6 \cdot M / (\mu \cdot (h - h'g)^2)$   
 $= 6 \cdot 15072430 / (0.2229 (440.0 - 6.0)^2)$   
 $= 215.4 \text{ N./mm}^2 \text{ (Should be } \leq 276.0)$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.  
 The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [ $H_{reqB}$ ]:  
 $= h + C_{ats} + C_{atc} = 387.1816 + 0 + 0 = 387.1816 \text{ mm.}$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [ $H_{req}$ ]:  
 $= \max(H_{reqB}; H_{reqS}) = \max(387.1816; 322.7339) = 387.1816 \text{ mm.}$

Tube Metal Area [ $A_t$ ]:  
 $= \pi (d_o - t) t$   
 $= \pi (25.4 - 2.77) 2.77$   
 $= 1.969 \text{ cm}^2$

#### Tube Weld Size Results per UW-20.3:

Tube Strength [ $F_t$ ]:  
 $= \pi \cdot t (d_o - t) S$   
 $= \pi \cdot 2.77 (25.4 - 2.77) 117.998$   
 $= 2369.519 \text{ Kgf}$

Groove Weld Strength [ $F_g$ ]:  
 $= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$   
 $= 0.85 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 118.0$   
 $= 2369.5188 \text{ Kgf}$

Weld Strength Factor [fw]:  
 = S / Sw where Sw = min( S and St )  
 = 117.998/117.998  
 = 1.000

Design Strength Ratio [fd]:  
 = 1 for full strength welds

Fillet Weld Strength [Ff]:  
 =  $0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw$  (but not > Ft)  
 =  $0.55 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 118.0$   
 = 1569.6356 Kgf

Minimum Length of Weld [ar]:  
 =  $2 \cdot \sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd) - 0.75 \cdot do}$   
 =  $2 \cdot \sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1.0 \cdot 1.0) - 0.75 \cdot 25.4}$   
 = 3.372 mm.

Minimum Required Fillet Weld Leg [afr]: 1.6859 mm.  
 Minimum Required Groove Weld Leg [agr]: 1.6859 mm.

Tube-Tubesheet Jt allowable, 2369.52 is >= tube strength 2369.52 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

#### Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

| Stress Description             | Actual   | Allowable                | Pass/Fail |
|--------------------------------|----------|--------------------------|-----------|
| Tubesheet bend. stress         | 215.4 <= | 276.0 N./mm <sup>2</sup> | Ok        |
| Tubesheet shear stress         | 81.0 <=  | 110.4 N./mm <sup>2</sup> | Ok        |
| Stress in Channel at Tubesheet | 244.7 <= | 484.0 N./mm <sup>2</sup> | Ok        |

#### Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

| Thickness (mm.)                  | Required | Actual  | P/F |
|----------------------------------|----------|---------|-----|
| Tubesheet Thickness :            | 387.182  | 440.000 | Ok  |
| Tube-Tubesheet Fillet Weld Leg : | 1.686    | 2.770   | Ok  |
| Tube-Tubesheet Groove Weld Leg : | 1.686    | 2.770   | Ok  |

#### U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

##### Results for 6 Load Cases:

| Case# | --Reqd. Thk. + CA<br>Tbsht Extnsn | ----       | Tubesheet<br>Bend | Stresses<br>Allwd | Shear | Allwd        | Case<br>Type | Pass/<br>Fail |
|-------|-----------------------------------|------------|-------------------|-------------------|-------|--------------|--------------|---------------|
| D1uc  | 406.372                           | 66.567     | 234               | 276               | 93    | 110          | Ps+Pt D1     | Ok            |
| D2uc  | 151.760                           | ...        | 37                | 276               | 12    | 110          | Ps+Pt D2     | Ok            |
| D3uc  | 387.182                           | ...        | 215               | 276               | 81    | 110          | Ps+Pt D3     | Ok            |
| D1c   | 407.193                           | 66.567     | 234               | 276               | 94    | 110          | Ps+Pt-c D1   | Ok            |
| D2c   | 153.254                           | ...        | 37                | 276               | 12    | 110          | Ps+Pt-c D2   | Ok            |
| D3c   | 388.051                           | ...        | 216               | 276               | 82    | 110          | Ps+Pt-c D3   | Ok            |
| Max:  | 407.1935                          | 66.567 mm. | 0.849             |                   | 0.850 | (Str. Ratio) |              |               |

#### Load Case Definitions:

[Ps & Pt]:  
 Shell-side and Tube-side Design or Operating Pressures  
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the  
 Load Case Tables

[c]:  
 With or Without Corrosion Allowance

[D1, D2, D3]:  
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:  
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

#### Shell and Channel Stress Summary:

| Case# | Shell | Channel | Shell Stress | Reqd. Thk. | Channel Stress | Stress Allwd | Pass Fail |
|-------|-------|---------|--------------|------------|----------------|--------------|-----------|
| D1uc  | ...   | ...     | ...          | ...        | 278            | 484          | Ok        |
| D2uc  | ...   | ...     | ...          | ...        | 33             | 207          | Ok        |
| D3uc  | ...   | ...     | ...          | ...        | 245            | 484          | Ok        |
| D1c   | ...   | ...     | ...          | ...        | 284            | 484          | Ok        |
| D2c   | ...   | ...     | ...          | ...        | 33             | 207          | Ok        |
| D3c   | ...   | ...     | ...          | ...        | 250            | 484          | Ok        |
| Max   | ...   | ...     | ...          | ...        | 0.586          |              |           |

Note:  
 Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

#### Summary of Thickness Comparisons for 6 Load Cases:

| Thickness (mm.)                         | Required | Actual  | P/F |
|-----------------------------------------|----------|---------|-----|
| Tubesheet thickness :                   | 407.193  | 440.000 | Ok  |
| Tubesheet thickness flanged extension : | 66.567   | 406.000 | Ok  |
| Tube thickness :                        | 1.931    | 2.770   | Ok  |
| Tube-tubesheet fillet weld leg :        | 1.686    | 2.770   | Ok  |
| Tube-tubesheet groove weld leg :        | 1.686    | 2.770   | Ok  |

Given Shell Thickness: 46.0000 mm.  
 Given Channel Thickness: 135.0000 mm.

Min Channel length of thk, (135.000) adj. to tubesheet: 874.900 mm.

Note: This is a full strength Tube to Tubesheet Joint.

#### Tubesheet MAWP used to Compute Hydrotest Pressure:

| Stress / Force Condition   | Tubeside MAWP | Ø shellside Stress Rat. | Shellside MAWP | Ø tubeside Stress Rat. |
|----------------------------|---------------|-------------------------|----------------|------------------------|
| Tubesheet Bending Stress   | 223.262       | 1.000                   | 185.487        | 0.999                  |
| Tubesheet Shear Stress     | 224.772       | 1.000                   | 224.772        | 1.000                  |
| Tubesheet-Channel Junction | 224.772       | 1.007                   | 224.772        | 1.236                  |
| Tube Pressure Stress       | 281.948       | 1.000                   | 187.794        | 1.000                  |
| Tubesheet Extension Stress | No Calc       | No Calc                 | 27.413         | ...                    |
| Minimum MAWP               | 223.262       |                         | 27.413         |                        |

#### Tubesheet MAPnc used to Compute Hydrotest Pressure:

| Stress / Force Condition   | Tubeside MAPnc | Ø shellside Stress Rat. | Shellside MAPnc | Ø tubeside Stress Rat. |
|----------------------------|----------------|-------------------------|-----------------|------------------------|
| Tubesheet Bending Stress   | 223.917        | 1.000                   | 187.541         | 1.000                  |
| Tubesheet Shear Stress     | 226.315        | 1.000                   | 226.316         | 1.000                  |
| Tubesheet-Channel Junction | 226.315        | 1.011                   | 226.316         | 1.232                  |
| Tube Pressure Stress       | 281.948        | 1.000                   | 187.794         | 1.000                  |
| Tubesheet Extension Stress | No Calc        | No Calc                 | 27.413          | ...                    |
| Minimum MAPnc              | 223.917        |                         | 27.413          |                        |

(\*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.



#### Tubesheet MDMT Calculations:

##### Channel Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):  
= max( tubesheet thk/4, min( tubesheet thk, channel thickness ) )  
= max( 440.0/4, min( 440.0, 135.0 ) )  
= 135.000 mm.

##### Warning:

*The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.*

##### Shell Side MDMT calculation:

Governing thickness for the shell side:  
= tubesheet thickness/4  
= 440.0/4  
= 110.000 mm.

##### Note:

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification -46 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:  
= max( pt/Tubeside MAPnc, ps/Shellside MAPnc ), must be  $\leq 1$   
= max( 191.0/223.92, 25.0/27.41 )  
= 0.912

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